Emma's Atlas of Numbers and Shapes

- A not-so-brief guide to all the math I know -

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2026-10-09

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Part I

Foundations

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Chapter 1

An Introduction to Mathematical Reasoning

Chapter 2

Sets

Almost all of modern mathematics can be put in the context of the study of ***sets*** and their ***elements***. Generally, given a collection of objects $a, b, c, d, \ldots$, they can be taken together as a set, generally written in curly brackets and represented by a capital letter:

$$M := \{a, b, c, d, e, \ldots\}$$

However, an approach where any collection of mathematical objects forms a set quickly runs into trouble - famously, ***Russel's paradox*** asks:

> *"Let R be the set of all sets that don't contain themselves. Does R contain itself?"*

If R doesn't contain itself, then by definition it must contain itself, but if it contains itself, then it cannot contain itself! If $R \in R$ is true, then it must be false! We arrive at a classic paradox. The conclusion, then, is that the set of all sets that don't contain themselves cannot exist - we cannot define sets to be any collection of things defined by any property, and must instead establish ground rules about which sets exist and exactly how we can use sets we already know to construct new ones.

These ground rules are given in the form of ***axioms***, which are statements that mathematicians take for granted, which can then be used as a starting point to then derive "the rest of mathematics" from. The following chapters contain a brief and hopefully easily understandable introduction to the ZFC axiom system, which was developed in the early 20th century by Ernst Zermelo and Abraham Fraenkel and is now the most widely accepted axiomatization of mathematics. Informally stated, the ZFC axioms are:

1. The empty set is a set, and the natural numbers are a set.
2. Sets are uniquely determined by their elements.
3. Any definable subset of a set is a set.
4. Given two sets, there exists a set whose elements are those two sets.
5. Given two sets, there exists a set containing the elements of both sets.
6. Given a set, there exists a set containing all subsets of that set as elements.

7. The image of a set under a definable function is a set.
8. There are no infinitely nested sets.
9. Given any set of nonempty sets, there exists a function mapping each sets to one of its elements.

2.1 Infinity: $\emptyset$ and $\mathbb{N}$

2.2 Extensionality

To say "x is an element of M", or equivalently "x contains M", we simply write:

$$x \in M$$

One of the most basic axioms of set theory is the ***Axiom of Extensionality***, which states:

Axiom 2.2.1: Axiom of Extensionality

Two sets A and B are equal if and only if they contain the same elements.

The Axiom of Extensionality establishes that sets are uniquely defined by the objects they contain and have no additional structure. In particular:

1. The order of elements in a set doesn't matter. $\{1, 2\}$ and $\{2, 1\}$ are the same set.
2. The number of times an element appears within the same set doesn't matter. $\{1\}$, $\{1, 1\}$, and $\{1, 1, 1\}$ are the same set.

Definition 2.2.2: Subsets

We call a set A a ***subset*** of a set B if and only if every element of A is contained in B. We write "A is a subset of B" as $A \subseteq B$.

The demand that "every element of A is contained in B" can be stated more formally as "$x \in A$ implies $x \in B$". Importantly, this means that every set is a subset of itself - we always have $A \subseteq A$.

This definition gives us another way of phrasing the Axiom of Extensionality - two sets are equal if and only if each is a subset of the other:

$$A = B \Leftrightarrow A \subseteq B \text{ and } B \subseteq A$$

This idea is incredibly important for writing out proofs in practice - if a problem asks you to establish that two sets A and B are the same, then the easiest approach is almost always to show that any element of A must be contained in B, and to then show that any arbitrary element of B must be contained in A.

The subset relation is frequently written as $A \subset B$. However, other authors reserve $A \subset B$ to mean "A is a subset of B that isn't equal to B" - we call such a

set a ***proper subset of B***. I will therefore attempt to only use the symbols $\subseteq$ and $\subsetneq$, which clear up this ambiguity.

2.3 Specification

2.4 Pairing, Union, and Power Sets

2.5 Relations

2.6 Functions

2.6.1 Injections, Surjections and Bijections

[TODO: Injectiveness and surjectiveness provide the prerequisites for inverting functions]

Definition 2.6.1

A function $f : X \to Y$ is called ***injective*** if it maps different elements of X to different elements of Y:

$$\forall x_1, x_2 \in X : \\ x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$$

Injective functions are also known as ***injections***. Some texts also refer to injections as ***one-to-one functions***. For clarity, injective functions are often written using a hooked arrow:

$$f : X \hookrightarrow Y$$

Note that we can equivalently define injectivity via the contrapositive of our condition:

$$\forall x_1, x_2 \in X : \\ f(x_1) = f(x_2) \implies x_1 = x_2$$

This version tends to be nicer for proving that a given function is injective, but I find that it isn't quite as intuitive.

Intuitively, an injection $f : X \hookrightarrow Y$ is a way of taking the entirety of X and placing it "into" Y, without having to "squeeze" X into a smaller size in any way. Later on, when we formally define "sizes of sets", also known as ***cardinalities***, we will formalize the idea that an injection $f : X \hookrightarrow Y$ can only exist if the set X is "smaller" than the set Y.

Definition 2.6.2. The ***canonical embedding of a set X into a superset Y*** is the identity function of X with its domain reinterpreted to be Y:

$$\begin{aligned} \iota_X : X &\hookrightarrow Y \\ x &\mapsto x \end{aligned}$$

The canonical embedding is generally written using the greek lower case iota ι, often with a subscript denoting the set we want to embed.

Theorem 2.6.3. Given a subset $V \subseteq Y$, we have

$$\iota_X^{-1}[V] = X \cap V$$

Proof. By the definition of preimages of sets, we have

$$\iota_X^{-1}[V] = \{x \in X \mid \iota_X(x) \in V\}$$

Since $\iota_X(x) = x$, this condition simplifies to

$$\iota_X^{-1}[V] = \{x \in X \mid x \in V\}$$

Thus, our preimage is the set of X also contained in V, which is exactly the definition of $X \cap V$. □

2.7 Cartesian Products

We can generalize the cartesian product to infinite products as follows:

Definition 2.7.1: Cartesian Product of sets

Let I be any set. Let $(X_i)_{i\in I}$ be a Family of nonempty sets indexed by I, i.e. there exists a function assigning to each element $i \in I$ an element of $(X_i)_{i\in I}$. Then the cartesian product of $(X_i)_{i\in I}$ is the unique set:

$$\prod_{i\in I} X_i := \left\{ x : I \to \bigcup_{i\in I} X_i \;\middle|\; x_i := x(i) \in X_i \text{ for all } i \in I \right\}$$

Written more simply: Each element of $\prod_{i\in I} X_i$ is a function assigning to each $i \in I$ an element $x(i) \in X_i$, where we generally write the argument in the index ($x_i := x(i)$), to stay consistent with the established notation in the countable case.

This definition is simply a reinterpretation of the definition of finite cartesian products: It amounts to viewing, for example, the tuple $T := (x, y, z)$ as a map:

$$T : \{1, 2, 3\} \to \{x, y, z\}$$

such that $T(1) = x$, $T(2) = y$, and $T(3) = z$.

Definition 2.7.2: Projections

Let $J \subset I$. Then we define the projection of I to J as the map

$$\begin{aligned} \pi_J := \pi_J^I : \prod_{i \in I} X_i &\to \prod_{j \in J} X_j \\ x &\mapsto \left(x|_J : J \to \bigcup_{j \in J} X_j \right) \end{aligned}$$

which restricts the domain of a given element x of the product to the subset J.

In particular, if J is a set consisting of a single element j, we have:

$$\begin{aligned} \pi_j := \pi_{\{j\}}^I : \prod_{i \in I} X_i &\to X_j \\ x &\mapsto x_j \end{aligned}$$

2.8 Replacement

2.9 Regularity and Well-Foundedness

2.10 Choice

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Part II

Algebraic Structures

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Chapter 3

$(\mathbb{N}, +)$ is a Monoid

Chapter 4

$(\mathbb{Z}, +, -)$ is a Group

4.1 Group Homomorphisms

Theorem 4.1.1. Let $\varphi : (G, \circ_G) \to (H, \circ_H)$ be a map that respects group operations, i.e:

$$\varphi(g_1 \circ_G h_2) = \varphi(g_1) \circ_H \varphi(g_2)$$

Then h also respects the neutral element and inverses, i.e:

$$\begin{aligned} \varphi(e_G) &= e_H \\ \varphi(g^{-1}) &= \varphi(g)^{-1} \end{aligned}$$

Proof. 1. For any $h \in H$, we have:

$$\begin{aligned} e_H &= e_H \circ_H e_H \\ &= \left(\varphi(e_G)^{-1} \circ_H \varphi(e_G)\right) \circ_H e_H \\ &= \varphi(e_G)^{-1} \circ_H \varphi(e_G) \\ &= \varphi(e_G)^{-1} \circ_H \varphi(e_G \circ_G e_G) \\ &= \varphi(e_G)^{-1} \circ_H \varphi(e_G) \circ_H \varphi(e_G) \\ &= e_H \circ_H \varphi(e_G) \\ &= \varphi(e_G) \end{aligned}$$

So φ respects the neutral element.

2. Since φ respects the neutral element, we get:

$$\begin{aligned} \varphi(g) \circ_H \varphi(g^{-1}) &= \varphi(g \circ_G g^{-1}) \\ &= \varphi(e_G) \\ &= e_H, \end{aligned}$$

i.e. $\varphi(g^{-1}) = \varphi(g)^{-1}$.

□

4.2 Permutations and the Symmetric Groups

4.3 Product Groups

Theorem 4.3.1

Let $G = G_1 \times G_2$ be a product group. Let $H \subseteq G$ be a subgroup. Then H isn't necessarily isomorphic to a product $H_1 \times H_2$ of subgroups of G_1 and G_2.

Proof. Let $G = S_3 \times S_3$. Let H be subgroup of elements (σ_1, σ_2) with $\operatorname{sgn}(\sigma_2\sigma_1) = 1$. We have $|G| = 36$, giving us $|H| = 18$.

The subgroups of S_3 are S_3 itself, C_1, $A_3 \cong C_3$ and C_2. If H was a product of subgroups, we would thus need to have $H \cong A_3 \times S_3$, since every other combination has the wrong order.

Now, notice that $A_3 \times S_3$ contains elements of order 6, such as $((1,2),(1,2,3))$. Since S_3 isn't cyclic, it doesn't contain elements of order 6, so an element of order 6 in H must be a pair consisting of an element σ_2 of order two (i.e. a transposition, which has odd order) and an element σ_3 of order three (i.e. a three-cycle, which has even order). But then $\operatorname{sgn}(\sigma_2\sigma_1) = -1$, so no such element can be contained in H. □

4.4 Group Actions

Definition 4.4.1

Let $(G, \circ, e)$ be a group. Let X be a set. Then a map $. : G \times X \to X$ is called a ***group action*** of G on X iff:

1. It respects the neutral element:

$$\forall x \in X :$$
$$e.x = x$$

2. It associates with the group operation:

$$\forall g, h \in G, x \in X :$$
$$(g \circ h).x = g.(h.x)$$

Exercise 4.4.2. *Show that a map $(g, x) \to g.x$ is a group operation if and only if the map $g \to (x \to g.x)$ is a group homomorphism from G into the symmetric group over X.*

Definition 4.4.3. Let $. : G \times X \to X$ be an action of a group G on a set X. Let $x \in X$. Then we call the set

$$Gx := \{g.x \mid g \in G\}$$

of all points that x can be taken to by an element of G the ***orbit*** of x under G.

Exercise 4.4.4. *Let $X = \mathbb{R}^2$ be the plane. Let G be the group of rotations around the origin. Sketch the orbits of different points $x \in X$.*

Definition 4.4.5. Let $. : G \times X \to X$ be an action of a group G on a set X. Then we call the set

$$X/G := \{Gx \mid x \in X\}$$

of all orbits under G the ***orbit space*** of G in X.

Exercise 4.4.6. *Show that the relation*

$$x \sim y \Leftrightarrow \exists g \in G : g.x = y$$

is an equivalence relation. Show that for any $x \in X$, we the equivalence class of x is exactly the orbit of x.

Definition 4.4.7. Let $. : G \times X \to X$ be an action of a group G on a set X. Then we call the action $.$ ***transitive*** if any point $x \in X$ can be taken to any point $y \in X$ by an element of G:

$$\forall x, y \in X :$$
$$\exists g \in G :$$
$$g.x = y$$

Exercise 4.4.8. *Show that a group action is transitive if and only if $Gx = X$ for all $x \in X$.*

Definition 4.4.9. Let $. : G \times X \to X$ be an action of a group G on a set X. Let $x \in X$. Then we call the set

$$G_x := \{g \in G \mid g.x = x\}$$

of all elements of G leaving x unchanged the ***isotropy group*** of x.

Exercise 4.4.10. *Show that the isotropy group G_x is always a subgroup of G.*

Exercise 4.4.11. *Show that if the action $(g, x) \mapsto g.x$ is transitive, the map*

$$\begin{aligned} \varphi_x : G/G_x &\to X \\ [g] &\mapsto g.x, \end{aligned}$$

where G/G_x is the quotient group of G by G_x, is well-defined and bijective.

We can also define the isotropy groups of points:

Definition 4.4.12. Let $. : G \times X \to X$ be an action of a group G on a set X. Let $A \subseteq X$. Then we define

$$g.A := \{g.a \mid a \in A\}$$

and call the set

$$G_A := \{g \in G \mid g.A = A\}$$

of all elements of G mapping elements of A to elements of A the ***isotropy group*** of A.

Chapter 5

$(\mathbb{Z}, +, -, \cdot)$ is a Ring

Chapter 6

$(\mathbb{Q}, +, -, \cdot, ^{-1})$ is a Field

Part III

Foundations of Analysis

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Chapter 7

The Real Numbers $\mathbb{R}$

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7.2 Rational Sequences

7.3 Dedekind Cuts

7.4 Uniqueness of $\mathbb{R}$

7.5 Cardinality of $\mathbb{R}$

Chapter 8

The Concept of Distance

8.1 Metric Spaces

A metric space is a set, alongside with a ***distance function***, more commonly known as a ***metric***, d which takes two points and outputs a distance between the points. Naturally, we want this "distance" to actually give us some usable structure on the set.

Now, what pieces of geometric intuition should we bake into our notion of distance? I would argue that the most fundamenal use of "distance" is to be able to ***compare distances between any two pairs of points***. Therefore, our metric should have as its output a ***totally ordered set***.

We also want to be able to add distances together as a way of comparing them - logically, the distance between two points x and y plus the distance between y and z gives us an upper bound to the distance between x and z, since we can always take this path to get from x to z, but there might be a shorter path we haven't found yet. As a formula, this is the ***triangle inequality***

$$d(x,z) \leq d(x,y) + d(y,z)$$

Finally, if we have an ordered group structure, we would really like for our order to be archimedean - the distance between two distinct points should be greater than zero, and it shouldn't be infinitesimal.

This leaves $\mathbb{R}_{\geq 0}$ as the obvious candidate for the domain of our distance function, since any archimedean ordered group is a subgroup of $\mathbb{R}_{\geq 0}$ anways. This shouldn't be too much of a surprise, even without this logical background - I assume that when most people read "distance between two points", they intuitively already assume that these distances will be given by positive real numbers.

Definition 8.1.1: Metric Spaces

A ***metric space*** is a set M together with a function $d : M \times M \to \mathbb{R}_{\geq 0}$, such that:

1. d is ***positive definite***, i.e. $d(x, y) = 0$ if and only if $x = y$.
2. d is ***symmetric***, i.e. $d(x, y) = d(y, x)$
3. d fulfills the ***triangle inequality***

$$d(x, z) \leq d(x, y) + d(y, z)$$

Theorem 8.1.2. Let $M^n := \prod_{i=1}^{n} M$ be a finite cartesian product of metric spaces. Then the ***Euclidean distance function***

$$d_\infty(x, y) := \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

gives a metric on M^n.

Theorem 8.1.3. Let $M^n := \prod_{i=1}^{n} M$ be a finite cartesian product of metric spaces. Then the ***Chebyshev distance function***

$$d_\infty(x, y) := \max_{i \in [1..n]} d(x_i - y_i)$$

gives a metric on M^n.

Definition 8.1.4: Balls

Let (M, d) be a metric space. Then the ***open ball of radius r around a point x*** is the set

$$B_r(x) := \{y \in M : d(x, y) < r\},$$

and the ***closed ball of radius r around x*** is the set

$$\overline{B}_r(x) := \{y \in M : d(x, y) < r\},$$

Definition 8.1.5: Open and Closed Sets

Let (M, d) be a metric space. Then we call a subset $U \subseteq M$ ***open*** (in M) if it contains an open ball (of any radius) around all of its points, i.e.

$$\forall x \in U :$$
$$\exists r \in \mathbb{R}_{\geq 0} :$$
$$B_r(x) \subseteq U.$$

We call U closed if its complement $M \setminus U$ is open.

Intuitively, a set is open if it contains "a little bit of wiggle room" around any point contained in it. This means that sets with a "clear border", such as our closed balls, aren't open. For example, the interval $[0, 1] \subset \mathbb{R}$ doesn't leave any

wiggle room around 0 and 1. Open sets are often denoted U, taken from the German word *Umgebung*, meaning "environment", "surroundings" or "neighbourhood".

Importantly, while it may seem unintuitive, there are in fact sets that are both closed and open at the same time. We call such sets ***clopen***.

Exercise 8.1.6. *There are exactly two clopen subsets of $\mathbb{R}$. What are they?*

Definition 8.1.7. Let (M, d) be a metric space. We call a subset $N \subseteq M$ ***dense in M*** if every nonempty open subset of M contains an element of N.

Lemma 8.1.8. Every nonempty open subset of $\mathbb{R}$ is uncountable.

Proof. Since every nonempty open subset of $\mathbb{R}$ contains an open interval (a, b), it suffices to show that open intervals are uncountable. We construct a series of simple bijections which show $(a, b) \cong \mathbb{R}$ when chained together.

1. The map
$$\begin{aligned} \varphi_1 : (a, b) &\to \left(-\frac{b-a}{2}, \frac{b-a}{2}\right) \\ x &\mapsto x - \frac{b+a}{2} \end{aligned}$$
is a bijection.

2. The map
$$\begin{aligned} \varphi_2 : \left(-\frac{b-a}{2}, \frac{b-a}{2}\right) &\to (-1, 1) \\ x &\mapsto \frac{2x}{b-a} \end{aligned}$$
is also a bijection.

3. Finally, the map
$$\begin{aligned} \varphi_3 : (-1, 1) &\to \mathbb{R} \\ x &\mapsto \frac{x}{1-|x|} \end{aligned}$$
is a bijection (the inverse is the map we get by replacing the $-$ with a $+$).

Therefore, $\varphi_3 \circ \varphi_2 \circ \varphi_1$ is a bijection $(a, b) \cong \mathbb{R}$, so any open interval is uncountable, so any open subset of $\mathbb{R}$ is uncountable. □

Theorem 8.1.9

The set of rational numbers $\mathbb{Q}$ is dense in $\mathbb{R}$. The set of irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ is also dense in $\mathbb{R}$.

Proof. 1. Since any nonempty open subset $U \subseteq \mathbb{R}$ contains an open interval (a, b), it suffices to show that any open interval contains a rational number. We assume without loss of generality that $0 \leq a < b$, since $a < 0 < b$ already implies $0 \in (a, b)$, while $a < b \leq 0$ implies that (a, b) contains a rational number if and only if $(-b, -a)$ contains a rational number.

Now, since the ordering on the reals is archimedean, we can find a natural number $n \in \mathbb{N}$ such that $\frac{1}{n} < b - a$. Now, the set

$$L = \left\{ k \in \mathbb{N} \;\middle|\; \frac{k}{n} > a \right\}$$

is nonempty - if $a = 0$, then $1 \in M$, and if $a \neq 0$, then the archimedean property tells us that there exists a $k \in \mathbb{N}$ such that

$$\frac{1}{k} < \frac{1}{na},$$

which implies $\frac{k}{n} > a$. Since M is a nonempty subset of $\mathbb{N}$, there must be a smallest element m. Since m is minimal, we have $m - 1 \neq M$, which implies

$$\frac{m-1}{n} \leq a,$$

which in turn implies

$$\frac{m}{n} = \frac{m-1}{n} + \frac{1}{n} < a + (b - a) = b$$

Therefore, $\frac{m}{n}$ is a rational number contained in (a, b), so $\mathbb{Q}$ is dense in $\mathbb{R}$.

2. Since any open subset $U \subseteq \mathbb{R}$ is uncountable, while the rationals are countable, U must contain an uncountable number of irrationals. Therefore, $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$.

□

Corollary 8.1.10

The set of rational numbers $\mathbb{Q}$ is neither closed nor open in $\mathbb{R}$.

Proof. Since both sets are dense in $\mathbb{R}$, every open ball containing a rational number must also contain an irrational number, and vice versa. □

8.2 Topologies

Topological spaces generalize metric spaces. By inheriting the concept of open and closed sets from metric spaces, they are able to still capture notions of closeness without putting a numerical value on exactly how close any given pair of points are. The notion of a topology is based on the following insight:

Exercise 8.2.1. *Let (M, d) be a metric space. Then:*

1. *The empty set is open in M.*
2. *Any union of sets that are open in M is open in M.*
3. *Any finite intersection of sets that are open in M is open in M.*
4. *M is open in M.*

We take this idea and make it a definition:

Definition 8.2.2

Let X be a set. Then a ***topology on X*** is a set $\mathcal{O}$ of subsets of X, called ***open sets***, such that:

1. The empty set is open.
2. Any union of open sets is open.
3. Any finite intersection of open sets is open.
4. X is open.

Most of the topological spaces we care about in practice are already metric spaces, but many proofs flow more elegantly when phrased in the generalized language of topological spaces, and there *are* topological spaces that aren't metric spaces.

Theorem 8.2.3: Extremal Topologies

Let X be an arbitrary set. Then the following two sets of subsets are always topologies on X:

1. The ***indiscrete topology***, where the only open sets are X and the empty set.
2. The ***discrete topology***, where every subset is open.

> *Proof Sketch.* Both topologies fulfill all requirements trivially. :) "□"

Definition 8.2.4. We say that a metric d on a space M ***generates*** a given topology $\mathcal{O}$ on M if they give the same open sets, i.e. the open sets in $\mathcal{O}$ are exactly the open sets in (M, d).

Exercise 8.2.5. 1. *Show that the discrete metric generates the discrete topology.*

2. *Show that there is no metric generating the indiscrete topology.*

Two different metrics can generate the same topology:

Exercise 8.2.6. *Show that the euclidean metric also generates the discrete topology on* $\mathbb{N}$.

Exercise 8.2.7. *Let* d *be a metric on a set* M. *Let* $c \in \mathbb{R}_{>0}$.

1. *Show that*

$$d_c(x, y) = c \cdot d(x, y)$$

   is a metric on M.

2. *Show that* d *and* d_c *generate the same topologies on* M.

This fact can be seen as a formalization of the intuition that changing the specific unit we choose to measure space with doesn't change its underlying structure.

Definition 8.2.8: Interior, Exterior and Boundary

Let X be a topological space. Let $A \subseteq X$. Then:

1. The ***interior*** $\operatorname{int} A$ of A is the set of all points that have an open neighbourhood entirely contained in A.
2. The ***exterior*** $\operatorname{ext} A$ of A is the interior of the complement $X \setminus A$, i.e. the set of all points that have an open neighbourhood entirely contained in $X \setminus A$.
3. The ***boundary*** ∂A of A is the set of all points $x \in X$ lying neither in $\operatorname{int} A$, nor in $\operatorname{ext} A$.

The interior of A is occasionally written as $\mathring{A}$. Intuitively, the interior is the largest open subset of A, and the exterior is the largest open set not containing any point of A.

Exercise 8.2.9. *Show that:*

1. *The interior of any set is open.*
2. *The exterior of any set is open.*
3. *The interior of* A *is the union of all open subsets of* A.
4. *The exterior of* A *is the union of all open subsets of* $X \setminus A$.
5. *The boundary of* A *is the set of all points* $x \in X$ *such that every neighbourhood of* x *contains both points in* A *and points in* $X \setminus A$.

Definition 8.2.10: Closure

The closure of A is the set $\overline{A} := A \cup \partial A$ containing A and all of its boundary points.

Exercise 8.2.11.

1. *Show that $\overline{A}$ is the set of all points $x \in X$ such that every neighbourhood of x contains points in A.*
2. *Show that $\overline{A}$ is closed.*
3. *Show $\overline{A}$ is the intersection of all closed sets containing A.*
4. *Show that if $U \subseteq X$ is open, and $A \subseteq X$ is an arbitrary set such that $U \cap A = \emptyset$, then we also have $U \cap \overline{A} = \emptyset$.*

Theorem 8.2.12. Let $B \subseteq \mathbb{R}$ be a bounded subset of $\mathbb{R}$. Then $\inf B$ and $\sup B$ are contained in $\overline{B}$.

Proof. Since B is bounded, completeness of the real numbers tells us that $\inf B$ and $\sup B$ are finite real numbers. If $\inf B \notin B$, then there would exists an open neighbourhood $U_{\inf}$ of $\inf B$ not containing any points of B. Since any neighbourhood of a real number x must contain numbers greater than x, this would give us a lower bound of B greater than $\inf B$, which contradicts the definition of $\inf B$. Similarly, if $\sup B \notin \overline{B}$, we could get a lesser upper bound of B, which is again a contradition. □

8.3 Convergence

Theorem 8.3.1: Detecting openness and closedness using sequences

Let M be a metric space. Let $A \subseteq M$. Let $x \in M$. Then:

1. $x \in \operatorname{int} A$ if and only if every sequence in M converging to x is eventually in A.
2. $x \in \overline{A}$ if and only if there exists a sequence in A converging to x.

Proof.

1. If $x \in \operatorname{int} A$, then there exists an $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq A$. If a sequence converges to x, it must thus be eventually in $B_\varepsilon(x)$, and thus eventually in A.

   Vice versa, if $x \notin \operatorname{int} A$, then every open ball $B_n := B_{\frac{1}{n}}(x)$ around A contains at least one point not in A, and so we can construct a sequence converging to x that isn't eventually in A by picking these points.

2. If there exists sequence of points in A converging to x, then every neighbourhood of x must contain at least one point in A, so $x \in \overline{A}$.

   Vice versa, if $x \in \overline{A}$, then every open ball around x must contain a point from A, so we can construct a sequence in A converging to x by taking the sequence $B_n := B_{\frac{1}{n}}(x)$ of open balls and picking a point $x_n \in A \cap B_n$ for each $n \in \mathbb{N}$.

□

Corollary 8.3.2.

1. A is closed in X if and only if it contains every limit of every convergent sequence of points in A.
2. A is open in X if and only if every sequence in X converging to a point of A is eventually in A.

8.4 Continuity

Definition 8.4.1. Let $f : M \to N$ be a function between metric spaces. Then we call f ***uniformly continuous*** if it is continuous and fulfills the additional requirement that, given an ε, we can pick the same δ for all points $x \in M$, i.e:

$$\begin{aligned}
&\forall \varepsilon \in \mathbb{R}_{>0} : \\
&\exists \delta \in \mathbb{R}_{>0} : \\
&\forall x, y \in M : \\
&\quad d(x, y) < \delta \implies d(f(x), f(y)) < \varepsilon
\end{aligned}$$

8.5 Compactness

Definition 8.5.1: Compact Sets

We call a subset $C \subseteq M$ of a metric space ***compact*** if every open covering of C has a finite subcovering.

Intuitively, compact sets are sets which only contain a finite amount of topological information. This means that as soon as compact sets are involved, potentially complicated problems involving infinitely many objects can often be turned into simple proofs involving a much more manageable finite number of objects. Later in this chapter, we will prove that every closed, bounded subset of $\mathbb{R}^n$ is compact.

The notion of compactness is a key ingredient of many incredibly useful results in analysis and topology. For example, early on into this chapter, we will prove the Maximum-Minimum-Theorem, which states that every function from a compact set into the real numbers attains a maximum and a minimum. More advanced applications will keep showing up throughout the later parts of this book - for example, compact manifolds tend to be much more well-behaved than their non-compact cousins, leading to nice classification theorems for compact surfaces and compact Lie groups.

Exercise 8.5.2. *Any closed subset of a compact set is compact.*

Exercise 8.5.3. *A compact subset of a metric space is always closed.*

Theorem 8.5.4: Compactness in Metric Spaces

A subset of a metric space is compact if and only if it is sequentially compact.

Theorem 8.5.5. A compact subset of a metric space is closed and bounded.

Proof. Let $K \subseteq M$ be compact. Since M is a metric space, K is closed. If we cover K using open balls, that cover must have a finite subcover. Thus, K must be contained in a finite union of open balls, which means K must be bounded. □

Theorem 8.5.6. Any closed interval $I = [a, b] \subset \mathbb{R}$ is compact.

Theorem 8.5.7. Let $f : M \to N$ be a continuous function between metric spaces. Let $K \subseteq M$ be compact. Then $f(K)$ is compact.

Proof. Let $K \subseteq M$ be compact. Let $\mathcal{U}$ be an open cover of $f(K)$. Since f is continuous, $f^{-1}[\mathcal{U}] := \left\{ f^{-1}[U] \,\middle|\, U \in \mathcal{U} \right\}$ is an open cover of N, which must have a finite subcover $\mathcal{V} \subseteq f^{-1}[\mathcal{U}]$. Since $f : K \to f(K)$ is surjective by definition, $f[\mathcal{V}]$ must be a finite cover of $f(K)$. □

Corollary 8.5.8: Maximum-Minimum-Theorem

Let $f : K \to \mathbb{R}$ be a function from a compact subset of a metric space into the real numbers. Then f attains a maximum and a minimum.

Proof. By the previous theorem, $f(K)$ is a compact subset of $\mathbb{R}$, i.e. closed and bounded by Theorem 8.5.5. But then Theorem 8.2.12 tells us that $\inf f(K)$ and $\sup f(K)$ are already contained in $\overline{f(K)} = f(K)$. □

Lemma 8.5.9. Let M, N be metric spaces. Let $K \subseteq M$ be compact. Let $f : K \to N$ be continuous. Then the preimage of any compact subset of M is a compact subset of K.

Proof. Since N is a metric space, compact subsets are closed. Since f is continuous, the preimage of a closed subset is closed. Since K is compact, any closed subset of K is compact. □

Theorem 8.5.10: Countable Metric Tychonoffs Theorem

Let M_1, M_2 be metric spaces. Let $K_1 \subseteq M_1, K_2 \subseteq M_2$ be compact. Then $K_1 \times K_2 \subseteq M_1 \times M_2$ is compact.

Proof. Using whichever bijection $\mathbb{N}^2 \cong \mathbb{N}$ you prefer, we can unroll a sequence s_n in $K_1 \times K_2$ into a sequence x_n in K_1 and a sequence y_n in K_2, such that the elements of s_n are in one-to-one correspondence with pairs (x_i, y_j) of elements of x_n and y_n. Since K_1 and K_2 are compact, x_n and y_n have convergent subsequences. Using our bijection, these can be reassembled into a subsequence of s_n, and since both components of this subsequence converge, the subsequence itself converges. Thus, every sequence in $K_1 \times K_2$ has a convergent subsequence. Since sequential compactness and compactness are equivalent in metric spaces by Theorem 8.5.4, $K_1 \times K_2$ is compact. □

Theorem 8.5.11: Heine-Borel Theorem

Every closed and bounded subset of $\mathbb{R}^n$ is compact.

Proof. A bounded set is a subset of a cube $C^n = \prod_{i=1}^{n} [-r, r]$. Such a cube is a product of closed intervals, which are compact by Tychonoff's Theorem 8.5.6. Every set that is closed and bounded is thus a closed subset of a

compact set, making them compact by Exercise 8.5.2. □

Theorem 8.5.12. Let M, N be metric spaces. Let $K \subseteq M$ be compact. Then every continuous function $f : K \to N$ is uniformly continuous.

Proof. Assume that $f : K \to N$ isn't uniformly continuous. Then there exists an $\varepsilon \in \mathbb{R}_{>0}$ such that for every $n \in \mathbb{N}$, we can find points $x_n, y_n \in K$ such that $d(x_n, y_n) < 1/n$, but $d(f(x_n) - f(y_n)) \geq \varepsilon$.
Since K is compact, it is sequentially compact by Theorem 8.5.4, so there exists a subsequence $(x_k) \subset (x_n)$ converging to a point p. Since $d(x_n, y_n) < 1/n$, there also exists a subsequence $x_k \subset y_n$ converging to p. If f was continuous, we would have

$$\lim_{k\to\infty} f(x_k) = \lim_{k\to\infty} f(y_k),$$

contradicting the fact that we have $d(f(x_k) - f(y_k)) \geq \varepsilon$ for all $k \in \mathbb{N}$. □

Lemma 8.5.13. (Lebesgue's Number Lemma:) Let X be a compact metric space. Let $\mathcal{U} = \{U_i \mid i \in I\}$ be an open covering of X. Then there exists a ***Lebesgue Number*** $\delta \in \mathbb{R}_{>0}$ such that

$$\operatorname{diam} A < \delta \implies \exists i \in I : A \subseteq U_i$$

for any subset $A \subseteq X$.

Proof. Since $\mathcal{U}$ is an open covering, every $x \in X$ must have an $\varepsilon(x) \in \mathbb{R}_{>0}$ such that $B_{2\varepsilon(x)}(x) \subseteq U_i$. Since all of the smaller open balls $B_{\varepsilon(x)}(x)$ must form a covering of X, and X is compact, we can pick a finite subcovering

$$\mathcal{B} := \left\{ B_{\varepsilon(x_k)}(x_k) \,\middle|\, k \in [1..n] \right\}.$$

Since $\mathcal{B}$ is an open covering, every $x \in X$ has an x_k such that $x \in B_{\varepsilon(x_k)}(x_k)$, i.e. $d(x, x_k) < \varepsilon(x_k)$.

Now, since $\mathcal{B}$ is finite, there exists a minimum

$$\delta = \min \{\varepsilon(x_k) \mid k \in [1..n]\}.$$

This means that, for any set $A \subseteq X$ with $\operatorname{diam} A < \delta$ and any $a_0 \in A$, we can pick the corresponding x_k, giving us

$$d(a, x_k) \leq \underbrace{d(a, a_0)}_{<\delta\leq\varepsilon(x_k)} + \underbrace{d(a_0, x_k)}_{\leq\varepsilon(x_k)} < 2\varepsilon(x_k),$$

i.e. $A \subseteq B_{2\varepsilon(x_k)}(x_k) \subseteq U_i$. □

8.6 Connectedness

Definition 8.6.1

A metric space is called ***connected*** if it can't be written as the disjoint union of two nonempty open subsets.

Exercise 8.6.2. *Show that $\mathbb{N}$ is not connected. Show that $\mathbb{Q}$ is not connected.*

Exercise 8.6.3. *Show that a metric space M is connected if and only if its only clopen subsets are M and $\emptyset$.*

Theorem 8.6.4. The connected subsets of $\mathbb{R}$ are the intervals $I \subseteq \mathbb{R}$.

Proof. 1. Let $I \subseteq \mathbb{R}$ be an interval. Assume we have a decomposition

$$I = I_1 \cup I_2$$

of I into disjoint nonempty open subsets. Without loss of generality, assume that there exists $x_1 \in I_1$ and $x_2 \in I_2$ such that $x_1 < x_2$ (otherwise swap the two subsets). Now, consider the set

$$S = \{x \in I_1 \mid x \leq x_2\}$$

S is nonempty, since $x_1 \in S$, and S has x_2 as an upper bound. Now, let $s = \sup S$. We have $x_1 \leq s \leq x_2$, so $s \in I$. By Theorem 8.2.12, we have $s \in \overline{S} \subseteq \overline{I}_1$.

Since we have $I_1 = I \setminus I_2$, we know that I_1 is closed in I, giving us $s \in \overline{I}_1 = I_1$.

Since I_1 is by definition also open in I, and $s \leq x_2$ cannot be an upper bound of I, there exists $\varepsilon \in \mathbb{R}_{>0}$ such that $s + \varepsilon \in I_1$, which contradicts $s = \sup I_1$. Therefore, all intervals must be connected.

2. Let $C \subseteq \mathbb{R}$ be connected. If C is empty or only contains one point, it vacuously qualifies as an interval (x, x) or $[x, x]$. Now, assume that C contains at least two points, but isn't an interval. Then there exist $c_1 \neq c_2 \in C$ and $c \in [c_1, c_2]$ such that $c \notin C$. But then

$$C_1 = (-\infty, c) \cap C$$
$$C_2 = (c, \infty) \cap C$$

is a decomposition of C into disjoint nonempty open subsets.

□

Theorem 8.6.5. Let $f : M \to N$ be a continuous map between metric spaces. Let M be connected. Then $f(M)$ is connected.

Proof. Assume $f(M)$ wasn't connected. Then we would have some decomposition $f(M) = U \cup V$ of $f(M)$ into disjoint open sets. Since f is continuous, $f^{-1}(U)$ and $f^{-1}(V)$ are open. But then $M = f^{-1}(U) \cup f^{-1}(V)$ can't be connected. □

Corollary 8.6.6: Intermediate Value Theorem

Let $f : [a, b] \to \mathbb{R}$ be a continuous map from a closed real interval into the real numbers. Let y be some number between $f(a)$ and $f(b)$ [1]. Then there exists $x \in [a, b]$ such that $f(x) = y$.

[1] i.e. $f(a) \leq y \leq f(b)$ or $f(b) \leq y \leq f(a)$

Proof. Since f is continuous, $f([a, b])$ is a connected subset of $\mathbb{R}$, i.e. an interval. Since $f([a, b])$ contains $f(a)$ and $f(b)$, it must therefore also contain all numbers inbetween the two. □

Corollary 8.6.7: Bolzano's Theorem

If a continuous function $f : M \to \mathbb{R}$ has a point $x \in M$ with $f(x) < 0$ and a point $y \in M$ with $f(y) > 0$, there exists a root $z \in M : f(z) = 0$ between x and y.

Exercise 8.6.8.

1. *Assuming that temperature varies continuously, show that there are always two points at a given latitude on earth at which the temperature is the same.*
2. *Assuming there are two points on earth that are a given distance apart from each other, are there always two points on earth that are that distance apart from each other and have the same temperature?*

8.7 Completeness

8.8 Function Spaces

Since we know that a sequence in a finite cartesian product converges if and only if all components converge, it seems natural to generalize this notion to arbitrary, potentially infinite cartesian products, i.e. arbitrary functions:

Definition 8.8.1: Pointwise Convergence

We say that a sequence of functions $f_n : M \to N$ converges ***pointwise*** to a given function $f : M \to N$ if and only if all components $f_n(x)$ converge in the usual sense, i.e:

$$\begin{aligned}&\forall x \in M : \\ &\forall \varepsilon \in \mathbb{R}_{>0} : \\ &\exists N \in \mathbb{N} : \\ &\forall n \geq N : \\ &\quad d(f_n(x) - f(x)) < \varepsilon\end{aligned}$$

We can't put a metric on N^M in which this form of converges matches regular metric convergence, but the intuition that this is the most natural form of convergence in N^M will be vindicated once we study abstract product spaces in point-set topology, at which point we confirm that in the natural product topology, an infinite cartesian product still converges if and only if every component converges individually.
However, there *is* a natural metric to put on N^M, which leads to a stronger form of convergence:

Definition 8.8.2. An ***extended metric space*** is a metric space where d can also take on the value ∞.

Theorem 8.8.3: Supremum Metric

The set N^M of functions from a metric space M to a metric space N can be metrized by the extended metric

$$d_\infty(f, g) := \sup_{x \in M} \left| f(x) - g(x) \right|.$$

Additionally, this function is a regular metric on the set $B(M, N) \subset N^M$ of bounded functions $M \to N$.

Note that is simply a direct generalization of the Chebyshev metric 8.1.3 to infinite cartesian products!

Theorem 8.8.4: Uniform Convergence

In (N^M, d_∞), a sequence of functions f_n converges to a given function f if and only if

$$\begin{aligned}&\forall \varepsilon \in \mathbb{R}_{>0} :\\ &\exists N \in \mathbb{N} :\\ &\forall n \geq N :\\ &\forall x \in M :\\ &\quad d(f_n(x) - f(x)) < \varepsilon,\end{aligned}$$

i.e. if the speed at which $f_n(x)$ converges to $f(x)$ doesn't depend on the specific x we choose. We call this form of convergence of a sequence of functions ***uniform convergence***.

Theorem 8.8.5: Uniform Limit Theorem

Let $(f_n)_{n\in\mathbb{N}}$ be a sequence of continuous functions $f_n : M \to N$ converging uniformly to a function $f : M \to N$. Then f is continuous.

Proof. Choose some $\varepsilon \in \mathbb{R}_{>0}$. Since $(f_n)_{n\in\mathbb{N}}$ converges to f uniformly, there exists an $N \in \mathbb{N}$ such that

$$\begin{aligned}&\forall x \in M :\\ &\quad n \geq N \implies d(f(x), f_n(x)) < \varepsilon/3\end{aligned}$$

Let $x_0 \in M$ be arbitrary. Since f_N is continuous, there exists a $\delta \in \mathbb{R}_{>0}$ such that

$$\begin{aligned}&\forall x \in M :\\ &\quad d(x, x_0) < \delta \implies d(f_N(x), f_N(x_0)) < \varepsilon/3\end{aligned}$$

Applying the triangle inequality twice gives

$$\begin{aligned}&d(f(x), f(x_0))\\ &\leq d(f(x), f_N(x)) + d(f_N(x), f_N(x_0)) + d(f_N(x_0), f(x_0))\end{aligned}$$

to which we can apply the first inequality, then the second, and then the first again, to get:

$$\begin{aligned}&d(f(x), f(x_0))\\ &\leq \underbrace{d(f(x), f_N(x))}_{<\varepsilon}\\ &+ \underbrace{d(f_N(x), f_N(x_0))}_{<\varepsilon}\\ &+ \underbrace{d(f_N(x_0), f(x_0))}_{<\varepsilon}\\ &< \varepsilon\end{aligned}$$

for all $x \in M$ with $d(x, x_0) < \delta$. □

Theorem 8.8.6. Let $U \subseteq (X, d)$ be open and path-connected. Then there exists a piecewise linear path $\gamma_n : [0, 1] \to U$ between any two $p, q \in U$.

Proof Sketch.

1. Since U is path connected, there exists a curve $\gamma : [0, 1] \to U$ connecting p and q.
2. We can approximate γ via a piecewise linear curve γ_n by setting
$$\gamma_n\left(\frac{x}{n}\right) := \gamma\left(\frac{x}{n}\right)$$
3. Since γ is continuous and defined on a compact set, it is uniformly continuous by Theorem 8.5.12. From this, it can be shown that the γ_n converge to γ uniformly.
4. Since $\gamma([0, 1]) \subseteq U$ is compact and $X \setminus U$ is closed and disjoint from U, we have $\varepsilon := d(\gamma([0, 1]), X \setminus U) > 0$.
5. Therefore, uniform convergence guarantees there is an n such that $d(\gamma_n([0, 1]), \gamma([0, 1])) < \varepsilon$, which implies $\gamma_n([0, 1]) \subseteq U$.

"□"

Chapter 9

Infinite Series

9.1 Achilles and the Turtle

9.2 Convergence Tests

9.3 Euler's Number e

9.4 The Exponential Function

Chapter 10

Euclidean Geometry

10.1 The Complex Numbers $\mathbb{C}$

10.2 Circles, Angles and Trigonometry

Part IV

Calculus in one Variable

Chapter 11

Derivatives

Chapter 12

Integrals

12.1 The Gamma Function

In this section, we discover some basic properties of the so-called gamma function. We will find that it is the canonical extension of the factorial function to positive real numbers. With some knowledge of complex analyis, it can easily be extended to arbitrary complex numbers, but that is beyond the scope of these notes.

Definition 12.1.1: Gamma function

For any $x \in \mathbb{R}_{>0}$, the gamma function is defined as the improper integral

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} \, dt$$

Theorem 12.1.2. The gamma function obeys the recurrence relation

$$\Gamma(x+1) = x \cdot \Gamma(x)$$

Proof. Integration by parts gives:

$$\begin{aligned}
\Gamma(x+1) &= \int_0^\infty t^x e^{-t} \, dt \\
&= \left[-t^x e^{-t}\right]_0^\infty + \int_0^\infty x t^{x-1} e^{-t} \, dt \\
&= 0 - \lim_{n\to\infty} (-t^x e^{-t}) + x \int_0^\infty t^{x-1} e^{-t} \, dt \\
&= x \int_0^\infty t^{x-1} e^{-t} \, dt \\
&= x \cdot \Gamma(x)
\end{aligned}$$

□

Corollary 12.1.3. For any $n \in \mathbb{N}$, we have

$$\Gamma(n+1) = n!$$

Proof. Our recurrence relation gives

$$\Gamma(n+1) = n! \cdot \Gamma(1)$$

and we have

$$\begin{aligned}
\Gamma(1) &= \int_0^\infty t^{1-1} e^{-t}\, dt \\
&= \int_0^\infty e^{-t}\, dt \\
&= \left[-e^{-t}\right]_0^\infty \\
&= \left(\lim_{n\to\infty} -e^{-n}\right) - (-e^0) \\
&= 0 - (-1) \\
&= 1
\end{aligned}$$

□

Lemma 12.1.4. We have:

$$\Gamma(x) = 2c^x \int_0^\infty t^{2x-1} e^{-ct^2}\, dt$$

Proof. Substituting $t := cu^2$, we get:

$$\begin{aligned}
\Gamma(x) &= \int_0^\infty t^{x-1} e^{-t}\, dt \\
&= \int_0^\infty (cu^2)^{x-1} e^{-cu^2} \cdot 2cu\, du \\
&= \int_0^\infty c^{x-1} u^{2(x-1)} e^{-cu^2} \cdot 2cu\, du \\
&= \int_0^\infty c^x u^{2x-1} e^{-cu^2}\, du \\
&= c^x \cdot \int_0^\infty u^{2x-1} e^{-cu^2}\, du \\
&= c^x \cdot \int_0^\infty t^{2x-1} e^{-ct^2}\, dt
\end{aligned}$$

□

Proposition 12.1.5. We have $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

Proof Sketch. By the last lemma, we have:

$$\begin{aligned}\Gamma\left(\frac{1}{2}\right) &= 2\int_0^\infty t^{2\cdot\frac{1}{2}-1}e^{-t^2}\,dt \\ &= 2\int_0^\infty e^{-t^2}\,dt \\ &= \int_{-\infty}^\infty e^{-t^2}\,dt\end{aligned}$$

The fact that $\int_{-\infty}^{\infty} e^{-t^2}\,dt = \sqrt{\pi}$ might be familiar if you have a basic knowledge of probability theory, since this is exactly the reason why π appears in the formula for a normal distribution:

$$\mathcal{N}(\mu,\sigma^2)(t) = \frac{1}{\sqrt{2\pi\sigma^2}}e^{\left(-\frac{(t-\mu)^2}{2\sigma^2}\right)}$$

Setting $\mu = 0$ and $\sigma^2 = \frac{1}{2}$, which was Gauss' original definition for the standard normal distribution, we get:

$$\mathcal{N}(0,1)(t) = \frac{1}{\sqrt{\pi}}e^{-t^2},$$

where the factor $\frac{1}{\sqrt{\pi}}$ appears to normalize the integral of the distribution to 1, since it wouldn't be a probability distribution otherwise.

We will formally prove this fact much later, when we introduce the substitution formula for multidimensional Lebesque integration. "□"

Part V

Linear Algebra

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Chapter 13

Analytic Geometry

13.1 Arithmetic in n dimensions

We now focus our attention on n-dimensional euclidean space, which is modelled by the cartesian product $\mathbb{R}^n$:

$$\mathbb{R}^n = \left\{ \vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \;\middle|\; x_1, \ldots, x_n \in \mathbb{R} \right\} = \left\{ \vec{x} : \{1, \ldots, n\} \to \mathbb{R} \right\}$$

Elements of $\mathbb{R}^n$ are known as ***vectors***[1]. In the context of linear algebra, they are usually written vertically, as ***column vectors***.

[1] We will see in the next chapter that $\mathbb{R}^n$ is an example of a ***vector space***, including the case $n = 1$ (and $n = 0$, if you want to get a bit pathological). Therefore, formally speaking and only in certain contexts, regular old real (or even rational) numbers can be vectors too. However, when encountering the term "vector" in the wild, it is generally implied that $n \geq 2$.

Vectors simultaneously represent both ***positions*** and ***directions*** in space - in the same way that the number 5 is both the point that lies 5 units away from 0 on the real number line *and* the distance between the numbers 3 and 8 on that same number line, the vector $\binom{1}{2}$ represents both

- The point on the cartesian plane that can be reached by going one unit in the direction of one coordinate axis and then two units in the direction of the other axis, *and*
- the distance between two points with coordinates $\binom{2}{3}$ and $\binom{3}{5}$, or any other pair of points that are one unit apart in one coordinate direction and two in the other.

As is expected for algebraic structures defined on cartesian products, addition is defined **componentwise**:

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

Given a real number c, we can scale elements of $\mathbb{R}^n$ by that number:

$$c \cdot \vec{x} = c \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} c \cdot x_1 \\ \vdots \\ c \cdot x_n \end{pmatrix}$$

The factor c is known as a ***scalar***, and the operator $\cdot : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ as ***scalar multiplication***. For the sake of readability, we differentiate scalars from vectors by writing a little arrow on top of a vector.

Arguably, it is these two operations - vector addition and scalar multiplication - that define linear algebra - next chapter, we will abstractly define ***vector spaces*** to be those structures where addition and scalar multiplication are well-behaved and the scalars come from a field, and ***modules*** to be their cousins whose scalars come from a ring.

In general, there is no well-behaved[2] vector multiplication that works for any n. Later on, we will prove this observation in the form of ***Frobenius' Theorem***, which roughly states:

Theorem 13.1.1: Frobenius' Theorem

The only dimensions in which an associative multiplication exists on $\mathbb{R}^n$ are $\mathbb{R}^1$, i.e. $\mathbb{R}$ itself, $\mathbb{R}^2$, where multiplication is given by treating two-dimensional vectors as complex numbers, and $\mathbb{R}^4$, where multiplication is given treating four-dimensional vectors as ***quaternions***.

We will take a closer look at the complex numbers $\mathbb{C} \cong \mathbb{R}^2$ and the quaternions $\mathbb{H} \cong \mathbb{R}^4$ in the following sections. In particular, we will see that quaternion multiplication isn't commutative, so if we require multiplication to be commutative, our options shrink even further, leaving just $\mathbb{R}$ and $\mathbb{C} \cong \mathbb{R}^2$.

[2] In this case, our criterion for multiplication being well-behaved is that multiplicative inverses should exist, i.e. that division should be possible.

13.2 The Inner Product: Angles and Lengths in $\mathbb{R}^n$

Another operation that is of central importance to euclidean geometry is the ***inner product***, also known as the ***dot product*** or ***scalar product***.

Definition 13.2.1: Euclidean Inner Product

Given two vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$, their ***Euclidean inner product***, written $\langle \vec{x}, \vec{y} \rangle$ or simply $\vec{x} \cdot \vec{y}$, is the sum of the products of the components of $\vec{x}$ with the components of $\vec{y}$:

$$\langle \vec{x}, \vec{y} \rangle = \sum_{i=1}^{n} x_i \cdot y_i$$

Theorem 13.2.2: Properties of Inner Products

The Euclidean inner product has the following properties:

1. ***(Positive definiteness):*** For every $\vec{x} \in \mathbb{R}^n$, we have $\langle \vec{x}, \vec{x} \rangle = 0$ if and only if $\vec{x} = \vec{0}$.
2. ***(Linearity in the first argument):*** For every $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^n$ and $a, b \in \mathbb{R}$, we have:
$$\langle ax + by, z \rangle = a \langle x, z \rangle + b \langle y, z \rangle$$
3. ***(Symmetry):*** For every $\vec{x}, \vec{y} \in \mathbb{R}^n$, we have:
$$\langle x, y \rangle = \langle y, x \rangle$$

Exercise 13.2.3. *Show that the Euclidean inner product fulfills the following version of the binomial formula:*

$$\langle \vec{x} + \vec{y}, \vec{x} + \vec{y} \rangle = \langle \vec{x}, \vec{x} \rangle + 2 \langle \vec{x}, \vec{y} \rangle + \langle \vec{y}, \vec{y} \rangle$$
$$\langle \vec{x} - \vec{y}, \vec{x} - \vec{y} \rangle = \langle \vec{x}, \vec{x} \rangle - 2 \langle \vec{x}, \vec{y} \rangle + \langle \vec{y}, \vec{y} \rangle$$

The importance of inner products comes from the fact they let us define ***lengths*** and ***angles***.

Definition 13.2.4: Euclidean Norm

The length of a vector $\vec{x} \in \mathbb{R}^n$, also known as its ***Euclidean norm***, written as $\|\vec{x}\|_2$, $\|\vec{x}\|$, or just $|x|$, is given by:

$$\|\vec{x}\|_2 = \sqrt{\langle \vec{x}, \vec{x} \rangle} = \sqrt{x_1^2 + \ldots + x_n^2}$$

Note that for $n = 1$, we simply get $\|x\|_2 = \sqrt{x^2} = |x|$, so any result concerning the Euclidean norm on $\mathbb{R}^n$ also holds for the absolute value function.

Theorem 13.2.5: Some properties of Norms

The Euclidean norm has the following properties:

1. ***(Non-negativity):*** For every $\vec{x} \in \mathbb{R}^n$, $\left\|\vec{x}\right\|_2 \geq 0$.
2. ***(Positive definiteness):*** For every $\vec{x} \in \mathbb{R}^n$, $\left\|\vec{x}\right\|_2 = 0$ if and only if $\vec{x} = \vec{0}$.
3. ***(Absolute homogeneity):*** For every $c \in \mathbb{R}$ and $\vec{x} \in \mathbb{R}^n$, we have

$$\left\|c \cdot \vec{x}\right\|_2 = |c| \cdot \left\|\vec{x}\right\|_2$$

The little subscript 2 in $\left\|\vec{x}\right\|_2$ differentiates the Euclidean norm from the many other norms we will meet over time - in general, for any $p \in \mathbb{N}_1$, $\left\|\vec{x}\right\|_p$ denotes the function

$$\left\|\vec{x}\right\|_p = \sqrt[p]{|x_1|^p + \ldots + |x_n|^p} \qquad \left(= \sqrt[p]{\sum_{i=1}^{n} |x_i|^p}\right),$$

known as the ***p-norm***. For now, we only really care about the Euclidean norm, but other values of p will start coming in handy once we start applying tools from analysis to vector spaces. The limiting case $p \to \infty$ is particularly useful: We will find that

$$\left\|\vec{x}\right\|_\infty = \max |x_i|.$$

Theorem 13.2.6: Cauchy-Schwartz Inequality

Let $\vec{x}, \vec{y} \in \mathbb{R}^n$. Then we have

$$\left|\langle \vec{x}, \vec{y} \rangle\right| \leq \left\|\vec{x}\right\|_2 \cdot \left\|\vec{y}\right\|_2$$

Proof. 1. Assume $\left\|\vec{x}\right\|_2 = 0$. Then we immediately get

$$\langle \vec{x}, \vec{y} \rangle = 0 = \left\|\vec{x}\right\|_2 \cdot \left\|\vec{y}\right\|_2.$$

2. Assume $\left\|\vec{x}\right\|_2 = \left\|\vec{y}\right\|_2 = 1$. Then we have $\left\|\vec{x}\right\|_2^2 + \left\|\vec{y}\right\|_2^2 = 2$, and thus:

$$\begin{aligned} 1 + \langle \vec{x}, \vec{y} \rangle &= \frac{1}{2}(\left\|\vec{x}\right\|_2^2 + \left\|\vec{y}\right\|_2^2) + \langle \vec{x}, \vec{y} \rangle \\ &= \frac{1}{2}(\left\|\vec{x}\right\|_2^2 + 2\langle \vec{x}, \vec{y} \rangle + \left\|\vec{y}\right\|_2^2) \\ &= \frac{1}{2}\left\|\vec{x} + \vec{y}\right\|_2^2 \\ &\geq 0 \end{aligned}$$

Thus we get $\langle \vec{x}, \vec{y} \rangle \geq -1$. Applying the same reasoning to $1 - \langle \vec{x}, \vec{y} \rangle$, we get

$$1 - \langle \vec{x}, \vec{y} \rangle = \frac{1}{2}\left\|\vec{x} - \vec{y}\right\|_2^2 \geq 0,$$

i.e. $\langle \vec{x}, \vec{y} \rangle \leq 1$. We thus have

$$\left|\langle \vec{x}, \vec{y} \rangle\right| \leq 1 = \left\|\vec{x}\right\|_2 \cdot \left\|\vec{y}\right\|_2,$$

as desired.

3. Now, let $\vec{x}, \vec{y} \in \mathbb{R}^n \setminus \{\vec{0}\}$ be arbitrary. Then $\frac{\vec{x}}{\|\vec{x}\|_2}$ and $\frac{\vec{y}}{\|\vec{y}\|_2}$ both have norm 1, letting us simply use our previous result:

$$\begin{aligned}
|\langle \vec{x}, \vec{y} \rangle| &= \left| \|\vec{x}\| \cdot \|\vec{y}\| \cdot \left\langle \frac{\vec{x}}{\|\vec{x}\|}, \frac{\vec{y}}{\|\vec{y}\|} \right\rangle \right| \\
&\leq \left| \|\vec{x}\| \cdot \|\vec{y}\| \cdot \left\| \frac{\vec{x}}{\|\vec{x}\|} \right\| \cdot \left\| \frac{\vec{y}}{\|\vec{y}\|} \right\| \right| \\
&= \|\vec{x}\| \cdot \|\vec{y}\|
\end{aligned}$$

□

Corollary 13.2.7: Triangle Inequality

For every $\vec{x}, \vec{y} \in \mathbb{R}^n$, we have

$$\|\vec{x} + \vec{y}\|_2 \leq \|\vec{x}\|_2 + \|\vec{y}\|_2$$

We had previously formalized the concept of an 'angle between two lines' via the complex exponential function, and then defined the trigonometric functions $\sin x$ and $\cos x$ as the complex and real parts of e^{ix}, respectively. If we take the cosine function as a given, then scalar products provide a convenient alternate way of defining angles, which works in any dimension:

Definition 13.2.8: Inner angle between two vectors

Given two vectors $\vec{x}, \vec{y} \in \mathbb{R}^n \setminus \{\vec{0}\}$, we define the ***inner angle*** between them to be:

$$\measuredangle(\vec{x}, \vec{y}) = \arccos\left(\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|_2 \cdot \|\vec{y}\|_2} \right)$$

Note that Cauchy-Schwartz gives us $\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|_2 \cdot \|\vec{y}\|_2} \in [-1, 1]$, showing that this expression actually is well-defined.

Exercise 13.2.9. *Verify that we have:*

1. $\measuredangle(\vec{x}, \vec{y}) = 0$ *if and only if there exists* $c \in \mathbb{R}_{>0}$ *such that* $\vec{x} = c\vec{y}$.
2. $\measuredangle(\vec{x}, \vec{y}) = \frac{\pi}{2}$ *if and only if* $\langle x, y \rangle = 0$.
3. $\measuredangle(\vec{x}, \vec{y}) = \pi$ *if and only if there exists* $c \in \mathbb{R}_{>0}$ *such that* $\vec{x} = -c\vec{y}$.

Notably, this suggests that our definition is sensible even in the case $n = 1$, since any pair of non-zero vectors fulfills either condition 1 or 2, implying that the angle between two vectors on the real number line is either 0 or π.

These inner angles match our earlier definition of angles in terms of complex numbers, as long as we are content with not being able to meaningfully talk about negative angles or angles greater than π anymore:

Exercise 13.2.10. *Earlier in this book, we defined angles in terms of complex numbers. Given a complex number $y = re^{i\varphi}$ we saw that the corresponding real vector is given by $\vec{y} = \binom{r\cos\varphi}{r\sin\varphi}$.*

Show that our two definitions match as long as $\varphi \in [0, \pi]$, i.e. that given a vector $\vec{x} = \binom{x}{0}$ lying on the x-axis, we have

$$\measuredangle(\vec{x}, \vec{y}) = \varphi$$

(Hint: Remember $\cos^2\varphi + \sin^2\varphi = 1$)

We have to be content with this restricted interval $[0, \pi]$, since the cosine function isn't injective on larger intervals, meaning the arccosine would no longer be well-defined.

More generally, any definition of an 'angle between two arbitrary vectors' will suffer from this - if we only specify two vectors, it's left ambiguous which side we should measure the angle on.

This problem can be solved by finding a way to take a tuple $(\vec{x}_1, \ldots, \vec{x}_n)$ of vectors and recognize whether it is 'positively ordered' or 'negatively ordered' - in $\mathbb{R}^2$, this would be asking whether the second vector lies 'clockwise' or 'counterclockwise' from the first, while in $\mathbb{R}^3$ the question is whether the vectors are 'left-handed' or 'right-handed'. A map that gives us this information is known as an ***orientation***. We will formalize this idea later, when we have introduced the necessary tools, those being vector spaces, their ***bases***, and ***determinant functions***.

In the complex case, we could dodge this issue because one of our vectors, call it $\vec{x}$, was always part of the x-axis, and thus we could obtain a canonical orientation by declaring that the angle between $\vec{x}$ and $\vec{y}$ was positive if $y_2 > 0$ and negative if $y_2 < 0$.

Exercise 13.2.11. *Prove the following theorems of classical Euclidean geometry:*

1. *The* **Law of Cosines***:*

$$\|\vec{x} - \vec{y}\|_2^2 = \|\vec{x}\|_2^2 + \|\vec{y}\|^2 - 2\|\vec{x}\|_2\|\vec{y}\|_2 \cdot \cos\big(\measuredangle(\vec{x}, \vec{y})\big)$$

2. *The* **Pythagorean Theorem***:*

$$\measuredangle(\vec{x}, \vec{y}) = \frac{\pi}{2} \implies \|\vec{x} - \vec{y}\|_2^2 = \|\vec{x}\|_2^2 + \|\vec{y}\|^2$$

13.3 The Cross Product

The ***cross product*** is an operation $\times : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$, such that:

1. $\vec{u} \times \vec{v}$ is orthogonal to both $\vec{u}$ and $\vec{v}$,
2. the length of $\vec{u} \times \vec{v}$ is the area of the parallelogram spanned by $\vec{u}$ and $\vec{v}$:

$$\left\|\vec{u} \times \vec{v}\right\|_2 = \left\|\vec{u}\right\|_2 \left\|\vec{v}\right\|_2 \sin\left(\measuredangle(\vec{u}, \vec{v})\right)$$

We also declare that our cross product should be ***anticommutative:*** given two vectors, we want to be able to change the sign of the cross product by switching the order in which we multiply:

$$\vec{v} \times \vec{w} := -\vec{w} \times \vec{v}$$

Notice how this immediately implies

$$\begin{aligned} & \vec{v} \times \vec{v} = -(\vec{v} \times \vec{v}) \\ \implies & \vec{v} \times \vec{v} = 0 \end{aligned}$$

Finally, notice that our geometric intuition immediately tells us that our product should be ***linear*** and ***distributive over vector addition***.

We can now derive an explicit formula for this operation on our own. First, note that the standard basis

$$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \vec{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

consists of orthogonal vectors of length one, so we can start by declaring

$$\begin{aligned} \vec{e}_1 \times \vec{e}_2 &:= \vec{e}_3 \\ \vec{e}_2 \times \vec{e}_3 &:= \vec{e}_1 \\ \vec{e}_3 \times \vec{e}_1 &:= \vec{e}_2 \end{aligned}$$

Note that once again, the direction in which these vectors point depends on a somewhat arbitrary choice of orientation - we could invert all of the directions and still end up with a valid definition.

These observations are already enough to calculate the cross product of two

arbitrary vectors: Using distributivity, we get:

$$\begin{aligned}
\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} &= (v_1\vec{e}_1 + v_2\vec{e}_2 + v_3\vec{e}_3) \times (w_1\vec{e}_1 + w_2\vec{e}_2 + w_3\vec{e}_3) \\
&= v_1w_1(\vec{e}_1 \times \vec{e}_1) \\
&+ v_1w_2(\vec{e}_1 \times \vec{e}_2) \\
&+ v_1w_3(\vec{e}_1 \times \vec{e}_3) \\
&\\
&+ v_2w_1(\vec{e}_2 \times \vec{e}_1) \\
&+ v_2w_2(\vec{e}_2 \times \vec{e}_2) \\
&+ v_2w_3(\vec{e}_2 \times \vec{e}_3) \\
&\\
&+ v_3w_1(\vec{e}_3 \times \vec{e}_1) \\
&+ v_3w_2(\vec{e}_3 \times \vec{e}_2) \\
&+ v_3w_3(\vec{e}_3 \times \vec{e}_3)
\end{aligned}$$

To which we can apply anticommutativity and our formulas for multiplying the basis vectors to get:

$$\begin{aligned}
&\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} \\
&\\
&= v_1w_1 \cdot \vec{0} \\
&+ v_1w_2 \cdot \vec{e}_3 \\
&+ v_1w_3 \cdot (-\vec{e}_2) \\
&\\
&+ v_2w_1 \cdot (-\vec{e}_3) \\
&+ v_2w_2 \cdot \vec{0} \\
&+ v_2w_3 \cdot \vec{e}_1 \\
&\\
&+ v_3w_1 \cdot \vec{e}_2 \\
&+ v_3w_2 \cdot (-\vec{e}_1) \\
&+ v_3w_3 \cdot \vec{0} \\
&\\
&= \begin{pmatrix} v_2w_3 - v_3w_2 \\ v_3w_1 - w_1v_3 \\ v_1w_2 - v_2w_1 \end{pmatrix}
\end{aligned}$$

Corollary 13.3.1: Cross Product

The cross product $\vec{v} \times \vec{w}$ of two vectors $\vec{v}, \vec{w} \in \mathbb{R}^3$ returns a vector whose direction is orthogonal to both $\vec{v}$ and $\vec{w}$ and whose length equals the area of the parallelogral spanned between them. Given coordinate representations of $\vec{v}$ and $\vec{w}$, the cross product can be calculated via the following formula:

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - w_1 v_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix}$$

Proposition 13.3.2. We have

$$\left\langle \vec{u} \times \vec{v}, \vec{w} \times \vec{t} \right\rangle = \left\langle \vec{u}, \vec{w} \right\rangle \left\langle \vec{v}, \vec{t} \right\rangle - \left\langle \vec{u}, \vec{t} \right\rangle \left\langle \vec{v}, \vec{w} \right\rangle .$$

Proposition 13.3.3. We have

$$\left\langle \vec{u} \times \vec{v}, \vec{w} \right\rangle = \left\langle \vec{v} \times \vec{w}, \vec{u} \right\rangle = \left\langle \vec{w} \times \vec{u}, \vec{v} \right\rangle$$

This expression is known as the ***box product*** of $\vec{u}, \vec{v}, \vec{w}$. Its absolute value corresponds to the volume of the parallelepiped spanned by the three vectors.

Proposition 13.3.4. We have

$$(\vec{u} \times \vec{v}) \times \vec{w} = \langle u, w \rangle \cdot v - \langle v, w \rangle \cdot u = w \times (v \times u).$$

This result is known as ***Grassmann's Identity***.

Proposition 13.3.5. We have

$$(\vec{u} \times \vec{v}) \times \vec{w} + (\vec{v} \times \vec{w}) \times \vec{u} + (\vec{w} \times \vec{u}) \times \vec{v} = 0$$

This result is known as ***Jacobi's Identity.***

13.4 The Quaternions $\mathbb{H}$

Definition 13.4.1: Quaternions

The Quaternions $\mathbb{H}$ consist of the set $\mathbb{R}^4$, with the following special notation:

$$\begin{aligned}\vec{1} &:= (1,0,0,0)\\ i &:= (0,1,0,0)\\ j &:= (0,0,1,0)\\ k &:= (0,0,0,1)\end{aligned}$$

Such that in addition to scalar multiplication and vector addition, which work as usual, we have an additional vector multiplication operation $\cdot : \mathbb{R}^4 \times \mathbb{R}^4 \to \mathbb{R}^4$ that behaves as follows:

1. $\vec{1}$ is a neutral element of vector multiplication:

$$\begin{aligned}&\forall x \in \mathbb{H} :\\ &\vec{1} \cdot x = x \cdot \vec{1} = x\end{aligned}$$

2. Vector multiplication is associative:

$$\vec{x}(\vec{y}\vec{z}) = (\vec{x}\vec{y})\vec{z}$$

3. Multiplicative inverses exist:

$$\begin{aligned}&\forall \vec{x} \in \mathbb{H}_{\neq 0} :\\ &\exists \vec{x}^{-1} \in \mathbb{H} :\\ &\vec{x} \cdot \vec{x}^{-1} = \vec{x}^{-1} \cdot \vec{x} = \vec{1}\end{aligned}$$

4. Vector multiplication distributes over vector addition, from both sides:

$$\begin{aligned}&\forall \vec{x}, \vec{y}, \vec{z} \in \mathbb{H} :\\ &\vec{x} \cdot (\vec{y} + \vec{z}) = \vec{x}\vec{y} + \vec{x}\vec{z}\\ &(\vec{x} + \vec{y}) \cdot \vec{z} = \vec{x}\vec{z} + \vec{y}\vec{z}\text{t}\end{aligned}$$

5. Vector multiplication is compatible with scalar multiplication:

$$\begin{aligned}&\forall a, b \in \mathbb{R}, \vec{x}, \vec{y} \in \mathbb{H} :\\ &a\vec{x} \cdot b\vec{y} = (ab)(\vec{x}\vec{y})\end{aligned}$$

6. The unit vectors fulfill the ***Quaternion multiplication formula***:

$$i^2 = j^2 = k^2 = ijk = -\vec{1}$$

Apart from the Quaternion multiplication formula, which is the defining feature of the Quaternions, all of our other demands together state that $\mathbb{H}$ should be an

associative real division algebra over $\mathbb{R}^4$.

One very important consequence of this definition is that vectors of the form $\vec{a} := (a, 0, 0, 0)$ behave exactly like real numbers - let $\vec{x} \in \mathbb{H}$, then we have:

$$\vec{a}\vec{x} = (a \cdot \vec{1}) \cdot \vec{x} = a \cdot (\vec{1} \cdot \vec{x}) = a\vec{x}$$

For this reason, it is common to write $\vec{1}$ simply as 1, and to then proceed as we did for the complex numbers and write:

$$(a, b, c, d) = a + bi + cj + dk$$

We can now already get most of the way to deriving multiplication formula for quaternions, by simply applying distributivity:

$$\begin{aligned}(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k)& \\ &= v_1u_1 + (v_1u_2)i + (v_1u_3)j + (v_1u_4)k \\ &+ (v_2u_1)i + (v_2u_2)i^2 + (v_2u_3)ij + (v_2u_4)ik \\ &+ (v_3u_1)j + (v_3u_2)ji + (v_3u_3)j^2 + (v_3u_4)jk \\ &+ (v_4u_1)k + (v_4u_2)ki + (v_4u_3)kj + (v_4u_4)k^2\end{aligned}$$

Now, using $i^2 + j^2 + k^2 = ijk = -1$, we can simplify slightly:

$$\begin{aligned}(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k)& \\ &= v_1u_1 + (v_1u_2)i + (v_1u_3)j + (v_1u_4)k \\ &+ (v_2u_1)i + (v_2u_2)i^2 + (v_2u_3)ij + (v_2u_4)ik \\ &+ (v_3u_1)j + (v_3u_2)ji + (v_3u_3)j^2 + (v_3u_4)jk \\ &+ (v_4u_1)k + (v_4u_2)ki + (v_4u_3)kj + (v_4u_4)k^2 \\ &= v_1u_1 + (v_1u_2)i + (v_1u_3)j + (v_1u_4)k \\ &+ (v_2u_1)i - (v_2u_2) + (v_2u_3)ij + (v_2u_4)ik \\ &+ (v_3u_1)j + (v_3u_2)ji - (v_3u_3) + (v_3u_4)jk \\ &+ (v_4u_1)k + (v_4u_2)ki + (v_4u_3)kj - (v_4u_4)\end{aligned}$$

All that's left is to figure out what the products of i, j and k should be.

We can derive ij and ji as follows:

1. Start with $ijk = -1$.
2. Multiply k from the right to get $ijk^2 = -k$, i.e. $-ij = -k$, i.e. $ij = k$.
3. Multiply ij from the left to get $ijij = ijk = -1$.
4. Multiply i from the left and j from the right to get $i(ijij)j = -ij$.
5. Simplify the left side to get $i(ijij)j = (ii)(ji)(jj) = ji$, i.e:

$$ij = -ji = k$$

Exercise 13.4.2. *Show that we also have $jk = -kj = i$ and $ki = -ik = j$.*

This means that Quaternion multiplication isn't commutative!

Our final multiplication table for $1, i, j$ and k is thus:

$\cdot$	1	i	j	k
1	1	i	j	k
i	i	-1	k	$-j$
j	j	$-k$	-1	i
k	k	j	$-i$	-1

Exercise 13.4.3. *Verify that this multiplication table doesn't violate associativity.*

Entering these new results into our partial multiplication formula, we get:

$$\begin{aligned}(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k)& \\ = v_1 u_1 + (v_1 u_2) i + (v_1 u_3) j + (v_1 u_4) k& \\ + (v_2 u_1) i - (v_2 u_2) + (v_2 u_3) ij + (v_2 u_4) ik& \\ + (v_3 u_1) j + (v_3 u_2) ji - (v_3 u_3) + (v_3 u_4) jk& \\ + (v_4 u_1) k + (v_4 u_2) ki + (v_4 u_3) kj - (v_4 u_4)& \\ = v_1 u_1 + (v_1 u_2) i + (v_1 u_3) j + (v_1 u_4) k& \\ + (v_2 u_1) i - (v_2 u_2) + (v_2 u_3) k - (v_2 u_4) j& \\ + (v_3 u_1) j - (v_3 u_2) k - (v_3 u_3) + (v_3 u_4) i& \\ + (v_4 u_1) k + (v_4 u_2) j - (v_4 u_3) i - (v_4 u_4)&\end{aligned}$$

which we can finally simplify to get:

$$\begin{aligned}(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k)& \\ = (v_1 u_1 - v_2 u_2 - v_3 u_3 - v_4 u_4)& \\ + (v_1 u_2 + v_2 u_1 + v_3 u_4 - v_4 u_3) i& \\ + (v_1 u_3 - v_2 u_4 + v_3 u_1 + v_4 u_2) j& \\ + (v_1 u_4 + v_2 u_3 - v_3 u_2 + v_4 u_1) k&\end{aligned}$$

Now, this is obviously a long and unreasonably complicated formula - is there any way to think of it differently? The complex numbers had a very nice and intuitive geometric definition, maybe the quaternions do too?

Well, the key trick here is to split the underlying set $\mathbb{R}^4$ into $\mathbb{R} \times \mathbb{R}^3$, thinking of each Quaternion as consisting of a ***real part*** $a \in \mathbb{R}$ and an ***imaginary part*** $\vec{v} \in \mathbb{R}^3$. Our multiplication formula now reads:

$$\begin{aligned}(a, \vec{v}) \cdot (b, \vec{u})& \\ = (ab - v_1 u_1 - v_2 u_2 - v_3 u_3)& \\ + (a u_1 + b v_1 + v_2 u_3 - v_3 u_2) i& \\ + (a u_2 - v_1 u_3 + b v_2 + v_3 u_1) j& \\ + (a u_3 + v_1 u_2 - v_2 u_1 + b v_3) k&\end{aligned}$$

which you might notice consists exclusively of familiar vector operations:

1. $ab - v_1u_1 - v_2u_2 - v_3u_3 = ab - \langle \vec{v}, \vec{u} \rangle$
2. $\begin{pmatrix} au_1 \\ au_2 \\ au_3 \end{pmatrix} = a\vec{u}, \begin{pmatrix} bv_1 \\ bv_2 \\ bv_3 \end{pmatrix} = b\vec{v}$
3. $\begin{pmatrix} v_2u_3 - v_3u_2 \\ v_1u_3 - v_3u_1 \\ v_1u_2 - v_2u_1 \end{pmatrix} = \vec{v} \times \vec{u}$

Putting everything together, we can now write out our multiplication formula in a much nicer form:

$$(a, \vec{v}) \cdot (b, \vec{u}) = (ab - \langle \vec{v}, \vec{u} \rangle, a\vec{u} + b\vec{v} + \vec{v} \times \vec{u}),$$

Exercise 13.4.4. *Verify that quaternion multiplication is associative and distributes over vector addition.*

Exercise 13.4.5. *Let $p = (a, \vec{v}) \in \mathbb{H}$. Then $pq = qp$ holds for all $q \in \mathbb{H}$ if and only if $\vec{v} \neq 0$.*

Similar to complex numbers, we can define ***quaternion conjugation***:

Definition 13.4.6: Quaternion Conjugation

Just like for $\mathbb{C}$, we define the ***conjugate*** of a quaternion $(a, \vec{v})$ to be the quaternion

$$\overline{(a, \vec{v})} = (a, -\vec{v})$$

An immediate application of quaternion conjugation is finding a formula for the inverse of a quaternion, which ones again ends up working out very similarly to the complex case - We have

$$\begin{aligned} \overline{(a, \vec{v})} \cdot (a, \vec{v}) &= (a, -\vec{v}) \cdot (a, \vec{v}) \\ &= (a^2 - \langle -\vec{v}, \vec{v} \rangle, a\vec{v} - a\vec{v} + (-\vec{v} \times \vec{v})) \\ &= \left(a^2 + \|\vec{v}\|^2, \vec{0}\right) \in \mathbb{R} \end{aligned}$$

And thus we can simply divide by this value to get our inverse formula:

Corollary 13.4.7: Inverse Quaternion

Let $(a, \vec{v}) \in \mathbb{H}_{\neq 0}$. Then $(a, \vec{v})$ has a multiplicative inverse, which is given by:

$$(a, \vec{v})^{-1} = \frac{(a, -\vec{v})}{a^2 + \|v\|^2}.$$

Definition 13.4.8. Let $q = (a, \vec{v}) \in \mathbb{H}$. Then

$$|q| := \sqrt{\overline{q} \cdot q} = a^2 + v_1^2 + v_2^2 + v_3^2 = \|q\|_2$$

Note how much this once again resembles both the real case $|x| = \sqrt{x^2} = \|x\|_2$ and the complex case $|z| = \sqrt{\overline{z}z} = \|z\|_2$.

13.5 Quaternions and $\mathbb{R}^3$

Theorem 13.5.1: Quaternions encode rotations

Let $\|\vec{v}\| = 1$ and

$$q = (\cos\varphi, \vec{v}\sin\varphi) \in \mathbb{H}$$

Then $qw\overline{q} \in \mathbb{R}^3$ is the vector obtained by rotating w around the axis $\{\lambda\vec{v} : \lambda \in \mathbb{R}\}$ by the angle 2φ.

Since 2π corresponds to a full rotation, φ and $\varphi + \pi$ describe the same rotation. Therefore, there are two different quaternions q and $-q$ associated with each rotation - in topological terms, we say that the quaternions form a ***double cover*** of the space of all rotations.

This phenomenon is actually closely related to the theory of ***spin*** in quantum physics. All known quantum particles that constitute ordinary matter, including protons, neutrons, electrons, neutrinos, and quarks, are ***spin-$\frac{1}{2}$***, meaning that they exhibit a curious behavior where a rotation of $360°$ leaves them in a different state than they were before, from which they return to their original state only after being rotated by another $360°$ around the same axis. This lets us mathematically model particles as quaternions, such that a $360°$ rotation transforms a particle from a state $q \in \mathbb{H}$ to the "rotationally equivalent" state $-q \in \mathbb{H}$.

Corollary 13.5.2. For every isometry $F : \mathbb{R}^3 \to \mathbb{R}^3$, there exist $u \in \mathbb{R}^3$ and $q \in \mathbb{H}$ such that we have either

1. $F(w) = u + qw\overline{q}$, if F preserves orientation, or
2. $F(w) = u + q\overline{wq}$, if F doesn't preserve orientation.

This representation turns out to be a lot more computationally efficient than other representations of the same rotation. Other, more intuitively "trivial" representations of rotation also run into issues such as ***gimbal lock***, where they may get "stuck" in certain orientations.

This makes quaternions surprisingly practical in many applied contexts, including game development and animation[1], robotics, and three-dimensional image processing.

[1] For an introduction to quaternions in game development, I highly recommend this excellent talk by Freya Holmer.

Chapter 14

Linear Maps

14.1 Vector Spaces

Intuitively, a vector space is any algebraic structure V equipped with a field $\mathbb{F}$, such that elements of V can be added to elements of V or multiplied by elements of $\mathbb{F}$, and these operations behave how vector addition and scalar addition behave in $\mathbb{R}^n$. More precisely:

Definition 14.1.1: Vector Spaces

A ***vector space V over a field of scalars*** $\mathbb{F}$ is a structure $(V, \mathbb{F}, +, .)$ consisting of a ***vector addition operation*** $+ : V \times V \to V$ and a ***scalar multiplication operation*** $. : \mathbb{F} \times V \to V$ such that:

1. vectors form a commutative group under vector addition, and
2. scalar multiplication is a homomorphism from $\mathbb{F}$ into the endomorphism ring of $(V, +)$.

The condition that $(V, +)$ is a commutative group should be simple to understand by now - it tells us that:

1. Vector addition is associative:

$$\begin{aligned} &\forall v, w, u \in V : \\ &(v + w) + u = v + (w + u) \end{aligned}$$

2. Vector addition is commutative:

$$\begin{aligned} &\forall v, w \in V : \\ &v + w = w + v \end{aligned}$$

3. There exists a vector $0_V \in V$ such that $v + 0_V = v$ for all $v \in V$.
4. For every vector $v \in V$, there exists an additive inverse $-v \in V$:

$$\begin{aligned} &\forall v \in V : \\ &\exists - v \in V : \\ &v + (-v) = 0_V \end{aligned}$$

As usual, we write $v - w := v + (-w)$ as an abbreviation.

The second condition might seem a bit more obtuse - however, all it tells us is that . assigns to each scalar $\lambda \in \mathbb{F}$ an endomorphism $\lambda. : V \to V$, such that:

1. Each multiplication operation is an endomorphism, i.e. scalar multiplication distributes over vector addition:

$$\begin{aligned}&\forall \lambda \in \mathbb{F} : \\ &\forall v, w \in V : \\ &\quad \lambda.(v + w) = \lambda.v + \lambda.w\end{aligned}$$

2. The map assigning operators to scalars is a homomorphism, i.e. scalar multiplication also distributes over scalar addition:

$$\begin{aligned}&\forall \lambda, \mu \in \mathbb{F} : \\ &\forall v \in V : \\ &\quad (\lambda + \mu).v = \lambda.v + \mu.v\end{aligned}$$

3. Scalar multiplication by the field identity $1_{\mathbb{F}}$ leaves vectors unchanged:

$$\begin{aligned}&\forall v \in V : \\ &\quad 1_{\mathbb{F}}.v = v\end{aligned}$$

Chapter 15

Affine Spaces

Affine spaces are structures similar to vector spaces, but where the notions of "points" and "directions" are separate.

Definition 15.0.1: Affine Space

A $\mathbb{k}$-***affine space*** is a tuple $(A, \vec{A})$, consisting of a set A of ***points*** and a $\mathbb{k}$-vector space $\vec{A}$ of ***translation vectors***, alongside a translation map $+ : A \times \vec{A} \to A$, such that:

1. Translation is associative:

$$\forall a \in A, \vec{v}, \vec{w} \in \vec{A} :$$
$$(a + \vec{v}) + \vec{w} = a + (\vec{v} + \vec{w})$$

2. For any pair of points, there exists a unique vector translating one point onto the other:

$$\forall a, b \in P_A :$$
$$\exists! \vec{v} \in \vec{A} : a + \vec{v} = b$$

The dimension of an affine space is simply the dimension of its associated vector space.

Any vector space V can be canonically turned into an affine space by taking V as both the set of points and the set of vectors. On the flipside, we can turn any affine space $(A, \vec{A})$ into a vector space by restricting our attention to translations of one "base point" $b \in A$, and then identifying any point $p \in A$ with the vector $\vec{v} \in \vec{A}$ translating b onto p.

For most practical purposes, it is therefore sufficient to work with a vector space as your base space for whatever math you are doing, but the non-canonicity of having to choose an arbitrary base point $b \in A$ to turn an affine space into a vector space means that some people prefer affine spaces for philosophical reasons - after all, what is the "base point" of our physical universe?

The distinction becomes a lot more important when considering affine subspaces,

which are essentialy vector subspaces translated by arbitrary points:

Definition 15.0.2: Affine subspaces

Let $(A, \vec{A})$ be an affine space. Then an affine subspace of $(A, \vec{A})$ is a tuple $(B, \vec{B})$ such that:

1. $B \subseteq A$.
2. $\vec{B}$ is a vector subspace of $\vec{A}$.
3. $\vec{B}$ is the set of translations between points in B, i.e. for any $b \in B$, $\vec{v} \in \vec{A}$, we have $b + \vec{v} \in B$ if and only if $\vec{v} \in \vec{B}$.

Thus, an affine subspace can always be written as

$$B = b + \vec{B} := \left\{ b + \vec{v} \;\middle|\; \vec{v} \in \vec{B} \subseteq \vec{A} \right\}$$

for an arbitrary base point $b \in B$.

An affine subspace of dimension 1 is known as a ***line***. An affine subspace of dimension 2 is known as a ***plane***. An affine subspace whose dimension is one lower than the space it resides in is known as an ***hyperplane***.

Definition 15.0.3. Two affine subspaces $(A, \vec{A})$, $(B, \vec{B})$ are called ***parallel*** if $\vec{A} = \vec{B}$.

Definition 15.0.4: Affine Maps

Let A, B be $\Bbbk$-affine spaces. Then a map $F : A \to B$ is called ***affine*** if there exists a linear map $\vec{F} : \vec{A} \to \vec{B}$ such that

$$\forall a \in P_a, \vec{v} \in \vec{A} : F(a + \vec{v}) = F(a) + \vec{F}(\vec{v})$$

As previously mentioned, if our affine space is a vector space, we can identify any point in P_A with the vector translating the origin onto that point, which lets us write:

$$F(\vec{v}) := F(0 + \vec{v}) = F(0) + \vec{F}(\vec{v}) := \vec{F}(\vec{v}) + \vec{w}$$

thus, an affine map between vector spaces is simply a linear map composed with a translation.

Theorem 15.0.5: Intercept Theorem

Let $v_1, v_2 \in \mathbb{R}$ be linearly independent. Let L_1 be an affine line through v_2 and v_1. Then if we have $\alpha, \beta \in \mathbb{R}_{\neq 0}$, and L_2 is a parallel line through αv_2 and βv_1, we must have $\alpha = \beta$.

Proof. Since L_1 and L_2 are parallel, their direction vectors must point in

the same direction, which means there must be a $\mu \in \mathbb{R}_{\neq 0}$ such that

$$\begin{aligned} & \mu(v_2 - v_1) = \mu(\alpha v_2 - \beta v_1) \\ \implies & \mu v_2 - \mu v_1 = \alpha v_2 - \beta v_1 \\ \implies & \mu v_2 - \alpha v_2 = \mu v_1 - \beta v_1 \\ \implies & (\mu - \alpha) v_2 = (\mu - \beta) v_1 \end{aligned}$$

Since v_2 and v_1 are linearly independent, this implies $\mu - \alpha = \mu - \beta = 0$, i.e. $\alpha = \beta = \mu$. □

Theorem 15.0.6. An injective map $\Phi : \mathbb{R}^2 \to \mathbb{R}^2$ is linear if and only if it maps $\vec{0}$ to $\vec{0}$ and lines to lines.

Proof. $\Rightarrow$ Let Φ be linear. Let $g := \left\{\vec{p} + \lambda \cdot \vec{v}\right\}$ be a line. Then we have

$$\begin{aligned} \Phi(g) &= \left\{\Phi(\vec{p} + \lambda \cdot \vec{v})\right\} \\ &= \left\{\Phi(\vec{p}) + \lambda \cdot \Phi(\vec{v})\right\}, \end{aligned}$$

i.e. $\Phi(g)$ is also a line.

$\Leftarrow$ Since Φ maps $\vec{0}$ to $\vec{0}$ and lines to lines, it must map lines through the origin to lines through the origin. Let

$$\begin{aligned} \Phi(e_1) &:= v_1 \\ \Phi[\lambda e_1] &:= L_1 = \{\lambda v_1\} \\ \Phi(e_2) &:= v_2 \\ \Phi[\lambda e_2] &:= L_2 = \{\lambda v_2\} \end{aligned}$$

Since Φ is injective, L_1 and L_2 can only intersect at $\{0\}$, and thus v_1 and v_2 must be linearly independent.

Now, let's take a closer look at images of individual points. Since $\Phi[\lambda e_1] = \{\lambda v_1\}$ and $\Phi[\lambda e_2] = \{\lambda v_2\}$, we must have

$$\begin{aligned} \Phi[\lambda e_1] &= \varphi_1(\lambda) v_1 \\ \Phi[\lambda e_2] &= \varphi_2(\lambda) v_2 \end{aligned}$$

for some $\varphi_1, \varphi_2 : \mathbb{R} \to \mathbb{R}$.

Now, notice that the line through λe_1 and λe_2 is parallel to the line through e_1 and e_2. Since Φ is injective, it must preserve parallelism, and so the line through v_1 and v_2 must be parallel to the line through $\varphi_1(\lambda) v_1$ and $\varphi_2(\lambda) v_2$. The intercept theorem now gives us $\varphi_1(\lambda) = \varphi_2(\lambda) = \mu$. We thus write $\varphi(\lambda) := \varphi_1(\lambda) = \varphi_2(\lambda)$.

Since Φ is injective, φ must be as well, since otherwise we would have some v with $\Phi(\lambda_1 v) = \Phi(\lambda_2 v)$ despite $\lambda_1 v \neq \lambda_2 v$. We have also have $\varphi(0) = 0$ and $\varphi(1) = 1$, since $\Phi(0) = 0$ and $\Phi(e_1) = v_1$.

Now, we want to show that $\Phi(v + w) = \Phi(v) + \Phi(w)$ holds for linearly independent v, w. Since Φ preserves parallelism, it must map parallelograms to parallelograms. Therefore, the parallogram spanned by v and w must be mapped to the parallogram spanned by $\Phi(v)$ and $\Phi(w)$. This means that the upper corner $v + w$ gets mapped to the upper corner $\Phi(v) + \Phi(w)$ which gives us $\Phi(v + w) = \Phi(v) + \Phi(w)$ as desired.

Putting our insights together, we now know that

$$\begin{aligned}\Phi(\lambda_1 e_1 + \lambda_2 e_2) &= \Phi(\lambda_1 e_1) + \Phi(\lambda_2 e_2) \\ &= \varphi(\lambda_1)\Phi(e_1) + \varphi(\lambda_2)\Phi(e_2)\end{aligned}$$

it only remains to be shown that φ is the identity. For this, consider that we have:

$$\begin{aligned}(\varphi(a) + \varphi(b))\Phi(e_1) &= \varphi(a)\Phi(e_1) + \varphi(b)\Phi(e_1) \\ &= \Phi(ae_1 + be_1 \\ &= \Phi((a + b)e_1) \\ &= \varphi(a + b)\Phi(e_1),\end{aligned}$$

which gives us $\varphi(a) + \varphi(b) = \varphi(a + b)$. For $a \neq 0$, the line through e_1 and ae_2 must point in the same direction as the line through be_1 and bae_2, so the line through $\Phi(e_1)$ and $\varphi(a)\Phi(e_2)$ must be parallel to the line through $\varphi(b)\Phi(e_1)$ and $\varphi(ab)\Phi(e_2)$, from which the intercept theorem gives us

$$\begin{aligned}\frac{\varphi(ab)}{\varphi(a)} &= \varphi(b) \\ \implies \varphi(ab) &= \varphi(a)\varphi(b)\end{aligned}$$

Since we have $\varphi(0) = 0$, $\varphi(1) = 1$, $\varphi(a + b) = \varphi(a) + \varphi(b)$, $\varphi(ab) = \varphi(a)\varphi(b)$, our map $\varphi : \mathbb{R} \to \mathbb{R}$ must be a field homomorphism, of which the identity is the only one. Thus, Φ is linear.

□

Exercise 15.0.7. *An injective map between real affine spaces of dimension ≥ 2 is affine if and only if it maps lines to lines.*

Chapter 16

Eigenvectors and Eigenvalues

Definition 16.0.1: Eigenvalues and Eigenvectors

Let k be a field, $F : V \to V$ an Endomorphism, and $\lambda \in k$. Then the ***λ-eigenspace*** of F is the set

$$\begin{aligned}\mathrm{Eig}(F, \lambda) &:= \left\{\vec{v} \in V : F(\vec{v}) = \lambda\vec{v}\right\} \\ &= \left\{\vec{v} \in V : (F - \lambda \,\mathrm{id}_V)(\vec{v}) = 0\right\} \\ &= \ker(F - \lambda \,\mathrm{id}_V)\end{aligned}$$

If $F(\vec{v}) = \lambda\vec{v}$, with $\vec{v} \neq \vec{0}$, we say that $\vec{v}$ is an ***eigenvector*** of F with ***eigenvalue*** λ.

Note that we exclude $\vec{0}$ from being an eigenvector, since $F(\vec{0}) = \lambda\vec{0}$ trivially holds for every endomorphism. Thus every λ would be an eigenvalue, which would make the definition pointless. We do, however, allow 0 as an eigenvalue - the 0-eigenspace of F is simply its kernel. Sometimes it ends up being useful to rephrase our definition of eigenspaces in terms of kernels:

$$\begin{aligned}\mathrm{Eig}(F, \lambda) &= \left\{\vec{v} \in V : F(\vec{v}) = \lambda\vec{v}\right\} \\ &= \left\{\vec{v} \in V : F(\vec{v}) - \lambda\vec{v} = 0\right\} \\ &= \left\{\vec{v} \in V : (F - \lambda \,\mathrm{id}_V)(\vec{v}) = 0\right\} \\ &= \ker(F - \lambda \,\mathrm{id}_V)\end{aligned}$$

This shows that $\mathrm{Eig}(F, \lambda)$ is a subspace of V - by definition, these subspaces must then be the maximal subspaces that are closed under application of F, i.e. $F(\mathrm{Eig}(F, \lambda)) = \mathrm{Eig}(F, \lambda)$.

Exercise 16.0.2. *In an informal, intuition-based way:*

1. *Find the real eigenvalues and eigenspaces of a reflection of $\mathbb{R}^2$ along an axis $\left\{\lambda\vec{v} : \lambda \in \mathbb{R}\right\}$*
2. *Find a rotation of $\mathbb{R}^2$ that has no real eigenvalues. Find two different rotations of $\mathbb{R}^2$ that do have real eigenvalues. Show that any rotation of $\mathbb{R}^2$ has complex eigenvalues.*
3. *Find the eigenvectors of the derivative map $\frac{d}{dx}$*

Our characterisation $\mathrm{Eig}(F, \lambda) = \ker(F - \lambda \,\mathrm{id}_V)$ means that, if F is finite-dimensional, we can find all eigenvectors of F with a given eigenvalue λ by solving $(F - \lambda \,\mathrm{id}_V)\vec{v} = 0$, which is simply a systen of linear equations. Finding the correct eigenvalues ends up being more tricky - in the section after this one, we will work towards proving the following key result:

Proposition 16.0.3. The eigenvalues of F are given by the zeroes of the ***characteristic polynomial*** $\chi_F(\lambda) := \det(F - \lambda \,\mathrm{id}_V)$.

However, before we do so, we want to familiarize ourselves with some of the more basic properties of eigenvectors and eigenvalues.

One important application is that finding eigenvectors lets us diagonalize endomorphisms, greatly easing computation afterwards:

Theorem 16.0.4. Let V be a vector space over a field k. Let $B = (b_1, \ldots, b_n)$ be a basis of V, and let F be an endomorphism represented in B by a matrix A. Then:

1. A basis vector b_i is an eigenvector of F iff. the j-th column of A has the form λe_j.
2. F is diagonalizable iff. there exists a basis of V consisting of eigenvectors of F.

Theorem 16.0.5. Let V be a vector space over a field k. Let $F : V \to V$ be an endomorphism. Then eigenvectors of F with different eigenvalues are linearly independent.

Corollary 16.0.6. Let V be an n-dimensional vector space over a field k. Let $F : V \to V$ be an endomorphism. Then:

1. F has no more than n distinct eigenvalues,
2. If $(\lambda_i)_{i \in I}$ are mutually distinct eigenvalues of F, the sum
$$\sum_{i \in I} \mathrm{Eig}(F, \lambda_i) \subseteq V$$
is direct.
3. F is diagonalizable if and only if V is a direct sum of eigenspaces of F.
4. If F has n distinct eigenvalues, it is diagonalizable.

Corollary 16.0.7. Let V, W be n-dimensional vector spaces over a field k. Let $F : V \to V$, $G : W \to W$ be endomorphisms. Then there exists a unique endomorphism $P : V \to W$ such that $P \circ F = G \circ P$ if and only if F and G have the same eigenvalues.

16.1 Characteristic Polynomials

Theorem 16.1.1: Characterstic Polynomial

Let V be a finite-dimensional vector space over a field $\Bbbk$. Let $F : V \to V$ be an endomorphism. Then $\lambda \in \Bbbk$ is an eigenvalue of F if and only if

$$\chi_F(\lambda) = \det(\lambda . \operatorname{id}_V - F) = 0_\Bbbk$$

We call χ_F the ***characteristic polynomial of F***.

Proof. We know that $\lambda \in \Bbbk$ is an Eigenvalue of F if and only if

$$\ker(F - \lambda \operatorname{id}_V) \neq \{0_V\},$$

i.e. if and only if $F - \lambda \operatorname{id}_V$ isn't injective.

We also know that $F - \lambda \operatorname{id}_V$ isn't injective if and only if it isn't surjective, i.e. if and only if it isn't invertible, i.e. if and only if

$$\det(\lambda . \operatorname{id}_V - F) = \chi_F(\lambda) = 0_\Bbbk.$$

□

Exercise 16.1.2. *Show that every endomorphism $F : V \to V$ of a finite-dimensional complex vector space V has at least one eigenvalue.*

Exercise 16.1.3. *Show that the constant term of a characteristic polynomial $\chi_F(\lambda)$ is given by $(-1)^n \cdot (\det F)$.*

16.2 Algebras, Cayley-Hamilton and Frobenius

Definition 16.2.1: Algebra over a Ring

An ***algebra over a commutative ring R*** is an R-module A equipped with a bilinear multiplication operation $\cdot : A \times A \to A$. We call A commutative, associative or unital if the multiplication operator $\cdot$ is commutative, associative, or has an identity element.

Exercise 16.2.2.

1. *Show that every commutative ring forms an algebra over itself.*
2. *Show that the set $C(M, \mathbb{R})$ of continuous functions $M \to \mathbb{R}$, where M is a metric space, forms a commutative associative unital algebra under pointwise addition and multiplication.*

Theorem 16.2.3: Polynomial Evaluation Map

Let R be a commutative ring, and let A be a commutative unital algebra over R. Then, for every $a \in A$, there exists a unique algebra homomorphism $\mathrm{ev}_a : R[X] \to A$ such that:

1. $\mathrm{ev}_a(r) = 1_A.r$ for all constant polynomials $r \in R[X]$,
2. $\mathrm{ev}_A(X) = X$.

We call ev_a the ***evaluation map of $R[X]$ at a***.

Informally, this means that we can take elements of an algebra A over R and plug them into polynomials that were originally only defined on R, and we won't run into issues. In particular, we can plug matrices into polynomials over an underlying ring! Since ev_a is unique, we will generally just write $P(a) := \mathrm{ev}_a(P)$ from now on.

Theorem 16.2.4: Cayley-Hamilton

Let R be a commutative unital ring. Let $A \in R^{n\times n}$. Then $\chi_A(A) = 0$.

16.3 Minimal Polynomials

Theorem 16.3.1: Minimal polynomial

Let F be an endomorphism of a vector space V over a field $\Bbbk$. Then there exists a unique polynomial $\mu_F \in \Bbbk[X]$ such that:

1. μ_F is either normalized or equal to $0_{\Bbbk[X]}$
2. μ_F is a divisor of every $P \in \Bbbk[X]$ with $P(F) = 0_{\mathrm{End}_\Bbbk(V)}$

We call μ_F the ***minimal polynomial of F***.

Chapter 17

Normal Forms of Endomorphisms

17.1 Decomposing Vector Spaces using Polynomials

Exercise 17.1.1. *Let G be an abelian group. Show that G becomes a $\mathbb{Z}$-module when equipped with the operation $n.g = \underbrace{g + \ldots + g}_{n \text{ times}}$.*

Exercise 17.1.2. *Let V be a vector space over a field $\Bbbk$. Show that V becomes an $\mathrm{End}(V)$-Module when equipped with the operation $F.v = F(v)$.*

Theorem 17.1.3. Any $\Bbbk[X]$-module is the same thing as a $\Bbbk$-vector space V equipped with a specified endomorphism F, and vice versa.

Proof.

1. Let V be a $\Bbbk$-vector space. We know from the previous exercise that V is an $\mathrm{End}(V)$-module. Thus, the polynomial evaluation map $\mathrm{ev}_F : \Bbbk[X] \to \mathrm{End}(V)$ turns V into $\Bbbk$-module via

$$P.v = \mathrm{ev}_F(P)(v) = P(F)(v)$$

2. Now let M be a $\Bbbk[X]$-module. Then M becomes a $\Bbbk$-module, i.e. a $\Bbbk$-vector space, if we restrict scalar multiplication to only accept constant polynomials. Then it can easily be seen that $G(v) = X.v$ is a linear transformation.

Now, note that if we start with a vector space (V, F), then use our construction to get a module M, and then turn it back into a vector space, we will obtain our original F again, since

$$G(v) = X._M v = X(F)(v) = F(v)$$

Conversely, if we start with a module M, then turn it into a vector space (V, F), and then back into a module M, we reobtain our original scalar multiplication, since

$$P._V v = P(G)(v) = P(X._M)(v) = P._M v$$

Therefore, the two directions of our construction are inverse to each other, i.e. we actually have a bijection. □

Thus, given a $\mathbb{k}[X]$-module M, we can write $M = (V, F)$

Exercise 17.1.4. *Let $M = (V, F)$ and $N = (W, G)$ be $\mathbb{k}[X]$-Modules. Then a $\mathbb{k}[X]$-linear map $H : M \to N$ is $\mathbb{k}$-linear and additionally fulfills $H \circ F = G \circ H$, and any $\mathbb{k}$-linear map H fulfilling these properties is $\mathbb{k}[X]$-linear.*

Let $M = (V, F)$ be a $\mathbb{k}[X]$-Module. Then, by definition, a subspace $U \subseteq V$ is a $\mathbb{k}[X]$-submodule of M if and only if U is closed under scalar multiplication with elements of $\mathbb{k}[X]$.

Exercise 17.1.5. *Show that U is closed under scalar multiplication with elements of $\mathbb{k}[X]$ if and only if $Xu \in U$, which itself holds if and only if $F(u) \in U$ holds for all $u \in U$, i.e. U is **F-invariant**.*

Exercise 17.1.6.
1. *Show that the scalar multiplication map $P. = P(F)$ is linear.*
2. *Conclude that $\ker(P(F))$ and $\operatorname{im}(P(F))$ are submodules of M.*
3. *Conclude that $\ker(P(F))$ and $\operatorname{im}(P(F))$ are F-invariant.*

Theorem 17.1.7: Structure theorem for modules over a Euclidean ring

Let M be a module over a euclidean domain R. Let $a = bc \in R$ such that b and c share no divisors and $a.m = 0$ for all $m \in M$. Then we have:

1. $\ker(b.) = \operatorname{im}(c.)$,
2. $\operatorname{im}(b.) = \ker(c.)$,
3. $M = \ker(b.) \oplus \ker(c.)$.

Proof.
1. Let $c.m \in \operatorname{im}(c.)$ for $m \in M$. Then we have $b.(c.m) = a.m = 0$, i.e. $\operatorname{im}(c.) \subseteq \ker(b.)$. Now, applying the Euclidean algorithm gives us $1 = k_1 b + k_2 c$. If we now let $m \in \ker(b.)$, we get:

$$\begin{aligned} m &= 1m \\ &= (k_1 b + k_2 c)m \\ &= k_1 \underbrace{(bm)}_{=0} + k_2(cm) \\ &= c(k_2 m) \in \operatorname{im}(c.) \end{aligned}$$

   Thus, we also have $\ker(b.) \subseteq \operatorname{im}(c.)$, i.e. $\ker(b.) = \operatorname{im}(c.)$.

2. $\ker(c.) = \operatorname{im}(b.)$ follows from the same argument by swapping the roles of b and c.

3. Now, let $m \in M$. Then, applying our identity from before, we get

$$\begin{aligned} m &= (k_1 b + k_2 c)m \\ &= b(k_1 m) + c(k_2 m) \\ &\in \operatorname{im}(b.) + \operatorname{im}(c.) \\ &= \ker(b.) + \ker(c.), \end{aligned}$$

which gives us $M = \ker(b.) + \ker(c.)$ as desired.

If we have $m \in \ker(b.) \cap \ker(c.)$, we get

$$m = k_1(b(m)) + k_2(c(m)) = 0,$$

so $\ker(b.) \cap \ker(c.) = \{0\}$, so our sum is direct.

□

Corollary 17.1.8. Let G be an abelian group. Let $n = mk \in \mathbb{Z}$ such that m and k share no divisors and $n \cdot g = 0$ for all $g \in G$. Then we have

$$G = \ker(m\cdot) \oplus \ker(k\cdot).$$

Corollary 17.1.9: Chinese Remainder Theorem

Corollary 17.1.10. Let V be a vector space over a field $\Bbbk$ and $F : V \to V$ be an Endomorphism. Let $P = QR \in \Bbbk[X]$ such that Q and R share no divisors and $P.v = P(F)(v) = 0$ for all $v \in V$. Then we have

$$V = \ker(Q(F)) \oplus \ker(R(F)).$$

Corollary 17.1.11: P-Generalized Eigenspace Decomposition

Let V be a vector space over a field $\Bbbk$ and $F : V \to V$ be an Endomorphism. Let $P \in \Bbbk[X]$ be a normed polynomial and let $P = P_1^{n_1} \cdot \ldots \cdot P_k^{n_k}$ be its prime factorization. Then we have

$$V = \bigoplus_{i=1}^{k} \ker(P_i^{n_i}(F))$$

17.2 The Jordan Normal Form

Corollary 17.2.1: Generalized Eigenspace Decomposition

Let V be a vector space over a field $\mathbb{k}$ and $F : V \to V$ be an Endomorphism. Assume that its minimal polynomial μ_F splits into linear factors over $\mathbb{k}$, i.e.

$$\mu_F(x) = \prod_{i=1}^{k} (x - \lambda_i)^{n_i}$$

with $\lambda_i \in \mathbb{k}$. Then we have

$$V = \bigoplus_{i=1}^{k} \ker((F - \lambda_i \operatorname{id}_V)^{n_i})$$

We call $\ker((F - \lambda_i \operatorname{id}_V)^{n_i})$ the ***generalized λ_i-Eigenspace of F*** and denote it $H(F, \lambda_i)$ (from the German "Hauptraum").

Exercise 17.2.2. *Show that $H(F, \lambda_i)$ contains all eigenvectors of F with eigenvalue λ_i.*

Exercise 17.2.3. *Show that $H(F, \lambda_i)$ is an F-invariant subspace of V, i.e. a subspace such that $F(H(F, \lambda_i)) \subseteq H(F, \lambda_i)$*

Exercise 17.2.4. *Let $V = V_1 \oplus V_2$ be a decomposition of V into F-invariant subspaces. Let B_1 be a basis of V_1, and let B_2 be a basis of V_2. Show that:*

1. *$B = B_1 \cup B_2$ is a basis of V*
2. *If the restriction $F|_{V_1}$ of F to V_1 is represented in basis B_1 as a matrix A_1, and the restriction $F|_{V_2}$ of F to V_2 is represented in basis B_2 as a matrix A_2, then F is represented in basis B as*

$$A = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

Exercise 17.2.5. *Let $F : V \to W$ be a linear map. Let A be a basis of $\ker F$, and let B be a tuple of vectors such that $F(B)$ is a basis of $\operatorname{im} F$. Then $A \cup B$ is a basis of V.*

We now want to make use of this to construct a Jordan basis B of a nilpotent Endomorphism N:

Corollary 17.2.6. Let $N : V \to V$ be a nilpotent endomorphism. Then there exists a basis B of V such that:

1. $B \cup \{0\}$ is F-invariant, i.e. if $b \in B$, then we have $F(b) \in B$ or $F(b) = 0$.
2. For every $b \in B$, there exists at most one element $b' \in B$ with $F(b') = b$.

We call such a basis a ***Jordan basis***.

> *Proof.* We prove this theorem in three steps: We construct a basis B, then show it is F-invariant, then show that each element of B has at most one preimage in B.

1. We construct our basis by induction over $n = \dim V$.

   If $n = 0$, then the empty set is a basis (since a sum over the empty set equals the additive identity, i.e. the zero vector) which (vacuously) fulfills our desired properties.

   Now, since the kernel of N is non-trivial, we have $\dim \operatorname{im} N < \dim V$. Therefore, by induction, we can assume we already have a Jordan basis A of $\operatorname{im} N$. We split this basis A into two parts:

   (a) $A_0 := \{a \in A \mid N(a) = 0\}$,

   (b) $A_{\neq 0} := \{a \in A \mid N(a) \neq 0\}$.

   A_0 is by definition a set of linearly independent vectors contained in $\ker A$. By the Steinitz exchange lemma, we can add a set C of Linearly independent vectors such that $A_0 \cup C$ is a basis of $\ker A$.

   $A_{\neq}0$ is the set of all A such that $N(A_{\neq 0}) \subseteq A$. We add a set D containing preimages of all the remaining elements of A, which must exist since $A \subseteq \operatorname{im} N$. By construction, we now have $F(A_{\neq 0} \cup D) = A$.

   Putting everything together, $A_0 \cup C$ is a basis of $\ker N$, and $F(A_{\neq 0} \cup D) = A$, which is a basis of $\operatorname{im} N$. Therefore, by exercise 17.2.5, $B = A \cup C \cup D$ is a basis of V.

2. Now, B must be F-invariant since A is F-invariant by definition, C is contained in the kernel of F and thus $F(B) = \{0\}$, and D consists of preimages of A, i.e. $F(D) = A \subseteq B$.

3. Finally, the preimages of A in B are contained in either $A_{\neq 0}$ or D, which are disjoint by construction.

□

Corollary 17.2.7. Let $F : V \to V$ be a matrix whose characteristic polynomial has $n = \dim(V)$ roots. Then there exists a basis B of V such that ${}_BF_B$ is a block diagonal matrix consisting of Jordan blocks.

Proof. Generalized Eigenspace decomposition 17.2.1 shows us that V is the direct sum of the generalized Eigenspaces of F. We know by exercise 17.2.3 that the generalized Eigenspaces of F are F-invariant. Therefore, exercise 17.2.4 tells us that if we find a basis B_i for each generalized Eigenspace $H(F, \lambda_i)$ such that $F|_{H(F,\lambda_i)}$ is represented by a matrix J_i in Jordan normal form, and we combine our bases into a basis $B = \bigcup_{i=1}^{n} B_i$ of V as a whole, then the matrix representation of F in basis B will be a block diagonal matrix consisting of these J_i, and thus F will also be in

Jordan normal form.

Now, let F_i be the restriction of $(F - \lambda_i \operatorname{id}_V)$ to $H(F, \lambda_i)$. By definition of the generalized Eigenspace, we have

$$H(F, \lambda_i) = \ker((F - \lambda_i \operatorname{id}_V)^{n_i})$$

therefore, by construction, F_i is nilpotent. Therefore, by theorem **??**, there exists a basis B_i such that ${}_{B_i}F_{iB_i}$ is in Jordan normal form, with zeroes on the diagonal. But then, since we have

$$F|_{H(F,\lambda_i)} = F_i + \lambda \mathrm{id}_{H(F,\lambda_i)}$$

the matrix representation of $F|_{H(F,\lambda_i)}$ is also in Jordan form, since we simply have to add λ to the diagonal of ${}_{B_i}F_{iB_i}$.

Therefore, the restriction to F to any of its generalized Eigenspaces is in Jordan normal form in basis B_i. As discussed at the beginning of this proof, this means F is in Jordan normal form in basis $B = \bigcup_{i=1}^{n} B_i$. □

Exercise 17.2.8. *Let F be an endomorphism, and let J be its Jordan normal form. Then:*

1. *The algebraic multiplicity of each eigenvalue of F corresponds to the number of times that eigenvalue appears on the diagonal of J.*
2. *The geometric multiplicity of any eigenvalue λ of F corresponds to the number of λ-Jordan blocks with that eigenvalue in J.*
3. *The exponent of $(X - \lambda)$ in the minimal polynomial of F corresponds to the size the of largest λ-Jordan block in J.*

17.3 The Weierstrass Normal Form

Chapter 18

Inner Product Spaces

18.1 Inner Products

Definition 18.1.1. Let $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Let V and W be vector spaces over $\mathbb{k}$. Then a map $\varphi : V \to W$ is called ($\mathbb{k}$-) ***semilinear*** or ***antilinear***, if we have

1. ***Addivity:*** $\forall u, v \in V : \varphi(u + v = \varphi(u) + \varphi(v)$
2. ***Antihomogeneity:*** $\forall v \in V, \lambda \in \mathbb{k} : \varphi(\lambda.v) = \overline{\lambda}.\varphi(v)$.

Notice that for $\mathbb{k} = \mathbb{R}$, there is no difference between linearity and antilinearity, since complex conjugation leaves real numbers unchanged.

Definition 18.1.2. Let $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Let V be a $\mathbb{k}$-vector space. Then a map $s : V \times V \to \mathbb{k}$ is called a ***sesquilinear form*** if it is antilinear in the first component and linear in the second component,i.e. the map $s_w : v \to s(v, w)$ is antilinear and the map $s_v : w \to s(v, w)$ is linear.

Exercise 18.1.3. *Let A be a matrix with entries in $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Show that the map*

$$s_A(v, w) := v^* A w$$

is sesquilinear. (Recall that v^* stands for the conjugated transpose of v.)

The latin prefix "sesqui-" means "one and a half" - a sesquilinear is a map that is linear in one component, and "sort of linear", i.e. "one-half-linear" in the other. Some sources require that a sesquilinear form is linear in the first component and antilinear in the second.

Definition 18.1.4. A sesquilinear form $s : V \times V \to \mathbb{k}$ is called a ***Hermitian form*** if it is ***conjugate symmetric***, i.e. $s(v, w) = \overline{s(w, v)}$.

Once again, notice that for $\mathbb{k} = \mathbb{R}$, this just means $s(v, w) = s(w, v)$.

Exercise 18.1.5. *Show that a matrix is Hermitian (i.e. fulfills $A = A^*$) if and only if its associated sesquilinear form $s_A(v, w) = v^* A w$ is Hermitian.*

Definition 18.1.6. A sesquilinear form $s : V \times V \to \mathbb{k}$ is called ***positive semidefinite*** if $s(v, v) \geq 0$ for all $v \in V$. It is called ***positive definite*** if it additionally fulfills $s(v, v) = 0$ if and only if $v = \vec{0}$.

Positive definiteness is the primary reason we are working with sesquilinear forms instead of bilinear forms - we need to be able to compare $s(v, v)$ to 0,

and there is no way to order all of $\mathbb{C}$ that is compatible with its field structure. However, if we have a sesquilinear Hermitian map, we get $s(v,v) = \overline{s(v,v)}$, i.e. $s(v,v) \in \mathbb{R}$, which means we can start comparing things again.

Definition 18.1.7. A ***scalar product*** or ***inner product*** on a vector space V over $\mathbb{R}$ or $\mathbb{C}$ is a positive definite Hermitian form. We call a real vector space equipped with a scalar product ***Euclidean***, and a complex vector space equipped with a scalar product ***unitary***.

Definition 18.1.8. The standard scalar product on $\mathbb{C}^n$ is given by

$$\langle v, w\rangle_{\mathbb{C}} = v^* w$$

Exercise 18.1.9. *Show that $\langle v, w\rangle_{\mathbb{C}}$ is actually a scalar product.*

Exercise 18.1.10. *Let $V = C^\infty([0,1],\mathbb{R})$ be the room of infinitely differentiable functions $[0,1] \to \mathbb{R}$. Show that the $\boldsymbol{L^2}$ scalar product*

$$\left\langle f, g\right\rangle_{L^2} = \int_0^1 f(t)g(t)\,dt$$

is actually a scalar product.

Definition 18.1.11. Let V be a vector space over $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Then a ***Norm*** is a map $\|\cdot\| : V \to \mathbb{k}$ fulfilling the following three axioms:

1. ***Positive definiteness:*** $\|v\| \geq 0$ and $\|v\| = 0$ iff. $v = \vec{0}$
2. ***Homogeneity:*** $\|\lambda.v\| = |\lambda|.\|v\|$
3. ***Triangle inequality:*** $\|v + w\| \leq \|v\| + \|w\|$

A norm is a way of assigning lengths to vectors. Once lengths of vectors are defined, one can easily define distances between vectors too:

Exercise 18.1.12. *Let $\|\cdot\|$ be a norm on a vector space V over a field $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Show that $d(v,w) := \|v - w\|$ is a metric on V.*

Theorem 18.1.13: Cauchy-Schwartz Inequality

Let s be a scalar product on a vector space V over a field $\mathbb{k}$. Then we have

$$|\langle v, w\rangle| \leq \sqrt{\langle v, v\rangle} \cdot \sqrt{\langle w, w\rangle}$$

One important property of scalar products is that they induce norms. Notice that the standard norm on $\mathbb{R}^n$ is just the square root of the standard scalar product on $\mathbb{R}^n$. This property generalizes to all scalar products:

Exercise 18.1.14. *Let $s : V \times V \to \mathbb{k}$ be a scalar product on a vector space V over a field $\mathbb{k}$. Let $\|v\|_s := \sqrt{s(v,v)}$. Show that:*

1. *$\|\cdot\|_s$ is a norm on V.*
2. *We have the* **parallelogram identity**

$$\|v + w\|_s^2 + \|v - w\|_s^2 = 2\|v\|_s^2 + 2\|w\|_s^2$$

3. *If $\mathbb{k} = \mathbb{R}$, we have the* **real polarisation formula**

$$s(v,w) = \frac{1}{4}\left(\|v + w\|_s^2 - \|v - w\|_s^2\right)$$

4. *If $\mathbb{k} = \mathbb{C}$, we have the* **complex polarisation formula**

$$s(v,w) = \frac{1}{4}\left(\|v+w\|_s^2 - \|v-w\|_s^2 - i\|v+i.w\|_s^2 + i\|v-i.w\|_s^2\right)$$

Exercise 18.1.15. *Find a norm on a vector space that cannot be constructed as the square root of a scalar product.*

Theorem 18.1.16. Let V be a vector space over $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Let $\|\cdot\|$ be a norm on V fulfilling the parallelogram identity. Then there exists a corresponding scalar product s such that $\|v\| = \sqrt{s(v,v)}$

18.2 Orthogonality

Definition 18.2.1. Let (V,s) be a vector space with scalar product over a field $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Then we call two vectors v,w ***orthogonal*** iff.

$$s(v,w) = 0$$

Exercise 18.2.2. *Let $B = (v_1,\dots,v_k)$ be a tuple of orthogonal vectors not containing the zero vector. Then B is linearly independent.*

Definition 18.2.3. A basis consisting of pairwise orthogonal vectors is called an ***orthogonal basis***. If the basis vectors are also normalized, i.e. have length 1, we call the basis an ***orthonormal basis***.

Exercise 18.2.4. *Let (V,s) be a vector space with scalar product. Show that $B = (b_1,\dots,b_n) \subset V$ is an orthonormal basis iff.*

$$s(b_i,b_j) = \delta_{ij}$$

(Recall that δ_{ij} is defined as the function that is 1 if $i = j$ and 0 otherwise).

Definition 18.2.5. Let (V,s) be a vector space with scalar product over a field $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Let $U \subseteq V$ be a subspace. Let $B_U := \{b_1,\dots,b_m\}$ be a basis of U. Then the **orthogonal projection of V onto U** is the map

$$\begin{aligned} p : V &\to U \\ v &\mapsto \sum_{i=1}^{m} g(b_i,v).b_i \end{aligned}$$

Exercise 18.2.6.

1. *Show that the orthogonal projection of V onto U leaves vectors in u unchanged.*
2. *Show that for each $v \in V$ and $u \in U$, u is orthogonal to $v - p(v)$.*
3. *Show that for each $v \in V$, $p(v)$ is the unique point in U for which $\left\|v - p(v)\right\|_s$ is minimal.*

Theorem 18.2.7. Let (V,s) be a vector space with scalar product over a field $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Let $(v_1,\dots,v_n)$ be a basis of V. Then there are unique vectors $(b_1,\dots,b_n) \subset V$ such that for all $p \in (1,\dots,n)$, we have:

1. $(b_1,\dots,b_p)$ is an s-orthonormal Basis of $\left\langle v_1,\dots,v_p\right\rangle \subseteq V$
2. $s(b_p,v_p) \in \mathbb{R}_{>0}$

These vectors b_p can be constructed inductively via $b_1 = \frac{v_1}{\|v_1\|}$ followed by projection onto the already constructed basis vectors, i.e.

$$w_p = v_p - \sum_{q=1}^{p-1} g(b_q, v_p).b_q$$
$$tb_p = \frac{w_p}{\|w_p\|}$$

Corollary 18.2.8. Every vector space with scalar product has an orthonormal basis.

18.3 Matrix Representations of Inner Products

Definition 18.3.1. Let s be a sesquilinear form on a finite dimensional vector space Vover a field $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Let $B = (b_1, \ldots, b_n)$ be a basis of V. Then the ***Gram matrix of B by g*** is defined as the matrix with entries

$$G_{ij} = s(b_i, b_j) \in \mathbb{k}^{n\times n}$$

Exercise 18.3.2. *Show that the Gram matrix of the standard basis of $\mathbb{R}^n$ by the scalar product is the identity matrix.*

Definition 18.3.3. A matrix $A \in \mathbb{k}^{n\times n}$ is called ***positive (semi-)definite*** if $s_A(v, w) = v^* Aw$ is positive (semi-)definite.

Exercise 18.3.4. *Let s be a sesquilinear form on a finite-dimensional vector space V over a field $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Let G be its Gram matrix. Then we have $s(v, w) = v^* Gy$ in basis B. In particular, s is a scalar product if and only if its Gram matrix is Hermitian and positive definite.*

Theorem 18.3.5. Let $A \in \mathbb{k}^{n\times n}$, where $\mathbb{k} = \mathbb{R}$ or $\mathbb{C}$. Then the following are equivalent:

1. A is hermitian and psitive definite.
2. ***Cholesky Decomposition:*** There exists an upper triangular Matrix B with positive real diagonal entries such that $A = B^*B$.
3. There exists an invertible Matrix B such that $A = B^*B$.
4. ***Sylvester-Hurwitz Criterion:*** A is Hermitian and and all of its square upper left submatrices A_r, known as its ***leading principal minors***, have strictly positive determinant.

18.4 Dual Spaces

Definition 18.4.1: Dual Space

Let V be a vector space over a field $\mathbb{k}$. Then we define $V^* = \mathrm{Hom}_{\mathbb{k}}(V, \mathbb{k})$. Elements of V^*, i.e. linear maps $V \to \mathbb{k}$, are known as ***linear forms***.

Theorem 18.4.2: Dual Basis

iven a basis $B = (b_1, \dots, b_n)$ of a vector space V, the set of linear forms $E = (\varepsilon_1, \dots, \varepsilon_n)$ given by

$$\varepsilon_p(e_q) = \delta_{pq}$$

is a basis of V^*, known as the ***dual basis*** of B.

Exercise 18.4.3. *Let $C^n(\mathbb{k}, \mathbb{k})$, $n \in 0, 1, \dots, \infty$ be the set of n-times continuously differentiable functions $\mathbb{k} \to \mathbb{k}$. Then for any $x_0 \in \mathbb{k}$, the map*

$$\cdot(x_0) : f \to f^{(n)}(x_0)$$

is a linear form on $C^n(\mathbb{k}, \mathbb{k})$ (where $f^{(n)}$ denotes the n-th derivative of f).

Exercise 18.4.4. *The map mapping an integrable function f to $\int_0^1 f(x)\,dx$ is a linear form. More generally, for any integrable $g : [0, 1] \to \mathbb{k}$, the map taking f to $\int_0^1 \overline{g(x)} \cdot f(x)\,dx$ is a linear form.*

Theorem 18.4.5. Let (V, g) be a $\mathbb{k}$-vector space with scalar product. Then g induces an injective antilinear map $g_1 : V \to V^*$ via

$$g_1(v) = g(v, \cdot)$$

A linear form $\alpha \in V^*$ is called ***representable*** by g if there exists a $v \in V$ such that $\alpha = g(v, \cdot)$.

Definition 18.4.6. Let V be a $\mathbb{k}$-vector space. Then a ***antilinear form*** is an antilinear forms $\gamma : V \to \mathbb{k}$. The ***antidual space*** $\overline{V}^*$ of V is the space of antilinear maps $V \to \mathbb{k}$.

Theorem 18.4.7. Let (V, g) be a finite-dimensional vector space over a field $\mathbb{k}$. Then the maps $g : V \to V^*$ and $\overline{g} : V \to V^*$ are bijective.

Corollary 18.4.8. Let (V, g) be a finite-dimensional vector space. Then all linear and antilinear forms on V are representable by g.

Definition 18.4.9. Let (V, g) and (W, h) be $\mathbb{k}$-vector spaces with scalar products. Then a linear map $F : V \to W$ is called ***adjointable*** if there exists a map $G : W \to V$ such that

$$g(G(w), v) = h(w, F(v)$$

for all $v \in V$ and $w \in W$. Such a map G is known as the ***adjoint of F***, and is written $G = F^*$. If $F = F^*$, we call F ***self-adjoint***.

Proposition 18.4.10. If it exists, the adjoint of a given map is unique.

Proposition 18.4.11. Adjoint maps are linear.

Proposition 18.4.12. If G is the adjoint of F, then F is the adjoint of G, i.e. we have $(F^*)^* = F$.

Proposition 18.4.13. If A is a matrix representation of F by two orthonormal bases, then the conjugate transpose A^* is a matrix representation of F^*.

Definition 18.4.14. Let V and W be $\Bbbk$-vector spaces, with $\Bbbk$ a field. Let $F : V \to W$ be a linear map. Then the ***dual*** of F is the map

$$\begin{aligned} F^\vee : W^* &\to V^* \\ \beta &\mapsto \beta \circ F \end{aligned}$$

i.e.

$$F^\vee(\beta)(\vec{v}) = \beta(F(\vec{v}))$$

for $\beta \in W^*$, $\vec{v} \in V$.

We can construct a dual of any linear map this way, while linear maps involving infinite-dimensional vector spaces are generally not adjoinable.

Exercise 18.4.15. *Verify the following properties of the dual map:*

1. $(\mathrm{id}_V)^\vee = \mathrm{id}_{V^*}$
2. *Let $F : V \to W$ and $G : U \to V$ be linear, then $(F \circ G)^\vee = G^\vee \circ F^\vee$*
3. *The dual of a linear map is linear.*
4. *If $F : V \to W$ is given by $F(\vec{v}) = A \cdot \vec{v}$ for some matrix A, then $F^\vee(\beta) = \beta \cdot A$, where elements $\beta \in W^*$ are represented as row vectors.*

Theorem 18.4.16. Let $U \subseteq V$ be vector spaces. Let $\iota : U \to V$ be the inclusion map. Let $q : V \to V/U$ be the quotient map. Then $(V/U)^* \cong \operatorname{im}\ q^\vee = \ker\ \iota^\vee$.

Proof. Since q maps V to V/U, we have $q^\vee : (V/U)^* \to \operatorname{im}\ q^* \subseteq V^*$. Since q is linear, so is $q^\vee$. By definition, any map surjectively maps onto its image. Therefore, all we need to show to obtain $(V/U)^* \cong \operatorname{im}\ q^\vee$ is injectivity.

So, let $\beta \in \ker\ q^\vee \subseteq (V/U)^*$, i.e. let $q^\vee(\beta) = \beta(q)$ be the zero map. Since q is surjective, $\beta(q(v)) = 0$ for all $v \in V$ already implies $\beta(w) = 0$ for all $w \in V/U$, so β must have already been the zero map.

Finally, we want to show $\operatorname{im}\ q^\vee = \ker\ \iota^\vee$.

First off, recall that $\iota^\vee : V^* \to U^*$. By the universal property of the quotient map, for every $\beta \in \ker \iota^\vee \subseteq V^*$, there exists $\overline{\beta} : V/U \to \Bbbk$ (i.e. $\overline{\beta} \in (V/U)^*$) with $\beta = \overline{\beta} \circ q = \overline{\beta}(q) \in \operatorname{im}\ q^\vee$, so we have $\ker \iota^\vee \subseteq \operatorname{im}\ q^\vee$.

Conversely, for any $\beta \in (V/U)^*$ and $v \in (V/U)$, we have

$$q^\vee(\beta)(v) = \beta(q(v)) = \beta(0) = 0,$$

so we also have $\operatorname{im} q^\vee \subseteq \ker \iota^\vee$. □

Theorem 18.4.17

Let (V, g) and (W, h) be $\mathbb{k}$-vector spaces with scalar products. Let $h_1 : W \to W^*$ and $g_1 : V \to V^*$ be the antilinear maps induced by g and h, i.e.

$$h_1(\vec{w}) = h(\vec{w}, \cdot)$$
$$g_1(\vec{v}) = g(\vec{v}, \cdot)$$

Let $F : V \to W$ be linear. Then $g_1 \circ F^* = F^\vee \circ h_1$.

Proof. Let $v \in V$, $w \in W$, then we have:

$$\begin{aligned} g_1(F^*(w))(v) &= g(F^*(w), v) \\ &= h(w, F(v)) \\ &= h_1(w)(F(v)) \\ &= F^\vee(h_1(w))(v) \end{aligned}$$

□

Corollary 18.4.18. If V is finite-dimensional, $F : V \to W$ is adjointable.

Proof. If V is finite-dimensional, g_1 is invertible. We get $F^* = g_1^{-1} \circ F^\vee \circ h_1$. □

Definition 18.4.19. Let V and W be $\mathbb{k}$-vector spaces. Let $F : V \to W$ be linear, and let $s : W \times W \to \mathbb{k}$ be a sesquilinear form. Then we define the ***pullback of s by F*** to be the map $F^*s : V \times V \to \mathbb{k}$ given by

$$(F^*s)(u, v) = s(F(u), F(v))$$

Exercise 18.4.20. *Verify that:*

1. *F^*s is a sesquilinear form.*
2. *If s is Hermitian, F^*s is Hermitian.*
3. *If s is positive semidefinite, F^*s is positive semidefinite.*
4. *If s is positive definite and F is injective, F^*s positive definite.*
5. *If we fix bases of V and W such that F is represented by a matrix A and s is represented by a Gram matrix G, then F^*s is represented by A^*GA.*
6. *Given $F : V \to W$ antilinear, we can achieve the same results by defining $F^*s(u, v) = s(F(v), F(u))$.*

Chapter 19

Normal Endomorphisms

Definition 19.0.1

Let (V, g) be a $\mathbb{k}$-vector space with scalar product. Let $F : V \to V$ be an adjointable endomorphism. Then we call F ***normal*** if $F^* \circ F = F \circ F^*$.

Exercise 19.0.2. *1. Show that self-adjoint maps are normal.*

2. *A linear map with $F^* = -F$ is called* **skew**. *Show that skew maps are normal.*

3. *A linear map represented in a given basis by a matric of the form*

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

is normal.

Definition 19.0.3

Let (V, g) and (W, h) be $\mathbb{k}$-vector spaces with scalar products. Then a linear map $F : V \to W$ is called an ***isometric embedding*** if $g(v, w) = h(F(v), F(w))$ for all $v, w \in V$.

Exercise 19.0.4. *Show that:*

1. *Isometric embeddings preserve lengths and distances.*

2. *Isometric embeddings preserve angles.*

3. *Isometric embeddings are injective.*

4. *If orthonormal bases are chosen for V and W, then the matrix representation A of F fulfills $A^*A = E_n$.*

5. *If $\dim V = \dim W$, F is invertible with $F^{-1} = F^*$. An invertible isometric embedding is called an* **isometry**.

6. *Isometries are normal.*

Theorem 19.0.5: Fundamental Theorem of Normal Endomorphisms

Let (V, g) be a finite-dimensional complex vector space with scalar product. Let $F : V \to V$ be an endomorphism. Then there exists an orthonormal basis of (V, g) consisting of eigenvectors of F if and only if F is normal.

Note that this only works for complex numbers, not for reals:

Exercise 19.0.6. *Recall that any endomorphism represented by a matrix of the form*

$$A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

is normal. Show that, if $b \neq 0$, A has no real eigenvalues.

Corollary 19.0.7: A Normal Form for Complex Normal Endomorphisms

Let (V, g) be a finite-dimensional real vector space with scalar product. Let $F : V \to V$ be an endomorphism. Then there exists an orthonormal basis of (V, g) in which F is represented by a block diagonal matrix consisting of 1×1 blocks and 2×2 blocks of the form

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

with $b > 0$ if and only if F is normal. If such a representation exists, it is unique up to permutation of the blocks.

Corollary 19.0.8: Spectral Theorem for Self-Adjoint Maps

Let (V, g) be a finite dimensional real or complex vector space with scalar product. Let $F : V \to V$ be an endomorphism. Then there exists an orthonormal basis of V in which F is represented by a diagonal matrix with real entries if and only if F is self-adjoint.

Exercise 19.0.9. *Derive an analogous spectral theorem for skew maps.*

Corollary 19.0.10. Let (V, g) be a finite-dimensional real or complex vector space with scalar product. Let s be a sesquilinear form on V. Then there exists a g-orthonormal Basis of V in which s is represented by a diagonal matrix with real entries if and only s is hermitian.

Corollary 19.0.11: Singular Value Decomposition

Let (V, g) and (W, h) be finite-dimensional real or complex vector spaces with scalar products. Let $F : V \to W$ be linear. Then there exist orthonormal bases of V and W in which F is represented by a diagonal matrix with uniquely determined real entries a_n such that $a_1 \geq a_2 \geq \ldots$ and $a_n = 0$ if $n \geq \operatorname{rk} F$. These entries are known as the ***singular values*** of F.

Singular value decomposition is fundamental to the JPG compression algorithm, which represents an image by a matrix, then calculates the singular value de-

composition of that matrix, and then sets all of the singular values below a given threshold to zero, leading to less data needing to be stored while still maintaining the singular values that 'noticably contribute' to the resulting image.

19.1 Linear Isometries

Corollary 19.1.1: Normal Form for Real Isometries

Let (V, g) be a finite-dimensional real vector space with scalar product g. Let $F : V \to V$ be a linear isometry. Then there exists an orthonormal basis of (V, g) in which F is represented by a block diagonal matrix consisting of 1×1 blocks and 2×2 blocks of the form

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

with $a^2 + b^2 = 1$.

Corollary 19.1.2: Normal Form for Complex Isometries

Let (V, g) be a finite-dimensional complex vector space with scalar product g. Let $F : V \to V$ be a linear isometry. Then there exists an orthonormal basis of (V, g) in which F is represented by a diagonal matrix whose entries have absolute value 1.

Corollary 19.1.3

Let (V, g) be a finite-dimensional complex vector space with scalar product g. Let $F : V \to V$ be a linear isometry. Then V can be decomposed into a direct sum of subspaces U of dimension one or two, such that F acts on the one-dimensional subspaces as the identity or reflection, and on the two-dimensional subspaces as a rotation around the origin.

Exercise 19.1.4. *A linear isometry is orientation-preserving if and only if the number of invariant subspaces on which it acts as a reflection is even.*

19.2 Affine Isometries

Theorem 19.2.1. Let $(A, \vec{A}, s))$, $(B, \vec{B}, g)$ be finite-dimensional normed affine spaces of the same dimension over vector spaces with scalar products. Then there exists an affine isometry $F : A \to B$.

Theorem 19.2.2. Let $(A, \vec{A}, g)$ be a finite-dimensional normed $\mathbb{k}$-affine space over a vector space with scalar product. Let $F : A \to A$ be an affine sometry. Then there exists a point $o \in A$ and a vector $x \in \mathrm{Eig}(\vec{F}, 1)$ such that

$$F(a) = o + x + \vec{F}(a - o)$$

x is uniquely determined by A, while o is an arbitrary point chosen from an affine-F-invariant subspace $(B, \mathrm{Eig}(\vec{F}, 1)) \subseteq (A, \vec{A})$.

Definition 19.2.3. We call $(B, \mathrm{Eig}(\vec{F}, 1))$ the ***axis set*** of F, and call lines of the form

$$\{b + \lambda.x \mid \lambda \in \Bbbk\} \subseteq B$$

with $b \in B$ the ***axes*** of F.

Our theorem thus states that any affine isometry can be represented as a linear map composed with a translation along the axis set of F.

Chapter 20

Revisiting Synthetic Geometry

20.1 Incidence Geometries

Definition 20.1.1: Incidence Geometries

An ***incidence geometry*** is a set X equipped with a set $G \subseteq \mathcal{P}(X)$ of subsets of X known as ***lines***, such that:

1. Each line contains at least two points,
2. For every pair of points, there exists exactly one line containing both.

Instead of "x is contained in $g \in G$", we often say "x lies on $g \in G$".

Exercise 20.1.2. *Let R be a division ring. Then $X := R^n$ can be turned into an incidence geometry by defining lines to be all sets of the form*

$$g = \{p + \lambda v : \lambda \in R\},$$

where $p, v \in R^n$.

Moving forward, this will be the canonical choice of incidence geometry on R^n.

Definition 20.1.3. A set of points lying on the same line is called ***colinear***.

Definition 20.1.4. A set of lines containing a common point is called ***confluent***.

Definition 20.1.5. A set of three non-colinear points is known as a ***triangle***.

Definition 20.1.6: Betweenness

A ***betweenness relation*** on an incidence geometry (X, G) is a subset $Z \subseteq X^3$ such that the following properties hold:

1. ***Induced Orderings on Lines:*** On every line $g \in G$, there exists an ordering $\leq$ such that $(x, y, z) \in Z$ if and only if $x \leq y \leq z$ or $x \geq y \geq z$.
2. ***Pasch's Axiom:*** Let $p, q, r \in X$. Let

$$[p, q] := \{x \in X : (p, x, q) \in Z\}$$

denote the ***line segment between p and q***. Then every line $g \in G$ either intersects none of the line segments $[p, q], [q, r], [r, p]$, or at least two.

An incidence geometry equipped with a betweenness relation is known as an ***ordered incidence geometry***.

Exercise 20.1.7. *Let K be an ordered field. Then:*

1. *The set of all tuples (x, y, z), where $x, y, z \in K^n$ fulfill $y = \lambda x + \mu z$ with $\lambda, \mu \in [0, 1] \subset K$ and $\lambda + \mu = 1$, gives a betweenness relation on K^n.*
2. *There exists an affine isomorphism $\varphi : K \cong (g \in G)$ between any line in G and K.*
3. *The ordering on K can be recovered from the betweenness relation on K^n by taking the unique ordering $\leq$ on $g \in G$ that fulfills $\varphi(0) \leq \varphi(1)$.*

Moving forward, this will be the canonical choice of betweenness relation on K^n.

Definition 20.1.8. Given an ordered incidence geometry (X, G, Z), a ***ray*** in X is a set $A \subseteq X$ of the form

$$A = \{x \in g \mid x \leq p\},$$

where $g \in G$ is a line and $p \in g$ is a point on that line.

Definition 20.1.9: Congruence Groups

Given an ordered incidence geometry (X, G, Z), a ***congruence group*** of (X, G, Z) is a group K of automorphism of (X, G, Z) such that every pair of rays in $A, B \subseteq X$ has exactly two elements $h, k \in K$ such that $h(A) = k(A) = B$.

Intuitively, one of these two elements will be a rotation and the other a rotation composed with reflection along an axis.

Exercise 20.1.10. *Let K be an ordered field. Let s be the standard scalar product in K^2. Then the set of s-invariant affine automorphisms of K^2 forms a congruence group of K^2.*

Moving forward, this will be the canonical choice of congruence group on K^2.

Definition 20.1.11. We say that an ordered incidence geometry has the ***least upper bound property*** if every nonempty subset $A \subset g \in G$ that is bounded from above has a least upper bound.

Definition 20.1.12. We call an ordered incidence geometry (X, G, Z) with congruence group K ***near-euclidean*** if (X, G, Z) fulfills the supremum property and G is nonempty.

Definition 20.1.13. We say that an incidence geometry (X, G) fulfills the ***parallel postulate*** iff. for every line $g \in G$ and every point $p \notin G$, there exists at most one line $h \in G$ such that $p \in h$ and $h \cap g = \emptyset$.

Exercise 20.1.14. *Let $K \subsetneq \mathbb{R}$ be a proper subfield of $\mathbb{R}$ containing square roots of every positive element. Then K^2 fulfills the parallel postulate.*

The primary goal of this chapter will be proving the following result:

Theorem 20.1.15: Classification of near-euclidean geometries

Let (X, G, Z) be a near-euclidean incidence geometry. Then:

1. If (X, G, Z) fulfills the parallel postulate, it is isomorphic to $\mathbb{R}^2$.
2. If (X, G, Z) doesn't fulfill the perallel postulate, it is isomorphic to the ***hyperbolic plane*** H.

20.2 Projection Maps

Definition 20.2.1. Let V be a vector space. Then a ***linear hyperplane*** is a vector subspace of W of codimension 1, i.e.

$$\operatorname{codim}(W \subseteq V) := \dim(V/W) = 1$$

Equivalently, $W \neq V$ such that V is generated by W and any vector $v \in V \setminus V$. If V is finite-dimensional, we have $\dim(V/W) = \dim V - \dim W$, and thus a linear hyperplane is a subspace W whose dimension is one lower than the dimension of V. An ***affine Hyperplane*** is defined entirely analogously as an affine subspace of codimension one.

Exercise 20.2.2.
1. *In any affine space, the intersection of an affine line with another affine subspace is either empty, a single point, or the entire line.*
2. *For any affine hyperplane H and any line g with directional vector $\vec{v} \notin \vec{H}$, the intersection of g and H consists of a single point.*

Definition 20.2.3. Given an affine space E, an affine hyperplane $H \subset E$, and a vector $\vec{v} \in \vec{E} \setminus \vec{H}$, the ***parallel projection of E onto H in direction $\vec{v}$*** is the map

$$\pi = \pi_{\vec{v}} = \pi_{\vec{v},H} : E \to H$$

assigning to each point e the intersection of the affine line $(e + \langle \vec{v} \rangle)$ with H.

Definition 20.2.4. Given an affine space E, an affine hyperplane $H \subset E$, and a point $q \in E \setminus H$, the ***central projection of E onto H centered at q*** is the map

$$\zeta = \zeta_q = \zeta_{q,H} : E \setminus (q + \vec{H}) \to H$$

assigning to each point $e \in E$ the intersection of the line though e and q with H.

Exercise 20.2.5. *The image of a line g under central projection is either:*

1. *A line,*
2. *a pointed line, i.e. a line with a single missing point (the* **vanishing point***),*
3. *a single point,*
4. *or the empty set.*

20.3 Projectivization

Definition 20.3.1: Projectivization

Let $\mathbb{k}$ be a field. Let V be a $\mathbb{k}$-vector space. Then the set of all lines through the origin in V is called the the ***projectivization of V*** and denoted by $\mathbb{P}V$.

In the finite-dimensional case, we have $\dim \mathbb{P}V = \dim V - 1$. It is therefore common to write

$$\mathbb{P}^n\mathbb{k} := \mathbb{P}(\mathbb{k}^{n+1})$$

Beware, this means $\mathbb{P}\mathbb{k} \neq \mathbb{P}^1\mathbb{k}$! The set $\mathbb{P}^1\mathbb{k}$ is called the ***projective line over*** $\mathbb{k}$, the set $\mathbb{P}^2\mathbb{k}$ is called the ***projective plane over*** $\mathbb{k}$.

Definition 20.3.2. Given a vector space V, we call a subset $g \subseteq \mathbb{P}V$ a ***projective line*** or ***$\mathbb{P}$-line*** if there exists a vector subspace $g' \subseteq V$ such that $\dim g' = 2$ and $g = \mathcal{P}g'$.

Lemma 20.3.3. For any vector space V, the set $\mathbb{P}V$ forms an incidence geometry with $\mathbb{P}$-lines as the lines.

Lemma 20.3.4. In a projective plane over any field, any two different $\mathbb{P}$-lines intersect in exactly one point.

Definition 20.3.5. An ***embedding of incidence geometries*** is an injective map $X \hookrightarrow Y$ such that points in X are colinear if and only if their images are colinear. A bijective embedding of incidence geometries is known as a ***colineation***.

Lemma 20.3.6. Any injective vector space homormophism $W \hookrightarrow V$ induces an embedding $\mathbb{P}W \hookrightarrow \mathbb{P}V$.

Definition 20.3.7

Let E be an affine space. Then we can equip the disjoint union

$$\mathbb{V}E := E \sqcup \mathbb{P}\vec{E}$$

with an incidence structure by designating lines to be:

1. Sets $\mathbb{V}g := g \sqcup \{\vec{g}\}$ for any affine line $g \in E$,
2. and $\mathbb{P}$-lines $l = \mathbb{P}l'$.

We call this structure the ***projective completion of E***. We call points in $\mathbb{P}\vec{E}$ ***points at infinity***.

Exercise 20.3.8. *If E is a two-dimensional affine space, then its projective completion $\mathbb{V}E$ contains exactly one line containing only infinitely distant points. We call this line the* **line at infinity**.

Exercise 20.3.9. *Given an affine space E, the trivial embeddings $E \hookrightarrow \mathbb{V}E$ and $\mathbb{P}\vec{E} \hookrightarrow \mathbb{V}E$ are embeddings of incidence geometries. Any injective affine map $F \hookrightarrow E$ of affine spaces is an embedding of incidence geometries and induces an embedding $\mathbb{V}E \hookrightarrow \mathbb{V}F$ of their projective completions.*

Theorem 20.3.10

Let V be a vector space. Let E be an affine hyperplane in V not containing the origin. Then

$$\mathbb{V}E \cong \mathbb{P}V$$

Corollary 20.3.11: Projective lines are projective completions

Let F be a field. Then we have $\mathbb{P}^1F \cong \mathbb{V}F = F \sqcup \{\infty\}$.

Exercise 20.3.12.

1. *For any field $\mathbb{F}$, we have*

$$\mathbb{P}^n\mathbb{F} \cong \bigsqcup_{i=0}^{n} \mathbb{F}^i$$

2. *For a finite Field $\mathbb{F}_q$, we have*

$$\left|\mathbb{P}^n\mathbb{F}_q\right| = \sum_{i=0}^{n} q^i$$

Chapter 21

Hyperbolic Geometry

21.1 Circle Inversions

Definition 21.1.1. Let E be a Euclidean plane. Then the corresponding ***extended Euclidean plane*** is the set

$$\hat{E} = E \cup \{\infty\},$$

adding a "point at infinity" to E.

Definition 21.1.2. We call a subset $L \subseteq \hat{E}$ a ***generalized circle*** if it is either a circle in E or the union of a line in E with $\{\infty\}$. We call the former ***real circles*** and the latter ***extended lines***.

Definition 21.1.3. Let E be a euclidean plane. Then we assign to each generalized circle $L \subseteq \hat{E}$ the ***circle inversion map***

$$s_L : \hat{E} \to \hat{E},$$

which is given by the ordinary reflection at L with the additional rule $s_L(\infty) = \infty$ if L is an extended line, and by

$$c + \vec{v} \mapsto \begin{cases} c + \left(\frac{r}{\|\vec{v}\|}\right)^2 \vec{v} & \vec{v} \neq \vec{0} \\ \infty & \vec{v} = \vec{0} \end{cases}$$

with $\infty \mapsto c$ if L is a circle with center c and radius r.

Exercise 21.1.4. *s_L is an involution, i.e. we have $s_L^2 = \mathrm{id}$.*

Exercise 21.1.5. *For any two points $p, q \in \hat{E}$, there exists a circle inversion s_L with $s_L(p) = q$ and $s_L(q) = p$*

Exercise 21.1.6. *Every circle inversion maps generalized circles to generalized circles.*

21.2 Möbius Transformations

Definition 21.2.1. A composition of circle inversions is known as a ***Möbius transformation***.

Exercise 21.2.2. *The set* $\mathrm{Möb}(\hat{E})$ *of Möbius transformations of a given extended Euclidean plane* $\hat{E}$ *forms a group under composition.*

Exercise 21.2.3. *Any two generalized circles in a generalized Euclidean plane can be mapped onto each other by a Möbius transformation.*

Theorem 21.2.4. Given an extended Euclidean plane E, its Möbius transformations are precisely the bijections $\varphi : \hat{E} \xrightarrow{\sim} \hat{E}$ mapping generalized circles to generalized circles.

Theorem 21.2.5. Any Möbius transformation that leaves a given generalized circle L pointwise unchanged is either the circle inversion s_L at that circle or the identity map.

Exercise 21.2.6. *Given a generalized Euclidean plane* $\hat{E}$*, a Möbius transformation* $\varphi : \hat{E} \to \hat{E}$ *and a generalized circle* $L \subseteq \hat{E}$*, we have* $\varphi \circ s_L \circ \varphi^{-1} = s_{\varphi(L)}$

Exercise 21.2.7. *Given a generalized Euclidean plane* $\hat{E}$*, a Möbius transformation* $\varphi : \hat{E} \to \hat{E}$*, and a generalized circle* $L \subseteq \hat{E}$*, we have* $\varphi(L) = L$ *if and only if* $\varphi \circ s_L = s_L \circ \varphi$*.*

Definition 21.2.8. Let $\hat{E}$ be a generalized Euclidean plane. Given a pair of generalized circles $L, M \subset \hat{E}$ we call L ***orthogonal to*** M, which we write $L \perp M$, if $L \neq M$ and $s_L(M) = M$.

Theorem 21.2.9. Let $L, M \subset \hat{E}$ be generalized circles in a generalized Euclidean plane $\hat{E}$ such that $L \perp M$. Let $\varphi : \hat{E} \to \hat{E}$ be a Möbius transformation. Then $\varphi(L) \perp \varphi(M)$.

Proposition 21.2.10. Let E be a Euclidean plane, $g \subset E$ a line, $H \subset E \setminus g$ an open Half-plane with g as its boundary, and

$$K_H = \mathrm{Möb}_H(\hat{E}) := \left\{ \varphi \in \mathrm{Möb}(\hat{E}) \;\middle|\; \varphi(H) = H \right\}$$

the set of Möbius transformations leaving H invariant. Then:

1. Any Möbius transformation that leaves H invariant also leaves the extended line $\hat{g}$ invariant.
2. $\mathrm{Möb}_H(\hat{E})$ operates transitively on H, i.e. any point of H can be mapped onto any other by a Möbius transformation $\varphi \in \mathrm{Möb}_H(\hat{E})$.
3. $\mathrm{Möb}_H(\hat{E})$ is generated by the set of circle inversions s_L with $L \perp \hat{g}$.
4. Different elements of K_H are still different when restricted to H.

21.3 Complex Möbius Geometry

Definition 21.3.1. We denote by γ the "double complex conjugation map"

$$\begin{aligned}\gamma : \mathbb{C}^2 &\to \mathbb{C}^2\\ (w,z) &\mapsto (\overline{w},\overline{z})\end{aligned}$$

Theorem 21.3.2. The set of all antilinear automorphisms of $\mathbb{C}^2$ is given by

$$\mathrm{GL}(\mathbb{C}^2)\gamma := \left\{F\circ\gamma \;\middle|\; F\in \mathrm{GL}(\mathbb{C}^2)\right\}$$

Definition 21.3.3. We denote by

$$\mathrm{GL}(\mathbb{C}^2)\langle\gamma\rangle := \mathrm{GL}(\mathbb{C}^2)\sqcup \mathrm{GL}(\mathbb{C}^2)\gamma$$

the set of all automorphisms of $\mathbb{C}^2$ that are either linear or antilinear.

Since $\mathbb{P}^1\mathbb{C}$ consists of subsets of $\mathbb{C}^2$, every map $\mathbb{C}^2\to\mathbb{C}^2$ mapping lines to lines acts on $\mathbb{P}^1\mathbb{C}$ in a trivial way. By theorem 20.3.11 $\mathbb{P}^1\mathbb{C}\cong\hat{\mathbb{C}}$, the map

$$\begin{aligned}\iota_{\mathbb{C}} : \hat{\mathbb{C}} &\xrightarrow{\sim} \mathbb{P}^1\mathbb{C}\hookrightarrow\mathbb{C}^2\\ z &\mapsto (z,1)\\ \infty &\mapsto (1,0)\end{aligned}$$

is a bijection, which means we also get an action on $\hat{\mathbb{C}}$.

Theorem 21.3.4: Action of $\mathrm{GL}(\mathbb{C}^2)\langle\gamma\rangle$ on $\hat{\mathbb{C}}$

The canonical action gives us a group homomorphism

$$\begin{aligned}\varphi : \mathrm{GL}(2;\mathbb{C})\langle\gamma\rangle &\to \mathrm{Ens}^{\times}(\hat{\mathbb{C}})\\ \gamma &\mapsto (z\mapsto\overline{z})\\ M := \begin{pmatrix}a & b\\ c & d\end{pmatrix} &\mapsto \left(z\mapsto\begin{cases}\frac{az+b}{cz+d} & z\neq\infty\wedge cz+d\neq 0\\ \frac{a}{c} & z=\infty\wedge c\neq 0\\ \infty & z\neq\infty\wedge cz+d=0\\ \infty & z=\infty\wedge c=0\end{cases}\right)\end{aligned}$$

Proof. 1. Let $z\in\mathbb{C}$. Translating directly from $\mathbb{C}$ to gives $\mathbb{C}^2$ $\iota_{\mathbb{C}}(z)=\binom{z}{1}$, which means $M(\iota_{\mathbb{C}}(z))=\binom{az+b}{cz+d}$.

   (a) If $cz+d$, then this point lies on the same line through the origin as $\binom{\frac{az+b}{cz+d}}{1}$, which is then translated back from $\mathbb{P}^1\mathbb{C}$ to $\hat{\mathbb{C}}$ as the point $\frac{az+b}{cz+d}$.

   (b) If $cz+d=0$, then this point lies on the same line through the origin as $\binom{1}{0}$, which is then translated back from $\mathbb{P}^1\mathbb{C}$ to $\hat{\mathbb{C}}$ as the point ∞.

2. If $z=\infty$, then $\iota_{\mathbb{C}}(\infty)=\binom{1}{0}$, which gets mapped to $\binom{a}{c}$, from which we can proceed as before.

□

Theorem 21.3.5. Our canonical group homomorphism $\varphi : \mathrm{GL}(\mathbb{C}^2)\langle y\rangle \to \mathrm{Ens}^{\times}(\hat{\mathbb{C}})$ is a surjective group homomorphism

$$\varphi : \mathrm{GL}(\mathbb{C}^2)\langle\gamma\rangle \twoheadrightarrow \mathrm{Möb}(\hat{\mathbb{C}})$$

whose kernel is the group consisting of all scalar multiplies of the identity.

Proof.

1. Since $\mathrm{GL}(\mathbb{C}^2)$ is generated by the diagonal matrices together with the matrices $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\mathrm{GL}(2,\mathbb{C})\langle\gamma\rangle$ is generated by these matrices along with γ. It therefore suffices to show that the images of these matrices are Möbius transformations. Now, according to our previous analysis of the maps generated by φ, these images are as follows:

   (a) A diagonal matrix with entries λ and μ gets mapped to $\frac{\lambda}{\mu}\cdot z$, which corresponds to a scaled rotation.

   (b) S gets mapped to $z \mapsto \frac{1}{z}$, which is the circle inversion at the unit circle composed with a reflection along the real axis.

   (c) T gets mapped to the translation $z \mapsto z+1$.

   (d) γ gets mapped to the conjugation $z \mapsto \overline{z}$, which is just the reflection along the real axis.

   All of these maps are known Möbius transformations.

2. We now know that the image of φ contains inversion at the unit circle, all translations, and all scalings. Therefore, it contains all inversions along proper circles. It also contains reflection along the extended real axes and all rotations at the origin, so it also contains reflections along extended lines. Thus´, the image contains the generators of $\mathrm{Möb}(\hat{\mathbb{C}})$, and therefore the entire group.

3. An antilinear automorphism cannot stabilize every one-dimensional subspace. Thus, antilinear automorphisms cannot be mapped the identity, i.e. they cannot be contained in the kernel.

   Meanwhile, the only linear automorphisms stabilizing every one-dimensional subspace are scalings, i.e. multiples of the identity.

□

Definition 21.3.6: Projective Linear Groups

Given a vector space V, the ***projective linear group*** $\mathrm{PGL}(V)$ is defined as the quotient group of $\mathrm{GL}(V)$ by the group $Z(V)$ of all scalar multiples of the identity. The ***projective special linear group*** $\mathrm{PSL}(V)$ is analogously defined as the quotient group of $\mathrm{SL}(V)$ by the group $SZ(V)$ of all multiples of the identity by a unit scalar.

Theorem 21.3.7: Möbius group as a projective linear group

$$\mathrm{Möb}^+(\hat{\mathbb{C}}) \cong \mathrm{PGL}\big(\mathbb{C}^2\big)$$

Definition 21.3.8. We denote by τ reflection along the axis $\langle e_1 \rangle$, i.e.

$$\begin{aligned} \tau : \mathbb{R}^2 &\to \mathbb{R}^2 \\ \begin{pmatrix} a \\ b \end{pmatrix} &\mapsto \begin{pmatrix} a \\ -b \end{pmatrix} \end{aligned}$$

Theorem 21.3.9: Action of $\mathrm{SL}(\mathbb{C}^2)$ on $\hat{\mathbb{C}}$

Our surjective group homomorphism $\varphi : \mathrm{GL}(\mathbb{C}^2)\langle\gamma\rangle \twoheadrightarrow \mathrm{Möb}\big(\mathbb{C}^2\big)$ induces a surjective group homomorphism

$$\mathrm{SL}(\mathbb{R}^2)\langle\tau\gamma\rangle \twoheadrightarrow \mathrm{Möb}_{\mathbb{H}}(\hat{\mathbb{C}})$$

whose kernel is $\pm \mathrm{Id}_{\mathbb{R}^2}$.

Theorem 21.3.10: Conjugations of $\mathbb{H}$ as a projective special linear group

$$\mathrm{Möb}^+_{\mathbb{H}}(\hat{\mathbb{C}}) \cong \mathrm{PSL}\big(\mathbb{R}^2\big)$$

Part VI

A First Taste of Category Theory

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Chapter 22

Categories, Functors and Natural Transformations

22.1 The Language of Categories

Definition 22.1.1: Categories

A category C consists of:

1. A class $\mathrm{ob}\, C$ of ***objects***,
2. and a class $\mathrm{mor}\, C$ of ***morphisms***, generally consisting of structure-preserving maps between objects,

such that:

1. Each morphism $f \in \mathrm{mor}\, C$ has specified objects as its ***domain*** and ***codomain***, with the notation $f : X \to Y$ denoting a morphism with X as its domain and Y as its codomain. The collection of all morphisms $X \to Y$ is called the ***hom-set from X to Y*** (often regardless of whether it ends up being an actual set or a proper class), and is ´generally denoted $\mathrm{Hom}(X, Y)$.
2. Each object $X \in \mathrm{ob}\, C$ has a designated ***identity morphism*** $\mathrm{id}_X : X \to X$.
3. For any pair of morphisms $f, g \in \mathrm{mor}\, C$ such that $\mathrm{cod}\, f = \mathrm{dom}\, g$, there exists a ***composite morphism*** $g \circ f : \mathrm{dom}\, f \to \mathrm{cod}\, g$:

$$\forall f : X \to Y, g : Y \to Z :$$
$$\exists g \circ f : X \to Z$$

4. Composition with identity morphisms leaves a morphism unchanged:

$$\forall f : X \to Y :$$
$$\mathrm{id}_Y \circ f = f \circ \mathrm{id}_X = f$$

5. Composition is associative:

$$\forall f : X \to Y, g : Y \to Z, h : Z \to W$$
$$(h \circ g) \circ f = h \circ (g \circ f) =: h \circ g \circ f$$

By taking objects as vertices and morphisms as edges, each category can be turned into a graph. This graph is directed, may contain parallel edges, and contains at least one loop at each vertex (given by the identity morphism). We call such a graph a ***reflexive quiver***.

Example 22.1.2. Many familiar notions of objects equipped with structure-preserving maps define categories:

1. The category `Set` has sets as its objects and functions as its morphisms.
2. The category `Poset` as partially ordered sets as its objects and order-preserving functions as morphisms.
3. The category `Top` has topological spaces as its objects and continuous functions as its morphisms.
4. The categories $\texttt{Set}_*$ and $\texttt{Top}_*$ have sets or topological spaces with a designated ***base point*** as objects and basepoint-preserving (continuous) functions as their morphisms.
5. The category `Group` has groups as its objects and group homomorphisms as its morphisms. Analogously, the categories `Ring` and `Field` have their respective homomorphism as their morphisms.
6. For a fixed field $\Bbbk$, $\texttt{Vec}_\Bbbk$ denotes the category of $\Bbbk$-vector spaces, with linear maps as their morphisms.
7. For a fixed ring R, ${}_R\texttt{Mod}$ and $\texttt{Mod}_R$ denote the categories of left R-modules and right R-modules, both equipped with R-module homomorphisms as their morphisms.

These sorts of categories, where objects have underlying sets and whose morphism are functions between those sets, are known as ***concrete categories*** - a precise definition of this concept will follow later. However, non-concrete categories are also prevalent:

Example 22.1.3.
1. For a ring R, $\texttt{Mat}_R$ is the category whose objects are positive integers and where a morphism from n to m is an $m \times n$-matrix with values in R.
2. Any monoid M defines a category $\texttt{B}M$ with a single object, with monoid elements as morphisms and the monoid operation as composition. Vice versa, applying this contruction in reverse allows us to view any single-object category as a monoid.
3. Any preorder $(P, \leq)$ can be viewed as a category, with elements of P as objects and a unique morphism $x \to y$ if and only if $x \leq y$. In this case, transitivity implies that composite morphisms exist and reflexivity implies that identity morphisms exist.

Definition 22.1.4. We call a category ***small*** if its class of morphisms forms a set, and ***locally small*** if the class of morphisms between any pair of objects forms a set.

Definition 22.1.5

1. An ***isomorphism*** in a category is a morphism $f : X \to Y$ which has an inverse, i.e. there exists a morphism $g : Y \to X$ such that $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$.
2. An ***endomorphism*** in a category is a morphism whose domain equals its codomain.
3. An endomorphism that is also an isomorphism is called an ***automorphism***.

Proposition 22.1.6.

1. The Isomorphisms in `Set` are precisely the bijections.
2. The Isomorphisms in `Group`, `Ring`, `Field` and ${}_R\mathtt{Mod}$ are the respective algebraic isomorphisms, which end up being exactly the bijective homomorphisms.
3. The Isomorphisms in `Top` are the homeomorphisms - in this case, not all bijective morphisms (i.e. continuous bijections) are isomorphisms.

Exercise 22.1.7. *In a partial order category $(P, \leq)$, the only isomorphisms are the identity morphisms.*

Definition 22.1.8. A category in which every morphism is an isomorphism is known as a ***groupoid***.

We saw earlier that monoids correspond to single-object categories. If we turn a group into a single-object category using the same construction, we find that the existence of group inverses guarantees that every morphism in this category is an isomorphism. Thus, a groupoid is the multi-object equivalent of a group. Category theorists with a sufficient amount of whimsy in their hearts may argue that categories are similarly multi-object monoids, and could thus be referred to as ***monoidoids***.

Proposition 22.1.9. Taking a category C and ignoring all morphisms that aren't isomorphisms defines a new category, called the ***maximal subgroupoid*** or ***core groupoid*** of C.

Chapter 23

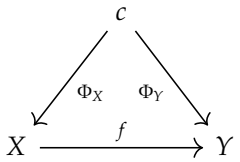
Limits and Colimits

Definition 23.0.1: Cones

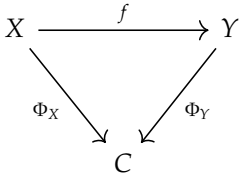
Given some commutative diagram D in a category $\mathcal{C}$:

$$X \xrightarrow{f} Y$$

A ***cone over D*** consists of an additional object c, alongside a family of morphisms Φ from c into every object in the diagram, such that the new diagram commutes:

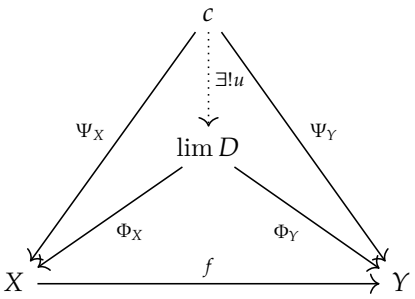


A ***cone under D*** is similarly defined, but with morphisms going into c:

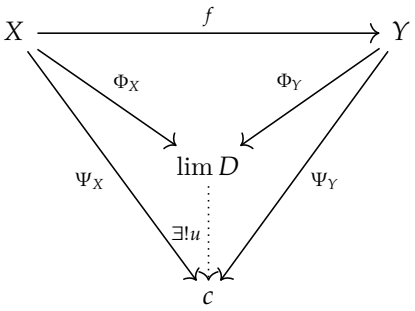


Definition 23.0.2: Limits and Colimits

A ***limit of a commutative diagram D*** is a ***universal cone over D***, i.e. a cone $(\lim D, \Phi)$ such that for every other cone $(c, \Psi$, there exists a unique morphism $u : c \to \lim F$ such that $\Phi_X \circ u = \Psi_x$ for all objects X in the diagram:

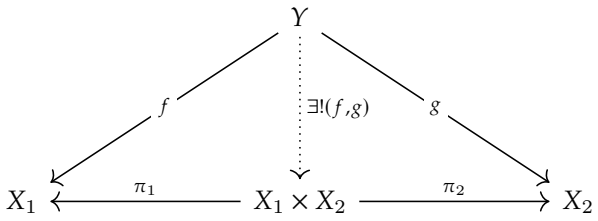


A ***colimit of D*** is the equivalent construction using a cone under D:



Exercise 23.0.3. *Show that limits and colimits of a given diagram are unique up to a unique isomorphism.*

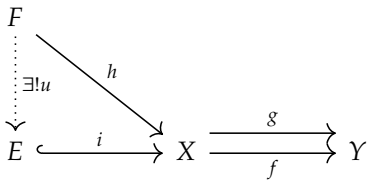
Definition 23.0.4. The limit of a discrete diagram consisting of objects without any morphisms between them is called the ***product object*** of the involved objects, and the given morphisms generalize projections from a cartesian product into its component sets:



Definition 23.0.5. Take a commutative diagram consisting of two objects with two morphisms from one into the other:

$$X \underset{f}{\overset{g}{\rightrightarrows}} Y$$

The limit of this diagram is called its ***equalizer*** E. Since cones have to commute, it suffices to specify only the morphism $i : E \to X$.



The universality of limits now tells us that any morphism $h : F \to X$ such that $f \circ h = g \circ h$ has a unique factorization u through i. Note that i has to be a monomorphism, since, given two morphisms $f_1, f_2 : F \to E$, uniqueness of u implies $i \circ f_1 = i \circ f_2 \implies u_1 = u_2 = f_1 = f_2$.

Example 23.0.6. We want to find the equalizer of a group homomorphism $\Phi : G \to H$ and the trivial homomorphism $\varepsilon : G \to H$. This means we need

$$e_H = \varepsilon(i(x)) = \Phi(i(x))$$

for any $x \in E$. This already tells us that we need $i(E) \subseteq \ker \Phi \subseteq X$. $i : E \to X$ is an injective group homomorphism, and since we can take $F = \ker \Phi$ and $h : \ker \Phi \hookrightarrow X$ the trivial embedding, h can only factor through i if $i(E) = \ker \Phi$. Thus, $i : E \to \ker \Phi$ is an isomorphism and our equalizer is given by $\ker \Phi$.

23.1 Quotient Objects

Definition 23.1.1: Quotient Object

The quotient object of a morphism f is the coequalizer of the kernel pair of f.

Part VII

Group Theory

Chapter 24

Discrete Symmetry Groups

24.1 Finite subgroups of the orthogonal groups

Theorem 24.1.1

The finite subgroups of $SO(3)$, up to isomorphism, are:

1. The cyclic groups C_n,
2. The dihedral groups D_n,
3. The tetrahedral group $T \cong A_4$,
4. The cube group $W \cong S_4$,
5. The icosahedral group $I \cong A_5$.

Corollary 24.1.2

The finite subgroups of $O(3)$, up to isomorphism, are:

1. The finite subgroups of $SO(3)$:
   (a) The cyclic groups C_n,
   (b) The dihedral groups D_n,
   (c) The tetrahedral group $T \cong A_4$,
   (d) The cube group $W \cong S_4$,
   (e) The icosahedral group $I \cong A_5$.
2. Products of these groups with C_2.

24.2 Lattice Groups

Definition 24.2.1. An n-dimensional real ***lattice*** is a subset of $\mathbb{R}^n$ of the form

$$\Lambda = \left\{ \sum_{i=1}^{n} k_i . v_i \;\middle|\; k_i \in \mathbb{Z} \right\},$$

where $v_1, \ldots, v_n$ is a basis of $\mathbb{R}^n$. We say that the lattice Λ is ***generated by the matrix*** $M = (v_1, \ldots, v_n)$.

Theorem 24.2.2: Crystallographic restriction theorem

Every rotational symmetry of a two- or three-dimensional lattice has order 1, 2, 3, 4, or 6.

24.3 The Two-Dimensional Point Groups

Definition 24.3.1. An n-dimensional ***crystallographic point group*** is a subgroup of $O(n)$ mapping an n-dimensional lattice to itself while leaving a given point unchanged.

Theorem 24.3.2

The two-dimensional crystallographic point groups are:

1. The cyclic groups C_1, C_2, C_3, C_4 and C_6,
2. The dihedral groups D_1, D_2, D_3, D_4 and D_6.

We have $D_1 \cong C_2$, so there are ten point groups up to equality and nine point groups up to equality.

Proof Sketch. Follows immediately from the classification of finite subgroups of $O(2)$ and the crystallographic restriction theorem. "□"

24.4 The Three-Dimensional Point Groups

Theorem 24.4.1

There are eighteen three-dimensional crystallographic point groups up to isomorphism, and they are:

1. Five cyclic groups:
   - C_1
   - $C_2 \cong D_1$
   - C_3
   - C_4
   - $C_6 \cong C_3 \times C_2$
2. Four non-cyclic dihedral groups:
   - $D_2 \cong C_2 \times C_2$
   - D_3
   - D_4
   - $D_6 \cong D_3 \times C_2$
3. Two remaining products of cyclic groups with C_2:
   - $C_4 \times C_2$
   - $C_6 \times C_2$
4. Three remaining products of dihedral groups with C_2:
   - $D_2 \times C_2$
   - $D_4 \times C_2$
   - $D_6 \times C_2$
5. Four remaining groups consisting of permutation groups and their products with C_2:
   - A_4
   - $A_4 \times C_2$
   - S_4
   - $S_4 \times C_2$

Theorem 24.4.2

The following table gives a full classification of three-dimensional crystallographic point groups up to identity:

Crystal System	Crystal Class	Example Figure	HM Notation	Isomorphism
triclinic	triclinic-pedial		1	C_1
	triclinic-pinacoidal		$\overline{1}$	C_2
monoclinic	monoclinic-sphenoidal		2	
	monoclinic-domatic		m	
	monoclinic-prismatic		$2/m$	D_2
orthorhombic	orthorhombic-disphenoidal		222	
	orthorhombic-pyramidal		$mm2$	
	orthorhombic-dipyramidical		mmm	$D_2 \times C_2$
tetragonal	tetragonal-pyramidal		4	C_4
	tetragonal-disphenoidal		$\overline{4}$	
	tetragonal-dipyramidal		$4/m$	$C_4 \times C_2$
	tetragonal-trapezoidal		422	D_4
	ditetragonal-pyramidal		$4mm$	
	tetragonal-scalenoidal		$\overline{4}2m$	
	ditetragonal-dipyramidal		$4/mmm$	$D_4 \times C_2$
trigonal	trigonal-pyramidal		3	C_3
	rhombohedral		$\overline{3}$	C_6
	trigonal-trapezoidal		32	D_3
	ditrigonal-pyramidal		$3m$	
	ditrigonal-scalenohedral		$\overline{3}m$	D_6
hexagonal	hexagonal-pyramidal		6	C_6
	trigonal-dipyramidal		$\overline{6}$	
	hexagonal-dipyramidal		$6/m$	$C_6 \times C_2$
	hexagonal-trapezohedral		622	D_6
	dihexagonal-pyramidal		$6mm$	
	ditrigonal-dipyramidal		$\overline{6}m2$	
	dihexagonal-dipyramidal		$6/mmm$	$D_6 \times C_2$
cubic	tetartoidal		23	A_4
	diploidal		$m\overline{3}$	$A_4 \times C_2$
	gyroidal		432	S_4
	hextetrahedral		$\overline{4}3m$	
	hexoctahedral		$m\overline{3}m$	$S_4 \times C_2$

24.5 Classification of Lattices

24.6 Wallpaper Groups

24.7 Three-Dimensional Space Groups

Part VIII

Multivariable Calculus

Chapter 25

Differentiation in $\mathbb{R}^n$

25.1 Differentials and Partial Derivatives

Definition 25.1.1: Fréchet Derivative

Let V, W be normed vector spaces. Let U be an open subset of V. Then $f : U \to W$ is called ***Fréchet differentiable*** or just ***differentiable*** at $p \in U$ if there exists a continuous linear map $Df(p) : V \to W$ such that

$$\lim_{h \to 0} \frac{\left\| f(p+h) - f(p) - Df(p)(h) \right\|_W}{\|h\|_V} = 0.$$

We call $Df(p)$ the ***differential of f at p***. We call f ***differentiable*** if it is differentiable at every point $p \in U$. We call f ***continuously differentiable*** if the map $Df : p \mapsto Df(p)$ is continuous.

Recall that linear maps are continuous if and only if they are bounded, which is always the case if V is finite-dimensional.

Exercise 25.1.2. *Verify that if we identify scalars $c \in \mathbb{R}$ with linear maps $x \mapsto c \cdot x$, this definition agrees with the usual definition for $f : \mathbb{R} \to \mathbb{R}$.*

Theorem 25.1.3. If $f : U \to W$ is differentiable, it is continuous.

Proof. If f is differentiable, we have

$$\lim_{\|h\|_V \to 0} \frac{\left\| f(p+h) - f(p) - Df(p)(h) \right\|_W}{\|h\|_V} = 0.$$

This can only be the case if

$$\lim_{\|h\|_V \to 0} \left\| f(p+h) - f(p) - Df(p)(h) \right\|_W = 0,$$

i.e. if

$$\lim_{\|h\|_V \to 0} f(p+h) = \lim_{\|h\|_V \to 0} f(p) + Df(p)(h).$$

Since $Df(p)$ is continous with $Df(0) = 0$, this simplifies to

$$\lim_{\|h\|_V \to 0} f(p+h) = f(p),$$

which is the definition of sequential continuity. □

Definition 25.1.4: Directional and Partial Derivates

Let V and W be normed vector spaces. Let $v \in V$. Let Let $U \subseteq V$ be open and $f : U \to W$. Then we call the limit

$$\frac{\partial}{\partial v} f(p) = \frac{d}{dh} f(p + hv)\bigg|_{h=0} = \lim_{h \to 0} \frac{f(p + hv) - f(p)}{h}$$

the ***directional derivative*** of f at p in direction v. If $(x_0, \ldots, x_n)$ is a basis of V, we call the directional derivatives

$$\partial_i f(p) := \frac{\partial}{\partial x_i} f(p)$$

the ***partial derivatives of f at p***. If all partial derivatives exist, we call f ***partially differentiable***. If all maps $\partial_i f : p \mapsto \partial_i f(p)$ are continuous, we call f ***continuously partially differentiable***.

Partial derivatives aren't actually any more complicated than standard one-dimensional derivatives - you can calculate partial derivatives by simply fixing all coordinates except one and then taking a regular derivative in the remaining coordinate. For example, if we have a function

$$\begin{aligned} f : \mathbb{R}^3 &\to \mathbb{R} \\ (x, y, z) &\mapsto f(x, y, z), \end{aligned}$$

we can define a function $f|_1 : \mathbb{R} \to \mathbb{R}$ by restriction to the first coordinate:

$$f|_1(x) := f(x, y, z).$$

Using this function, we can simplify the partial derivative:

$$\begin{aligned} \partial_1 f(x, y, z) &= \lim_{h \to 0} \frac{f((x, y, z) + he_1) - f(x, y, z)}{h} \\ &= \lim_{h \to 0} \frac{f(x + h, y, z) - f(x, y, z)}{h} \\ &= \lim_{h \to 0} \frac{f|_1(x + h) - f(x)}{h} \\ &= \frac{d}{dx} f|_1(x) \end{aligned}$$

This tells us that, for example, the function $f(x, y, z) = x \cdot y \cdot z$ has partial derivatives

$$\begin{aligned} \partial_1 f(x, y, z) &= \frac{d}{dx} xyz = yz \\ \partial_2 f(x, y, z) &= \frac{d}{dy} xyz = xz \\ \partial_3 f(x, y, z) &= \frac{d}{dz} xyz = xy. \end{aligned}$$

Exercise 25.1.5. 1. *Show that if f is differentiable, then all directional derivatives exist and fulfill $\frac{\partial}{\partial v} f(p) = Df(p)(v)$.*

2. *Conclude that $Df(p)$ is represented in basis $(x_0, \dots, x_n)$ by the* **Jacobian matrix**

$$Jf(p) = \left(\frac{\partial}{\partial x_1} f(p), \dots, \frac{\partial}{\partial x_n} f(p) \right) = \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(p) & \dots & \frac{\partial}{\partial x_n} f_1(p) \\ \vdots & & \vdots \\ \frac{\partial}{\partial x_1} f_n(p) & \dots & \frac{\partial}{\partial x_n} f_n(p) \end{pmatrix}$$

If the image of f is one-dimensional, the Jacobian matrix is simply a row vector. We call the transpose of this vector the ***gradient of f at p*** and denote it

$$\nabla f(p) := Jf(p)^\top = \begin{pmatrix} \frac{\partial}{\partial x_1} f(p) \\ \vdots \frac{\partial}{\partial x_n} f(p) \end{pmatrix}$$

The transposition comes from the fact that we would like to be able to get the directional derivatives of f at p by taking a scalar product with the gradient, which gives us

$$\begin{aligned} \left\langle \nabla f(p), v \right\rangle &= \nabla f(p)^\top v \\ &= Jf(p) \cdot v \\ &= Df(p)(v) \\ &= \frac{\partial}{\partial v} f(p) \end{aligned}$$

There is one very important pitfall to watch out for when it comes to differentiability: We have just shown that if f is differentiable, we can get the differential by construction the Jacobian matrix from the partial derivatives. This makes it look like every partially differentiable function is differentiable, which turns out to not be true in general.

For example, there exist functions which are partially differentiable, but not continuous - we showed earlier that all differentiable functions are continuous, so these functions are quick counterexamples.

One example is

$$\begin{aligned} f : \mathbb{R}^2 &\to \mathbb{R} \\ (x, y) &\mapsto \begin{cases} \frac{xy}{(x^2+y^2)^2} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases}. \end{aligned}$$

This function has $\partial_1 f(0) = \partial_2 f(0) = 0$, since $f(h, 0) = f(0, h) = 0$ for all $h \in \mathbb{R}$, but isn't continuous, since $\lim_{n\to\infty} f\left(\frac{1}{n}, \frac{1}{n}\right) = \lim_{n\to\infty} \frac{n^2}{4} = \infty \neq f(0)$.

For a more involved counterexample, consider the function

$$\begin{aligned} f : \mathbb{R}^2 &\to \mathbb{R} \\ (x, y) &\mapsto \begin{cases} \frac{y^3}{x^2+y^2} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases} \end{aligned}$$

This function is continuous - for $(x, y) \neq 0$, it is a composition of continuous functions, and thus continuous, and since $x^2 + y^2 \geq y^2$, we additionally have

$$\begin{aligned}\left|f(x,y)\right| &= \frac{\left|y^3\right|}{\left|x^2+y^2\right|} \\ &= \frac{\left|y^3\right|}{x^2+y^2} \\ &\leq \frac{\left|y^3\right|}{y^2} \\ &= \left|y\right|,\end{aligned}$$

so $f(x, y)$ converges to $f(0, 0) = 0$ as (x, y) converges to 0, implying that f is also continuous at the origin.

f is also directionally differentiable in all directions:

$$\begin{aligned}\partial_v f(0) &= \lim_{h\to 0} \frac{f(h\cdot v) - f(0)}{h} \\ &= \lim_{h\to 0} \frac{f(h\cdot v)}{h} \\ &= \lim_{h\to 0} \frac{\left(\frac{h^3 v_2^3}{h^2 v_1^2 + h^2 v_2^2}\right)}{h} \\ &= \lim_{h\to 0} \frac{h^3 v_2^3}{h^3 v_1^2 + h^3 v_2^2} \\ &= \lim_{h\to 0} \frac{v_2^3}{v_1^2 + v_2^2} \\ &= \frac{v_2^3}{v_1^2 + v_2^2} \\ &= f(v)\end{aligned}$$

In particular, $\partial_1 f(0) = 0$ and $\partial_2 f(0) = 1$. However, f is not differentiable at 0, since otherwise our representation $Df(0) = Jf(0)$ would give us:

$$\lim_{\vec{h}\to 0} \frac{\left|f(h) - f(0) - Df(0)(\vec{h})\right|}{\left\|\vec{h}\right\|_2} = \lim_{\vec{h}\to 0} \left|\frac{f(h) - Df(0)(h)}{\left\|\vec{h}\right\|_2}\right|$$

$$= \lim_{\vec{h}\to 0} \left|\frac{f(h) - \partial_1 f(0)\cdot h_1 + \partial_2 f(0)\cdot h_2}{\sqrt{h_1^2 + h_2^2}}\right| = \lim_{\vec{h}\to 0} \left|\frac{f(h) - h_2}{\sqrt{h_1^2 + h_2^2}}\right|$$

$$= \lim_{\vec{h}\to 0} \left|\frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2^3}{h_1^2 + h_2^2} - h_2\right)\right| = \lim_{\vec{h}\to 0} \left|\frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2^3 - h_2(h_1^2 + h_2^2)}{h_1^2 + h_2^2}\right)\right|$$

$$= \lim_{\vec{h}\to 0} \left|\frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2^3 - h_2 h_1^2 - h_2^3}{h_1^2 + h_2^2}\right)\right| = \lim_{\vec{h}\to 0} \left|\frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left(\frac{h_2 h_1^2}{h_1^2 + h_2^2}\right)\right| = \lim_{\vec{h}\to 0} \left|\frac{h_2 h_1^2}{\sqrt{h_1^2 + h_2^2}^3}\right|$$

Since we are only interested in whether this expression converges to 0, we can

square it to simplify slightly:

$$\left|\frac{h_2 h_1^2}{\sqrt{h_1^2+h_2^2}^3}\right|^2 = \left|\frac{h_2^2 h_1^4}{(h_1^2+h_2^2)^3}\right|$$

If we now set $h := \left(\frac{1}{n}, \frac{1}{n}\right)$, we get:

$$\left|\frac{h_2^2 h_1^4}{(h_1^2+h_2^2)^3}\right| = \left|\frac{\frac{1}{n^6}}{\left(\frac{2}{n^2}\right)^3}\right| = \left|\frac{\frac{1}{n^6}}{\frac{8}{n^6}}\right| = \frac{1}{8} \neq 0$$

The important takeaway here is that directional differentiability only guarantees "differentiability along straight paths", while full differentiability also requires "differentiability along curved paths".

Theorem 25.1.6

$f : V \to W$ is continuously differentiable if and only if it is continuously partially differentiable.

Proof. $\Rightarrow$: We have $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m) \cong \mathbb{R}^{mn}$. If the differential $Jf : \mathbb{R}^n \mapsto \mathbb{R}^{mn}$ exists, it is continuous if and only if every component $(Jf)_i : \mathbb{R}^n \mapsto \mathbb{R}$ is continuous. By the definition of Jf, these are the partial derivatives.

$\Leftarrow$: Since every normed complex vector space is isomorphic to a normed real vector space, we can assume without loss of generality that W is real. Since all norms are equivalent, and since a sequence in W converges iff. all components converge, we can additionally assume that W is one-dimensional, i.e. $W \cong \mathbb{R}$.

We need to show that

$$\lim_{h \to \vec{0}} \frac{f(p+h) - f(p) - Jf(p)h}{\|h\|_2} = \vec{0}$$

Let $h := \sum_{i=1}^n h_i e_i$. We decompose the path from p to $p+h$ into its partial steps, taking one coordinate direction at a time:

$$\forall k \in [0..n] :$$

$$p_k = p + \sum_{i=1}^{k} h_i e_i$$

Now, by the mean value theorem, there exists an $s_k \in [0,1]$ such that

$$\begin{aligned} f(p_k) - f\left(p_{k-1}\right) &= f(p_{k-1} + h_k e_k) - f\left(p_{k-1}\right) \\ &= \partial_k f(p_{k-1} - s_k h_n e_n) h_k \end{aligned}$$

Now, since

$$Jf(p)(h) = \sum_{i=1}^{n} \partial_i f(p) h_i,$$

we get:

$$\begin{aligned}
&\frac{\left|f(p+h)-f(p)-Jf(p)(h)\right|}{\|h\|_2}\\
=&\frac{1}{\|h\|_2}\left|f(p_n)-f(p_0)-\sum_{i=1}^{n}\partial_i f(p)h_i\right|\\
=&\frac{1}{\|h\|_2}\left|\sum_{i=1}^{n}f(p_i)-f(p_{i-1})-\partial_i f(p)h_i\right|\\
=&\frac{1}{\|h\|_2}\left|\sum_{i=1}^{n}\partial_i f(p_{i-1}+s_ih_ie_i)h_i-\partial_i f(p)h_i\right|\\
=&\frac{1}{\|h\|_2}\left|\sum_{i=1}^{n}\left(\partial_i f(p_{i-1}+s_ih_ie_i)-\partial_i f(p)\right)h_i\right|\\
=&\frac{1}{\|h\|_2}\left|\sum_{i=1}^{n}\left(\partial_i f\left(p+\sum_{j=1}^{i-1}h_je_j+s_ih_ie_i\right)-\partial_i f(p)\right)h_i\right|\\
\leq&\frac{1}{\|h\|_2}\sum_{i=1}^{n}\left|\left(\partial_i f\left(p+\sum_{j=1}^{i-1}h_je_j+s_ih_ie_i\right)-\partial_i f(p)\right)h_i\right|\\
=&\sum_{i=1}^{n}\left|\partial_i f\left(p+\sum_{j=1}^{i-1}h_je_j+s_ih_ie_i\right)-\partial_i f(p)\right|\cdot\frac{|h_i|}{\|h\|_2}\\
\leq&\sum_{i=1}^{n}\left|\partial_i f\left(p+\sum_{j=1}^{i-1}h_je_j+s_ih_ie_i\right)-\partial_i f(p)\right|
\end{aligned}$$

Since $s_i \in [0,1]$, we have

$$\lim_{h\to 0} p+\sum_{j=1}^{i-1}h_je_j+s_ih_ie_i = p$$

And thus, since the partial derivatives $\partial_i f$ are continuous, the right side converges to zero, so f is differentiable.

□

Exercise 25.1.7. *Let V be an inner product space over a field $\mathbb{F}=\mathbb{R}$ or $\mathbb{F}=\mathbb{C}$. Let $v \in V$. Then the function*

$$\begin{aligned}\langle\cdot, v\rangle : V &\to \mathbb{F}\\ w &\mapsto \langle w, v\rangle\end{aligned}$$

is differentiable, with $D\langle\cdot, v\rangle(w) = v^\top$ for all $w \in V$.

Exercise 25.1.8. *Let b be a symmetric bilinear form. Let*

$$q(v) := b(v,v)$$

be the corresponding quadratic form. Then q is differentiable, with

$$Dq(v)(w) = 2b(v,w)$$

25.2 Rules of Differentiation

Exercise 25.2.1. *Show that the differential operator D is linear, i.e. that we have*

$$D(\alpha f + \beta g)(p) = \alpha D f(p) + \beta D g(p)$$

for f, g differentiable at p.

Theorem 25.2.2: Chain Rule in $\mathbb{R}^n$

Let $f : U \to \mathbb{R}^m$ and $g : V \to \mathbb{R}^r$ with $U \subset \mathbb{R}^n$, $V \subset \mathbb{R}^m$ open and $f(U) \subseteq V$. Let f be differentiable at a point $p \in U$ and let g be differentiable at $f(p)$. Then

$$D(g \circ f)(p) = Dg(f(p)) \circ Df(p)$$

Exercise 25.2.3. *Verify that this version of the chain rule agrees with the familiar chain rule for $f, g : \mathbb{R} \to \mathbb{R}$.*

Proof. By definition of the differential, we have:

$$\begin{aligned} f(x) &= f(p) \\ &\quad + Df(p)(x-p) \\ &\quad + \varepsilon_f(x)\|x-p\| \\ g(f(x)) &= g(f(p)) \\ &\quad + Dg(f(p))(f(x)-f(p)) \\ &\quad + \varepsilon_g(f(x))\|f(x)-f(p)\| \end{aligned}$$

For some functions $\varepsilon_f : U \to \mathbb{R}^m$, $\varepsilon_g : U \to \mathbb{R}^r$ which are continuous in p and $f(p)$ and fulfill $\varepsilon_f(p) = 0$ and $\varepsilon_g(f(p)) = 0$.
The second equation tells us:

$$\begin{aligned} g(f(x)) &= g(f(p)) \\ &\quad + Dg(f(p))(f(x)-f(p)) \\ &\quad + \varepsilon_g(f(x))\|f(x)-f(p)\| \end{aligned}$$

Plugging in our equation for $f(x)$ now lets us regroup our terms into an

equation of the same form as the definition of the differential:

$$
\begin{aligned}
g(f(x)) &= g(f(p)) \\
&\quad + Dg(f(p))\left[\overbrace{f(p) + Df(p)(x-p) + \varepsilon_f(x)\|x-p\|}^{f(x)} - f(p)\right] \\
&\quad + \varepsilon_g(f(x))\left\|\overbrace{f(p) + Df(p)(x-p) + \varepsilon_f(x)\|x-p\|}^{f(x)} - f(p)\right\| \\
&= g(f(p)) \\
&\quad + Dg(f(p))\left[Df(p)(x-p) + \varepsilon_f(x)\|x-p\|\right] \\
&\quad + \varepsilon_g(f(x))\left\|Df(p)(x-p) + \varepsilon_f(x)\|x-p\|\right\| \\
&= g(f(p)) \\
&\quad + Dg(f(p))Df(p)(x-p) \\
&\quad + Dg(f(p))\varepsilon_f(x)\|x-p\| \\
&\quad + \varepsilon_g(f(x))\left\|Df(p)(x-p) + \varepsilon_f(x)\|x-p\|\right\| \\
&= g(f(p)) \\
&\quad + Dg(f(p))Df(p)(x-p) \\
&\quad + \underbrace{\left(\varepsilon_f(x)Dg(f(p)) + \varepsilon_g(f(x))\left\|Df(p)\left(\frac{x-p}{\|x-p\|}\right) + \varepsilon_f(x)\right\|\right)}_{:=\varepsilon_{g\circ f}(x)}\|x-p\| \\
&= g(f(p)) \\
&\quad + (Dg(f(p)) \circ Df(p))(x-p) \\
&\quad + \varepsilon_{g\circ f}(x)\|x-p\|
\end{aligned}
$$

All that remains is to show that $\varepsilon_{g\circ f}(x)$ is continuous if we set

$$\varepsilon_{g\circ f}(p) = 0.$$

Since we have $\varepsilon_f(p) = 0$ and $\varepsilon_g(f(p)) = 0$, we have our desired result if we can show that

$$\left\|Df(p)\left(\frac{x-p}{\|x-p\|}\right)\right\| = \frac{\|Df(p)(x-p)\|}{\|x-p\|}.$$

is bounded. Since $\mathbb{R}^n$ is finite-dimensional, we can find a constant c such that

$$\|Df(p)(x-p)\| \leq c \cdot \|x-p\|,$$

which means that our quotient is indeed bounded, completing the proof.

□

Theorem 25.2.4: Product Rule in $\mathbb{R}^n$

Let $U \subseteq \mathbb{R}^n$. Let $f, g : U \to \mathbb{R}$ at a point $p \in U$. Then

$$D(f \cdot g)(p) = Df(p) \cdot g(p) + f(p) \cdot Dg(p).$$

Exercise 25.2.5. *Verify that this version of the product rule agrees with the familiar product rule for $f, g : \mathbb{R} \to \mathbb{R}$.*

Proof. We define the multiplication function $m(x, y) := x \cdot y$. The partial derivatives are $\partial_1 m(x, y) = y$ and $\partial_2 m(x, y) = x$, i.e.

$$Dm(x, y) = \begin{pmatrix} y & x \end{pmatrix}$$

Applying the chain rule now gives

$$\begin{aligned} D(f \cdot g)(p) &= D(m \circ (f, g))(p) \\ &= Dm((f, g)(p)) \circ D(f, g)(p) \\ &= Dm(f(p), g(p)) \circ (Df(p), Dg(p)) \\ &= Dm(f(p), g(p)) \cdot (Df(p), Dg(p)) \\ &= \begin{pmatrix} g(p) & f(p) \end{pmatrix} \cdot \begin{pmatrix} Df(p) \\ Dg(p) \end{pmatrix} \\ &= Df(p) \cdot g(p) + f(p) \cdot Dg(p). \end{aligned}$$

□

Exercise 25.2.6. *Let $f : V \to W$ be differentiable. Let $\gamma : [0,1] \to V$ be differentiable. Then $f \circ \gamma$ is differentiable with*

$$(f \circ \gamma)'(t) = Df(\gamma'(t)) \cdot \gamma(t)$$

Theorem 25.2.7. Let $U \subseteq V$ be open. Let $f : U \to W$ be differentiable with $Df = 0$. Then f is locally constant.

Corollary 25.2.8. Let $U \subseteq V$ be open and connected. Let $f : U \to W$ be differentiable with $Df = 0$. Then f is constant.

25.3 Higher Partial Derivatives

Theorem 25.3.1: Schwartz's Theorem

Let $f : U \subseteq V \to W$ be twice continuously partially differentiable. Then the order of partial derivation does not matter, i.e.

$$\partial_i \partial_j f = \partial_j \partial_i f$$

for all $i, j \in [1..n]$.

Theorem 25.3.2: First-Order Taylor Expansion in multiple Variables

Let $f : U \subseteq V \to W$. Let f be continuously differentiable. Let $x \in U$, $h \in V$ fulfill $[x, x + h] \subseteq V$. Then

$$f(x + h) = f(x) + Df(x)(h) + r(h),$$

with $r \in o(\|h\|)$.

25.4 Differential Operators

Chapter 26

The Two Queens of Multivariable Calculus

26.1 The Inverse Function Theorem

Theorem 26.1.1: Inverse Function Theorem

Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \to \mathbb{R}^n$ be continuously differentiable. Let $x \in U$ be a point such that $Df(x) : \mathbb{R}^n \to \mathbb{R}^n$ is invertible. Then there exists an open neighbourhood V of x such that:

1. $f(V)$ is an open neighbourhood of $f(x)$.
2. $f|_V$ is a C^1-diffeomorphism, i.e. invertible with continuously differentiable inverse.

In short: if a continuously differentiable function has an invertible differential at a given point, there exists an open neighbourhood around that point in which the function is invertible:

Corollary 26.1.2. Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \to \mathbb{R}^n$ be continuously differentiable. Let $Df(x)$ be invertible for all $x \in \mathbb{R}^n$. Then $f(U)$ is open.

26.2 The Implicit Function Theorem

Theorem 26.2.1: Implicit Function Theorem

Let $U \subseteq \mathbb{R}^m \times \mathbb{R}^k$ be open. Let $f : U \to \mathbb{R}^k$ be continuously differentiable. Let $(x, y) \in U$ with $f(x, y) = z$. Let $Df|_{\mathbb{R}^k}(x, y) : \mathbb{R}^k \to \mathbb{R}^k$ be invertible. Then there exist

1. open neighbourhoods V of x and W of y,
2. and a continuously differentiable function $g : V \to W$,

such that the set of values in $V \times W$ for which f takes the value z is the graph of g, i.e.

$$\{(v, w) \in V \times W \mid f(v, w) = z\} = \{(v, g(v)) \mid v \in V\}$$

In short: under mild assumptions, the level sets of a continuously differentiable function are graphs of continuously differentiable functions:

Chapter 27

Multiple Integrals

Lemma 27.0.1. Let $K \subseteq \mathbb{R}^n$ be compact. Let $f : K \times [a,b] \to \mathbb{R}$ be continuous. Then the function

$$x \mapsto \int_a^b f(x,t)\,dt$$

is uniformly continuous.

Proof. $K \times [a,b]$ is a finite product of compact sets, i.e. compact by Theorem 8.5.10. Thus, f is uniformly continuous by Theorem 8.5.12, which means that for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for any $x, y \in \mathbb{R}, t \in \mathbb{R}^n$

$$d((x,t),(y,t)) < \delta \implies d(f(x,t),f(y,t)) < \varepsilon.$$

This tells us that for any $x, y \in \mathbb{R}$,

$$\begin{aligned} d(x,y) < \delta &\implies d((x,t),(y,t)) < \varepsilon \\ &\implies \left|f(x,t) - f(y,t)\right| = d(f(x,t),f(y,t)) < \varepsilon \\ &\implies d\left(\int_a^b f(x,t)\,dt, \int_a^b f(y,t)\,dt\right) \\ &\quad = \left|\int_a^b f(x,t)\,dt - \int_a^b f(y,t)\,dt\right| \\ &\quad \le \int_a^b \left|f(x,t) - f(y,t)\right| dt \\ &\quad \le (b-a)\varepsilon, \end{aligned}$$

since ε was arbitrary, our function is uniformly continuous as desired. □

Definition 27.0.2. Given a compact cuboid $Q = \prod_{i=1}^n [a_i, b_i] \subseteq \mathbb{R}^n$ and a continuous function $f : Q \to \mathbb{R}$, we declare the ***integral of f over Q*** to be the nested integral

$$\int_Q f(x)\,d^n x := \int_{a_n}^{b_n} \dots \int_{a_1}^{b_1} f(x_1, \dots, x_n) dx_1 \dots dx_n$$

By Lemma 27.0.1, all involved functions are continuous, and thus their integrals are indeed well-defined. This definition will become a fully justified theorem once we begin our study of the Lebesgue integral later on.

Exercise 27.0.3. *Verify that linearity and monotonicity carry over from the single-variable case, i.e show that for any $f, g : Q \to \mathbb{R}$ and $\lambda \in \mathbb{R}$, we have:*

1. $\int_Q f(x) + g(x)\, d^n x = \int_Q f(x)\, d^n x + \int_Q g(x)\, d^n x$
2. $\int_Q \lambda f(x)\, d^n x = \lambda \int_Q f(x)\, d^n x$
3. $f \leq g \implies \int_Q f(x)\, d^n x \leq \int_Q g(x)\, d^n x$
4. $\left|\int_Q f(x)\, d^n x\right| \leq \int_Q \left|f(x)\right| d^n x$

Definition 27.0.4. We define

$$\text{vol } Q = \int_Q 1 = \prod_{i=1}^{n} |b_i - a_i|$$

Definition 27.0.5. By partitioning the intervals $[a, b] \times [c, d] := Q \subseteq \mathbb{R}^2$ into r equidistant chunks with boundaries

$$a = a_1 \leq a_2 \leq \ldots \leq a_r = b$$
$$c = c_1 \leq c_2 \leq \ldots \leq c_r = d,$$

we partition Q into r^2 equally sized smaller rectangles

$$Q_{ij} := [a_i, c_{i+1}] \times [a_j, c_{j+1}]$$

which must each have area $\text{vol } Q/r^2$. We can thus define ***two-dimensional Riemann sums*** analogously to how we did it in one dimension:

$$S_r^2(f) := \sum_{i=0}^{r-1} \sum_{j=0}^{r-1} f(a_i, c_j) \frac{\text{vol } Q}{r^2}$$

Lemma 27.0.6: Multivariable Integration via Riemann Sums

Given a compact cuboid $Q \subseteq \mathbb{R}^2$ and a continuous function $f : Q \to \mathbb{R}$, we have

$$\int_Q f(x)\, d^2 x = \lim_{r \to \infty} S_r^2(f)$$

Proof. Applying the single-variable version of this theorem twice, we get:

$$\begin{aligned}
\int_Q f(x)\, d^2 x &= \int_c^d \int_a^b f(x_1, x_2)\, dx_1\, dx_2 \\
&= \lim_{r \to \infty} S_r \left(\int_a^b f(x_1, x_2)\, dx_1 \right) \\
&= \lim_{r \to \infty} S_r^2 \big(f(x_1, x_2) \big)
\end{aligned}$$

□

Part IX

Point-Set Topology

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Chapter 28

Topological Spaces

28.1 Topologies

Definition 28.1.1: Accumulation points and isolated points

Let X be a topological space. Let $A \subseteq X$. Then we call $x \in X$ an ***accumulation point*** of A if every neighbourhood of x contains a point in A ***other than x itself***.
If a point $x \in A$ isnt an accumulation point of A, we call x an ***isolated point***.

Definition 28.1.2. Let X be a topological space. Let $A \subseteq X$. Then the set of all accumulation points of A is denoted A', and is known as the ***derived set of A***.

Exercise 28.1.3. *Show that* $A' \subseteq \overline{A}$.

28.2 Basic Separation Properties

A very common axiom is the Hausdorff separation axiom, which says that any two distinct points can be housed orf from each other by disjoint open sets.

"Topology via Logic",
Steven Vickers

Definition 28.2.1. Let $(X, \mathcal{O})$ be a topological space. Let $x, y \in X$. Then we call x and y ***topologically distinguishable*** iff. $\mathcal{U}(x) \neq \mathcal{U}(y)$, i.e. one of the points has a neighborhood not containing the other.

Definition 28.2.2: $T0$ **Space**

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***T0*** if every pair of points is topologically distinguishable.

Definition 28.2.3: $T1$ **Space**

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***T1*** if for every pair of points, both of them have neighbourhoods not containing the other:

$$\begin{aligned} &\forall x, y \in X : \\ &\exists U_x \in \mathcal{U}(x), U_y \in \mathcal{U}(y) : \\ &\quad x \notin U_y, y \notin U_x. \end{aligned}$$

Exercise 28.2.4.
1. *Find a topological space that isn't T0.*
2. *Show that every T1 space is T0.*
3. *Find a space that is T0, but not T1.*

Exercise 28.2.5. *Show that $(X, \mathcal{O})$ is T1 if and only if every singleton set is closed.*

Exercise 28.2.6. *Show that $(X, \mathcal{O})$ is T1 if and only if every subset of X is the intersection of all of its neighborhoods:*

$$\forall Y \subseteq X : Y = \bigcap \{U \mid U \in \mathcal{U}(Y)\}$$

Definition 28.2.7: Hausdorff Space

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***T2*** or ***Hausdorff*** if every pair of points can be housed orf from each other, i.e. they have disjoint neighbourhoods:

$$\begin{aligned} &\forall x \neq y \in X : \\ &\exists U_x \in \mathcal{U}(x), U_y \in \mathcal{U}(y) : \\ &\quad U_x \cap U_Y = \emptyset \end{aligned}$$

Exercise 28.2.8. *Show that every Hausdorff space is T1.*

Exercise 28.2.9. *Let $(X, \mathcal{O})$ be Hausdorff. Then every convergent sequence in X has a unique limit.*

Exercise 28.2.10. *Let X be a Hausdorff space. Let x be an accumulation point of $A \subseteq X$. Then every neighbourhood of x contains infinitely many points of A.*

Lemma 28.2.11. Let X be a Hausdorff space. Then the set of accumulation points of any subset is closed.

Proof. Let $A \subseteq X$. Let A' be the set of accumulation points of A. Let x be an accumulation point of A'. Since X is Hausdorff, every neighbourhood of x must contain infinitely many points $x_i \in A'$. But since these neighbourhoods are also neighbourhoods of the x_i, they must also all contain infinitely many points in A.

Therefore, x is already an accumulation point of A, i.e. $x \in A'$. Therefore, A' contains all of its accumulation points, and is thus closed. □

Exercise 28.2.12. *Find a space X and a subset A such that the set of accumulation points of A isn't closed.*

Theorem 28.2.13. Let $(X, \mathcal{O})$ be a topological space. Then $(X, \mathcal{O})$ is Hausdorff if and only if every point is the intersection of the closures of all of its neighbourhoods:

$$\forall x \in X : \{x\} = \bigcap \left\{ \overline{U} \;\middle|\; U \in \mathcal{U}(x) \right\}$$

Proof. $\Rightarrow$ Let $(X, \mathcal{O})$ be Hausdorff. Let $x, y \in X$ and $y \notin \{x\}$, i.e. $y \neq x$. Then, by Hausdorff, there exist disjoint neighborhoods U_x and U_y of x and y. Since U_x and U_y are disjoint, we have $U_x \subseteq X \setminus U_y$. But then, by monotonicity of the closure operator, we get

$$\overline{U}_x \subseteq \overline{X \setminus U_y} = X \setminus U_y,$$

i.e. $y \notin \overline{U}_x$, i.e.

$$y \notin \bigcap \left\{ \overline{U} \;\middle|\; U \in \mathcal{U}(x) \right\}.$$

$\Leftarrow$ Let $x \neq y$. Let $\{x\} = \bigcap \left\{ \overline{U} \;\middle|\; U \in \mathcal{U}(x) \right\}$. Then there must be an open neighborhood U_x of x whose closure doesn't contain y. But then the complement $U_y := X \setminus \overline{U}_x$ of that closure is an open neighborhood of y that is disjoint from U_x.

□

28.3 The Subspace Topology

Theorem 28.3.1

Let $(X, \mathcal{O})$ be a topological space, and let $S \subseteq X$ be any subset. Then the collection

$$\mathcal{O}_S := \{U \subseteq S \mid \exists V \in \mathcal{O} : U = S \cap V\},$$

consisting of all intersections of open sets of X with S, forms a topology, which we call the ***subspace topology***.

Exercise 28.3.2. *Let $S \subseteq X$. Show that $B \subseteq S$ is closed in S if and only if it is equal to the intersection of S with some closed subset of X.*

Exercise 28.3.3. *Let X be a topological space. Let $U \subseteq S \subseteq X$.*

1. *If U is open in X, then U is open in S.*
2. *If U is closed in X, then U is closed in S.*

Exercise 28.3.4. *Let X be a topological space. Let $U \subseteq S \subseteq X$.*

1. *If U is open in S, and S is open in X, then U is open in X.*
2. *If U is closed in S, and S is closed in X, then U is closed in X.*

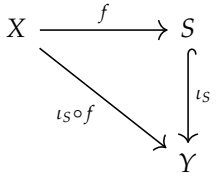
If $U \subseteq S \subseteq X$, the notations $\overline{U}$, $\operatorname{int} U$, and $\operatorname{ext} U$ will always refer to the closure, interior, and exterior of U in X, not in S.

Exercise 28.3.5. *Let X be a topological space. Let $U \subseteq S \subseteq X$.*

1. *The closure of U in S is equal to $\overline{U} \cap S$.*
2. *Find an example where the interior of U in S is not equal to $\operatorname{int} U \cap S$.*
3. *Show that $\operatorname{int} U \cap S$ is a subset of the interior of U in S.*

Theorem 28.3.6: Universal Property of the Subspace Topology

Let X and Y be topological spaces. Let $S \subseteq Y$. Then $f : X \to S$ is continuous if and only if the embedded map $\iota_S \circ f : X \to Y$ is continuous.



Proof. $\Rightarrow$: Suppose that $f : X \to S$ is continuous. Let $V \subseteq Y$ be open. Then we have

$$\begin{aligned}(\iota_S \circ f)^{-1}(V) &= f^{-1}(\iota_S^{-1}(V)) \\ &= f^{-1}(S \cap V) \qquad (2.6.3)\end{aligned}$$

which is open in X since $S \cap V$ is open in S.

$\Leftarrow$:

□

Corollary 28.3.7. Let $S \subseteq X$. Then the inclusion map $\iota_S : S \hookrightarrow X$ is continuous.

Corollary 28.3.8. Let X and Y be topological spaces. Let $f : X \to Y$ be continuous. Then:

1. If $S \subseteq X$, the restriction
$$f|_S : S \to Y$$
is continuous.
2. If $f(X) \subseteq T \subseteq Y$, the map
$$f : X \to T$$
is well-defined and continuous.
3. If $Y \subseteq Z$, then the embedded function
$$\iota_Y \circ f : X \to Z$$
is continuous.

Exercise 28.3.9. *Let X be a topological space. Let $R \subseteq S \subseteq X$. Then the subspace topologies of R in S and X agree.*

Exercise 28.3.10.

1. *Every subspace of a To space is To.*
2. *Every subspace of a T1 space if T1.*
3. *Every subspace of a Hausdorff space is Hausdorff.*

Definition 28.3.11

A map that is a homeomorphism onto its image is called a ***topological embedding***, or just ***embedding***.

Exercise 28.3.12. *A homeomorphism is a surjective topological embedding.*

Exercise 28.3.13. *Let X be a topological space. Let $S \subseteq X$. Then the inclusion map $\iota_S : S \hookrightarrow X$ is an embedding.*

Exercise 28.3.14. *Every embedding is continuous and injective.*

Exercise 28.3.15. *Every injective continuous map that is open or closed is an embedding.*

Exercise 28.3.16. *Find an embedding that is neither open nor closed.*

Theorem 28.3.17

Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \to \mathbb{R}^k$ be continuous. Then the graph $\Gamma(f)$ of f is homeomorphic to U.

Proof. Let $\Phi_f : U \to \mathbb{R}^{n+k}$ be the canonical map

$$\Phi_f(x) = (x, f(x))$$

taking U to its graph.
By definition of the graph $\Gamma(f)$, $\Phi_f(x) : U \to \Gamma(f)$ is surjective.
Since Φ_f is a composition of continuous maps, it is continuous.
Since the first n components of Φ_f are the identity, Φ_f is injective.
Finally, the projection $p : \mathbb{R}^{n+k} \to \mathbb{R}^n$ is continuous, so the restriction $p|_{\Gamma(f)}$ is a continuous inverse of Φ_f. □

28.4 Bases and Countability

Exercise 28.4.1. *Let $(X, \mathcal{O})$ be a topological space. Let $S \subseteq X$. Then if $\mathcal{B}$ is a basis of $\mathcal{O}$, the collection*

$$\mathcal{B}_S = \{B \cap S \mid B \in \mathcal{B}\}$$

is a basis of the subspace topology $\mathcal{O}_S$.

Definition 28.4.2

We call a topological space ***first countable*** if it has a countable neighbourhood basis around every point x, i.e. a set of neigbourhoods B_i of x such that every neighbourhood of x contains some B_i.

Exercise 28.4.3. *Every metric space is first countable.*

It turns out that first countable spaces are exactly the spaces where sequences are sufficient to maintain our intuitions about how sequence convergence interacts with topological properties:

Theorem 28.4.4: Detecting openness and closedness using sequences

Let X be first countable. Let $A \subseteq X$. Let $x \in X$. Then:

1. $x \in \operatorname{int} A$ if and only if every sequence in X converging to x is eventually in A.
2. $x \in \overline{A}$ if and only if there exists a sequence in A converging to x.

Proof. Copy the proof of the metric case, replacing open balls with our countable neighbourhood basis. □

Corollary 28.4.5. Let X be first countable. Let $A \subseteq X$. Let $x \in X$. Then:

1. A is closed in X if and only if it contains every limit of every convergent sequence of points in A.
2. A is open in X if and only if every sequence in X converging to a point of A is eventually in A.

Definition 28.4.6

We call a topological space ***separable*** if it contains a countable dense subset.

Exercise 28.4.7.

1. *Every countable space is separable.*
2. *$\mathbb{R}^n$ is separable for every $n \in \mathbb{N}$.*

Definition 28.4.8

We call a topological space ***second countable*** if it has a countable basis.

Exercise 28.4.9. *$\mathbb{R}^n$ is second countable for every $n \in \mathbb{N}$.*

Exercise 28.4.10. *Let X be second countable. Then:*

1. *X is first countable.*
2. *X is separable.*

Theorem 28.4.11. A metric space is second countable if and only if it is separable.

Exercise 28.4.12. *Let $H = \left\{(0, y) \,\middle|\, y \in \mathbb{R}_{\geq 0}\right\} \subseteq \mathbb{R}^2$ be the closed upper half plane. We define a topology $\mathcal{O}$ on H by taking as a basis the open subsets of $\mathbb{R}^2$ that are contained in H and intersect the boundary of H in at most one point. Show that $(H, \mathcal{O})$ is separable, but not second countable.*

Chapter 29

Compactness

The familiar notions of open covers and compactness generalize directly from metric spaces to topological spaces. The most major difference is that sequential compactness is generally weaker than "covering compactness", and has to be replaced with a notion of compactness based on ultrafilter convergence.

Definition 29.0.1. Let $(X, \mathcal{O})$ be a topological space. Then we call a set $\mathcal{U} \subseteq \mathcal{O}$ of open subsets an ***open covering*** of X if every point of X is contained in the union of all elements of $\mathcal{U}$, i.e. if

$$\bigcup_{U_i \in \mathcal{U}} U_i = X$$

Definition 29.0.2: Compactness

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***compact*** if every open covering of X contains a finite open covering as a subset. We call a subset $K \subseteq X$ compact if it is compact as a topological space equipped with the subspace topology.

Exercise 29.0.3. *Every closed subset of a compact space is compact.*

Exercise 29.0.4. *Let $(X, \mathcal{O})$ be compact and $f : X \to Y$ be continuous. Then $f[X]$ is compact.*

Exercise 29.0.5. *Let $(X, \mathcal{O})$ be a nonempty topological space. Then the following are equivalent:*

1. *$(X, \mathcal{O})$ is compact, i.e. every open covering has a finite subcovering.*
2. *Every set $\mathcal{A}$ of closed sets whose intersection is the empty set has a finite subset whose intersection is the empty set.*
3. *If $\mathcal{A}$ is a set of closed sets such that the intersection over any finite subset of $\mathcal{A}$ is non-empty, the intersection of $\mathcal{A}$ is non-empty.*
4. *Every filter on $(X, \mathcal{O})$ has a limit point.*
5. *Every ultrafilter on $(X, \mathcal{O})$ is convergent.*

Theorem 29.0.6. ***(Alexander Subbase Theorem)*** Let $(X, \mathcal{O})$ be a topological space. Let $\mathcal{S}$ be a subbasis. Then X is compact if every open covering of X by sets of $\mathcal{S}$ has a finite subcovering.

Proof. Assume that X is not compact. Then there exists a non-convergent ultrafilter $\mathcal{F}$, i.e. we have

$$\begin{aligned}&\forall x \in X : \\ &\exists U_x \in \mathcal{U}(x) : \\ &\quad (X \setminus U_x) \in \mathcal{F}\end{aligned}$$

By definition of subbases, every $U_x \in \mathcal{U}(x)$ contains an open subset O_x of the form

$$O_x = \bigcap_{k=0}^{n} S_{k,x}$$

for some $S_{k,x} \in \mathcal{S}$ and $n \in \mathbb{N}$. We therefore have

$$\begin{aligned}X \setminus U_x &\subseteq X \setminus O_x \\ &= X \setminus \bigcap_{k=0}^{n} S_{k,x}\end{aligned}$$

which implies $X \setminus \bigcap_{k=0}^{n} S_{k,x} \in \mathcal{F}$ by the filter axioms. Since the intersection of this set with $\bigcap_{k=0}^{n} S_{k,x} \in \mathcal{F}$ is empty, and filters are closed under finite intersection, $\mathcal{F}$ cannot contain every $S_{k,x}$. Therefore, since $\mathcal{F}$ is an ultrafilter, it must contain at least one set $X \setminus S_{k,x}$ for every x. We will refer to these $S_{k,x}$ with $X \setminus S_{k,x} \in X$ as S_x.

Since every S_x is an open neighborhood of x, and thus contains x, the set of all S_x is an open convering of X. By compactness, there must be a finite subcovering

$$\bigcup_{x \in X'} S_x = X.$$

Now, since $\mathcal{F}$ contains every $X \setminus S_x$, it also contains

$$\bigcap_{x \in X'} (X \setminus S_x) = X \setminus \bigcup_{x \in X'} S_x = X \setminus X = \emptyset.$$

which contradicts $\mathcal{F}$ being a filter. □

Lemma 29.0.7. Let $(X, \mathcal{O})$ be a Hausdorff space. Let $K \subseteq X$ be compact. Then K can be separated from any point $x \in X \setminus K$ by open neighborhoods, i.e:

$$\begin{aligned}&\forall x \in X : \\ &\exists U_x \in \mathcal{U}(x), U_K \in \mathcal{U}(K) : \\ &\quad U_x \cap U_K = \emptyset\end{aligned}$$

Exercise 29.0.8. *Let $(X, \mathcal{O})$ be compact and Hausdorff. Let $A, B \subseteq X$ be closed. Then A and B can be separated from each other by disjoint open neigborhoods.*

Exercise 29.0.9. *Every compact subset of a Hausdorff space is closed.*

Exercise 29.0.10. *Every continuous function from a compact space into a Hausdorff space is closed.*

Exercise 29.0.11. *Every continuous injective function from a compact space into a Hausdorff space is an embedding.*

Definition 29.0.12. We call a topological space X ***Lindelöf*** if every open cover of X has a countable subcover.

Theorem 29.0.13. Every second-countable space is Lindelöf.

Definition 29.0.14. We call a topological space X ***countably compact*** if every countable open cover of X has a finite subcover.

Exercise 29.0.15. *A countably compact space is compact if and only if it is Lindelöf.*

Theorem 29.0.16. A metric space is Lindelöf if and only if it is second countable.

Theorem 29.0.17. Every sequentially compact space is countably compact.

Theorem 29.0.18. A first-countable space is countably compact if and only if it is sequentially compact.

Theorem 29.0.19. In a second countable Hausdorff space, compactness, countable compactness and sequential compactness are all equivalent to each other.

Chapter 30

New Spaces from Old

30.1 Initial Topologies and Product Spaces

Lemma 30.1.1. Let $(X, \mathcal{O})$ be a topological space. Then $(X, \mathcal{O})$ is Hausdorff if and only if the diagonal

$$\Delta := \{(x, x) \mid x \in X\}$$

is closed in the product topology.

Proof. $\Rightarrow$ Assume $(X, \mathcal{O})$ is Hausdorff. Let $(x, y) \notin \Delta$. Then $x \neq y$ by definition of Δ. Therefore, there exist disjoint neighbourhoods U_x, U_y of x and y. Since these neighborhoods are disjoint, we have

$$\Delta \cap (U_x \times U_y) = \emptyset$$

. Therefore, every point $(x, y) \notin \Delta$ has an open neighborhood $(U_x \times U_y)$ that doesn't intersect Δ, i.e. Δ is closed.

$\Leftarrow$ Let $x \neq y$. Assume that Δ is closed. Then $(x, y) \notin \Delta$ has a basic open neighborhood $U_x \times U_y$ that doesn't intersect Δ. Since $U_x \times U_y$ doesn't intersect Δ, U_x and U_y must be disjoint.

□

30.2 Final Topologies and Quotient Spaces

Definition 30.2.1: Final Topologies

Let X be a topological space. Let Y be any set. Let $\mathcal{F} = \left\{ f_i : X \to Y \;\middle|\; i \in I \right\}$ be a collection of functions. Then the ***final topology on Y with respect to $\mathcal{F}$*** is the finest topology such that all functions are continuous, i.e. $U \subseteq Y$ is open in Y if and only if $f_i^{-1}[U]$ is open in X for all $i \in I$.

A final topology with respect to a single surjective function $q : X \to Y$ is known as a ***quotient topology***, and a space Y equipped with a quotient topology is known as a ***quotient space***.

The name "quotient space" comes from the fact that if $q : X \to Y$ is surjective, then each $y \in Y$ can be viewed as an equivalence class $y = \left\{ x \in X \;\middle|\; q(x) = y \right\}$ of the equivalence relation $x_1 \sim x_2 \leftrightarrow q(x_1) = q(x_2)$, i.e. every surjective functions $q : X \to Y$ can be viewed as the projection function of some equivalence relation. Vice versa, every projection function is surjective by definition.

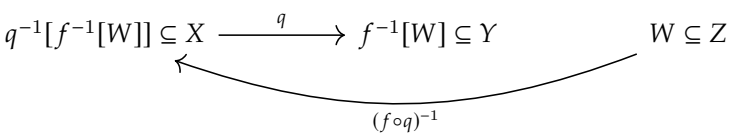
Definition 30.2.2. A continuous surjection $q : X \to Y$ between topological spaces is called a ***quotient map*** or an ***identification map*** if the topology on Y corresponds to the quotient topology induced by q.

Note that this means that we need that V is open in Y if and only if $q^{-1}(V)$ is open in X. We already get one direction for free - since q is continuous, V being open in Y implies that $q^{-1}(V)$ is open in X. Therefore, a continuous surjection is a quotient map if and only if openness of $q^{-1}(V)$ implies openness of V.

Theorem 30.2.3. A surjection $q : X \to Y$ is a quotient map if and only if for all functions $f : Y \to Z$, continuity of $f \circ q : X \to Z$ implies continuity of g and vice versa.

Proof. $\Rightarrow$: Let $q : X \to Y$ be a quotient map. Then q is continuous by definition. Thus, if f is also continuous, then the composition $f \circ q$ must also be continuous.

Vice versa, let $f \circ q$ be continuous. Then, starting with an open set $W \subseteq Z$, $q^{-1}[f^{-1}[W]] = (f \circ q)^{-1}$ is open in Y, since $f \circ q$ is continuous. Since q is a quotient map, a set $q^{-1}[U]$ is open in Y if and only if U is open in X, so $q^{-1}[f^{-1}[W]]$ can only be open in Y if $f^{-1}[W]$ was already open in Y. Thus, f is continuous.



$\Leftarrow$: Let $(Y, \mathcal{O}_Y)$ be the topological space Y equipped with whatever topology it comes packed with.

First of all, note that $q : X \to (Y, \mathcal{O}_Y)$ must be continuous, since we

can simply take $f = 1_Y : (Y, \mathcal{O}_Y) \to (Y, \mathcal{O}_Y)$, so that continuity of 1_Y implies continuity of $q = 1_Y \circ q$.

Now, let $(Y, \mathcal{O}_q)$ be the topological space we get by equipping the set Y with the quotient topology induced by q.

Let $f = 1_Y : (Y, \mathcal{O}_Y) \to (Y, \mathcal{O}_q)$ be the identity function between our two different realizations of Y as a topological space. Then the composition $f \circ q : X \to (Y, \mathcal{O}_q)$ is just $q : X \to (Y, \mathcal{O}_q)$, which is continuous by definition, since $\mathcal{O}_q$ is the quotient topology induced by q. Thus, f must also be continuous.

Finally, the function f^{-1} is continuous, since $f^{-1} \circ q$ is just $q : X \to (Y, \mathcal{O}_Y)$. Therefore, $f : (Y, \mathcal{O}_Y) \cong (Y, \mathcal{O}_q)$ is a homeomorphism, i.e. $\mathcal{O}_Y$ already carries the quotient topology induced by q, i.e. q is a quotient map.

□

Definition 30.2.4: Collapsing a Subspace

Let X be a topological space. Let $A \subseteq X$. Then X/A denotes the quotient space obtained by "collapsing A into a single point", i.e. by the equivalence relation whose equivalence classes are A and all of the points $\{x\} \in X \setminus A$.

Theorem 30.2.5. Let X be regular. Let $A \subseteq X$ be closed. Then $X \setminus A$ is Hausdorff.

Theorem 30.2.6. Let X be normal. Let $A \subseteq X$ be closed. Then $X \setminus A$ is normal.

Theorem 30.2.7. The result of taking the mantle $S^n \times [0,1]$ of an $n+1$-dimensional cylinder and collapsing its bottom is homeomorphic to an $n+1$-dimensional closed ball:

$$S^n \times [0,1]/S^n \times \{0\} \cong \overline{B^{n+1}}$$

Proof. Take the continuous function

$$\begin{aligned} f : S^n \times [0,1] &\to \overline{B^{n+1}} \\ (x,t) &\mapsto tx \end{aligned}$$

We have $f(S^n \times \{0\}) = 0$. Therefore, we can factor our map through our equivalence relation to get a well-defined bijection

$$\begin{aligned} \overline{f} : S^n \times [0,1]/S^n \times \{0\} &\to \overline{B^{n+1}} \\ [x,t] &\mapsto tx. \end{aligned}$$

We have $\overline{f} = f \circ q$, so $\overline{f}$ is a composition of continuous functions, i.e. continuous. Furthermore, $\overline{B^{n+1}}$ is a subspace of $\mathbb{R}^{n+1}$, i.e. Hausdorff, and $S^n \times [0,1]$ is a product of compact sets, i.e. compact. Thus, $\overline{f}$ is a continuous bijection from a compact space into a Hausdorff space, i.e. a Homeomorphism. □

Theorem 30.2.8. Taking a closed n-dimensional ball $\overline{B^n}$ and collapsing its boundary gives an n-sphere:

$$\overline{B^n}/S^{n-1} \cong S^n$$

Definition 30.2.9: Saturated Sets

Let $q : X \to Y$ be a map between topological spaces. Then we call a subset $U \subseteq X$ ***saturated with respect to q*** if

$$\exists V \subseteq Y : U = q^{-1}(V).$$

By extension, given an equivalence relation $\sim$ on X, we call a subset $U \subseteq X$ ***saturated with respect to*** $\sim$ if it is saturated with respect to the induced quotient map, i.e. if

$$U = q^{-1}(V) = \bigcup_{u \in U} [u]_\sim.$$

We call the set $\bigcup_{u\in U}[u]_\sim$ the ***saturation of*** U.

Theorem 30.2.10. A continuous surjection $q : X \to Y$ is a quotient map if and only if it takes saturated open subsets to open subsets or saturated closed subsets to closed subsets.

Proof. 1. Assume that $q : X \to Y$ is a quotient map, i.e. $V \subseteq Y$ is open if and only if $q^{-1}(V) \subseteq X$ is open.

Then if $U \subseteq X$ is saturated, we get $q^{-1}(q(U)) = U$. This immediately tells us that images of saturated open sets are open. If $C \subseteq X$ is closed and saturated, then its complement must be open and saturated, so surjectivity gives us

$$\begin{aligned} q^{-1}(Y \setminus q(C)) &= q^{-1}(q(X) \setminus q(C)) \\ &= q^{-1}(q(X \setminus C)) \\ &= X \setminus C, \end{aligned}$$

i.e. $Y \setminus q(C)$ is open, i.e. $q(C)$ is closed.

2. We have just shown that a continuous surjection taking saturated open sets to open sets also takes saturated closed sets to closed sets. Therefore, assume without loss of generality that q is a continuous surjection taking saturated open sets to open sets.

   If $V \subseteq Y$ is open, continuity tells us directly that $q^{-1}(V) \subseteq X$ is open. On the other hand, given an arbitrary subset $V \subseteq Y$, $q^{-1}(V)$ is saturated by definition, so if it is additionally open, then $q(q^{-1}(V))$ must be open as well. Since q is surjective, we have $q(q^{-1}(V)) = V$, so V is open.

□

Theorem 30.2.11. Let $q : X \to Y$ be a quotient map. Let $U \subseteq X$ be saturated with respect to q and open or closed. Then $q|_U : U \to q(U)$ is a quotient map.

Proof. Since U is saturated and open, we can write $U = q^{-1}(V)$ for some open subset $V \subseteq Y$. The problem then becomes showing that

$$p := q|_{q^{-1}(V)} : q^{-1}(V) \to V$$

is a quotient map, i.e. that p takes saturated open sets to open sets.
Thus, let

$$p^{-1}(W) = q^{-1}(W) \cap q^{-1}(V) \subseteq q^{-1}(V)$$

be an arbitrary open subset of $q^{-1}(V)$ that's saturated with respect to p.
Since preimages play nicely with intersection, we have

$$p^{-1}(W) = q^{-1}(W) \cap q^{-1}(V) = q^{-1}(V \cap W),$$

so $p^{-1}(W)$ is also saturated with respect to q.
Since $q^{-1}(V)$ is open in X, its open subsets are open in X as well, including $q^{-1}(W \cap V)$. Since q is a quotient map, this implies that $W \cap V$ is open in Y, and since $q^{-1}(W \cap V) \subseteq q^{-1}(V)$, we get $W \cap V \subseteq V$, so $W \cap V$ is open in V as well. □

Definition 30.2.12. Let $A \subseteq X$ be topological spaces. Then a continuous map $f : X \to A$ such that $f(a) = a$ for all points $a \in A$ is called a ***retraction***. If such a map exists, A is called a ***retract of X***.

30.3 Adjunction Spaces

Definition 30.3.1: Attaching Spaces along a Subset

Let X and Y be topological spaces, let $A \subseteq X$ be closed, and $f : A \to Y$ be continuous. Then we denote by

$$Y \cup_f X = (X \amalg Y)/\sim$$

the quotient of the disjoint union $X \amalg Y$ by the equivalence relation generated by $a \sim f(a)$ for all $a \in A$. Such a quotient space is called an ***adjunction space***, and is said to be formed by ***attaching Y to X along the attaching map f***.

Note that this equivalence relation identifies each point $y \in Y$ with the entire preimage $f^{-1}(y) \subseteq A$. If $A = \emptyset$, then our relation is empty and we get $Y \cup_f X = X \amalg Y$. If $Y = \{y\}$ only contains a single point, we get $Y \cup_f X = X/A$.

Theorem 30.3.2. Let $Y \cup_f X$ be an adjunction space. Let

$$q : X \amalg Y \to Y \cup_f X$$

be the associate quotient map. Then:

1. $q|_Y : Y \to Y \cup_f X$ is an embedding onto a closed subspace,
2. $q|_{X \setminus A} : X \setminus A \to Y \cup_f X$ is an embedding onto an open subspace,
3. $Y \cup_f X = q(Y) \amalg q(X \setminus A)$.

Proof. 1. Let $B \subseteq Y$ be closed. To show that $q(B) \subseteq Y \cup_f X$ is closed, we need to show that $q^{-1}(q(B))$ is closed in $X \amalg Y$, i.e. that $q^{-1}(q(B)) \cap X$ is closed in X and $q^{-1}(q(B)) \cap Y$ is closed in Y.

Since no two distinct elements of Y can be related to each other, we get $q(B) = B$, and since every element of Y is identified with itself and with its preimages, we get $q^{-1}(q(B)) = B \cup f^{-1}(B)$. Intersecting with X and Y gives

$$q^{-1}(q(B)) \cap Y = B$$
$$q^{-1}(q(B)) \cap X = f^{-1}(B).$$

The former is closed by definition. It follows that $f^{-1}(B)$ is closed in A, and since A is closed in X so is $f^{-1}(B)$. $q|_Y$ is clearly injective, since no points in Y are identified with each other. Thus, $q|_Y$ is a continuous closed injection, and thus an embedding. Note that $q|_Y$ being closed implies that $q(Y)$ is closed.

2. Every element of $X \setminus A$ is exclusively related to itself, so $X \setminus A$ is saturated. Therefore, $q|_{X \setminus A}$ is a quotient map onto its image by Theorem 30.2.11. Since every equivalence class contains only one element, this map is additionally injective, and since $q^{-1}(q(x)) = x$ for all $x \in X \setminus A$, this map is clopen, and thus an embedding.

□

Theorem 30.3.3. If $f : X \to Y$ is a quotient map and K is a locally compact Hausdorff space, then the function $f \times \mathrm{id}_K : X \times K \to Y \times K$ is a quotient map.

Definition 30.3.4. Let $f : X \to Y$ be a continuous function. Then the ***mapping cylinder*** of f is the space

$$M_f = Y \cup_{f_0} (X \times [0, 1]),$$

where $f_0 : X \times \{0\} \to Y$ is the function $f_0(x, 0) = f(x)$.

By abuse of notation, we identify the subset $X \times \{1\} \subseteq M_f$ with X itself.

Definition 30.3.5. Let $f : X \to Y$ be a continuous function. Then the ***mapping cone*** of f is the space

$$C_f = M_f/(X \times \{1\})$$

Theorem 30.3.6. A function $M_f \to Z$ is continuous if and only if the induced functions $X \times I \to Z$ and $Y \to Z$ are both continuous.

Chapter 31

Topological Manifolds

Definition 31.0.1

We call a topological space X ***locally Euclidean of dimension n*** if every point $x \in X$ has a neighbourhood that is homeomorphic to an open subset of $\mathbb{R}^n$.

Exercise 31.0.2. *If a topolocical space X is a locally euclidean of dimension n, then:*

1. *Every point $x \in X$ has a neighbourhood homeomorphic to an open ball in $\mathbb{R}^n$, or*
2. *Every point $x \in X$ has a neighbourhood homeomorphic to $\mathbb{R}^n$, or*

Exercise 31.0.3. *Show that a topological space is locally Euclidean of dimension 0 if and only if it is discrete.*

Definition 31.0.4

A topological space is called an ***n-dimensional topological manifold***, or just ***topological n-manifold***, if it is:

1. locally euclidean of dimension n,
2. second countable,
3. and Hausdorff.

The latter two requirements exclude some pathological spaces - the long line would be a topological manifold if it was second countable, and the bug-eyed line would be a topological manifold if it was Hausdorff. Most interesting results about topological manifolds require both of these properties, so little would be gained by omitting them.
Intuitively, it sounds very sensible that a nonempty topological space cannot be both a topological m-manifold and a topological n-manifold. This statement turns out to be true - however, we will save this proof for later, since it requires a substantial amount of machinery borrowed from algebraic topology.

Lemma 31.0.5. Every open subset of a topological n-manifold is a topological n-manifold.

Exercise 31.0.6. *A topological space is a topological 0-manifold if and only if it is countable and discrete.*

Theorem 31.0.7. Any separable metric space that is locally Euclidean of dimension n is a topological n-manifold.

Definition 31.0.8

We call a topological space an ***n-dimensional manifold with boundary*** if it is second countable, Hausdorff, and every open point has a neighbourhood homeomorphic to an open subset of either $\mathbb{R}^n$, or to the n-dimensional upper half-space

$$\mathbb{H}^n = \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$$

Definition 31.0.9

Let M be a topological n-manifold with boundary. Then a ***coordinate chart for M*** is a pair (U, φ), consisting of an open subset $U \subseteq M$, called a ***coordinate domain***, and a homeomorphism φ from U to an open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$, called a ***coordinate map***. If U is homeomorphic to an open ball, we call U a ***coordinate ball***.

Note that if we can assign coordinate maps for some covering of a topological space X, then that space must by definition be locally euclidean.

Definition 31.0.10: Manifold interior and manifold boundary

If $\varphi(U)$ is an open subset of $\mathbb{R}^n$, we call φ an ***interior chart*** and points $p \in U$ ***interior points***. If $\varphi(U)$ is a subset of $\mathbb{H}^n$ with $\varphi(U) \cap \partial\mathbb{H}^n \neq \emptyset$, we call φ a ***boundary chart***. If there exists a boundary chart taking p to $\partial\mathbb{H}^n$, we call p a ***boundary point of M***.

Every manifold is a manifold with boundary, but the opposite might not hold. We sometimes use the redundant term "manifold without boundary" to refer to a manifold with boundary which has the empty set as its boundary. The word "manifold", without further qualifiers, always means a manifold without boundary.
The definition of a manifold's boundary only agrees with the topological definition once we find the right ambient space to embed M into, since M always has empty topological boundary when viewed as a topological space of its own.
For example, the manifold boundary of the closed unit disk $\overline{\mathbb{B}}^2$ is the unit circle, which agrees with the topological boundary if the disk is viewed as a subset of $\mathbb{R}^2$, but not with the topological boundary of the disk when viewed as its own topological space (which is trivially empty, as stated), or with the topological boundary of the disk viewed as a subset of $\mathbb{R}^3$ (which is the entire disk).

Theorem 31.0.11. If M is a topological n-manifold with boundary, then $\operatorname{int} M$ is both an open subset of M and a topological n-manifold without boundary.

Proof. Since $\operatorname{int} M$ is a union of coordinate domains, it is open. Second-countability and Hausdorffness are trivially inherited by subspaces. By definition, $\operatorname{int} M$ consists only of points homeomorphic to an open subset of $\mathbb{R}^n$, so $\operatorname{int} M$ is additionally locally euclidean of dimension n. □

Proposition 31.0.12. Let M be a topological manifold with boundary. Then a point $x \in M$ cannot be both a boundary point and an interior point.

The general proof once again requires some tricky algebraic topology, so we save it for later.

Corollary 31.0.13. If M is a nonempty topological n-manifold with boundary, then ∂M is a closed subset of M, and M is a topological n-manifold if and only if $\partial M = \emptyset$.

Proof. Since $\operatorname{int} M$ is an open subset of M, $\partial M = M \setminus \operatorname{int} M$ is closed. If M is a manifold, then every point is an interior point, so $M = \int M$, so $\partial M = \emptyset$, and vice versa. □

31.1 Manifolds from Embeddings

Theorem 31.1.1

Let $U \subseteq \mathbb{R}^n$ be open. Let $f : U \to \mathbb{R}^k$ be continuous. Then the graph $\Gamma(f)$ of f is a topological n-manifold.

Proof. By 28.3.17, $\Gamma(f)$ is homeomorphic to $U \subseteq \mathbb{R}^n$, making $\Gamma(f)$ a trivial topological n-manifold coverable by a single coordinate domain. □

Theorem 31.1.2

The unit n-spheres $\mathbb{S}^n$ are topological n-manifolds.

Proof. For each $i \in [1...n+1]$, let U_i^+ and U_i^- denote the sets

$$U_i^+ := \left\{x \in \mathbb{R}^{n+1} \;\middle|\; x_i > 0\right\}$$
$$U_i^- := \left\{x \in \mathbb{R}^{n+1} \;\middle|\; x_i < 0\right\},$$

which are open in $\mathbb{R}^{n+1}$.
Since $\mathbb{S}^n$ doesn't contain the origin, every point on $\mathbb{S}^n$ must have at least one non-zero coordinate, so these sets cover $\mathbb{S}^{n+1}$.
On these sets, $\|x\| = 1$ has a unique solution, so we can find the intersection $x \in \mathbb{S}^n \cap U_i^\pm$ by solving for x_i, which gives us

$$\begin{aligned}
&\sqrt{\sum_{j=1}^{n+1} x_j^2} = 1 \\
\Leftrightarrow &\sum_{j=1}^{n+1} x_j^2 = 1 \\
\Leftrightarrow &\; x_i^2 = 1 - \sum_{j \neq i} x_j^2 \\
\Leftrightarrow &\; x_i = \begin{cases} \sqrt{1 - \sum_{j \neq i} x_j^2} & x \in U_i^+ \\ -\sqrt{1 - \sum_{j \neq i} x_j^2} & x \in U_i^- \end{cases}
\end{aligned}$$

We can thus describe the intersections of $\mathbb{S}^n$ with our sets $U_i^\pm$ by taking every component except the i-th to be arbitrary, and setting the i-th component to the value taken by these continuous square root functions. Swapping the i-th and the $n+1$-st components is continuous, so our intersections are the graphs of these functions. Since these intersections cover $\mathbb{S}^n$, and our functions are $\mathbb{R}^n \to \mathbb{R}$, $\mathbb{S}^n$ is a topological n-manifold.
□

Theorem 31.1.3. Let $p \in \mathbb{S}^n$. Let $H := \langle p \rangle^\perp$ be the hyperplane given as the orthogonal complement of the line through p and the origin. Then the ***stereographic projection map*** σ_p taking a point $x \in \mathbb{S}^n \setminus \{p\}$ to the point where the line $\vec{px}$ intersects H, is a homeomorphism $\mathbb{S}^n \setminus \{p\} \cong H (\cong \mathbb{R}^n)$

Proof. Let $N := e_{n+1} = (0, \ldots, n+1)^\top$ be the "north pole" of the sphere. Take a rotation R in the plane spanned by p and N mapping p to N. Since rotations in the origin preserve $\mathbb{S}^n$, we get

$$\sigma_p = R^{-1} \circ \sigma_N \circ R.$$

Since rotations are orthogonal maps, and thus homeomorphisms, it suffices to show that σ_N is a homeomorphism.

In this case, the hyperplane H orthogonal to $\langle N \rangle$ is simply given by $x_{n+1} = 0$. Therefore, the intersection $u := \sigma_N(x)$ of the line $\vec{Nx} = \{\lambda.(x - N) + N \mid \lambda \in \mathbb{R}\}$ can be found by finding the value of λ that makes the $n+1$st component zero. We get:

$$\begin{aligned} &\lambda.(x_{n+1} - 1) + 1 = 0 \\ \Leftrightarrow\ &\lambda.(x_{n+1} - 1) = -1 \\ \Leftrightarrow\ &\lambda = \frac{1}{1 - x_{n+1}} \end{aligned}$$

For $i \neq n+1$, we have $N_i = 0$, so $u_i = \lambda.x_i$. Substituting our λ into this equation gives

$$\sigma_N(x) = \frac{x_1, \ldots, x_n, 0}{1 - x_{n+1}}$$

This map is continuous for $x_{n+1} \neq 1$, i.e. $x \in \mathbb{S}^n \setminus \{N\}$.
Solving our equations for x instead gives

$$\begin{aligned} x_i &= \frac{u_i}{\lambda}, \\ x_{n+1} &= \frac{\lambda - 1}{\lambda}, \end{aligned}$$

And since our preimage needs to lie on $\mathbb{S}^n$, we substitute into $\|x\|^2 = 1$

and get

$$\begin{aligned}
1 &= \sum_{i=1}^{n+1} x_i^2 \\
&= \left(\sum_{i=1}^{n} \left(\frac{u_i}{\lambda}\right)^2\right) + \left(\frac{\lambda - 1}{\lambda}\right)^2 \\
&= \frac{\sum_{i=1}^{n} u_i^2}{\lambda^2} + \frac{\lambda^2 - 2\lambda + 1}{\lambda^2} \\
&= \frac{\|u\|^2 + \lambda^2 - 2\lambda + 1}{\lambda^2} \\
\Leftrightarrow \lambda^2 &= \|u\|^2 + \lambda^2 - 2\lambda + 1 \\
\Leftrightarrow 2\lambda &= \|u\|^2 + 1 \\
\Leftrightarrow \lambda &= \frac{\|u\|^2 + 1}{2}
\end{aligned}$$

Substituting back into our two equations gives

$$\sigma_N^{-1}(u_1, \ldots, u_n) = \frac{(2u_1, \ldots, 2u_n, \|u\|^2 - 1)^\top}{\|u\|^2 + 1},$$

which is continuous. Therefore, $\sigma_N : \mathbb{S}^n \setminus \{N\} \cong H \cong \mathbb{R}^n$ is indeed a homeomorphism, implying that every σ_p is a homeomorphism. □

σ_N and σ_S (where $S = -N$ is the "south pole") give an alternate coordinate atlas on $\mathbb{S}^n$, which is sometimes easier to work with than our original atlas given by square root functions.

Chapter 32

Topologies on Algebraic Structures

32.1 Topological Groups

Definition 32.1.1

Let $(G, \circ)$ be a group. Let $\mathcal{O}$ be a topology on G. Then we call $(G, \circ, \mathcal{O})$ a ***topological group*** iff. the maps $^{-1} : g \to g^{-1}$ and $\circ : G \times G \to G$ are continuous.

Exercise 32.1.2. *Show that $(G, \circ, \mathcal{O})$ is a topological group if and only if the map*

$$\varphi : (g, h) \mapsto g \circ h^{-1}$$

is continuous.

Exercise 32.1.3. *Let $(G, \circ, \mathcal{O})$ be a topological groups. Then for any $g \in G$, the left and right group operation maps $l_g : h \mapsto g \circ h$ and $r_g : h \mapsto h \circ g$ are homeomorphisms.*

Exercise 32.1.4. *Let $(G, \circ, \mathcal{O}_G)$ be a topological group. Let $(H, \circ, \mathcal{O}_H)$ be a normal subgroup equipped with the supspace topology, and let $(G/H, \circ, \mathcal{O}_q)$ be the quotient group, equipped with the quotient topology. Then the quotient map $p : G \to G/H$ is continuous and open.*

Corollary 32.1.5. If H is closed in G and G is Hausdorff, then G/H is Hausdorff. If G/H is T1, then H is closed.

Chapter 33

Connectedness

Theorem 33.0.1. Let $\{C_i\}_{i\in I}$ be a collection of connected sets in a topological space X such that no two of the C_i are disjoint. Then $\bigcup_{i\in I} C_i$ is connected.

Theorem 33.0.2. Let C be connected. Then every set D with $C \subseteq D \subseteq \overline{C}$ is connected. In particular, $\overline{C}$ is connected.

> *Proof.* Assume that D isn't connected. Then we can write $D = U \cap V$, with U and V disjoint and open. Since $C \subseteq D$, we then have $C = U' \cap V'$ for $U' = U \cap C$ and $V' = V \cap C$. Both sets are open in C by definition of the subspace topology, and since every neighbourhood of a point in $D \subseteq \overline{C}$ contains a point in C, neither of the two are empty. Therefore, C cannot be connected. □

Corollary 33.0.3. Connected components of a topological space are closed.

Chapter 34

Filters

34.1 Filter Convergence

It would be nice to find a generalization of sequence convergence to non-first-countable spaces that shares many of its useful properties.... It turns out that we can! The correct objects for the task are ***filters***.

Exercise 34.1.1. *Let $(X, \mathcal{O})$ be a topological space. Then $(X, \mathcal{O})$ is Hausdorff if and only if every convergent filter on X has a unique limit.*

Theorem 34.1.2. Let X be a topological space. Let $U \subseteq X$. Then U is open iff. U is contained in every ultrafilter converging to some point in U.

Proof. $\Rightarrow$: By definition, if U is an open neighborhood of x, any filter convering to x must contain U.

$\Leftarrow$: Let $U \subseteq X$ and assume $U \in \mathcal{U}$ for every ultrafilter converging to a point $u \in U$. Assume U is not open. Then there exists a point $u \in U$ such that every open neighborhood of u contains at least one point in $X \setminus U$. But then any two members of the family

$$\{X \setminus U\} \cup \{U(u)\}$$

have nonempty intersection. Therefore, this family can be extended to an Ultrafilter $\mathcal{U}_u$. Since $\mathcal{U}_u$ contains all open neighborhoods of u by construction, it converges to u, and must thus contain U. Since it also contains $X \setminus U$, we arrive at a contradiction.

□

34.2 Tychonoff's Theorem

Definition 34.2.1. Let $\mathcal{F}$ be a filter on X. Let $f : X \to Y$. Then the ***pushforward Filter of f on $\mathcal{F}$*** is the set

$$f_*(\mathcal{F}) := \left\{ A \subseteq Y \;\middle|\; p_i^{-1}[A] \in \mathcal{F} \right\},$$

Exercise 34.2.2. *Verify that the pushforward of a filter is a filter, and that the pushforward of an ultrafilter is an ultrafilter.*

Intuitively, the pushforward of a filter under a function generalizes the sequence $f(x_n)$ for some sequence x_n and function f. By extension, the following theorem generalizes the fact that in metric spaces, continuity and sequential continuity are equivalent:

Theorem 34.2.3: Characterization of continuity in terms of ultrafilters

Let $f : X \to Y$. Then f is continuous if and only if $f_*(\mathcal{U})$ converges to $f(x)$ for every ultrafilter $\mathcal{U}$ that converges to $x \in X$.

Proof. 1. Assume f is continuous. Let $\mathcal{U}$ be an ultrafilter on X converging to $x \in X$.

Let $U \subseteq Y$ be an open neighbourhood of $f(x)$. Then, since f is continuous, $f^{-1}(U)$ is an open neighbourhood of x. We thus have $f^{-1}(U) \in \mathcal{U}$, which gives us $U \in f_*(\mathcal{U})$ by the definition of the pushforward filter.

Thus, every open neighbourhood of $f(x)$ is contained in $f_*(\mathcal{U})$, so $f_*(\mathcal{U})$ converges to $f(x)$.

2. Assume that the pushforward of any ultrafilter converging to x converges to $f(x)$. Let $U \subseteq Y$ be open.

Let $\mathcal{U}$ be an ultrafilter on X converging to some point $x \in f^{-1}(U)$. Then, by our hypothesis, $f_*(\mathcal{U})$ converges to $f(x)$. Since U is an open neighborhood of $f(x)$, we have $U \in f_*(\mathcal{U})$, and thus $f^{-1}(U) \in \mathcal{U}$. Thus, by theorem 34.1.2, $f^{-1}(U)$ is open, so f is continuous.

□

This theorem generalizes the fact that a sequence in a metric product space converges if and only if every individual component converges.

Theorem 34.2.4. Let $(X_i, \mathcal{O}_i)_{i \in I}$ be a family of topological spaces. Let $\mathcal{F}$ be a filter on $\prod_{i \in I} X_i$. Then $\mathcal{F}$ converges to a point $x \in \prod_{i \in I} X_i$ if and only if each pushforward filter $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$.

Proof. $\Rightarrow$: Assume that $\mathcal{F}$ converges to $x \in \prod_{i \in I} X_i$. Since all projections p_i are continuous, each $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$ by theorem 34.2.3.

$\Leftarrow$: Assume that each $(p_i)_*(\mathcal{F})$ converges to $p_i(x) = x_i$. By definition of the product topology, the sets $S_i := p_i^{-1}(U_i)$ for open $U_i \in X_i$ form a subbase of $\prod_{i \in I} X_i$. Therefore, any open neighborhood $U \in \mathcal{U}(x)$ contains an open neighborhood $B = \bigcap_{i \in J \subseteq I} p_i^{-1}(U_i)$ for some open neighborhoods $U_i \in \mathcal{U}(x_i)$.

Since each $(p_i)_*(\mathcal{F})$ converges to x_i, we have $U_i \in (p_i)_*(\mathcal{F})$ for each i. Therefore, by definition of the pushforward filter, we have $p_i^{-1}(U_i) \in \mathcal{F}$ for all i. Since filters are closed under intersection, we get $B \in \mathcal{F}$,

and since $B \subseteq U$ we also get $U \in \mathcal{F}$ as desired.

□

Theorem 34.2.5: Tychonoff's Theorem

Let $(X_i, \mathcal{O}_i)_{i \in I}$ be a family of topological spaces. Then if all X_i are compact, $\prod_{i \in I} X$ is compact.

Proof.

1. If there exists an $i \in I$ such that $X_i = \emptyset$, then $\prod X_i = \emptyset$ is compact.
2. Now, assume $\prod_{i \in I} X_i \neq \emptyset$. Then there exists an ultrafilter $\mathcal{U}$ on $\prod_{i \in I} X_i$.

   Since each X_i is compact, each pushforward $(p_i)_*(\mathcal{U})$ converges to an $x_i \in X_i$. Thus, by theorem 34.2.4, $\mathcal{U}$ converges to $(x_i)_{i \in I}$ and $\prod_{i \in I} X_i$ is compact.

□

Chapter 35

Higher Separation Properties

35.1 Regular Spaces

Definition 35.1.1: Regular Space

Let (X, O) be a topological space. Then we call (X, O) ***regular*** if every point can be housed orf from any closed set not containing it:

$$\begin{aligned}&\forall (X \setminus C) \in O : \\ &\forall x \notin C : \\ &\exists U_x \in \mathcal{U}(x), U_C \in \mathcal{U}(C) : \\ &\quad U_x \cap U_C = \emptyset\end{aligned}$$

We call (X, O) ***T3*** or ***regular Hausdorff***, if it is regular and Hausdorff.

Exercise 35.1.2. *Find a topological space that is regular, but not T0.*

Exercise 35.1.3. *Show that a topological space is regular Hausdorff (T_3) if and only if it is regular and T_1.*

Theorem 35.1.4. Every compact Hausdorff space is regular.

Proof. Let $C \subseteq X$ be closed, and let $y \in X \setminus C$. Since X is Hausdorff, there exist disjoint open neighbourhoods $U_x \ni x \in C$ and $V_x \ni y$ for each $x \in C$. Let $\mathcal{U} := \{U_x \mid x \in C\}$. Since C is a closed subset of the compact space X, C is also compact. Therefore, the open cover $\mathcal{U}$ of C has a finite subcover

$$\mathcal{U}' = \left\{U_{x_i} \,\middle|\, i \in [1..n]\right\}.$$

If we now set

$$\mathcal{V}' = \left\{V_{x_i} \,\middle|\, i \in [1..n]\right\},$$

we have our desired separation of x and C via

$$\begin{aligned}U &= \bigcup \mathcal{U}' \\ V &= \bigcap \mathcal{V}'.\end{aligned}$$

□

Theorem 35.1.5. A Hausdorff space is regular if and only if the closed neighbourhoods of any point form a neighbourhood basis of that point.

Definition 35.1.6: Completely Regular Space

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***completely regular*** if every point can be housed orf from any closed set not containing it by a continuous function:

$$\begin{aligned}&\forall (X \setminus C) \in \mathcal{O} : \\ &\forall x \notin C : \\ &\exists f \in C(X, [0,1]_{\mathbb{R}}) : \\ &\quad f(x) = 1,\ f[A] \subseteq \{0\}\end{aligned}$$

We call $(X, \mathcal{O})$ ***T3a***, ***T3$\frac{1}{2}$***, or ***completely regular Hausdorff***, if it is completely regular and Hausdorff.

Exercise 35.1.7. *Show that every completely regular space is regular.*

Exercise 35.1.8. *Find a space that is completely regular, but not T0.*

Exercise 35.1.9. *Show that a space if completely regular Hausdorff if and only if it is completely regular and T1.*

Theorem 35.1.10. Every zero-dimensional space $(X, \mathcal{O})$ is completely regular.

Proof. Since X is zero-dimensional, it has a basis consisting of clopen sets. Given any clopen set U, the function

$$f_U : x \mapsto \begin{cases} 1 & x \in U \\ 0 & x \in X \setminus U \end{cases}$$

is continuous. Now, let C be a closed set not containing x. Then $X \setminus C$ is an open neighbourhood of x. We can now choose a basic (i.e. clopen) neighbourhood $U \subseteq X \setminus C$ of x, at which point f_U is our desired function separating x and C. □

Theorem 35.1.11. Let $(X, \mathcal{O})$ be completely regular. Then the set

$$\left\{ f^{-1}[U] \,\middle|\, f \in C(X, [0,1]), U \in \mathcal{O}_{\mathbb{R}} \right\}$$

is a basis of X.

Proof.

1. Since the f are continuous, every basis set is open.
2. Let $x \in X$. Let U be an open neighbourhood of x. Since X is completely regular, there exists a continuous function f separating $\{x\}$ and $X \setminus U$. Then the basic set $f^{-1}[(0,1]]$ must be a subset of U, since every element outside of U gets mapped to 0. Thus, every open neighhourhood contains a basic neighbourhood.

□

Theorem 35.1.12. Every completely regular Hausdorff space $(X, \mathcal{O})$ can be embedded into the cube

$$\prod_{f \in C(X,[0,1])} [0,1]$$

Proof. We choose the trivial embedding $\iota : x \mapsto (f(x))_{f \in C(X,[0,1])}$.

1. ι is continuous, since the projections of ι are exactly the elements of $C(X, [0,1])$, i.e. continuous.

2. Let $x \neq y$. By T1, all singletons are closed. Therefore, by complete regularity, there exists a function f with $f(x) = 1$ and $f(y) = 0$. Therefore, we have $\iota(x)_f \neq \iota(y)_f$, i.e. $\iota(x) \neq \iota(y)$.

3. By 35.1.11, we only need to show openness for the basis sets

$$\left\{ f^{-1}[U] \;\middle|\; f \in C((X, \mathcal{O}), [0,1]_{\mathbb{R}}), U \in \mathcal{O}_{\mathbb{R}} \right\}.$$

   For these sets, we get:

$$\begin{aligned}
\iota\left[f^{-1}[U]\right] &= \iota[(p_f \circ \iota)^{-1}[U]] \\
&= \iota[\iota^{-1}(p_f^{-1}[U])] \\
&= \iota\left[\iota^{-1}\left[U \times \prod_{g \neq f} [0,1]\right]\right] \\
&= \left(U \times \prod_{g \neq f} [0,1]\right) \cap \iota[X]
\end{aligned}$$

   Where $\left(U \times \prod_{g \neq f} [0,1]\right)$ is open, since every component besides the f-component is open.

□

35.2 Normal Spaces

Definition 35.2.1: Normal Space

Let $(X, \mathcal{O})$ be a topological space. Then we call $(X, \mathcal{O})$ ***normal*** if every pair of disjoint closed sets can be housed orf from each other by disjoint neighbourhoods:

$$\begin{aligned} &\forall X \setminus C_1, X \setminus C_2 \in \mathcal{O} : \\ &\quad C_1 \cap C_2 = \emptyset \implies \\ &\exists U_1 \in \mathcal{U}(C_1), U_2 \in \mathcal{U}(C_2) : \\ &\quad U_1 \cap U_2 = \emptyset \end{aligned}$$

We call $(X, \mathcal{O})$ ***T4***, or ***normal Hausdorff***, if it is normal and Hausdorff.

Exercise 35.2.2. *Find a topological space that is normal, but not To.*

Exercise 35.2.3. *Show that every compact Hausdorff space is normal. (Hint: Our proof of Theorem 35.1.4 (every compact Hausdorff space is regular) gets us most of the way there and only needs a small alteration.)*

Lemma 35.2.4: Urysohn's Lemma

A topological space is normal if and only if any two disjoint closed subsets can be separated by a continuous function.

Corollary 35.2.5. Every normal Hausdorff space is completely regular Hausdorff.

Theorem 35.2.6. A second-countable regular topological space is normal.

Proof. Let A, B be disjoint closed sets in a second-countable regular topological space X. Let $\{U_i\}_{i \in I}$ be the collection of open sets such that $U \cap A \neq \emptyset$ and $\overline{U} \cap B = \emptyset$. Since X is regular, this collection forms an open covering of A, and X is second-countable, we can take a countable subcovering $\{U_n\}_{n \in \mathbb{N}}$.
Analogously, we can construct a countable open covering $\{V_n\}_{n \in \mathbb{N}}$ of B fulfilling $\overline{V_n} \cap A = \emptyset$.
Now, in order to make these two coverings disjoint from each other as well, we take

$$Y_n := U_n \cap \bigcap_{k=1}^{n} X \setminus \overline{V}_k$$

$$Z_n := V_n \cap \bigcap_{k=1}^{n} X \setminus \overline{U}_k$$

By construction of our U_n and V_n, our Y_n form an open cover of A and our Z_n form an open cover of B. Furthermore, we have $Y_m \cap Z_n = \emptyset$ for

all $m, n \in \mathbb{N}$, since $m \geq n$ implies

$$\begin{aligned} Y_m \cap Z_n &\supseteq Y_m \cap V_n \\ &\supseteq Y_m \cap \overline{V}_n \\ &\supseteq U_m \cap (X \setminus \overline{V}_n) \cap \overline{V}_n \\ &= \emptyset, \end{aligned}$$

and $n \geq m$ implies the same, but with Y, Z and U, V swapped.
Therefore, $\bigcup_{m\in\mathbb{N}} Y_m$ and $\bigcup_{n\in\mathbb{N}} Z_n$ are disjoint open sets containing A and B, completing the proof that X is normal. □

35.3 Urysohn's Metrization Theorem

Corollary 35.3.1: Urysohn's Metrization Theorem

Every second-countable completely regular Hausdorff space X can be embedded into the Hilbert Cube

$$\prod_{n\in\mathbb{N}}[0,1]$$

Proof. Since X is second-countable completely regular Hausdorff, X is normal Hausdorff by Theorem 35.2.6. Now, we define a set

$$S = \left\{(U,V) \in \mathcal{B}^2 \;\middle|\; \overline{U} \subseteq V\right\}.$$

Urysohn's Lemma 35.2.4 tells us that for each $s = (U,V) \in S$, there exists a continuous function $g_s : X \to [0,1]$ such that $g_s[\overline{U}] = 1$ and $g_s[X \setminus V] = 0$. This lets us find a canonical continuous map $X \to \prod_{s\in S}[0,1]$:

$$\begin{aligned} g : X &\to \prod_{s\in S}[0,1] \\ x &\mapsto (s \mapsto g_s(x)) \end{aligned}$$

This map is injective, since for $x \neq y$, we can use complete regularity to find a pair $s = (U,V) \in S$ with $x \in U$ and $y \notin V$ such that $g_s(x) = 1$ and $g_s(y) \in g_s[X \setminus V] = 0$, i.e. $g(x) \neq g(y)$.

It remains to be shown that g is open. If W is an open neighbourhood of $x \in X$, we want the set $g(W)$ to be open in $\prod_{s\in S}[0,1]$, which follows if we can find for each $g(y) \in g(W)$ a subbasis neighbourhood $\pi^{-1}(S)$ of the product topology such that

$$g(y) \in g(X) \cap \pi^{-1}(S) \subseteq g(W)$$

Now, as earlier, we find basic neighbourhoods $V \subseteq W \in \mathcal{B}$ and $s = (U,V) \in S$ such that $g_s(x) \in g_s[\overline{U}] = 1$ and $g_s[X \setminus V] = 0$, i.e.

$$g_s(y) \neq 0 \implies y \in V,$$

i.e. $g_s^{-1}[(0,1]] \subseteq V$. Since we have $g_s = \pi_s \circ g$, we get $g_s^{-1} = g^{-1} \circ \pi_s^{-1}$. Fiddling around with images and preimages for a bit now gives us:

$$\begin{aligned} g_s^{-1}[(0,1]] \cap X = g^{-1}(\pi_s^{-1}[(0,1]]) \cap X \subseteq V \\ &\implies g(g^{-1}(\pi_s^{-1}[(0,1]]) \cap X) \subseteq g[V] \\ &\overset{g \text{ inj.}}{\implies} g(g^{-1}(\pi_s^{-1}[(0,1]])) \cap g[X] \subseteq g[V] \\ &\implies \pi_s^{-1}[(0,1]] \cap g[X] \subseteq g[V] \end{aligned}$$

□

35.4 Local Compactness and Compactification

Definition 35.4.1

We call a topological space ***locally compact*** if every point has a compact neighbourhood.

Theorem 35.4.2. Let X be a locally compact Hausdorff space. Then:

1. X is completely regular.
2. Every neighbourhood of $x \in X$ contains a compact neighbourhood of x.

Proof. Let C be a compact neighbourhood of x. Let U be an arbitrary neighbourhood of x. Let $V \subseteq C \cap U$ be open with $x \in V$. Then $\overline{V} \subseteq C$ is a compact Hausdorff space, and thus normal by Exercise 35.2.3 and completely regular by Urysohn's Lemma 35.2.4. Since every point has a completely regular neighbourhood, X is completely regular.

Since $V \subseteq \overline{V}$ is also completely regular, the closed neighbourhoods of any point $x \in V$ form a neighbourhood basis by Theorem 35.1.5.

Thus, there exists a neighbourhood $N \subseteq V$ of $x \in C$ which is closed in $\overline{V}$, and thus closed in X. Since $N \subseteq C$ is closed and C is compact, N is also compact. Since N is a neighbourhood of x in $\overline{V}$ and $N \subseteq V$,< N is also a neighbourhood of x in the open set V, and thus in all of X. □

Theorem 35.4.3: One-Point Compactification

Let $(X, \mathcal{O})$ be a locally compact Hausdorff space. Let $X^+ = X \cup \{\infty\}$ be the set consisting of X and an additional "point at infinity" not originally contained in X.

Define the collection of open sets in X^+ to be

$$\mathcal{O}^+ := \mathcal{O} \cup \left\{ X^+ \setminus C \;\middle|\; C \subseteq X \text{ compact} \right\},$$

i.e. the original open sets in X plus the complements of compact sets in X with the additional point ∞ added. Then $(X^+, \mathcal{O}^+)$ is a compact Hausdorff space called the ***one point compactification*** of X. Furthermore, $\mathcal{O}^+$ is the only topology making the set X^+ a compact Hausdorff space with X as a subspace.

35.5 Paracompactness

Definition 35.5.1. Let $\mathcal{U}$ and $\mathcal{V}$ be coverings of a topological space X. We call $\mathcal{U}$ a ***refinement*** of $\mathcal{V}$ if each element of $\mathcal{U}$ is a subset of some element of $\mathcal{V}$.

Definition 35.5.2. Let $\mathcal{U}$ be a collection of subsets of a topological space X. We call $\mathcal{U}$ ***locally finite*** if each point $x \in X$ has a neighbourhood N which only intersects a finite number of elements of $\mathcal{U}$.

Theorem 35.5.3. Let $\mathcal{U} = \{U_i \mid i \in I\}$ be a locally finite collection of subsets of a topological space X. Then

$$\overline{\bigcup_{i\in I} U_i} = \bigcup_{i\in I} \overline{U}_i$$

Proof. $\supseteq$: Follows trivially, since

$$\begin{aligned} &\forall i \in I : \bigcup_{i\in I} U_i \supseteq U_i \\ \implies &\forall i \in I : \overline{\bigcup_{i\in I} U_i} \supseteq \overline{U}_i \\ \implies &\overline{\bigcup_{i\in I} U_i} \supseteq \bigcup_{i\in I} \overline{U}_i \end{aligned}$$

$\subseteq$: It suffices to show that $\bigcup_{i\in I} \overline{U}_i$ is closed: $\bigcup_{i\in I} \overline{U}_i$ trivially contains $\bigcup_{i\in I} U_i$, which means that $\overline{\bigcup_{i\in I} U_i}$ is contained in the closure of $\bigcup_{i\in I} \overline{U}_i$.

Thus, we pick $x \notin \bigcup_{i\in I} \overline{U}_i$. Local finiteness tells us that there is a neighbourhood $N \ni x$ which intersects only finitely many U_i, which we denote $U_{i_1}, \dots, U_{i_n}$. This lets us construct an open neighbourhood

$$N' = N \setminus \bigcup_{k=1}^{n} \overline{U}_{i_k},$$

which still contains x but doesn't intersect any U_i. Since N' is open, it also doesn't intersect any $\overline{U}_i$ (otherwise there would have to be a point $y \in N'$ which doesn't have a anyneighbourhood in N').

□

This equality doesn't hold for arbitrary collections, since an infinite union of closed sets may not be closed - consider for example $I = \mathbb{N}$ and

$$U_i := [1/n, 1 - 1/n].$$

Definition 35.5.4: Paracompact Spaces

We call a topological space X ***paracompact*** if every open covering of X has a locally finite open refinement.

Exercise 35.5.5. *Every closed subset of a paracompact space is paracompact.*

Theorem 35.5.6. Every paracompact Hausdorff space is regular.

Proof. We start by constructing disjoint neighbourhoods $U_x \ni x \in C$, $V_x \ni y \in X \setminus C$ as we did in the compact case 35.1.4. Then $\mathcal{U} = \{U_x \mid x \in C\}$ is an open covering of C. Let $\mathcal{U}' = \{U_i \mid i \in I\}$ be an open locally finite refinement, and let $U' = \bigcup_{i \in I} U_i$.

Since $\mathcal{U}'$ is locally finite, we have

$$\overline{U'} = \bigcup_{i \in I} \overline{U}_i$$

by Theorem 35.5.3.

Since y isn't contained in any U_x, and $y \in V_x$, which is an open subset disjoint from U_x, y isn't contained in any $\overline{U}_x$, and therefore can't be contained in any $\overline{U}_i \subseteq \overline{U}_x$ and especially not in their union $\bigcup_{i \in I} \overline{U}_i = \overline{U'}$.

Thus, $U' \supseteq C$ and $X \setminus \overline{U'} \ni y$ are disjoint open sets containing C and y respectively. □

Corollary 35.5.7. Every paracompact Hausdorff space is normal.

Proof. Let $C_1, C_2 \subseteq x$ be disjoint closed sets. Since every paracompact Hausdorff space is regular, we can apply the same construction as in the regular case, but set $x \in C_1$ and have each V_x be a neighbourhood of the entirety of C_2. □

Thus, every second countable paracompact space is metrizable by Urysohn's Lemma 35.2.4.

Proposition 35.5.8. Every metric space is paracompact.

Proof Sketch. The proof makes use of some set-theoretic trickery involving ordinals and the well-ordering theorem. If you're interested, a short version is given by Mary Ellen Rudin in *A new proof that metric spaces are paracompact*, AMS (1968) "□"

Definition 35.5.9

Let $f : X \to \mathbb{R}$ be a continuous map from a topological space into the real numbers. Then the ***support of f*** is the smallest closed set on which $f(x) \neq 0$:

$$\operatorname{supp} f = \overline{\left\{x \in X \;\middle|\; f(x) \neq 0\right\}}$$

Definition 35.5.10. Let $\mathcal{U} = \{U_i \mid i \in I\}$ be an open covering of the space X. Then a ***partition of unity*** subordinate to this covering is a collection of maps

$$\left\{f_j : X \to [0,1] \;\middle|\; j \in J\right\}$$

such that:

1. There exists a locally finite open refinement $\left\{V_j \;\middle|\; j \in J\right\}$ of $\mathcal{U}$ such that

$$\forall j \in J : \operatorname{supp} f_j \subseteq V_j$$

2. The functions "partition unity" in the sense that they add up to the constant function $\mathbb{1}_X$:

$$\forall x \in X : \sum_{j \in J} f_j(x) = 1$$

Theorem 35.5.11. Let X be a paracompact topological space. Let $\mathcal{U}$ be an open covering of X. Then there exists a partition of unity subordinate to $\mathcal{U}$.

Corollary 35.5.12. Let X be paracompact. Let $\{U_i \mid i \in I\}$ be a locally finite open covering of X. Then there exists an open covering $\{V_i \mid i \in I\}$ such that for each $i \in I$, we have $\overline{V}_i \subseteq U_i$.

Definition 35.5.13. We call a topological space ***σ-compact*** if it is the union of countably many compact subspaces.

Theorem 35.5.14. A locally compact Hausdorff space is paracompact if and only if it is the disjoint union of open σ-compact subsets.

Theorem 35.5.15. Let X be a locally compact second countable Hausdorff space. Then X is σ-compact and paracompact.

In particular, X is a second countable completely regular Hausdorff space by Corollary 35.5.7, and thus metrizable by Urysohn's metrization theorem 35.3.1.

Proof. Let X be a locally compact second countable Hausdorff space. Let $\mathcal{B}$ be a countable basis of X. Let $C \subseteq X$ be compact. By local compactness, every point $x \in C$ has a compact neighbourhood N. Since N is a neighbourhood, there exists a basis element $U_x \in \mathcal{B}$ such that $x \in U_x \subseteq N$. Since $U_x \subseteq N$, each closure $\overline{U}_x$ is compact, so the closures of these basis elements give a countable covering of X by compact sets, i.e. X is σ-compact. By Theorem 35.5.14, X is also paracompact. □

Theorem 35.5.16. If X is a locally compact second countable Hausdorff space, then its one-point compactification X^+ is metrizable.

Proof. Continuing the construction from the previous proof, we can cover C by the our basis elements U_x and pick a finite subcover $\{U_{x_k} \mid k \in [1..n]\}$.

Then $\bigcup_{k=1}^{n} \overline{U_{x_k}}$ is a finite union of compact sets, i.e. compact, so the set

$$V_C = X^+ \setminus \bigcup_{k=1}^{n} \overline{U_{x_k}}$$

is an open neighbourhood of ∞ contained in an arbitrary neighbourhood of the form $X^+ \setminus C$.

Each set V_C is uniquely defined by a finite subset of the countable set $\mathcal{B}$, so there are only countably many V_C. Thus, $\mathcal{B} \cup \{V_C \mid C \text{ compact}\}$ is a countable basis of X^+. Furthermore, X^+ is a compact Hausdorff space by construction, and thus completely regular. Metrizability follows, as usual, from Urysohn's metrization theorem 35.3.1. □

Chapter 36

Lattice Theory

36.1 Lattices

Definition 36.1.1. A partially ordered set $(L, \leq)$ is called a ***lattice*** if every pair of elements $x, y \in L$ has a least upper bound $x \wedge y$, also known as a ***join***, and a greatest lower bound $x \vee y$, also known as a ***meet***. A ***complete lattice*** is a lattice in which every subset $A \subseteq L$ has a join $\sup A$ and a meet $\inf A$.

Definition 36.1.2. Given a complete lattice L, the ***bottom element*** is the element $0 := \inf L$, and the ***top element*** is the element $\infty := \sup L$.

Definition 36.1.3. Let L be a complete latice. Let $x, y \in L$. Then we say that x is ***well above*** y, denoted $x \succ y$, iff for any subset $A \subseteq L$,

$$y \geq \inf A \implies \exists a \in A : x \geq a$$

Lemma 36.1.4. Let L be a complete lattice. Let $x, y, z \in L$. Then:

1. $x \succ y$ implies $x \geq y$,
2. $x \succ y$ and $y \geq z$ implies $x \geq z$,
3. $x \geq y$ and $y \succ z$ implies $x \succ z$.

Proof.
1. We have $y \geq \inf \{y\}$, so $x \succ y$ implies $x \geq y$.
2. Follows from applying transitivity to the first part of the lemma.
3. $z \geq \inf A \implies \exists a \in A : x \geq y \geq a$.

□

Lemma 36.1.5. Let L be a complete lattice. Let $x \in L$ and $A \subseteq L$. Then

$$x \succ \inf A \Leftrightarrow \exists a \in A : x \succ a$$

36.2 Value Quantales

36.3 Generalized metrics metrize every topological space

Part X

Differential Geometry

Chapter 37

Smooth Manifolds

37.1 Abstract Smooth Manifolds

Definition 37.1.1: Smooth Manifolds via Charts

An ***n-dimensional smooth manifold*** is a second-countable Hausdorff space M together with a collection of maps called "charts" such that:

1. Each chart is a homeomorphism $\Phi : U \to U' \subseteq \mathbb{R}^n$, where U is open in M and U' is open in $\mathbb{R}^n$.
2. Each point $x \in M$ is in the domain of some chart,
3. Given charts $\Phi : U \to U'$ and $\Psi : V \to V'$, the coordinate change function $\Phi\Psi^{-1} : \Psi(U \cap V) \to \Phi(U \cap V)$ is smooth (i.e. infinitely continuously differentiable)
4. The collection of charts is maximal, i.e. if a map fulfills conditions (1) and doesn't violate condition (3) when taken together with preexisting charts, then that new map is a chart.

We call our any collection of charts satisfying conditions $1, 2$ and 3 an ***atlas*** of M.

Note that this definition directly generalizes topological manifolds, but with additional smoothness demands on our charts. Since most manifolds can't be equipped with the structure of a vector space, differentiability isn't well-defined for our charts $\Phi : U \to U \subseteq \mathbb{R}^n$, so instead we move that demand to the coordinate change functions $\Phi\Psi^{-1}$, which are functions mapping $\mathbb{R}^n$ into itself.

We also want to construct a second, slightly more abstract definition of a smooth manifold, based on the notion of ***functional structures***.

Definition 37.1.2

Let $(X, \mathcal{O})$ be a topological space. A ***fuctional structure*** on X is a function F_X defined on the collection $\mathcal{O}$ of open sets in X, such that:

1. $F_X(U)$ is a subalgebra of the algebra $C(U, \mathbb{R})$ of continuous real-valued functions on U[1],
2. $F_X(U)$ contains all constant functions,
3. If $V \subseteq U$ and $f \in F_X(U)$, then $f|_V \in F_X(V)$, and
4. If $U = \bigcup_{i \in I} U_i$ and $f|_{U_i} \in F_x(U_i)$ for all i, then $f \in F_X(U)$.

The pair (X, F) is called a ***functionally structured space***.

[1] Originally defined in Exercise 16.2.2.

In particular, we care about the following functional structures on $X = \mathbb{R}^n$:

1. $F_X(U) = C(U, \mathbb{R})$,
2. $F_X(U) = C^k(U, \mathbb{R})$,
3. $F_X(U) = C^\infty(U, \mathbb{R})$,
4. $F_X(U) = A(U, \mathbb{R})$ (= the set of analytic functions $U \to \mathbb{R}$).

Exercise 37.1.3. *If (X, F_X) is a functionally structured space, we can turn (U, F_U) into a functionally structured space by setting $F_U(V) = F_X(V)$ for all subsets $V \subseteq U$.*

Definition 37.1.4

A ***morphism of functionally structured spaces***

$$(X, F_X) \to (Y, F_Y)$$

is a continuous function $\Phi : X \to Y$ such that precomposition pulls back elements of $F_Y(U)$ to elements of $F_X(\Phi^{-1}(U))$:

$$f \in F_Y(U) \implies f \circ \Phi \in F_X(\Phi^{-1}(U))$$

An isomorphism, as always, is a morphism whose inverse exists and is a morphism.

Definition 37.1.5: Smooth Manifolds via Functional Structure

An n-dimensional differentiable manifold is a second countable functionally structured Hausdorff space (M, F) which is locally isomorphic to $(\mathbb{R}^n, C^\infty)$, i.e. each point in M has a neighborhood U such that $(U, F_U) \cong (V, C^\infty_V)$.

Theorem 37.1.6. Our two definitions of smooth manifolds are equivalent.

Proof. $\Rightarrow$: Let M be a smooth manifold in the sense of Definition 37.1.1. We need to define the sets $F(U)$ for all open sets U. Let U be the domain of a chart $\Phi : U \to U' \subseteq \mathbb{R}^n$. Then we can set

$$F(U) = \left\{ f \circ \Phi \,\middle|\, f \in C^\infty(U', \mathbb{R}) \right\}$$

Then F inherits the properties defining a functional structure from $C^\infty(U', \mathbb{R})$, and Φ is an isomorphism between F and C^∞ by definition of F.

$\Leftarrow$: Let (M, F) be a smooth manifold in the sense of Definition 37.1.5. We take our charts $\Phi : U \to V$ to be isomorphisms $(U, F_U) \cong (V, C^\infty_V)$.

It remains to be shown that the coordinate change function $\theta := \Phi\Psi^{-1}$ of any pair of charts is C^∞, and that our atlas is maximal. Since θ is a morphism, $f \circ \theta$ is C^∞ for any $f \in C^\infty$. Therefore, $\pi_i \circ \theta$ is C^∞ for all $i \in [1..n]$, so θ is also C^∞.

At the same time, if Φ and Ψ are homeomorphisms and θ is C^∞, then θ is clearly an isomorphism, since $\theta \in C^\infty$ implies $f \circ \theta \in C^\infty$ for all $f \in C^\infty$ and $\theta^{-1} \in C^\infty$ exists and fulfills the same conditions. Therefore, our atlas is maximal.

□

Proposition 37.1.7. These two constructions are inverse to each other.

From now on, we will implicitly assume that every functional structure induced by a chart structure and every chart structure induced by a functional structure is created via this construction, where charts correspond to local isomorphisms of the functional structures.

This lets us generalize smoothness from vector spaces to manifolds:

Definition 37.1.8

A map between manifolds is called ***smooth*** or $\boldsymbol{C^\infty}$ if it is a morphism between their respective functional structures

Theorem 37.1.9

A map $f : M \to N$ between two smooth manifolds is smooth in the sense of being a morphism between functional structures if and only if, for any charts $\Phi : U \to U' \subseteq \mathbb{R}^m$ on M and $\Psi : V \to V' \subseteq \mathbb{R}^n$ on N, the function $\Psi \circ f \circ \Phi^{-1}$ is smooth.

Proof. $\Rightarrow$: Let $f : (M, F_M) \to (N, F_N)$ be a morphism. Let $\pi_i : \mathbb{R}^n \to \mathbb{R}$ be the i-th coordinate projection function. Since charts are isomorphisms, we get:

$$\begin{aligned} & \pi_i \in C^\infty(\mathbb{R}^n, \mathbb{R}) \\ \implies & \pi_i \circ \Psi \in F_N(V) \\ \implies & \pi_i \circ \Psi \circ f \in F_M(f^{-1}(V)) \\ \implies & \pi_i \circ \Psi \circ f \circ \Phi^{-1} \in C^\infty(\Phi(f^{-1}(V)), \mathbb{R}) \end{aligned}$$

Since every component of $\Psi \circ f \circ \Phi^{-1}$ is smooth, the entire function is smooth.

$\Leftarrow$: Let $\Psi \circ f \circ \Phi^{-1}$ be smooth for all charts Φ, Ψ. Since Φ is bijective, the set on which this function is well-defined is

$$\Phi(f^{-1}(V)) \cap \Phi(U) = \Phi(f^{-1}(V) \cap U).$$

Let $g \in F_N(U)$. Then we have $g = h \circ \Psi$ for some chart $\Psi : V \to \mathbb{R}^n$ and some smooth real-valued function $h \in C^\infty(\mathbb{R}^n, \mathbb{R})$.

We get:

$$\begin{aligned} & g := h \circ \Phi \in F_N(V) \\ \implies & h \in C^\infty(\mathbb{R}^n, \mathbb{R}) \\ \implies & h \circ \Psi \circ f \circ \Phi^{-1} \in C^\infty(\Phi(f^{-1}(V) \cap U), \mathbb{R}) \\ \implies & (g \circ f)|_{f^{-1}(V) \cap U} = h \circ \Psi \circ f \circ \Phi^{-1} \circ \Phi \in F_M(f^{-1}(V) \cap U, \mathbb{R}) \end{aligned}$$

Picking different charts $\Phi_i : U_i \to \mathbb{R}^n$ whose domains cover $f^{-1}(V)$ gives $g \circ f \in F_M(f^{-1}(V))$ as desired.

□

Another concept that manifolds inherit from vector spaces is orientation:

Definition 37.1.10

Let M be an n-manifold. Then an atlas A which is maximal with regards to the property that, for any pair of charts Φ, Ψ in the atlas, the Jacobian of the coordinate change function $\Phi\Psi^{-1}$ has positive determinant at all points in its domain, is called an ***orientation*** of M. An n-manifold having such an atlas is called ***orientable***.

Theorem 37.1.11. Every orientable manifold has at least two orientations.

Proof Sketch. Given an orientation of M, we can pick one coordinate of each chart and reverse the sign, leading to a new orientation. "□"

Theorem 37.1.12. Every connected manifold has at most two orientations.

Proof. Since charts are bijective, their Jacobians never have determinant zero.

Now, $\mathcal{A}$ and $\mathcal{B}$ be two orientations. Define a map $s(p) : M \to \{1, -1\}$ by setting $s(p) = 1$ if all coordinate change functions $\Phi\Psi^{-1}$ involving one chart from each orientation have positive determinant (in which case all of them have positive determinant, since $\Phi\psi^{-1} = \Phi\Psi^{-1}\Psi\psi^{-1}$), and $s(p) = -1$ otherwise. Since Jacobian determinants vary continuously, $s(p)$ is locally constant. Since M is connected, $s(p)$ must be constant, leaving two options:

1. If $s(p) = 1$, then $\mathcal{A}$ and $\mathcal{B}$ are compatible, and must thus already contain each others charts, since orientations are maximal.
2. If $s(p) = -1$, let $\mathcal{C}$ be another orientation for which the analogously defined function s_C comparing $\mathcal{C}$ to $\mathcal{A}$ is negative. Then, given charts Φ_A, Φ_B, Φ_C from each orientation whose domains match, we get

$$\Phi_B\Phi_C^{-1} = (\Phi_B\Phi_A^{-1}) \circ (\Phi_A\Phi_C^{-1})$$

since the Jacobians of $(\Phi_B\Phi_A^{-1})$ and $(\Phi_A\Phi_C^{-1})$ have negative signs, the Jacobian of their composition must have positive sign, so $\mathcal{B}$ and $\mathcal{C}$ must be compatible, and thus identical.

□

Definition 37.1.13. As with topological manifolds, smooth manifolds with boundary are defined by taking smooth manifolds and replacing the domain of our charts with the upper half-space $\mathbb{H}^n := \{x \in \mathbb{R}^n \mid x_n \geq 0\}$.

37.2 Induced Functional Structures

Example 37.2.1. Consider the torus T^2 defined as the quotient space of $\mathbb{R}^2$ under the equivalence relation identifying points whose coordinates differ by integer amounts (this is fundamentally the same definition as taking the unit square $[0, 1]^2$ with opposite sides identified). Let $q : \mathbb{R}^2 \to T^2$ be the corresponding quotient map.

Finding an atlas on T^2 isnt too complicated: For any open set $U \subseteq \mathbb{R}^2$ with diameter at most one, $q : U \subseteq \mathbb{R}^2 \to q(U) \subseteq T^2$ is a homeomorphism whose inverse we can take as a chart. Since these small diameter sets U still collectively cover $\mathbb{R}^2$, our charts domains $q(U)$ cover T^2.

If $\Phi = (q|_U)^{-1} : q(U) \to U \subseteq \mathbb{R}^n$ and $\Psi = (q|_V)^{-1} : q(V) \to V \subseteq \mathbb{R}^n$ are two charts defined in this way, then we have

$$\Phi\Psi^{-1} : \Psi(q(U) \cap q(V)) \to \Phi(q(U) \cap q(V))$$

Since charts are injective, they respect intersection, giving us

$$\Phi(q(U) \cap q(V)) = \Phi(q(U)) \cap \Phi(q(V)) = U \cap \Phi(q(V))$$
$$\Psi(q(U) \cap q(V)) = \Psi(q(U)) \cap \Psi(q(V)) = \Psi(q(U)) \cap V$$

Thus, Φ^{-1} takes the set $U \cap \Phi(q(V)) \subseteq U \subseteq \mathbb{R}^2$ to its quotient $q(U) \cap q(V)$ in T^2 and then Ψ takes that quotient to the set $\Psi(q(U)) \cap V$. By construction, out original set and our image set must be equivalent, and must thus be translated copies of each other. This tells us that our coordinate change function $\Phi\Psi^{-1}$ are simply translations, and thus smooth.

TODO: Sketch

To find a sensible functional structure on T^2, we can take the following natural definition:

Definition 37.2.2. Let F_X be a functional structure on a topological space X. Let $\Phi : X \to Y$ be continuous. Then the ***induced functional structure*** on Y is given by the finest functional structure such that Φ is a morphism:

$$F_Y(V) = \left\{ f \in C^0(V, \mathbb{R}) \;\middle|\; f \circ \Phi \in F_X(\Phi^{-1}[V]) \right\}$$

Note that this is fundamentally the same idea as the definition of the final topology induced by a continuous function Φ.

We cannot expect this definition to always give a smooth structure on Y - for example, if Y isnt even a *topological* manifold, then any such attempt will obviously fail immediately. For now, we won't concern ourselves with attempts to show when exactly this does end up working out, and instead prove it separately in each application.

In our current example, we want to take the structure induced by q and show that $q|_U : U \subseteq \mathbb{R}^2 \to q[U] \subseteq T^2$ is an isomorphism of functionally structured spaces for sufficiently small open U.

Each $q|_U$ is already a morphism by definition of the induced functional structure, so all we need is a local inverse of $q|_U$.

It suffices to simply take the actual inverse $q|_U^{-1}$, since

$$\begin{aligned} \tau : q^{-1}[q[U]] &\to U \\ x &\mapsto q|_U^{-1}(q(x)) \end{aligned}$$

corresponds to the map collapsing the set $q^{-1}[q[U]]$ of translated copies of U back to the set U itself. Since this map acts as a translation, and thus smoothly, on every connected component of $q^{-1}[q[U]]$, we have $\tau \in C^\infty(q^{-1}[q[U]], U)$.

Thus, for every $f \in C^\infty(U, \mathbb{R})$, we have

$$(f \circ q|_U^{-1}) \circ q = f \circ \tau \in C^\infty(q^{-1}[q[U]], \mathbb{R}),$$

confirming that $f \circ q|_U^{-1} \in F_{T^2}(q[U])^*$, and thus that $q|_U^{-1}$ is a morphism.

*Note that $q[U] = (q|_U^{-1})^{-1}[U]$

37.3 Tangent Spaces

Part XI

Algebraic Topology

Chapter 38

Homotopy

Definition 38.0.1

Let X, Y be topological spaces. Let $f, g : X \to Y$ be continuous. Then f is called ***homotopic*** to g via a map h iff. h is a "continuous interpolation between f and g", i.e:

1. h is a continous map $X \times [0,1] \to Y$,
2. For all $x \in X$, we have $h(x,0) = f(x)$,
3. For all $x \in X$, we have $h(x,1) = g(x)$.

We call h a ***free homotopy***, or simply a ***homotopy***, between f and g. We write $f \sim_h g$ if f is a homotopy between f and g. We write $f \sim g$ if there exists a homotopy between f and g.

Exercise 38.0.2. *Show that the property of being homotopic is an equivalence relation.*

Exercise 38.0.3. *Let $f_0, f_1 : X \to Y$ and $g_0, g_1 : Y \to Z$ be continuous. Let h_0 be a homotopy between f_0 and f_1 and let h_1 be a homotopy between f_1 and g_1. Then*

$$H : (x,t) \mapsto h_1(h_0(x,t),t)$$

is a homotopy between $g_0 \circ f_0$ and $g_1 \circ f_1$.

Definition 38.0.4

Let $f : X \to Y$ be continuous. Then f is called a ***homotopy equivalence*** between X and Y iff. there exists a map $g : Y \to X$ such that $g \circ f$ is homotopic to id_X and $f \circ g$ is homotopic to id_Y. We write $X \simeq_{f,g} Y$, $X \simeq_f Y$, or simply $X \simeq Y$. If X and Y are homotopy equivalent, we sometimes say that they have ***the same homotopy type.***

Exercise 38.0.5. *Show that the property of being homotopy equivalent is an equivalence relation.*

Exercise 38.0.6. *Show that homeomorphism implies homotopy equivalence.*

Definition 38.0.7. A topological space X is called ***contractible*** iff. X is homotopy equivalent to a one-point space.

Exercise 38.0.8. 1. *Show that*

$$h : (x, t) \mapsto x - tx$$

is a homotopy between $\mathrm{id}_{\mathbb{R}^n}$ *and* $f : \mathbb{R}^n \mapsto \{0\}$.

2. *Conclude that* $\mathbb{R}^n$ *is contractible.*

3. *Conclude that homotopy equivalence does not imply homeomorphism.*

Exercise 38.0.9. *A topological space* X *is contractible if and only if there exists a point* $x_0 \in X$ *such that* id_X *is homotopic to the constant map* $X \to x_0$.

Definition 38.0.10. Let X be a subset of a topological vector space. Then we call X ***star-shaped*** iff. there exists a point $x_0 \in X$ such that X contains the connecting line between x_0 and any other point $x \in X$, i.e.

$$\begin{aligned} &\forall x \in X : \\ &\forall t \in [0,1] : \\ &(1-t)x_0 + tx \in X \end{aligned}$$

Exercise 38.0.11. *Show that any star-shaped set is contractible. Conclude that any convex set is contractible.*

Lemma 38.0.12. Let X be a topological space. Let $A \subseteq X$. Then A and X are homotopy equivalent if there exists a homotopy between id_X and $\mathrm{id}_X|_A$, i.e. a continuous map $h : X \times [0,1] \to X$ such that for all $x \in X$:

1. $h(x, 0) = x$,
2. $h(x, 1) \in A$,
3. $x \in A \implies h(x, 1) = x$

Definition 38.0.13. Let $A \subseteq X$ be topological spaces. Then A is called a ***deformation retract of X*** if there is a homotopy $h : X \times [0,1] \to X$ such that

1. $h(\cdot, 0) = \mathrm{id}_X$:

$$\forall x \in X : h(x, 0) = x$$

2. $h(\cdot, 1) : X \to A$:

$$\forall x \in X : h(x, 1) \in A$$

3. $h(\cdot, 1)|_A = \mathrm{id}_A$:

$$\forall a \in A : h(a, 1) = a$$

If the last equation can be strengthened to

$$\forall a \in A, t \in [0,1] : h(a, t) = a,$$

we call A a ***strong deformation retract of X***.

Exercise 38.0.14. *Any deformation retract* A *of a topological space* X *is homotopy equivalent to* X.

Exercise 38.0.15. *If $f : X \to Y$ is a continuous function, then the canonical map $r : M_f \to Y$ is a strong deformation retraction.*

38.1 Relative Homotopy

Definition 38.1.1. We say that f and g are ***homotopic relative to a subset $A \subseteq X$*** and write $f \sim_{\mathrm{rel}\, A} g$, $f \sim_{h,\mathrm{rel}\, A} g$ or $f \sim g \,\mathrm{rel}\, A$ iff:

1. $f \sim_h g$,
2. For all $t \in [0,1]$, we have $h(t,\cdot)|_A = f|_A = g|_A$

I personally find the designation "homotopic relative to A" quite unintuitive, given that the property of being homotopic relative to a given subset is stronger than the property of being homotopic.

Exercise 38.1.2. *Show that $f \sim_h g$ is equivalent to $f \sim_{h,\mathrm{rel}\,\emptyset} g$.*

Theorem 38.1.3. Let $\Phi_1, \Phi_2 : [0,1] \to [0,1]$ be homotopies such that $\Phi_1(0) = \Phi_2(0) \in \{0,1\}$ and $\Phi_1(1) = \Phi_2(1) \in \{0,1\}$. Let $F : X \times [0,1] \to Y$ be a homotopy. Let $G_1(x,t) = F(x, \Phi_1(t))$ and $G_2(x,t) = F(x, \Phi_2(t))$. Then we have

$$G_1 \sim_{\mathrm{rel}\, X \times \{0,1\}} G_2$$

Definition 38.1.4. We denote by C a constant homotopy that makes sense in the given context.

Proposition 38.1.5. Given two homotopies $F, G : X \times [0,1] \to Y$ with $F(\cdot,1) = G(\cdot,0)$, their ***concatenation***

$$(F * G)(x,t) = \begin{cases} F(x,2t) & 0 \le t \le 1/2, \\ G(x,2t-1) & 1/2 \le t \le 1. \end{cases}$$

is again a homotopy $F * G : X \times [0,1] \to Y$.

Note that the order here doesn't match the order in function composition - we go through F first, then through G.

Theorem 38.1.6. We have $C * F \sim F * C \sim F \,\mathrm{rel}\, X \times \{0,1\}$

Definition 38.1.7. Let $h : X \times [0, 1 \to Y]$ be a homotopy. We define the ***inverse of h*** to be the homotopy that "moves through h backwards", i.e.

$$h^{-1}(x,t) = \overline{h}(x,t) := h(x, 1-t)$$

Theorem 38.1.8. Given a homotopy h, we have

$$h * h^{-1} \sim C \,\mathrm{rel}\, X \times \{0,1\},$$

where C is the constant homotopy $C(x,t) = h(x,0)$

38.2 The Fundamental Group

Definition 38.2.1. Let X be a topological space. Then a continuous map $\gamma[0,1] \to X$ with $\gamma(0) = \gamma(1)$ is known as a ***loop***. We denote the set of all loops in X with $\gamma(0) = \gamma(1) = x_0$ as $\Omega(X, x_0)$.

Lemma 38.2.2. Let $\gamma : [0,1] \to X$ be a loop. Let $\varphi : [0,1] \to [0,1]$ be a continuous map such that $\gamma(0) = 0$ and $\gamma(1) = 1$. Then $\gamma \circ \varphi$ and γ are homotopic relative to $\{0,1\}$.

Proof. The map

$$h : (x,t) \mapsto \gamma((1-t)x + t\varphi(x))$$

is a homotopy between γ and $\gamma \circ \varphi$. □

Definition 38.2.3. Let $f, g \in \Omega(X, x_0)$. Then we define the ***concatenation of f and g*** as the map

$$(f * g)(t) = \begin{cases} f(2t) & 0 \le t \le 1/2, \\ g(2t-1) & 1/2 \le t \le 1. \end{cases}$$

Definition 38.2.4. Let $\gamma \in \Omega(X, x_0)$. We define the ***inverse loop*** of γ to be the loop that "moves through gamma backwards", i.e.

$$\overline{\gamma}(t) := \gamma(1-t)$$

Exercise 38.2.5. *Verify that* $\overline{\overline{\gamma}} = \gamma$.

Exercise 38.2.6. *Let* $\gamma \in \Omega(X, x_0)$. *Then* $\gamma * \overline{\gamma}$ *is homotopic to* $e_{x_0} : X \to \{x_0\}$ *relative to* $\{0,1\}$.

Theorem 38.2.7: Existence of the fundamental group

Let X be a topological space. Let $x_0 \in X$. Then the quotient space

$$\pi_1(X, x_0) = \Omega(X, x_0)/\sim_{\mathrm{rel}(0,1)}$$

of loops modulo homotopy becomes a group, known as the ***fundamental group of*** (X, x_0), when equipped path concatenation

$$[f] * [g] = [f * g],$$

as the group operation. The neutral element of this group is $e_{x_0} := (x \mapsto x_0)$.

Theorem 38.2.8. If we additionally define

$$\pi_1(f) : \pi_1(X, x) \mapsto \pi_1(Y, f(x))$$
$$[\gamma] \mapsto [f \circ \gamma]$$

for all continuous $f : X \to Y$, the map π_1 becomes a covariant functor from the category of topological spaces into the category of groups.

Definition 38.2.9. Whenever we need to make the base point clear, we write $\pi_1(f, x) := \pi_1(f) : \pi_1(X, x) \mapsto \pi_1(Y, f(x))$

Theorem 38.2.10: Fundamental groups at different base points

Let X be a topological space. Let $x, y \in X$. Then every continuous path $\delta : [0,1] \to X$ induces a group isomorphism

$$J_\delta : \pi_1(X, \delta(0)) \cong \pi_1(X, \delta(1))$$
$$[\gamma] \mapsto \left[\overline{\delta} * \gamma * \delta\right]$$

Proof.

1. Since loop concatenation preserves homotopy, J_δ sends different representatives of the same equivalence class to the same equivalence class.

2. J_δ goes from $\pi_1(X, x)$ to $\pi_1(X, y)$, since for any $\gamma : [0,1] \to X$ with $\gamma(0) = \gamma(1) = x$, $\overline{\delta}$ traces out a path from $\delta(1)$ to $\delta(0)$, then γ traces out a loop from $\delta(0)$ to $\delta(0)$, and then δ traces out a path from $\delta(0)$ back to $\delta(1)$, i.e. $\overline{\delta} * \gamma * \delta$ is a loop from $\delta(1)$ to $\delta(1)$.

3. J_δ is a group homomorphism, since

$$\begin{aligned} J_\delta([\gamma_1 * \gamma_2]) &= \left[\overline{\delta} * \gamma_1 * \gamma_2 * \delta\right] \\ &= \left[\overline{\delta} * \gamma_1 * e_{x_0} * \gamma_2 * \delta\right] \\ &= \left[\overline{\delta} * \gamma_1 * \delta * \overline{\delta} * \gamma_2 * \delta\right] \\ &= \left[\overline{\delta} * \gamma_1 * \delta\right] * \left[\overline{\delta} * \gamma_2 * \delta\right] \\ &= J_\delta([\gamma_1]) * J_\delta([\gamma_2]) \end{aligned}$$

   (Recall that any map that respects the group operation also respects the neutral element and inverses, and is thus a homomorphism.)

4. J_δ is bijective with $J_\delta^{-1} = J_{\overline{\delta}}$:

$$J_\delta(J_{\overline{\delta}}(\gamma)) = \left[\overline{\delta} * \left(\delta * \gamma * \overline{\delta}\right) * \delta\right] = [\gamma]$$
$$J_{\overline{\delta}}(J_\delta(\gamma)) = \left[\delta * \left(\overline{\delta} * \gamma * \delta\right) * \overline{\delta}\right] = [\gamma]$$

□

Corollary 38.2.11. If X is path connected, $\pi_1(X, x) \cong \pi_1(X, y)$ for any $x, y \in X$.

Definition 38.2.12. Let $\gamma_0 \in \Omega(X, x)$ an $\gamma_1 \in \Omega(X, y)$. Then we call $h : [0,1] \times [0,1] \to X$ a ***free homotopy of loops***, or simply ***loop homotopy***, between γ_0 and γ_1 iff. h is a homotopy between γ_0 and γ_1 such that $h_t(x) := h(x, t)$ is a loop for any fixed $t \in [0,1]$.

Exercise 38.2.13. *1. Verify that h is a loop homotopy iff. h is a homotopy and $h_t(0) = h_t(1)$ for any $t \in [0,1]$.*

2. *Show that any $\{0,1\}$-homotopy is a loop homotopy.*

The additional constraint that and $h(\cdot, t)$ is a loop may seem superfluous, but there are in fact homotopies between loops that aren't loop homotopies:

Exercise 38.2.14. 1. *Let $\gamma_0 = \gamma_1 : [0,1] \to \{0\}$ be constant loops. Let $h(x,t) := xt(1-t)$. Then h is a homotopy between γ_0 and γ_1, but fails to be a loop homotopy.*

2. *Let $\gamma_0(x) = e^{2\pi i x}$. Let $\gamma_1(x) = e^{4\pi i x}$. Then $h(x,t) := e^{2\pi i(1+t)x}$ is a homotopy between γ_0 and γ_1, but fails to be a loop homotopy.*

3. *Assume that any loop $\gamma : [0,1] \to S^1$ with 1 as its base point is homotopic a loop of the form $\gamma(x) := e^{2\pi n i x}$ for some $n \in \mathbb{Z}$ (we will prove this later). Does 2. imply that the fundamental group $\pi_1(S^1, 1)$ is trivial?*

Definition 38.2.15. A loop $\gamma \in \Omega(X,x)$ is called ***contractible in X*** iff. there exists an $x \in X$ and a loop homotopy h such that $\gamma \sim_h e_x$.

Lemma 38.2.16. $\gamma \in \Omega(X,x)$ is contractible in X if $\gamma \sim_{\mathrm{rel}\,0,1} e_{\gamma(0)}$.

Theorem 38.2.17. Let $\gamma \in \Omega(X,x)$ be a loop. Let $S^1 \cong I/\{0,1\}$ be equipped with the quotient topology. Then γ is contractible if and only if $\gamma \circ p : S^1 \to X$ has a continuous extension to the closed unit ball $\overline{B}^2$.

Theorem 38.2.18. Let h be a homotopy between $f, g : X \to Y$. Let $x_1 \in X$. Then

$$\begin{aligned} h(x_1, \cdot) : [0,1] &\to Y \\ t &\mapsto h(x_1, t) \end{aligned}$$

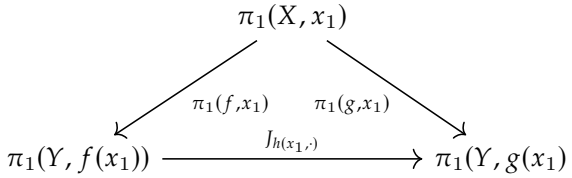
is a path from $f(x_1)$ to $g(x_1)$ which fulfills

$$\pi_1(g, x_1) = J_{h(x_1,\cdot)} \circ \pi_1(f, x_1)$$

Recall that $\pi_1(g, x_1)$ is the map $[\gamma] \mapsto [g \circ \gamma]$, and that J_δ is the map $[\gamma] \mapsto [\overline{\delta} * \gamma * \delta]$, so this theorem states that

$$[g \circ \gamma] = \left[\overline{h}(x_1, \cdot) * (f \circ \gamma) * h(x_1, \cdot)\right]$$

for any $\gamma \in \pi_1(X, x_1)$, or, equivalently, that the diagram



commutes.

Proof. Define

$$H_t(x) := h(x_1, t + (1-t)x).$$

We have $(f \circ \gamma) \in \Omega(X, f(x_1)$ and $(g \circ \gamma) \in \Omega(X, g(x_1))$, since

$$\begin{aligned} f(\gamma(0)) &= f(x_1) = f(\gamma(1)) \\ g(\gamma(0)) &= g(x_1) = g(\gamma(1)) \end{aligned}$$

Then a homotopy between $g \circ \gamma$ and $\overline{h}(x_1, \cdot) * (f \circ \gamma) * h(x_1, \cdot)$ is given by:

$$k(x,t) := (\overline{H_t} * (h(\cdot, t) \circ \gamma) * H_t)(x),$$

since we get:

$$\begin{aligned} k(x,0) &= (\overline{H_0} * (h(\cdot, 0) \circ \gamma) * H_0)(x) \\ &= (\overline{H_0} * (f \circ \gamma) * H_0)(x) \\ &= (\overline{h}(x_1, \cdot) * (f \circ \gamma) * h(x_1, \cdot))(x) \end{aligned}$$

and

$$\begin{aligned} k(x,1) &= (\overline{H_1} * (h(\cdot, 1) \circ \gamma) * H_1)(x) \\ &= (h(x_1, (1-1) + (1-(1-1))(\cdot)) * (g \circ \gamma) * h(x_1, 1 + (1-1)(\cdot)))(x) \\ &= (h(x_1, \cdot) * (g \circ \gamma) * h(x_1, 1))(x) \\ &= (e_{g(x_1)} * (g \circ \gamma) * e_{g(x_1)})(x) \\ &= (g \circ \gamma)(x), \end{aligned}$$

where the second to last step follows since the loop $(g \circ \gamma)$ ends on the base point $g(x_1)$. □

Lemma 38.2.19. Let $\gamma : [0,1] \to X$ be continuous. Let $f : X \to Y$. Then

$$\pi_1(f, \gamma(1)) \circ J_\gamma = J_{f \circ \gamma} \circ \pi_1(f, \gamma(0)),$$

i.e. the diagram

$$\begin{array}{ccc} \pi_1(X, \gamma(0)) & \xrightarrow{\pi_1(f, \gamma(0))} & \pi_1(Y, f(\gamma(0))) \\ \downarrow{\scriptstyle J_\gamma} & & \downarrow{\scriptstyle J_{f \circ \gamma}} \\ \pi_1(X, \gamma(1)) & \xrightarrow{\pi_1(f, \gamma(1))} & \pi_1(Y, f(\gamma(1))) \end{array}$$

commutes.

Proof. Let $\sigma \in \Omega(X, \gamma(0))$. Then we have

$$\begin{aligned} \pi_1(f, \gamma(1)) \circ J_\gamma([\sigma]) &= \pi_1(f, \gamma(1))([\overline{\gamma} * \sigma * \gamma]) \\ &= \left[f \circ (\overline{\gamma} * \sigma * \gamma)\right] \\ &= \left[(f \circ \overline{\gamma}) * (f \circ \sigma) * (f \circ \gamma)\right] \\ &= \left[\overline{(f \circ \gamma)} * (f \circ \sigma) * (f \circ \gamma)\right] \\ &= J_{f \circ \gamma}([f \circ \sigma]) \\ &= J_{f \circ \gamma} \circ \pi_1(f, \gamma(0))([\sigma]) \end{aligned}$$

□

Theorem 38.2.20. Let $f : X \to Y$ be a homotopy equivalence. Let $x_0 \in X$. Then the map $\pi_1(f, x_0)$ is a group isomorphism $\pi_1(X, x_0) \cong \pi_1(Y, f(x_0))$.

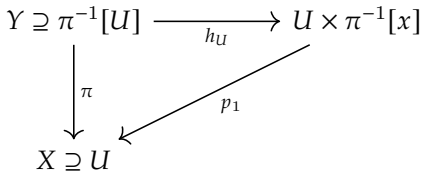
Chapter 39

Fibrations and Coverings

Definition 39.0.1: Locally Trivial Fibration

Let $\pi : Y \to X$ be continuous. Then π is called a ***locally trivial fibration*** iff. for every $x \in X$, there exists an open neighborhood U such that there exists a homeomorphism $h_u : \pi^{-1}[U] \cong U \times \pi^{-1}[x]$ fulfilling $p_1 \circ h_U = \pi|_{\pi^{-1}[U]}$.

$\pi^{-1}[x]$ is called the ***Fiber of π at x***, and sometimes written F_x.



Definition 39.0.2. Let $\pi : Y \to X$ be a locally trivial fibration. Then:

1. If all $\pi^{-1}[x]$ are homeomorphic, we call π a ***fiber bundle***.
2. If every $\pi^{-1}[x]$ is discrete, we call π a ***covering map***.
3. If we can pick $U = X$, π is called a ***trivial fibration***.

Exercise 39.0.3. *If we replace $\pi^{-1}[x]$ in the definition by an arbitrary topological space F_U, we get $F_U \cong \pi^{-1}[x]$. Thus, both definitions are equivalent.*

Lemma 39.0.4. Let $\pi : Y \to X$ be a locally trivial fibration. Let X be connected. Then π is a trivial fibration.

Proof. Let $x_0 \in X$. Let $V = \{x \in X \mid F_x \cong F_{x_0}\}$ be the set of all points whose fibers are homeomorphic to the fiber of x_0. Let U be a neighborhood as in the definition. Then the fibers of all $x \in U$ must be homeomorphic by definition. Thus, every point in V has a neighborhood contained in V, so V is open. We have

$$X \setminus V = \bigcup_{y \notin V} \left\{x \in X \,\middle|\, F_x \cong F_y\right\} := \bigcup_{y \notin V} V_y$$

Since every V_y is open by the same argument, $X \setminus V$ must be open. But since X is connected with $(X \setminus V) \cup V = X$ and V nonempty, $X \setminus V$ must

be empty. □

Theorem 39.0.5. A continuous function $\pi : Y \to X$ is a covering map if and only if every $x \in X$ has an open neighborhood U such that $\pi^{-1}[U]$ can be decomposed into disjoint open sets $V_i \subseteq Y$ such that $\pi|_{V_i}$ is a homeomorphism $V_i \cong U$.

Proof. $\Rightarrow$: Let π be a covering map. Let $I = F_U$. Since h_U is bijective, the sets $V_i := h_U^{-1}[U \times \{i\}]$ are disjoint. Since h_U is a homeomorphism and $I = F_U$ is discrete, the V_i are also open. Since $p_1|_{U\times\{i\}}$ is a homeomorphism $U \times \{i\} \cong U$, the composition

$$\pi|_{V_i} = (p_1|_{U\times\{i\}}) \circ h|_{V_i}$$

must be a homeomorphism $V_i \mapsto U$.

$\Leftarrow$: Define

$$\begin{aligned} h_U : \pi^{-1}[U] &\cong U \times I \\ y \in V_i &\mapsto (\pi(y), i). \end{aligned}$$

Then the restriction of h_U to each V_i is by definition a homeomorphism $V_i \cong U \cong U \times \{i\}$, and so the whole of h_U is a homeomorphism $\pi^{-1}[U] = \bigsqcup_{i\in I} V_i \cong \bigsqcup_{i\in I} U \times \{i\} = U \times I$.

□

Definition 39.0.6: Lifting

Let $\pi : Y \to X$ be a locally trivial fibration. Let $f : Z \to X$ be continuous. Then a continuous map $\hat{f} : Z \to Y$ is called a ***lift from f to Y*** if $\pi \circ \hat{f} = f$.

Any path can be lifted:

Theorem 39.0.7. Let $\pi : Y \to X$ be a locally trivial fibration. Let $\gamma : [0, 1] \to XX$ be continuous. Let $\pi(y_0) = \gamma(0)$. Then there exists a lift $\hat{\gamma} : [0, y] \to Y$ with $\hat{\gamma}(0) = y_0$ such that $\pi \circ \hat{f} = f$.

And lifts respect fundamental groups:

Theorem 39.0.8. Let $\pi : Y \to X$ be a locally trivial fibrations. Let $f : Z \to X$. Let $\hat{f} : Z \to Y$ be a lift of f. Then we have:

$$\begin{aligned} &\forall z_0 \in Z : \\ &\exists y_0 \in Y : \\ &\quad \hat{f}(z_0) = y_0, \\ &\quad \pi_1(f)[\pi_1(Z, z_0)] \subseteq \pi_1(\pi)[\pi_1(Y, y_0)] \end{aligned}$$

Proof.

$$\begin{aligned}\pi_1(f)[\pi_1(Z, z_0)] &= (\pi_1(\pi \circ \hat{f}))[\pi_1(Z, z_0)] \\ &= (\pi_1(\pi) \circ \pi_1(\hat{f}))[\pi_1(Z, z_0)] \\ &= \end{aligned}$$

□

Part XII

Numerical Mathematics

Chapter 40

Numerical Linear Algebra

40.1 Operator Norms and Conditioning

Definition 40.1.1. Let $x \in V$ be an element of a normed vector space. Let $\widetilde{x} \in V$ be an approximation of x. Then the ***relative error*** of $\widetilde{x}$ is

$$\varepsilon(x, \widetilde{x}) := \frac{\|x - \widetilde{x}\|}{\|x\|}.$$

Definition 40.1.2. A function $\Phi(x) : V \to W$ is called ***poorly conditioned*** (at $x \in X$) if a small pertubation in the input can lead to a large pertubation in the output, i.e. there exists $\widetilde{x}$ such that

$$\varepsilon_\Phi := \varepsilon(\Phi(x), \Phi(\widetilde{x})) \gg \varepsilon(x, \widetilde{x}) =: \varepsilon_x.$$

$\Phi(x)$ is called ***well-conditioned*** if there exists a 'reasonable' constant c such that

$$\varepsilon_\Phi \leq c \cdot \varepsilon_x$$

for all $x, \widetilde{x}$.

Lemma 40.1.3. Multiplication $\Phi(x, y) := x \cdot y$ of two numbers is well conditioned, with the approximation

$$\varepsilon_\Phi \leq \varepsilon_x \varepsilon_y + \varepsilon_x + \varepsilon_y$$

Proof.

$$\begin{aligned}
\varepsilon_\Phi &= \frac{\left\|\widetilde{x\widetilde{y}} - xy\right\|}{\left\|xy\right\|} \\
&= \frac{\left\|\widetilde{x\widetilde{y}} - x\widetilde{y} + x\widetilde{y} - xy\right\|}{\left\|xy\right\|} \\
&= \frac{\left\|(\widetilde{x} - x)\widetilde{y} + x(\widetilde{y} - y)\right\|}{\left\|xy\right\|} \\
&\leq \frac{\left\|(\widetilde{x} - x)\widetilde{y}\right\|}{\left\|xy\right\|} + \frac{\left\|x(\widetilde{y} - y)\right\|}{\left\|xy\right\|} \\
&= \frac{\|\widetilde{x} - x\|}{\|x\|}\frac{\left\|\widetilde{y}\right\|}{\|y\|} + \frac{\left\|\widetilde{y} - y\right\|}{\|y\|} \\
&= \frac{\|\widetilde{x} - x\|}{\|x\|}\frac{\left\|\widetilde{y} - y + y\right\|}{\|y\|} + \frac{\left\|\widetilde{y} - y\right\|}{\|y\|} \\
&\leq \frac{\|\widetilde{x} - x\|}{\|x\|}\frac{\left\|\widetilde{y} - y\right\|}{\|y\|} + \frac{\|\widetilde{x} - x\|}{\|x\|}\frac{\|y\|}{\|y\|} + \frac{\left\|\widetilde{y} - y\right\|}{\|y\|} \\
&= \frac{\|\widetilde{x} - x\|}{\|x\|}\frac{\left\|\widetilde{y} - y\right\|}{\|y\|} + \frac{\|\widetilde{x} - x\|}{\|x\|} + \frac{\left\|\widetilde{y} - y\right\|}{\|y\|} \\
&= \varepsilon_x\varepsilon_y + \varepsilon_x + \varepsilon_y
\end{aligned}$$

□

Definition 40.1.4. An algorithm is called ***unstable*** if rounding errors significantly increase the relative error beyond what the conditioning of the problem would imply, i.e. if the function Φ is also being approximated by a function $\widetilde{\Phi}$ and

$$\varepsilon_{\widetilde{\Phi}} := \varepsilon(\widetilde{\Phi}(\widetilde{x}), \Phi(x)) \gg \varepsilon_\Phi$$

A function is stable if and only if every individual step of the calculation is well-conditioned.

Theorem 40.1.5. Subtraction of approximately equal real numbers x and $y := (1 + \varepsilon_y)x$ is poorly conditioned.

Proof. Let $\widetilde{x} := (1 + \varepsilon_x)x$. Then

$$\varepsilon_\Phi := \frac{\left\|(\widetilde{x} - y) - (x - y)\right\|}{\left\|x - y\right\|} = \frac{\|\varepsilon_x x\|}{\left\|-\varepsilon_y x\right\|} = \frac{\varepsilon_x}{\varepsilon_y}$$

If ε_y's order of magnitude is equal or smaller than that of ε_y, this relative error will be significantly larger than 1. □

Corollary 40.1.6. The function

$$\widetilde{\Phi}_1(x) := \left(\frac{1}{x}\right) - \left(\frac{1}{x+1}\right)$$

is unstable, since it is poorly conditioned even though the mathematically equivalent function

$$\widetilde{\Phi}_2(x) := \frac{1}{x(x+1)}$$

is well-conditioned.

Definition 40.1.7: Induced Operator Norm

Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed vector spaces. Then the operator norm on $\mathrm{Hom}(V, W)$ is defined by

$$\|F\|_{V,W} = \sup\{\|F(x)\|_W \mid x \in V : \|x\|_V = 1\}$$

The operator norm of F thus tells us the largest norm that any normalized vector ends up with after being transformed.
Equivalently, it tells us the radius of the smallest W-circle around the image of the normalized V-circle.

Theorem 40.1.8. $\|\cdot\|_{V,W}$ is a norm.

Exercise 40.1.9. *Show that* $\|\mathrm{id}_V\|_{V,V} = 1$ *for any norm* $\|\cdot\|_V$ *on a vector space* V.

Exercise 40.1.10. *Show that, given any eigenvalue* λ *of* $F \in \mathrm{Hom}(V, V)$ *and any norm* $\|\cdot\|_V$ *on* V,

$$\|F\|_{V,V} \geq |\lambda|$$

Theorem 40.1.11. Let V, W be $\mathbb{F}$-vector spaces. Then

$$\|F(x)\|_{V,W} = \inf\{c \in \mathbb{F} \mid \forall x \in V : \|F(x)\|_W \leq c \cdot \|x\|_V\}$$

> *Proof Sketch.* The largest factor by which $\|x\|_V$ can increase is equivalently the smallest upper bound on such factors. "□"

Note that this means $\|F(x)\|_W \leq \|F\|_{V,W} \cdot \|x\|_V$ for all $x \in V$.

Corollary 40.1.12. Let $F \in \mathrm{Hom}(V, W)$ and $G \in \mathrm{Hom}(W, U)$. Then

$$\|G \circ F\|_{V,U} \leq \|G\|_{W,U} \cdot \|F\|_{V,W}.$$

Lemma 40.1.13. The infimum is a minimum, i.e. there exists an x attaining it.

Lemma 40.1.14. Let $F \neq 0$. Let $x \in V$ with $\|x\|_V \leq 1$. Then $\|F(x)\|_W = \|F\|_{V,W}$ implies $\|x\| = 1$.

Proposition 40.1.15. The l^1-norm

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|$$

on $\mathbb{R}^n$ and $\mathbb{R}^m$ induces the ***column sum norm***

$$\|A\|_1 = \max_{j=1,\dots,n} \sum_{i=1}^{m} \left|a_{ij}\right|$$

on $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$.

Proposition 40.1.16. The l^∞-norm

$$\|x\|_\infty = \max\{|x_i| \mid i = 1, \dots, n\}$$

on $\mathbb{R}^n$ and $\mathbb{R}^m$ induces the ***row sum norm***

$$\|A\|_\infty = \max_{j=1,\dots,m} \sum_{j=1}^{n} \left|a_{ij}\right|$$

on $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$.

Definition 40.1.17

The ***spectral radius*** $\rho(F)$ of an endomorphism $F \in \mathrm{Hom}(V, V)$ of a finite-dimensional vector space is the absolute value of the largest eigenvalue of F.

Proposition 40.1.18. The l^2-norm

$$\|x\|_2 = \sqrt{\sum_{i=1}^{n} |x_i|^2}$$

on $\mathbb{R}^n$ and $\mathbb{R}^m$ induces the ***spectral norm***

$$\|A\|_2 = \sqrt{\rho(A^\top A)}$$

on $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$.

Definition 40.1.19. The ***Frobenius norm*** on $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$ is the natural norm given by $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m) \cong \mathbb{R}^{mn}$:

$$\|A\|_F := \sqrt{\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2}$$

Exercise 40.1.20. *Show that the Frobenius norm is not an operator norm.*

Theorem 40.1.21. Let $\|\cdot\|$ be an operator norm on $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$. Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ be invertible. Let $x, \widetilde{x}, b, \widetilde{b} \in \mathbb{R}^n \setminus \{0\}$ such that $Ax = b$ and $A\widetilde{x} = \widetilde{b}$. Then

$$\varepsilon_x \leq \|A\| \left\|A^{-1}\right\| \varepsilon_b.$$

This means that the product $\|A\| \left\|A^{-1}\right\|$ tells us how much our accumulated relative error increases when we solve a system of linear equations. We want to give this product a special name:

Definition 40.1.22

Let $\|\cdot\|$ be an operator norm on $\mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$. Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ be invertible. Then we define the ***condition number*** of A to be

$$\mathrm{cond}(A, \|\cdot\|) = \|A\| \left\|A^{-1}\right\|$$

If $\|\cdot\|_p$ is a p-norm, we write $\mathrm{cond}(A, \|\cdot\|_p) =: \mathrm{cond}_p(A)$.

Exercise 40.1.23. *Show that* $\mathrm{cond}(A, \|\cdot\|) \geq 1$ *for all* $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$ *and any operator norm* $\|\cdot\|$.

Exercise 40.1.24. *Show that if A is symmetric with eigenvalues $\lambda_1, \dots, \lambda_n$, then*

$$\operatorname{cond}_2(A) = \frac{\max |\lambda_i|}{\min |\lambda_j|}$$

Exercise 40.1.25. *Show that the linear map given by*

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1+\varepsilon \end{pmatrix}$$

is poorly conditioned by approximating $\operatorname{cond}_2(A)$.

Definition 40.1.26: Generalized Condition Number

The condition number of an arbitrary function $\Phi : X \to Y$ between normed real vector spaces is defined as

$$\kappa_{\Phi(x)} = \inf \{\kappa \in \mathbb{R}_{\geq 0} \mid \varepsilon_\Phi \leq \kappa \cdot \varepsilon_x\}$$

Exercise 40.1.27. *Show that* $\operatorname{cond}(A) = \sup \{\kappa_{Ax} \mid x \in V\}$.

Exercise 40.1.28. *Show that if Φ is differentiable, we have* $\kappa_\Phi(x) = \frac{\|D\Phi(x)\|\|x\|}{\|\Phi(x)\|}$

Definition 40.1.29: Stability Indicator

The ***stability indicator*** of a numerical algorithm $\widetilde{\Phi}_\varepsilon$ with rounding accuracy ε is

$$\inf \left\{ \sigma \geq 0 \;\middle|\; \exists \delta > 0 : \forall \varepsilon \leq \delta : \varepsilon_{\widetilde{\Phi}_\varepsilon} \leq \sigma \kappa_{\Phi(x)} \varepsilon \right\} \cdot \varepsilon$$

40.2 Triangular Decompositions

Solving a system of linear equations corresponding to a triangular matrix is trivial. Thus, if we can write a given matrix as a product of two triangular matrices, further computations become significantly easier.

Exercise 40.2.1. *Let U, V be upper triangular matrices. Then $U \cdot V$ is an upper triangular matrix.*

Exercise 40.2.2. *Let U be an upper triangular matrix. Then if U is invertible, U^{-1} is an upper triangular matrix whose diagonal entries are the inverses of the diagonal entries of U.*

Theorem 40.2.3. Let $A \in \operatorname{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ be an upper triangular matrix such that all upper left submatrices are invertible. Then there exists a unique decomposition

$$A = LU$$

such that U is an invertible upper triangular matrix and L is an invertible lower triangular matrix with normalized diagonal entries.

Proof. For $n = 1$, we simply set $L = 1$ and $U = A$.
A decomposition $A_{n-1} = L_{n-1}U_{n-1}$ for the upper left block of size $n-1$ in A can be extended to a decomposition $A = LU$ by solving a system of linear equations, whose solvability is guaranteed by the invertibility of L_{n-1} and U_{n-1}:

$$\begin{aligned} & A = LU \\ \iff & \begin{pmatrix} A_{n-1} & a_U \\ a_L^\top & a_{nn} \end{pmatrix} = \begin{pmatrix} L_{n-1} & 0 \\ l_L & 1 \end{pmatrix} \cdot \begin{pmatrix} U_{n-1} & u_U \\ 0 & u_{nn} \end{pmatrix} \\ & = \begin{pmatrix} L_{n-1} \cdot U_{n-1} & L_{n-1}u_U \\ (U_{n-1}^\top l_L)^\top & l_L^\top u_U + u_{nn} \end{pmatrix} \end{aligned}$$

Since we've already solved the upper left block $A_{n-1} = L_{n-1}U_{n-1}$, we simply need to solve the equations

$$\begin{aligned} a_U &= L_{n-1}u_U \\ a_L &= U_{n-1}l_L \\ a_{nn} &= l_L^\top u_U + u_{nn} \end{aligned}$$

Since L_{n-1} and U_{n-1} are invertible, the first two equations are uniquely solvable (and obtaining these solutions is trivial, since both matrices are triangular). The solutions u_U and l_L then give a unique solution for u_{nn} as well. □

Exercise 40.2.4. *If A is positive definite, it has an LU-decomposition.*

Definition 40.2.5. We call a matrix $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ ***diagonally dominant*** if the diagonal entries of A are at least as large as the sum over the other entries in a given row of A:

$$\forall i \in [1..n] : \sum_{j=1, j\neq i}^{n} |a_{ij}| \leq |a_{ii}|$$

If this equality is strict, we call A ***strictly diagonally dominant***.

Exercise 40.2.6. *Show that diagonally dominant matrices are invertible, and thus have an LU-decomposition.*

Note that $\mathrm{cond}(A) \leq \mathrm{cond}(L)\,\mathrm{cond}(U)$, meaning that LU-decomposition can lead to a decrease in numerical stability. However, in practical applications, this decrease is generally not noticable.

If A is symmetrical, then A is uniquely defined by the entries on the diagonal and below - thus, a natural followup question would be to ask if we can find a decomposition $A = LL^\top$ of A into the product of a lower diagonal matrix with its transpose.

Exercise 40.2.7. *Show that such a decomposition $A = LL^\top$ can only exist if A is positive definite.*

Exercise 40.2.8. *Let A be symmetric and positive definite. Then there exists a unique decomposition $A = LL^\top$, known as the* **Cholesky decomposition** *of A.*

40.3 Linear Regression

Many practical problems lead to systems of linear equations with more equations than variables - for example, we might try to find a line that approximates messy data linearly.
In such tasks, an exact solution is generally not achievable. Instead, we seek to find an x that minimizes the error $\|Ax - b\|$.

Theorem 40.3.1: Gauß Normal Equation

1. The minimizers of $\|Ax - b\|$ are exactly the solutions of
$$A^\top Ax = A^\top b.$$
2. The solution set is non-empty.
3. For any two solutions x, y, we have $Ax = Ay$.

The Gauß normal equation is incredibly useful for establishing theoretical results, but we have $\mathrm{cond}_2(A^\top A) = \mathrm{cond}_2(A)^2$, so solving a linear regression problem by numerically solving the normal equation is generally impractical.

Exercise 40.3.2. *$A^\top A$ is invertible if* $\ker A = \{0\}$.

Since orthogonal matrices preserve the euclidean norm, we try to construct an orthogonal matrix Q such that QA is a (possibly non-square) upper triangular matrix, in which case we have

$$\|QAx - Qb\|_2 = \|Ax - b\|_2.$$

Since orthogonal matrices preserve norms, we have $\mathrm{cond}_2(QA) = \mathrm{cond}_2(A)$. This will end up leading us to a more numerically stable solution to linear regression.

Theorem 40.3.3

Given a normalized vector $n \in \mathbb{R}^n$, reflection along the hyperplane $x^\top n = 0$ is given by
$$P_n = \mathrm{id}_{\mathbb{R}^n} - 2nn^\top.$$

Proof. We have $P_n(x) = x - 2nn^\top x = x - 2\langle n, x\rangle .x$, i.e. P_n orthogonally projects x onto our hyperplane, and then applies the same translation another time, mapping x onto its reflection. □

Lemma 40.3.4. Let $x \in \mathbb{R}^n \setminus \langle e_i\rangle$. Define
$$v = x + (\mathrm{sgn}(x_1)\|x\|_2).e_1,$$
declaring $\mathrm{sgn}(0) = 1$. Let $w = v/\|v\|_2$. Then
$$P_w(x) = (-\mathrm{sgn}(x_1)\|x\|_2).e_i.$$

Theorem 40.3.5: QR-decomposition

Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$ with $m \geq n$ and $\ker A = \{0\}$. Then there exist $Q \in \mathrm{O}(m)$ and an invertible upper triangular matrix $R \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$ such that $A = QR$.

Proof Sketch. Using our Lemma, we can bring A into upper triangular form R by repeated application of hyperplane reflections to its lower right submatrices, at which point Q is given by the product of these reflections. "□"

QR-decomposition is $O(n^3)$, which is asymptotically the same as LU-decomposition. However, the actual runtime of QR-decomposition is generally about twice as long.

Theorem 40.3.6

Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$ with $m \geq n$ and $\ker A = \{0\}$. Then QR-decomposition of A gives a solution x of the linear regression problem $\|Ax - b\|_2$ via

$$R_U x = b_U,$$

where

$$\begin{pmatrix} b_U \\ b_L \end{pmatrix} = Q^\top b, \quad \begin{pmatrix} R_U \\ 0 \end{pmatrix} = R = Q^\top A$$

Theorem 40.3.7: Singular Value Decomposition

Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^m)$. Then there exist orthonormal bases B_n, B_m of $\mathbb{R}^n$ and $\mathbb{R}^m$ such that the diagonal of ${}_{B_m}A_{B_n}$ contains the square roots of the non-zero eigenvalues of $A^\top A$, known as the ***singular values*** of A, ordered by size and followed by some amount of zeroes.

Definition 40.3.8. Let $A = U\Sigma V^\top$ be a singular value decomposition of A. Let Σ^+ be constructed from Σ by inverting all of the singular values. Then $A^+ := V\Sigma^+ U^\top$ is known as the ***pseudoinverse*** or ***Moore-Penrose-inverse*** of A.

Exercise 40.3.9. *Show that $A^+ = A^{-1}$ if A is square.*

Proposition 40.3.10. A^+ is the unique matrix fulfilling all of the following equations:

1. $AA^+A = A$,
2. $A^+AA^+ = A^+$,
3. $(AA^+)^\top = AA^+$,
4. $(A^+A)^\top = A^+A$.

Theorem 40.3.11. A^+b is a minimal solution of the linear regression problem $Ax \cong b$.

40.4 Eigenvalue Problems

40.4.1 Localization

Theorem 40.4.1. Every eigenvalue of a matrix $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ is contained in a set of the form

$$K_i := \left\{ z \in \mathbb{C} \;\middle|\; |z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}| \right\}.$$

These sets are known as ***Gerschgorin circles***.

Notice that K_i is a circle of radius $\sum_{j \neq i} |a_{ij}|$ centered on a_{ii}. Importantly, this tells us that if the off-diagonal entires of A are small, the diagonal entries will always give us a good approximation of the eigenvalues.

Proof. Let $x \in \mathbb{C}^n \setminus \{0\}$ be an eigenvector of A with eigenvalue λ. Let x_i be the entry of x with the largest absolute value. Then we have

$$\lambda x_i = (Ax)_i = \sum_{j=1}^{n} a_{ij} x_j.$$

Dividing by $x_i \neq 0$ gives

$$\lambda = (Ax)_i = \sum_{j=1}^{n} a_{ij} \frac{x_j}{x_i},$$

subtraction of a_{ii} gives

$$\lambda - a_{ii} = (Ax)_i = \sum_{j \neq i}^{n} a_{ij} \frac{x_j}{x_i}.$$

Applying the triangle equality gives

$$|\lambda - a_{ii}| \leq (Ax)_i = \sum_{j \neq i}^{n} \left| a_{ij} \frac{x_j}{x_i} \right|,$$

and since x_i is maximal, we have $|x_j|/|x_i| \leq 1$, i.e.

$$|\lambda - a_{ii}| \leq (Ax)_i = \sum_{j \neq i}^{n} |a_{ij}|.$$

□

Theorem 40.4.2. Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ be symmetric. Then the maximal and minimal eigenvalues of A are the maximum and minimum of the ***Rayleigh quotient***

$$\begin{aligned} R(x) : \mathbb{R}^n \setminus \{0\} &\to \mathbb{R} \\ x &\mapsto \frac{x^\top A x}{\|x\|_2^2} \end{aligned}$$

Proof Sketch. By the spectral theorem, we can choose an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A. In this basis, we find that

$$x^\top Ax = \sum_{i=1}^{n} \lambda_i x_i^2$$

We can bound this expression from above and below as

$$\lambda_{\min} \sum_{i=1}^{n} x_i^2 \leq x^\top Ax \leq \lambda_{\max} \sum_{i=1}^{n} x_i^2$$

which gives us our desired result, since

$$\lambda_{\min} \sum_{i=1}^{n} x_i^2 = \|x\|_2^2.$$

"□"

Theorem 40.4.3. Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ be diagonalizable with $A = VDV^{-1}$. Let $\lambda_B \in \mathbb{C}$ be an eigenvalue of $A + B$ for any $B \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$. Then there exists an eigenvalue $\lambda_A \in \mathbb{C}$ of A such that

$$|\lambda_B - \lambda_A| \leq \mathrm{cond}_2(V)\|B\|_2.$$

Corollary 40.4.4. If A is normal, we get

$$|\lambda_B - \lambda_A| \leq \|B\|_2.$$

This means that finding eigenvalues of normal matrices is a well-conditioned problem - adding a small amount of noise, represented by another matrix B, to a matrix A won't change the eigenvalues much. This doesn't hold for arbitrary matrices - for example, the ***Frobenius matrix***

$$A = \begin{pmatrix} 0 & & & & -a_0 \\ 1 & 0 & & & -a_1 \\ & \ddots & \ddots & & \vdots \\ & & 1 & 0 & -a_{n-2} \\ & & & 1 & -a_{n-1} \end{pmatrix}$$

has characteristic polynomial $p(t) = (-1)^n \sum_{i=0}^{n} a_i t^i$, with $a_n = 1$. In general, adding a small constant to an arbitrary polynomial can change the roots quite significantly, so finding the eigenvalues of A is generally not well-conditioned.

40.4.2 Power Iteration

Theorem 40.4.5: Von Mises iteration

Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$. Let $x_0 \in \mathbb{R}^n \setminus \{0\}$. Then the sequence

$$x_{n+1} = \frac{Ax_n}{\|Ax_n\|_2}$$

converges to a normalized eigenvector of A corresponding to the eigenvalue with the largest absolute value, as long as x_0 is not orthogonal to that eigenvector.
More specifically, let v_i be normed eigenvalues of A corresponding to eigenvalues λ_i, ordered by absolute value. Let $x_0 = \sum_{i=1}^{n} \alpha_i v_i$ be the representation of x_0 by our eigenvalues. Then there exists a constant c such that

$$\|\|Ax_K\|_2 - |\lambda_1\| \leq 4c \cdot \|A\|_2 \left|\frac{\lambda_2}{\lambda_1}\right|^k$$

Proposition 40.4.6.

$$x_n = \frac{A^k x_0}{\left\|A^k x\right\|_2}$$

Proposition 40.4.7. Under "suitable conditions", applying von Mises iteration to $(A - \mu I_n)^{-1}$ gives the eigenvalue closest to μ.

Theorem 40.4.8: QR Iteration

Let A_0 be diagonalizable. Then the sequence

$$A_k = Q_k R_k$$
$$A_{k+1} = R_k Q_k,$$

obtained by repeatedly finding the QR decomposition of A_k and then swapping the factors, converges to a diagonal matrix containing the eigenvalues of A_0, ordered by absolute value.

In praxis, this iteration generally converges even if A_0 is not diagonalizable. Each step of this iteration is $O(n^3)$.
This can be decreased to a one-time cost of $O(n^3)$ followed by $O(n^2)$ for every subsequent step by transforming A into *Hessenberg form* (an upper triangular matrix $\hat{A}$ with one extra diagonal below the main diagonal), whose QR decomposition is then cheaper to calculate.

40.4.3 The Jacobi Eigenvalue Algorithm

The Gerschogorin circle theorem tells us that we can find the eigenvalues of A by making A 'almost diagonal', i.e. by transforming A so that the off-diagonal entries become negligible. This is the idea behind the ***Jacobi eigenvalue algorithm*** (Not to be confused with the *Jacobi Method*, which is an algorithm for

approximating the solution to a system of linear equations that we will meet later).

Definition 40.4.9. Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$. We define $N(A)$ to be the Euclidean norm of the off-diagonal entries of A, i.e.

$$N(A) = \sum_{i \neq j} a_{ij}^2$$

Proposition 40.4.10. For every diagonal entry a_{ii}, there exists an eigenvalue λ such that

$$|\lambda - a_{ii}| \leq \sqrt{N(A)}.$$

Lemma 40.4.11. Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ be symmetrical. Let $R_{ij}(\theta)$ be a rotation of θ degrees in the plane spanned by the basis vectors b_i and b_j. Then $B = R_{ij}(\theta)^\top A R_{ij}(\theta)$ fulfills

$$\begin{aligned} N(B) &= N(A) - 2(a_{ij}^2 - b_{ij}^2), \\ b_{ij} = b_{ji} &= \cos(2\theta) a_{ij} + \frac{1}{2}\sin(2\theta)(a_{ii} - a_{jj}) \end{aligned}$$

Lemma 40.4.12. Let $a_{ij} \neq 0$. Let

$$\theta = \frac{1}{2}\arctan\left(\frac{2a_{ij}}{a_{jj} - a_{ii}}\right)$$

Then $B = R_{ij}(\theta)^\top A R_{ij}(\theta)$ fulfills $b_{ij} = b_{ji} = 0$.

Proof. We saw in the last lemma that

$$b_{ij} = \cos(2\theta) a_{ij} + \frac{1}{2}\sin(2\theta)(a_{ii} - a_{jj})$$

Setting $b_{ij} = 0$ gives

$$\begin{aligned} 0 &= \cos(2\theta) a_{ij} + \frac{1}{2}\sin(2\theta)(a_{ii} - a_{jj}) \\ \iff \tan(2\theta) &= \frac{\sin(2\theta)}{\cos(2\theta)} = \frac{2a_{ij}}{a_{jj} - a_{ii}} \end{aligned}$$

□

Note that performing this trick multiple times may return entries we previously transformed to zero back to being non-zero. However, if we whack the off-diagonal entries in descending order of absolute value, we do see convergence:

Theorem 40.4.13. If we choose a_{ij} to be the off-diagonal entry of A with the largest absolute value and define $R_{ij}(\theta)$ and B as in the previous lemma, we get

$$N(B) \leq \left(1 - \frac{2}{n(n-1)}\right) N(A)$$

Proof. Since a_{ij} is the element with the largest absolute value, we have

$$N(A) \leq n(n-1)a_{ij}^2,$$

giving us

$$N(B) = N(A) - 2a_{ij}^2 \leq \left(1 - \frac{2}{n(n-1)}\right)N(A)$$

□

Repeated iterations of this procedure is known as the ***Jacobi eigenvalue algorithm***. The algorithm requires $O(n^2 \log(1/\varepsilon))$ iterations to guarantee $N(A_{k+1}) \leq \varepsilon$. Searching for the largest off-diagonal entries can be impractical, so instead one often simply iterates over all off-diagonal entries.

40.5 Contraction Methods

Theorem 40.5.1. Let $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$. Then the spectral radius $\rho(A)$ is the greatest lower bound of all operator norms of A.

Corollary 40.5.2. If $\rho(A) < 1$, any affine map $\Phi : x \mapsto Ax + b$ is a contraction.

We can decompose any matrix A into a sum

$$A = L + D + U$$

of a strictly lower triangular matrix, a diagonal matrix, and a strictly upper triagonal matrix. Since D, $D + U$, and $D + L$ are easy to invert (as long as they *are* invertible), this approach leads to an entire family of iterative algorithms for solving systems of linear equations.
The ***Jacobi method*** iterates via

$$\begin{aligned} Lx^k + Dx^{k+1} + Ux^k &= b \\ \Leftrightarrow x^{k+1} &= -D^{-1}(A - D)x^k + D^{-1}b \\ &:= M^J + c^J. \end{aligned}$$

The ***Gauß-Seidel method*** iterates via

$$\begin{aligned} Lx^{k+1} + Dx^{k+1} + Ux^k &= b \\ \Leftrightarrow x^{k+1} &= -(L + D)^{-1}Ux^k + (L + D)^{-1}b \\ &:= M^{GS}x^k + c^{GS}. \end{aligned}$$

We want to find sufficient conditions that A needs to fulfill in order for these iterations to be contractions, and thus to converge.

Exercise 40.5.3. *Show that both methods converge if A is strictly diagonally dominant.*

Theorem 40.5.4. If A is symmetric and positive definite, the Gauß-Seidel method iterates over a contraction, and thus converges.

Definition 40.5.5. We call a matrix $A \in \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}^n)$ ***reducible*** if there exists a decomposition of $[1, ..n]$ into disjoint index sets I, J such that $a_{ij} = 0$ for all $(i, j) \in I \times J$.

Theorem 40.5.6. If A is reducible, we can decompose the system of linear equations $Ax = b$ into two systems $A_{II}x_I = b_I$ and $A_{JJ}x_J = b_J - A_{JI}x_I$.

Theorem 40.5.7. If A is reducible and diagonally dominant, it is invertible with non-zero diagonal entries.

Theorem 40.5.8. If A is irreducible and diagonally dominant, the Jacobi and Gauß-Seidel methods iterate over contractions, and thus converge.

Chapter 41

Numerical Analyis

41.1 Polynomial Interpolation

Definition 41.1.1. The ***Lagrange interpolation*** problem consists of finding a polynomial $p \in \mathcal{P}_n$ such that $p(x_i) = y_i$ for some $a \leq x_0 < \ldots < x_n \leq b$ and $y_0, \ldots, y_n$.

A convenient solution to this problem would be finding a set of polynomials L_i such that $L_i(x_j) = \delta_{ij}$, since those would let us solve any interpolation problem using the right linear combination. A first idea is

$$L_i(x) := \prod_{i \neq j} (x - x_j)$$

which gives us $L_i(x_j) = 0$ if $i \neq j$ and $L_i(x_i) \neq 0$. Now, we only need to normalize these polynomials such that $L_i(x_i) = 1$. This can be done by setting

$$L_i(x) := \prod_{i \neq j} \frac{x - x_j}{x_i - x_j}$$

Theorem 41.1.2

Given some $x_0 < x_1 \ldots x_n$, the polynomials

$$L_i(x) := \prod_{i \neq j} (x - x_j)$$

fulfill $L_i(x_j) = \delta_{ij}$. They are known as ***Lagrange polynomials***.

Corollary 41.1.3. The polynomial

$$p(x) = \sum_{i=0}^{n} y_i \cdot L_i(x)$$

is the unique solution of the Lagrange interpolation problem.

Exercise 41.1.4. *Show that* $\sum_{i=0}^{n} L_i(x) = 1$ *for all* $x \in [a, b]$.

Theorem 41.1.5. Let $f \in C^{n+1}([a, b])$ such that $f(x_i) = y_i$. Let p be the solution of the corresponding Lagrange polynomial interpolation problem. Then for every $x \in [a, b]$, there exists a $\xi \in [a, b]$ such that

$$f(x) - p(x) = \frac{1}{(n+1)!} \cdot f^{(n+1)}(\xi) \cdot \prod_{j=0}^{n} (x - x_j).$$

Corollary 41.1.6.

$$\|f - p\|_{\sup} \leq \frac{\|f^{(n+1)}\|_{\sup}}{(n+1)!} \cdot (b-a)^{n+1}$$

Corollary 41.1.7. Increasing the number of nodes x_i or decreasing $b - a$ leads to uniform convergence of the interpolation polynomials to f, as long as the derivatives of f don't grow rapidly.

In practice, this leads to uniform convergence for most f. Given uniformly spaced nodes, one counterexample is given by ***Runge's function*** $f(x) = \frac{1}{1+25x^2}$.

However, evaluating the interpolation polynomial can be difficult and unstable in practice. We now focus our attention on deriving a scheme that allows a stable evaluation of $p(x)$ in $O(n^2)$ time.

Definition 41.1.8. Given nodes $x_0, \ldots, x_n$ and values $y_0, \ldots, y_n$ and $j \in [0..n]$, we define $p_{i,j}$ as the Lagrange interpolation polynomial fulfilling $p_{i,j}(x_k) = y_k$ for all $k \in [i..i+j]$

Note that $p_{i,0}(x) = y_i$ and $p_{0,n}(x) = p(x)$.

Theorem 41.1.9: Neville Scheme

The polynomials $p_{i,j}$ are given by $p_{i,0}(x) = y_i$ and

$$p_{i,j}(x) = \frac{(x - x_i) \cdot p_{i+1,j-1}(x) - (x - x_{i+j}) p_{i,j-1}(x)}{x_{i+j} - x_i}$$

Recursive implementations of the Neville scheme are highly inefficient. A simple successive "forwards" evaluation scheme leads to a runtime in $O(n^2)$.

Definition 41.1.10. Given $x_0, \ldots, x_{n-1}$, the ***Newton basis*** $q_0, \ldots, q_n$ of $\mathcal{P}_n$ is given by

$$q_j(x) = \prod_{k=0}^{j-1} (x - x_k)$$

Theorem 41.1.11: Divided Differences

Given $p \in \mathcal{P}_n$ and $x_0, \ldots, x_n \in [a, b]$, we can find the representation $p(x) = \sum_{j=0}^{n} \lambda_j q_j(x)$ of p in the corresponding newton basis by setting $\lambda_{i,0} = y_i$ and iterating

$$\lambda_{i,j} = \frac{\lambda_{i+1,j-1} - \lambda_{i,j-1}}{x_{i+j} - x_i}$$

p can then be evaluated easily using the Horner Scheme

$$p(x) = \lambda_0 + (x - x_0) \cdot (\lambda_1 + (x - x_1) \cdot (\lambda_2 + \ldots (\lambda_{n-1} + (x - x_{n-1} \lambda_n))))$$

This approach has the advantage of allowing us to easily add more nodes later.

Now, we shift our attention to the problem of lowering interpolation error. One way of doing this is by cleverly choosing our nodes x_i.

Definition 41.1.12. The ***Chebyshev-Polynomials*** on $t \in [-1, 1]$ are given by

$$T_n(t) = \cos(n \cdot \arccos(t)).$$

The roots of a Chebyshev polynomial are known as ***Chebyshev nodes***.

Lemma 41.1.13. We have $|T_n(t)| \leq 1$ for all $t \in [-1, 1]$.

Lemma 41.1.14. The Chebyshev polynomials are given by the recurrence relation

$$\begin{aligned} T_0(t) &= 1 \\ T_1(t) &= t \\ T_{n+1}(t) &= 2tT_n(t) - T_{n-1}(t) \end{aligned}$$

Corollary 41.1.15. We have

$$T_{n+1}(x) = 2^n \prod_{j=0}^{n} (x - t_j)$$

Lemma 41.1.16. The roots of T_n are given by

$$t_j = \cos\left(\frac{(j + \frac{1}{2})\pi}{n}\right)$$

for $j \in [0..n-1]$.

Lemma 41.1.17. The extreme values of T_n are given by

$$s_j = \cos\left(\frac{j\pi}{n}\right)$$

for $j \in [0..n]$.

Theorem 41.1.18. Among all possible choices of nodes for polynomial interpolation, Chebyshev nodes minimize the supremum norm of the interpolation polynomial. If Chebyshev nodes are used, the interpolation error becomes

$$\left\|f - p\right\|_{\sup} \leq 2^{-n} \frac{\left\|f^{(n+1)}\right\|_{\sup}}{(n+1)!}$$

Definition 41.1.19. The ***Hermite interpolation problem*** consists of finding a polynomial $p \in \mathcal{P}_N$ such that $p^{(k_i)}(x_i) = y_i^{(k_i)}$, where $N + 1 = \sum_{i=0}^{n}(k_i + 1)$.

Proposition 41.1.20. Hermite interpolation has a unique solution. Given only a single node, the solution is the corresponding taylor polynomial.

41.2 Spline Interpolation

Definition 41.2.1

A function $s : [a, b] \to \mathbb{R}$ is known as a ***Spline of degree m and order k*** if $s \in C^k([a, b])$ is a piecewise combination of polynomials of degree m over some partition I_n of $[a, b]$ into n nodes. The space of m, k-splines on P is written $\mathcal{S}^{m,k}(I_n)$. If no order is given, it is generally assumed that $k = m - 1$.

Proposition 41.2.2. For $k \geq m$, we have $\mathcal{P}_m|_{[a,b]} = \mathcal{S}^{m,k}(I_n)$.

Theorem 41.2.3. Linear spline interpolation is solved by $s = \sum_{i=0}^{n} y_i \varphi_i$, where φ_i are the ***hat functions***, given by linearly interpolating $\varphi_i(x_j) = \delta_j$

Theorem 41.2.4.

$$\dim \mathcal{S}^{m,m-1}(I_n) = n + m$$

Definition 41.2.5. Given a cubic spline interpolation problems, we generally want to fulfill one of the following boundary conditions:

1. ***Natural boundary conditions***, given by

$$s''(a) = s''(b) = 0$$

2. ***Complete boundary conditions*** or ***Hermite boundary conditions***, given by

$$\begin{aligned} s'(a) &= \alpha, \\ s'(b) &= \beta \end{aligned}$$

   for some $\alpha, \beta \in \mathbb{R}$.

3. ***Periodic boundary conditions*** $s'(a) = s'(b)$, $s''(a) = s''(b)$

If a thin, elastic board has its position fixed at certain points, the shape that the board takes is exactly the solution of the corresponding cubic spline interpolation problem.

Theorem 41.2.6. Given a spline s and any other function g interpolating the same points and fulfilling the same boundary condtions, the curvature of s is a lower bound to the curvature of g.

Theorem 41.2.7. Let s_1 be the linear spline interpolating $f \in C^2([a,b])$. Then

$$\left\| f - s_1 \right\|_{L^2} \leq \frac{h^2}{2} \left\| f'' \right\|_{L^2}$$

with $h = \max \{x_i - x_{i-1} \mid i \in [1..n]\}$.

Theorem 41.2.8. Let s_3 be a cubic spline interpolating $f \in C^2([a,b])$. Then

$$\left\| f - s_3 \right\|_{L^2} \leq \frac{h^4}{4} \left\| f^{(4)} \right\|_{L^2}$$

with $h = \max \{x_i - x_{i-1} \mid i \in [1..n]\}$.

41.3 Discrete Fourier Transform

Definition 41.3.1: Real Trigonometric Interpolation

Given an even number of evenly spaces nodes $x_0, \dots, x_{n-1}$ with $n = 2m$ even, the ***real trigonometric ineterpolation problem*** consists of finding $a_k, b_k \in \mathbb{R}$, $k \in [1..m-1]$ and $a_0, a_m \in \mathbb{R}$ such that

$$T(x) = \frac{a_0}{2} + \sum_{k=1}^{m-1} (a_k \cos(kx) + b_k \sin(kx)) + \frac{a_m}{2} \cos(mx)$$

fulfills $T\left(\frac{2\pi j}{n}\right) = y_j$ for all j.

Trigonometric interpolation can be written much more compactly in the complex case:

Definition 41.3.2: Real Trigonometric Interpolation

Given an even number of evenly spaces nodes $x_0, \dots, x_{n-1}$ with $n = 2m$ even, the ***real trigonometric ineterpolation problem*** consists of finding $c_k \in \mathbb{C}$, $k \in [1..n-1]$ such that

$$T(x) = \sum_{k=0}^{n-1} c_k e^{ikx}$$

fulfills $T\left(\frac{2\pi j}{n}\right) = y_j$ for all j.

Theorem 41.3.3. A set of coefficients c_k solves the complex trigonometric interpolation problem iff. the coefficients a_k, b_k defined by

$$\begin{aligned} a_0 &= 2c_0 \\ a_k &= c_k + c_{2m-k} \\ b_k &= i(c_k - c_{2m-k}) \\ a_m &= 2c_m \end{aligned}$$

solve the corresponding real trigonometric interpolation problem.

If we write $y = (y_0, \dots, y_{n-1})^\top$ and $\vec{\omega}_n^k = (\omega_n^{jk})_{j\in[1..n]}$, the complex trigonometric interpolation problem is given by the system of linear equations

$$y = \sum_{k=0}^{n-1} \beta_k \vec{\omega}_n^k$$

Theorem 41.3.4

The family $(\vec{\omega}_n^k)_{k\in[1..n]}$ is an orthogonal basis of $\mathbb{C}^n$ fulfilling

$$\left\langle \vec{\omega}_n^k, \vec{\omega}_n^l \right\rangle = n\delta_{kl}$$

known as the ***Fourier basis***.

Complex trigonometric interpolation thus comes down to performing a change of basis from the standard basis into the Fourier basis. We know from linear algebra that the matrix $T_n = (\vec{\omega}_n^0, \dots, \vec{\omega}_n^n)$ is the basis change matrix from the Fourier basis into the standard basis. Thus, its inverse must be the basis change matrix from the standard basis into the Fourier basis.

Exercise 41.3.5. *Show* $T_n^{-1} = \frac{1}{n} T_n^*$

Thus, we get:

Theorem 41.3.6: Discrete Fourier Transform

1. Change of basis from the standard basis of $\mathbb{C}^n$ into the Fourier basis, known as the ***discrete Fourier transform***, is given by
$$\beta = \frac{1}{n} T_n^* y$$
2. Change of basis from the Fourier basis of $\mathbb{C}^n$ into the standard basis, known as the ***inverse discrete Fourier transform*** or as ***Fourier synthesis***, is given by
$$y = T_n \beta$$

41.4 Numerical Integration

The goal of numerical integration, also known as ***quadrature***[3] is numerical approximation of the integral operator

$$I(f) = \int_a^b f(x)\, dx.$$

[3] Based on the ancient Greek practice of finding the area of a shape by constructing a square of equal area.

Definition 41.4.1. A ***quadrature formula*** on an interval $[a, b]$ is a linear map $Q : C^0([a, b]) \to \mathbb{R}$ of the form

$$Q(f) = \sum_{i=0}^{n} w_i . f(x_i)$$

Definition 41.4.2. The number

$$\|Q\| = \frac{1}{b-a} \sum_{i=0}^{n} |w_i|$$

is known as the ***stability indicator*** of Q.

The Riemann integral is defined as the limit of the quadrature formula

$$Q(f) = \sum_{k=0}^{n-1} (x_{i+1} - x_i) . f(x_i)$$

as the distance between the nodes approaches zero.

Definition 41.4.3. A quadrature formula Q is called ***exact of degree d*** if $Q(p) = I(p)$ for all $p \in \mathcal{P}_d$.

Exercise 41.4.4. *A quadrature formula is exact of degree* 0 *if and only if* $\sum_{i=0}^{n} w_i = b - a$.

Lemma 41.4.5. Given $n = 2m$ equidistant nodes, a quadrature formula on these n nodes that is exact of degree $2d$ is also exact of degree $2d + 1$

Theorem 41.4.6

The ***Newton-Cotes-Formula***

$$Q(f) = \sum_{i=0}^{n} \left(\int_a^b L_i(x)\, dx \right) . f(x_i)$$

is exact of degree n. The case $n = 0$ is known as the ***middle point rule***, the case $n = 1$ is known as the ***trapeze rule***, and the case $n = 2$ is known as the ***Simpson rule***.

Theorem 41.4.7. The trapeze rule is given in closed form by

$$T(f) = \frac{b-a}{2}(f(a) + f(b))$$

Theorem 41.4.8. The Simpson rule is given in closed form by

$$T(f) = \frac{b-a}{6}\left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right).$$

since we chose equidistant nodes x_0, x_1, x_2, it is exact of degree 3.

We can increase the accuracy of a quadrature formula by partitioning our interval $[a, b]$ into smaller intervals, and applying a quadrature formula to each one separately. We write this as

$$Q^n(f) = \sum_{k=1}^{n} Q_k(f|_{(x_{j-1}, x_k)})$$

A summed quadrature formula is called exact of degree d if

$$\left| Q^n(f) - I(f) \right| = O\left(\left(\frac{b-a}{n} \right)^d \right)$$

Proposition 41.4.9. The summed trapeze rule is exact of degree 2. The summed Simpson rule is exact of degree 4.

Lemma 41.4.10. A quadrature formula with $n + 1$ weights has exactness degree at most $2n + 1$.

Proof. Let $Q(f) = \sum_{i=0}^{n} w_i \cdot f(x_i)$ be our quadrature formula. Let $p(x) = \prod_{i=0}^{n} (x - x_i)^2$. Then we have $Q(p) = 0$, but $I(p) > 0$. Since $p \in \mathcal{P}_{2n+2}$, Q cannot have exactness $2n + 2$. □

Lemma 41.4.11. A quadrature formula with $n + 1$ weights is exact of degree n if and only if

$$w_i = \int_a^b L_i(x)\, dx$$

Proof. 1. If Q is exact of degree n, it must integrate the lagrange polynomials exactly. We get:

$$\begin{aligned}\int_a^b L_i(x)\,dx &= \sum_{j=0}^{n} w_j L_i(x_j) \\ &= \sum_{j=0}^{n} w_j \delta_{ij} \\ &= w_i\end{aligned}$$

2. Any quadrature formula Q that integrates the Lagrange polynomials exactly must also integrate any linear combination of Lagrange polynomials exactly. Since the Lagrange polynomials at $n+1$ nodes form a basis of $\mathcal{P}_n$, it follows that Q integrates any polynomial of degree n exactly.

□

Lemma 41.4.12. If a quadrature formula is exact of even degree $n = 2m$, we have $w_i > 0$ for all i.

Proof Sketch.

$$w_i = \int_a^b L_i(x)^2 > 0$$

"□"

We will now devote our attention to deriving the ***Gauß quadrature***, which chooses $n+1$ quadrature points such that the maximum possible exactness degree is achieved.

Proposition 41.4.13. Let $\omega \in C^0([a,b])$ with $\omega(x) > 0$. Then $\langle f, g \rangle_\omega = \int_a^b f(x)g(x)\omega(x)\,dx$ is a scalar product on $C^0([a,b])$.

Using Gram-Schmidt starting with the standard monomial basis, we can construct an ω-orthonormal base $(\pi_i)_{i\in[0..n]}$ of $\mathcal{P}_n$ such that $\pi_i \in \mathcal{P}_i$ for all i.

Proposition 41.4.14. Using $\omega(x) = (1-x^2)^{-\frac{1}{2}}$ on $(-1,1)$ gives the Chebyshev polynomials.

Proposition 41.4.15. Using $\omega(x) = 1$ on $[-1,1]$ gives the Legendre polynomials.

Lemma 41.4.16. Every orthonormal polynomial π_i has i roots of order 1 lying in the given interval (a,b).

Theorem 41.4.17

Let x_i be the roots of π_{n+1} for some $\omega \in C^0([a,b])$. Then the quadrature formula with roots x_i and weights

$$w_i = \int_a^b L_i(x)\omega(x)\,dx$$

is exact of degree $2n+1$.

If the error of a summed quadrature formula $T(h)$ is of order h^γ for a $\gamma \in \mathbb{N}$, we have

$$T(h) = T(0) + \varphi(h)$$

with $\varphi(0) = 0$ and $\left|\varphi(h)\right| \leq ch^\gamma$ for some $c \in \mathbb{R}$. Taylor expansion of φ at 0 gives

$$\varphi(h) = c_1 h^\gamma + c_2 h^{\gamma+1} + o(h^{\gamma+1})$$

for some $c_1, c_2 \in \mathbb{R}$ and, i.e.

$$T(h) = T(0) + c_1 h^\gamma + c_2 h^{\gamma+1} + o(h^{\gamma+1})$$

If we evaluate at $h/2$, we get

$$T(h/2) = T(0) + \frac{c_1}{2^\gamma}h^\gamma + \frac{c_2}{2^{\gamma+1}}h^{\gamma+1} + o\left(\frac{h^{\gamma+1}}{2^{\gamma+1}}\right)$$

This lets us eliminate the $c_1 h^\gamma$-term by setting

$$T^*(h) := \frac{T(h) - 2^\gamma T(h/2)}{1-2^\gamma} = T(0) + c_2\frac{1-2^{-1}}{1-2^\gamma}\cdot h^{\gamma+1} + o(h^{\gamma+1}),$$

which is a new summed quadrature formula with an error of $h^{\gamma+1}$, instead of h^γ, which can be a significant improvement for $\gamma \ll 1$. Given a non-polynomial function $f \in C^k([a,b])$ whose exact integral $I(f)$ is known, we can investigate the errors

$$e_h = \left|I(f) - Q^N(f)\right|$$

for various $h > 0$. Since $e_h \approx c_1 h^\gamma$, we can evauate at two different step sizes $h, H > 0$ and get

$$c_1 \approx \frac{e_h}{h^\gamma} \approx \frac{e_H}{H^\gamma}$$

and thus

$$\gamma \approx \frac{\log(e_h/e_H)}{\log(h/H)} = \frac{\log(e_h) - \log e_H}{\log(h) - \log(H)}$$

We can now try out various pairs h, H, and know that we have obtained the correct $\gamma \in h$ if these different pairs all give nearly identical answers.

41.5 Nonlinear Optimization

Let $U \subseteq \mathbb{R}^n$. Then typical problems include:

1. Find $x^* \in U$ such that $f(x^*) = 0$ for some $f : U \to \mathbb{R}^m$.
2. Find $x^* \in U$ such that $g(x^*) = \min_{x \in U} g(x)$ for some $g : U \to \mathbb{R}$.

In general, neither of these problems have closed form solutions. Instead, approximate solutions are found by iterative algorithms, i.e. we find $(x_k)_{k \in \mathbb{N}}$ with $x_k \to x^*$.

Definition 41.5.1. A numerical algorithm which returns a sequence $(x_k)_{k \in \mathbb{N}}$ approximating a solution x^* given some initial vector $x_0 \in U$ is known as:

1. ***Globally convergent***, if $x_k \to x^*$ for all $x_0 \in U$.
2. ***Locally convergent***, if there exists some $\varepsilon > 0$ such that $x_k \to x^*$ for all $x_0 \in U$ with $d(x_0, x^*) < \varepsilon$.

Definition 41.5.2. If $x_k \neq x^*$ for all k, we can define the ***rate of convergence*** of x_k by saying that a locally convergent algorithm has ***convergence order $\alpha \geq 1$*** if there exists a $q \in \mathbb{R}$ such that the errors $\delta_k = d(x^* - x_k)$ fulfill

$$\limsup_{k \to \infty} \frac{\delta_{k+1}}{\delta_k^\alpha} = q.$$

If $\alpha = 1$, we additionally demand $q \leq 1$.

Proposition 41.5.3. If $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ is a contraction, the basic fixed point iteration $x_{k+1} = \Phi(x_k)$ is globally linearly convergent.

Proposition 41.5.4. If $\Phi \in C^1(\mathbb{R}, \mathbb{R})$ has a fixed point x^*, then the fixed point iteration $x_{k+1} = \Phi(x_k)$ is locally convergent if $|\Phi'(x^*)| < 1$ and divergent if $|\Phi'(x^*)| > 1$.

41.6 Computing Roots Numerically

Theorem 41.6.1. ***Bisection Algorithm:*** Let $f \in C^0([a, b], \mathbb{R})$ with $f(a)f(b) \leq 0$. Set $\varepsilon > 0$. Consider the following algorithm:

1. Set $(a_0, b_0) = (a, b)$ and $k = 0$.
2. Set $c_k = \frac{a_k + b_k}{2}$ and
$$(a_{k+1}, b_{k+1}) = \begin{cases} (a_k, c_k) & \text{if } f(a_k)f(c_k) \leq 0 \\ (c_k, b_k) & \text{otherwise} \end{cases}$$
3. Return b_{k+1} if $d(b_{k+1}, a_{k+1}) \leq \varepsilon$. Otherwise set $k \leftarrow k + 1$ and return to step 2.

Then iteration stops after $J \leq 1 + \log_2\left(\frac{b-1}{\varepsilon}\right)$ steps, the interval $[a_J, b_J]$ contains a root, and if $\varepsilon = 0$, then the sequence $(c_k)_{k \in \mathbb{N}}$ converges to a root linearly with $q = \frac{1}{2}$.

Proposition 41.6.2. The bisection algorithm converges linearly.

Theorem 41.6.3. ***Secant Method:*** Let $f \in C^0([a, b] \to \mathbb{R})$. Then the following algorithm converges to a root of f, if one exists:

1. Let $\varepsilon > 0$, $x_0 = a$, $x_1 = b$, and $k = 1$.

2. If $f(x_k) \neq f(x_{k-1})$, set

$$x_{k+1} = x_k - \frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})} f(x_k)$$

3. Return x_{k+1} if $d(x_{k+1}, x_k) \leq \varepsilon$. Otherwise set $k \leftarrow k + 1$ and return to step 2.

Proposition 41.6.4. The secant method converges superlinearly, but subquadratically, with convergence order $\varphi = \frac{1+\sqrt{5}}{2}$ (the golden ratio!)

If $f \in C^1(U \subseteq \mathbb{R}^n, \mathbb{R}^n)$, Taylor approximation gives

$$0 = f(x^*) = f(x) + Df(x)(x^* - x) + \varphi(d(x^*, x)) \approx f(x) + Df(x)(x^* - x),$$

from which we can set $x := x_k$ to get a new iteration

$$x_{k+1} = x_k - Df(x_k)^{-1} f(x_k) \approx x^*$$

as long as $Df(x_k)$ is regular. This is the (multidimensional analogue of) the Newton method, and is generally much more efficient than the secant method.

Theorem 41.6.5. Let $f \in C^2(U, \mathbb{R}^n)$. Let $x^* \in U$ be a root of f such that $Df(x^*)$ is invertible. Then there exists an $\varepsilon > 0$ such that the Newton method converges for any $x_0 \in U$ with $d(x^*, x_0) < \varepsilon$. Additionally, there exists a constant $c \geq 0$ such that the iterates $(x_k)_{k \in \mathbb{N}}$ fulfill

$$\|x^* - x_{k+1}\| \leq c\|x^* - x_k\|^2,$$

i.e. the Newton method converges quadratically.

If $d(x^*, x_0) \geq \varepsilon$, the Newton method might diverge. In this case, introducing a damping factor $\omega \in (0, 1)$ and iterating

$$x_{k+1} = x_k - \omega Df(x_k)^{-1}$$

can lead to convergence, but makes convergence asymptotically slower (no longer quadratic in general).

Part XIII

Measure Theory

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Chapter 42

Measure Theory

42.1 The Measure Problem

The most basic goal of measure theory is to establish a generalized notion of a "measure function", which assigns a "volume" to a given set. In particular, we would like to establish a function that assigns volume functions to subsets of $\mathbb{R}^n$ and has the following properties:

1. When given a subset with an easily intuitively definable volume, the volume function should agree with that volume. In particular, the volume of a cuboid should be the product of the lengths of its sides, and the length of a real interval (a, b) should be $b - a$.
2. The volume of a countable disjoint union of sets should be the sum of the individual volumes. This property is generally referred to as *σ-additivity*.
3. The volume should be invariant under isometries, i.e. functions like rotations, translations, and reflections should not change the volume of a set.
4. We want our volume to be a positive real number.

We call a σ-additive function a *measure* - we want to eventually find a meausure on $\mathbb{R}^n$ that also fulfills the other properties, but we want measures to be far more generally applicable, and as such we want to be able to define measures much more general spaces that may not have a defined notion of an interval or an isometry.

It will turn out that there exists exactly one function on $\mathbb{R}^n$, called the *Lebesgue measure* λ^n, that fulfills these conditions for a very large family of sets - enough to include every "somewhat reasonable" subset of $\mathbb{R}^n$. However, there are still counterexamples.

Proposition 42.1.1. Every subset of $\mathbb{R}^n$ being Lebesgue-measurable is consistent with ZF (without the axiom of choice).

Theorem 42.1.2. Assuming the axiom of choice, there exist subsets of $\mathbb{R}^n$ that cannot be assigned a volume without arriving at a contradiction.

It turns out that this result crucially relies on the full axiom of choice, and in particular is not implied by commonly used weaker forms of the axiom of choice such as the axiom of dependent choice.

The following two subsections deal with two different ways of proving this theorem by construction *non-measurable sets*: The *Vitali Sets*, and the decomposition of a sphere given in the *Banach-Tarski-Paradox*.

42.1.1 Vitali Sets

Proposition 42.1.3. The relation $x \sim y \Leftrightarrow x - y \in \mathbb{Q}$ is an equivalence relation on the real numbers.

Theorem 42.1.4. There exist sets $V \subset [0,1]$ such that for each $r \in \mathbb{R}$, there exists exactly one number $v \in V$ such that $v - r$ is rational. We call such a set a **Vitali Set**.

> *Proof.* Consider the aforementioned equivalence relation $x \sim y$ on $\mathbb{R}$. Each equivalence class must contain at least one representative also contained in $[0,1]$, since if $x - y = q \in \mathbb{Q}$, we have $x - (y + q) \in \mathbb{Q}$ for all $q \in \mathbb{Q}$, letting us pick a q such that $x - (y + q) \in [0,1]$. Being equivalence classes, they must also be disjoint.
>
> This means we can use the axiom of choice to pick exactly one element of each equivalence class of $\sim$, giving us our set V. □

Lemma 42.1.5. Let $q_1, q_2, \ldots$ be an enumeration of $\mathbb{Q} \cap [-1,1]$. Then for $j \neq k$, we have

$$(q_j + V) \cap (q_k + V) = \emptyset$$

> *Proof.* Assume the intersection is non-empty. Then there exist $v_1, v_2 \in V$ such that $q_j + v_1 = q_k + v_2$, meaning $v_1 - v_2 \in \mathbb{Q}$. Therefore, v_1 must be in the same equivalence class as v_2. Since V contains exactly one element of each equivalence class, we have $v_1 = v_2$, and therefore we also have $q_j = q_k$, i.e. $j = k$. □

Lemma 42.1.6. We have

$$[0,1] \subset \bigcup_{k \in \mathbb{N}} (q_k + V) \subset [-1,2]$$

> *Proof.*
> 1. $\bigcup_{k \in \mathbb{N}} (q_k + V) \subset [-1,2]$ follows trivially from $q_k \in [-1,1]$ and $V \subset [0,1]$.
> 2. $[0,1] \subset \bigcup_{k \in \mathbb{N}} (q_k + V)$ follows from the definition of V, since for every

$y \in [0,1]$ we have a unique $v \in V$ such that $y - v := q \in \mathbb{Q}$, and since $y \in [0,1]$ and $v \in [0,1]$, we have $q \in [-1,1]$, i.e. q is contained in our enumeration.

□

Corollary 42.1.7. Vitali sets are not measurable by a translation-invariant normed measure.

Proof. Assume that λ is translation-invariant, countably additive and that $\lambda([a,b]) = b - a$. Then we have:

$$1 = \lambda([0,1]) \leq \lambda\left(\bigcup_{k\in\mathbb{N}}(q_k + V)\right) \leq \lambda([-1,2]) = 3$$

Since our measure is countably additive and invariant under isometries, we can translate each individual set and preserve the measure:

$$\begin{aligned}\lambda\left(\bigcup_{k\in\mathbb{N}}(q_k + V)\right) &= \sum_{k\in\mathbb{N}} \lambda\left(q_k + V\right) \\ &= \sum_{k\in\mathbb{N}} \lambda(V)\end{aligned}$$

Now, if $\lambda(V) \leq 0$, we wouldn't have $1 \leq \sum_{k\in\mathbb{N}} \lambda(V)$, and if $\lambda(V) > 0$, we wouldn't have $\sum_{k\in\mathbb{N}} \lambda(V) \leq 3$. Therefore every possible measure we could assign to V leads to a contradiction. □

42.1.2 The Banach-Tarski Paradox

Theorem 42.1.8. Given any two sets $A, B \subset \mathbb{R}^n$, with $n \geq 3$, such that both A and B have a nonempty interior, there exist disjoint decompositions $A_1 \sqcup \ldots \sqcup A_k = A$ and $B_1 \sqcup \ldots \sqcup B_k = B$ such that for each i, A_i and B_i can be transformed into each other by an isometry.

Corollary 42.1.9. The unit sphere can be transformed into two copies of the unit sphere by a finite disjoint decomposition followed by an isometry. The subsets of the decomposition therefore violate the countable union condition we expect from measure functions, and can therefore not be a assigned a meaningful volume.

42.2 Lattices and Boolean Algebras

We have just seen that we cannot define our desired measure function on the full power set of $\mathbb{R}^n$. This means we will have to work on smaller systems of subsets of a given set. Naturally, we want to find the largest such systems that are still well-behaved enough to allow us to define a sensible notion of measure.

We will arrive at different algebraic structures on subsets of power sets, which will serve as the domains of our measure functions.

In order to gain a full birds-eye view of these definitions, we will first introduce the more general notion of *boolean algebras*:

42.2.1 Boolean Algebras

Definition 42.2.1. A **boolean algebra** is a set X, equipped with two binary operations $\wedge$ and $\vee$, a unary operation $\neg$, and two elements 0 and 1, such that:

1. $\wedge$ and $\vee$ are commutative,
2. 1 is a neutral element of $\wedge$, and 0 is a neutral element of $\vee$,
3. $\wedge$ distributes over $\vee$ and $\vee$ distributes over $\wedge$,
4. $x \wedge \neg x = 0$, and $x \vee \neg x = 1$.

Corollary 42.2.2. Any boolean algebra also has the following properties:

1. $\wedge$ and $\vee$ are associative,
2. $\wedge$ and $\vee$ have the following **absorption property**:
$$a \wedge (a \vee b) = a$$
$$a \vee (a \wedge b) = a,$$
3. $a = b \wedge a$ if and only if $a \vee b = b$.

There are three "central" boolean algebras, from which most of the terminology describing them is descended:

Theorem 42.2.3. The set of **propositional formulas** forms a boolean algebra, where 0 is the logical falsum ($\bot$, an unfulfillable formula), 1 is the logical verum ($\top$, a tautological formula), and $\wedge$, $\vee$ and $\neg$ are logical "and", "or" and "not".

In computer science and circuit engineering, one often considers the subalgebra of this boolean algebra where every formula is directly evaluated to "0" or "1".

Theorem 42.2.4. (Power set algebra): The **power set** $\mathcal{P}(X)$ of any set X forms a boolean algebra, where $0 = \emptyset$, $1 = X$, $\wedge$ is the set intersection operation $\cap$, $\vee$ is the set union operation $\cup$, and $\neg$ is the set complement operation $M \to X \setminus M$.

Theorem 42.2.5. (Restrictions of Boolean algebras): Let $\mathcal{B}$ be a Boolean algebra on a set X, and let Y be a subset of X. Then the **restriction of $\mathcal{B}$ to Y**, defined as
$$\mathcal{B}|_Y := \{E \cap Y \mid E \in \mathcal{B}\},$$
is a boolean Algebra on Y.

Theorem 42.2.6. If $Y \in \mathcal{B}$, then

$$\mathcal{B}|_Y = \mathcal{B} \cap \mathcal{P}(Y) = \{E \subset Y \mid E \in \mathcal{B}\}$$

Theorem 42.2.7. (Atomic algebra): Let X be partioned into a union

$$X = \bigcup_{\alpha \in I} A_\alpha$$

of disjoint sets A_α, which we refer to as *atoms*. Then this partition forms a Boolean algebra

$$\mathcal{A}((A_\alpha)_{\alpha \in I}) := \left\{ E \;\middle|\; E = \bigcup_{\alpha \in J} A_\alpha, J \subset I \right\}$$

of all the sets that can be represented as a union of atoms.

The power set Algebra on X is exactly the atomic algebra where X is partitioned into singleton atoms.

Theorem 42.2.8. Atomic algebras are uniquely determined by their atoms, up to relabeling. More precisely: Let $(A_\alpha)_{\alpha \in I}$ and $(B_\beta)_{\beta \in J}$ be two partitions of a set X. Then

$$\mathcal{A}((A_\alpha)_{\alpha \in I}) = \mathcal{A}((B_\beta)_{\beta \in J})$$

if and only if there exists a bijection $\varphi : I \to J$ such that $B_{\varphi(\alpha)} = A_\alpha$ for all $\alpha \in I$.

Theorem 42.2.9. Every finite Boolean algebra is an atomic algebra.

Corollary 42.2.10. Every finite Boolean algebra has cardinality 2^n, where $n \in \mathbb{N}$.

Corollary 42.2.11. There is a one-to-one correspondence, up to relabeling, between finite Boolean algebras on a set X and finite partitions of X into non-empty sets.

Theorem 42.2.12. (Dyadic algebras): Let $n, i_1, \ldots, i_d \in \mathbb{Z}$. The **dyadic algeba** $\mathcal{D}_n(\mathbb{R}^d)$ at scale 2^{-n} in $\mathbb{R}^d$ is the atomic algebra generated by the products of the half-open dyadic intervals

$$I_j := \left[\frac{i_j}{2^n}, \frac{i_j + 1}{2^n} \right)$$

of length 2^{-n}.

This algebra consists exactly of the "grid figures" made up of a finite number of "pixels" of length 2^{-n}.

Theorem 42.2.13. (Intersection of Boolean Algebras): The intersection of a family $(\mathcal{B}_\alpha)_{\alpha \in I}$ of Boolean algebras on a set X is again a Boolean algebra, assuming the convention that, if I is empty, the intersection is the full power set. Furthermore, this Intersection is the finest Boolean algebra that is coarser than every $\mathcal{B}_\alpha$.

Definition 42.2.14. Let $\mathcal{F}$ be any family of subsets of a set X. Then we define $\langle \mathcal{F} \rangle_{\mathcal{B}}$ to be the intersection of all Boolean algebras that contain $\mathcal{F}$. We call this the **Boolean algebra generated by $\mathcal{F}$**.

Equivalently, $\langle \mathcal{F} \rangle_{\mathcal{B}}$ is the smallest Boolean algebra containing $\mathcal{F}$.

Theorem 42.2.15. $\mathcal{F}$ is a Boolean algebra if and only if $\langle \mathcal{F} \rangle_{\mathcal{B}} = \mathcal{F}$.

42.2.2 Lattices

Boolean algebras themselves turn out to be specific instances of *lattices*, which play an important role in order theory and universal algebra.

Definition 42.2.16. A **lattice** is an algebraic structure $(L, \vee, \wedge)$, consisting of a set L, an operation $\vee$, called **join**, and an operation $\wedge$, called **meet**, such that the absorbtion laws $a \vee (a \wedge b) = a$ and $a \wedge (a \vee b) = a$.

Proposition 42.2.17. Equivalently, a partially ordered set $(L, \leq)$ is a lattice if every pair of elements has a least upper bound $\sup(a, b) := a \vee b \in L$ and a greatest lower bound $\inf(a, b) := a \wedge b \in L$.

Definition 42.2.18. We call a lattice **bounded** if there exists a **least element** 0, i.e. 0 fulfills $a \vee 0 = a$, and a **greatest element** 1, which fulfills $a \wedge 1 = a$.

Corollary 42.2.19. A boolean algebra is a bounded lattice such that meet and join are distributive over each other and such that complements exist.

42.3 Set Algebras

Definition 42.3.1. Let X be a set and $\mathcal{A} \subset \mathcal{P}(X)$. Then we call $\mathcal{A}$ a **set algebra** if it has the following properties:

1. $\emptyset \in \mathcal{A}$
2. For any $A \in \mathcal{A}$, we have $X \setminus A \in \mathcal{A}$ ($\mathcal{A}$ is closed under the operation of taking complements).
3. For any $F, G \in \mathcal{A}$, we have $F \cup G$ in $\mathcal{A}$ ($\mathcal{A}$ is closed under binary unions).

Corollary 42.3.2. If $\mathcal{A}$ is a set algebra on X, it also fulfills the following:

1. $X \in \mathcal{A}$,
2. For any $F, G \in \mathcal{A}$, we have $F \cap G$ in $\mathcal{A}$,
3. For any $A_1, \ldots, A_n \in \mathcal{A}$, we have $\bigcup_{i=1}^{n} A_i \in \mathcal{A}$,
4. For any $A_1, \ldots, A_n \in \mathcal{A}$, we have $\bigcap_{i=1}^{n} A_i \in \mathcal{A}$.

Thus, we obtain the following more concise (but less readable) definition of set algebras:

Corollary 42.3.3. A set algebra is a subalgebra of the power set boolean algebra on X.

Corollary 42.3.4. A topology on X is "simply" a set algebra on X that is closed under arbitrary unions.

Theorem 42.3.5. (Stone's Representation Theorem for Boolean Algebras): Every boolean algebra is isomorphic to a set algebra.

42.3.1 Set Rings

A very important weakening of the concept of set algebras is given by *set rings*, which contain the empty set and are closed under intersection and union, but don't have to contain the full set X or be closed under complements.

Theorem 42.3.6. Let $\mathcal{A}$ be a set ring. Then it is closed under finite symmetric difference, and forms a ring in the algebraic sense, with symmetric difference as addition and intersection as multiplication. If it contains the full set X, it forms a ring with identity.

Definition 42.3.7. Let $I \subset \mathbb{R}$. We call I an **interval** if there exist $a, b \in \mathbb{R}$ such that $(a, b) \subset I \subset [a, b]$.

Theorem 42.3.8. The set of subsets of the real numbers which can be written as a finite union of intervals forms a set ring.

Theorem 42.3.9. Let $\mathcal{R}$ and $\mathcal{S}$ be set rings. Then the set of finite unions of cartesian products of elements of R_i and S_i, i.e. of elements of the form,

$$\bigcup_{i \in \mathbb{N}} R_i \times S_i,$$

is also a set ring, which we will denote by $\mathcal{R} \boxtimes \mathcal{S}$.

Corollary 42.3.10. The set of finite unions of cuboids in $\mathbb{R}^n$, where we define a cuboid to be a product of arbitrary intervals, i.e. we don't care if the boundary on any particular side is open or closed, forms a set ring. We call sets of this form **Elementary Sets**.

42.3.2 Set Semirings

Theorem 42.3.11. Let $\mathcal{S} \subset \mathcal{P}(X)$. We call $\mathcal{S}$ a **set semiring**, or **semiring of sets**, if:

1. $\emptyset \in \mathcal{S}$,
2. $\mathcal{S}$ is closed under finite intersections,
3. For $A, B \in \mathcal{S}$, there exist disjoint sets $S_1, \ldots, S_n \in \mathcal{S}$ such that $A \setminus B = \bigcup_{i=i}^{n} S_i$.

This means that a set semiring is a weakened form of a set ring where complements are not necessarily contained in the semiring, but can still be "constructed" from elements of the ring. Any set ring is therefore immediately also a set semiring.

Sadly, unlike with rings of sets, there is absolutely no connection between set semirings and the algebraic notion of a semiring - a semiring of sets is exclusively a (semi)(ring of sets), and *not* a (semiring)(of sets). This makes it tempting for me to use an alternative name which makes this distinction more clear, but since I haven't encountered any good alternative names anywhere else (and because I already know I will forget to stick with this convention moving forwards) I will stick with the less than perfect established name.

Set semirings are of fundamental importance to measure theory because the set of cuboids in $\mathbb{R}^n$ forms a set semiring, and we will end up defining our Lebesgue measure by approximating sets through coverings of the set with cuboids. Of course, we first have to establish a basic theory of set semirings and prove this claim.

Theorem 42.3.12. The set $\mathcal{I}$ of real intervals forms a set semiring.

Theorem 42.3.13. The product of two set semirings is again a set semiring.

Corollary 42.3.14. The set $\mathcal{Q}$ of cuboids in $\mathbb{R}^n$ (once again with both open and closed sides allowed) forms a set semiring.

42.4 σ-Algebras

The most important type of set algebra for the purposes of measure theory is the *σ-Algebra*, on which we will eventually define the notion of a "measure" in our desired final form.

Definition 42.4.1: σ-Algebra

Let X be an arbitrary set and $\mathcal{A} \subset \mathcal{P}(X)$. We call $\mathcal{A}$ a **σ-Algebra on X** if:

1. $X \in \mathcal{A}$
2. For all $A \in \mathcal{A}$, we have $X \setminus A \in \mathcal{A}$ ($\mathcal{A}$ is closed under the operation of taking a complement).
3. For all $A_i \in \mathcal{A}$, we have $\bigcup_{i \in \mathbb{N}} \in \mathcal{A}$ ($\mathcal{A}$ is closed under the operationg of taking countable unions).

If $\mathcal{A}$ is a σ-algebra on X, we call $(X, \mathcal{A})$ a **measure space**, and any set $A \in \mathcal{A}$ **$\mathcal{A}$-measurable**.

The "σ" here once again stands for "countable sum", as it also did for σ-additivity and F_σ sets.

Corollary 42.4.2. Any σ-algebra contains the empty set and is closed under countable intersection.

Corollary 42.4.3. A σ-algebra can be more concisely defined as a set algebra that is closed under *countable* union and intersection, not just finite ones.

This also trivially makes every σ-algebra a Boolean algebra.

Theorem 42.4.4. Every atomic algebra is a σ-algebra.

Theorem 42.4.5: Induced σ-algebra

Just like for Boolean algebras, the restriction $\mathcal{A}|_Y$ of a σ-algebra $\mathcal{A}$ on X to a subset $Y \subset X$, defined by

$$\mathcal{A}|_Y = \{A \cap Y \mid A \in \mathcal{A}\},$$

is again a σ-algebra on Y.

Theorem 42.4.6. We have $\mathcal{A}|_Y \subset \mathcal{A}$ if and only if $Y \in \mathcal{A}$.

Proof.

1. Assume $\mathcal{A}|_Y \subset \mathcal{A}$. We have $X \cap Y = Y \in \mathcal{A}|Y$, and thus we get $Y \in \mathcal{A}$.
2. Assume $Y \in \mathcal{A}$. Then since σ-algebras are closed under intersection, we have $A \cap Y \in \mathcal{A}$ for all $A \in \mathcal{A}$, i.e. we have $\mathcal{A}|_Y \subset \mathcal{A}$.

□

Theorem 42.4.7. Let $X \neq \emptyset$, let $(Y, \mathcal{B})$ be a measurable space and let $f : X \to Y$ be an arbitrary map. Then

$$f^{-1}(\mathcal{B}) = \left\{ f^{-1}(B) \,\middle|\, B \in \mathcal{B} \right\}$$

is a σ-algebra.

Note however that we don't necessarily have $f^{-1}(\mathcal{B}) = \mathcal{A}$, or even $f^{-1}(\mathcal{B}) \subset \mathcal{A}$! We call maps that fulfill the latter property *measurable*. Measurable maps play a key role in measure theory and especially in integration theory, entirely analogous to the role continuous functions play in topology. We will study them in more detail in a later chapter.

Theorem 42.4.8. The intersection of arbitrarily many σ-algebras is on the same set X is once again a σ-algebra.

Corollary 42.4.9: Generated σ-algebra

Let $\mathcal{E} \subset \mathcal{P}(X)$. Then $\langle \mathcal{E} \rangle_\sigma$ denotes the intersection of all σ-algebras containing $\mathcal{E}$.

In particular, note that if $\mathcal{A}$ is a σ-Algebra containing $\mathcal{E}$, we have

$$\langle \mathcal{E} \rangle_\sigma \subseteq \mathcal{A}.$$

Theorem 42.4.10. We have $\langle \mathcal{F} \rangle_\mathcal{B} = \langle \mathcal{F} \rangle_\sigma$ if and only if $\langle \mathcal{F} \rangle_\mathcal{B}$ is a σ-algebra.

Theorem 42.4.11. Let X be any set. Let $\mathcal{E}_i \subset \mathcal{P}(X)$ be a family of sets of subsets of X indexed by $i \in I$. Then we have:

$$\left\langle \bigcup_{i \in I} \langle \mathcal{E}_i \rangle_\sigma \right\rangle_\sigma = \left\langle \bigcup_{i \in I} \mathcal{E}_i \right\rangle_\sigma$$

42.4.1 The Borel σ-Algebra

Definition 42.4.12. Let X be a topological space. The **Borel σ-algebra $\mathcal{B}[X]$ of X** is the σ-Algebra generated by the open subsets of X.

Theorem 42.4.13. The Borel σ-Algebra $\mathcal{B}[\mathbb{R}^d]$ is equivalently generated by any of the following:

1. The closed subsets of $\mathbb{R}^d$,
2. The compact subsets of $\mathbb{R}^d$,
3. The open balls of $\mathbb{R}^d$,
4. The boxes in $\mathbb{R}^d$,
5. The elementary sets in $\mathbb{R}^d$.

42.5 Dynkin Systems

Definition 42.5.1: Dynkin System

Let $\mathcal{D} \subset \mathcal{P}(X)$. Then we call $\mathcal{D}$ a **Dynkin system** iff:

1. $X \in \mathcal{D}$
2. $\mathcal{D}$ is closed under complements $B \setminus A$ as long as $A \subset B$
3. $\mathcal{D}$ is closed under coutable pairwise disjoint union

Theorem 42.5.2

A Dynkin system is a σ-algebra iff. it is closed under intersection.

Theorem 42.5.3. Let $\mathcal{E} \subset \mathcal{P}(X)$ be closed under intersection. Then $\langle \mathcal{E} \rangle_{\mathcal{D}} = \langle \mathcal{E} \rangle_{\sigma}$.

42.6 Monotone Classes

Definition 42.6.1: Monotone classes

Let $\mathcal{M} \subset \mathcal{P}(X)$. Then we call $\mathcal{M}$ a **monotone class** if:

1. For any family $(A_i)_{i\in\mathbb{N}} \subset \mathcal{M}$ such that $A_i \subset A_{i+1}$ holds for all $i \in \mathbb{N}$, we have
$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{M}$$
2. For any family $(A_i)_{i\in\mathbb{N}} \subset \mathcal{M}$ such that $A_i \supset A_{i+1}$ holds for all $i \in \mathbb{N}$, we have
$$\bigcap_{i=1}^{\infty} A_i \in \mathcal{M}$$

We denote the smallest monotone class containing a set $\mathcal{S} \subset \mathcal{P}(X)$ as $\langle \mathcal{S} \rangle_{\mathcal{M}}$.

Corollary 42.6.2. Every σ-algebra is a monotone class.

Corollary 42.6.3. Given $\mathcal{F} \subset \mathcal{P}(X)$, we have $\langle \mathcal{F} \rangle_{\mathcal{M}} \subseteq \langle \mathcal{F} \rangle_{\sigma}$

Recall that a *set algebra* is a set $\mathcal{A} \subset \mathcal{P}(X)$ that contains the empty set and is closed under complements and binary unions.

Theorem 42.6.4. Let $\mathcal{F} \subset \mathcal{P}(X)$ be a set algebra and a monotone class. Then $\mathcal{F}$ is a σ-algebra.

Proof. We need to show that $\mathcal{F}$ is closed under countable unions. Let $(F_j)_{j\in\mathbb{N}} \subset \mathcal{F}$ be a family of elements of $\mathcal{F}$. Since $\mathcal{F}$ is closed under finite union, the increasing sequence of sets
$$A_i := \bigcup_{j=1}^{i} F_j$$
must be contained in $\mathcal{F}$. Therefore, since $\mathcal{F}$ is a monotone class, the union
$$\bigcup_{i\in\mathbb{N}} A_i = \bigcup_{i\in\mathbb{N}} \left(\bigcup_{j=1}^{i} F_j \right) = \bigcup_{j\in\mathbb{N}} F_j$$
must also be contained in $\mathcal{F}$. Therefore, $\mathcal{F}$ is closed under countable union, making it a σ-algebra. □

The relevance of monotone classes is given by the following theorem, which gives us an additional way of verifying whether a given subset of a power set forms a σ-algebra:

Theorem 42.6.5: Monotone class theorem

Let $\mathcal{A} \subset \mathcal{P}(X)$ be a set algebra. Then

$$\langle \mathcal{A} \rangle_{\mathcal{M}} = \langle \mathcal{A} \rangle_{\sigma}$$

Proof. We have already observed that, since any σ-algebra is a monotone class, we have $\langle \mathcal{A} \rangle_{\mathcal{M}} \subset \langle \mathcal{A} \rangle_{\sigma}$. Therefore, it suffices to show that we also have $\langle \mathcal{A} \rangle_{\mathcal{M}} \supset \langle \mathcal{A} \rangle_{\sigma}$. This holds automatically if $\langle \mathcal{A} \rangle_{\mathcal{M}}$ is a σ-algebra, and since we already know that any monotone class is a σ-algebra if it is a set algebra, it suffices to show that $\langle \mathcal{A} \rangle_{\mathcal{M}}$ is a set algebra.

For any $E \in \langle \mathcal{A} \rangle_{\mathcal{M}}$, let $\mathcal{M}_E$ be the system of "good subsets" of X which can be combined with E as desired, i.e.

$$\mathcal{M}_E := \left\{ F \subset X \;\middle|\; \begin{array}{c} F \setminus E \in \langle \mathcal{A} \rangle_{\mathcal{M}} \\ E \setminus F \in \langle \mathcal{A} \rangle_{\mathcal{M}} \\ E \cup F \in \langle \mathcal{A} \rangle_{\mathcal{M}} \end{array} \right\}$$

1. First, we want to show that $\mathcal{M}_E$ is a monotone class.

2. Next, we want to show that for every $E \in \mathcal{A}$, we have $\langle \mathcal{A} \rangle_{\mathcal{M}} \subset \mathcal{M}_E$. Since $\mathcal{A}$ is a set algebra, every element of $\mathcal{A}$ combines with E as desired, giving us $\mathcal{A} \subset \mathcal{M}_E$. Therefore, since $\mathcal{M}_E$ is a monotone class, we have

$$\langle \mathcal{A} \rangle_{\mathcal{M}} \subset \langle \mathcal{M}_E \rangle_{\mathcal{M}} = \mathcal{M}_E.$$

3. Next, we want to show that for every $F \in \langle \mathcal{A} \rangle_{\mathcal{M}}$, we have $\langle \mathcal{A} \rangle_{\mathcal{M}} \subset \mathcal{M}_F$. We just showed that $\langle \mathcal{A} \rangle_{\mathcal{M}} \subset \mathcal{M}_F$, which means we have $F \in \mathcal{M}_E$ for any $E \in \mathcal{A}$. Since the definition of $\mathcal{M}_E$ is entirely symmetric in E and F, we also have $E \in \mathcal{M}_F$. Therefore, since all $E \in \mathcal{A}$ are contained in $\mathcal{M}_F$, we have $\mathcal{A} \subset \mathcal{M}_F$, implying

$$\langle \mathcal{A} \rangle_{\mathcal{M}} \subset \langle \mathcal{M}_F \rangle_{\mathcal{M}} = \mathcal{M}_F.$$

4. We have just shown that, for any $E, F \in \langle \mathcal{A} \rangle_{\mathcal{M}}$, we have $E \in \mathcal{M}_F$, implying that $\langle \mathcal{A} \rangle_{\mathcal{M}}$ is closed under complementation and binary union. Since $X \in \mathcal{A}$, we also have $X \in \langle \mathcal{A} \rangle_{\mathcal{M}}$. Therefore, $\langle \mathcal{A} \rangle_{\mathcal{M}}$ is a set algebra.

Therefore, $\langle \mathcal{A} \rangle_{\mathcal{M}}$ is indeed both a set algebra and a monotone class, making it a σ algebra. □

42.7 Measures

With our different subset systems in place, we can finally give a general formal definition of a measure. Along the way, we will encounter outer measures, contents, and premeasures, which are "weakened" measures that we can use to generate proper ones.

Definition 42.7.1: Outer measure

Let X be a set and μ be a function $\mathcal{P}(X) \to [0, \infty]$. We call μ an **outer measure on X** if it is σ-subadditive.

Theorem 42.7.2. Let X be any set. Then the constant function $\mu = 0$ is an outer measure on X, which is known as the **trivial measure**.

Theorem 42.7.3. Let X be any set. Let $A \subset X$ and $x \in X$. Then the function

$$\delta_x(A) := \begin{cases} 1 & x \in A \\ 0 & x \in X \setminus A \end{cases}$$

is an outer measure on X, which is known as the **Dirac measure**.

Theorem 42.7.4: Counting measure

Let X be any set. Let $A \subset X$. Then the function

$$\operatorname{card}(A) := \begin{cases} |A| & |A| \text{ is finite} \\ \infty & |A| \text{ is infinite} \end{cases}$$

is an outer measure on X, which is known as the **counting measure**.

Definition 42.7.5: Measurable set

Let μ be an outer measure on a set X. Then we call a subset $A \subset X$ **μ-measurable**, or just **measurable**, if for all $S \subseteq X$ we have

$$\mu(S) = \mu(S \cap A) + \mu(S \setminus A)$$

The system of all μ-measurable sets is sometimes denoted $\mathcal{M}(\mu)$.

By this definition, μ-measurable sets are exactly the sets for which μ is σ-additive. This means that if we restrict μ to its measurable sets, we get a proper measure.

Note that by the subadditivity of outer measures, we already get that the left side is at most as large as the right side, meaning that this condition can equivalently be weakened to

$$\mu(S) \geq \mu(S \cap A) + \mu(S \setminus A).$$

Theorem 42.7.6: Importance of σ-algebras

Let μ be an outer measure. Then the set of μ-measurable sets forms a σ-Algebra.

Since the restriction of an outer measure to its measurable sets forms a proper measure, this theorem tells us that σ-algebras give us the most sensible definition for which properties the domain of a measure should fulfill. Therefore, we finally get a proper definition of a measure:

Definition 42.7.7: Measure

Let $\mathcal{A}$ be a σ-Algebra. Then we call a σ-additive function $\mu : \mathcal{A} \to [0, \infty]$ a **measure**.

Definition 42.7.8. If $\mathcal{A}$ is a σ-algebra on X, and μ is a measure on $\mathcal{A}$, we call the triple $(X, \mathcal{A}, \mu)$ a **measure space**.

Definition 42.7.9. Let $(X, \mathcal{A}, \mu)$ be a meausure space. Then we call this space a **probability space**, and μ a **probability measure**, iff $\mu(X) = 1$.

Definition 42.7.10: Content, Premeasure

Let $\mathcal{S}$ be a set semiring. Then we call a finitely additive function $\mathcal{S} \to [0, \infty]$ a **content**, and a σ-additive function $\mathcal{S} \to [0, \infty]$ a **premeasure**.

In effect, a premeasure is a measure whose domain might not be as big as it could be. Every measure is trivially also a content and a premeasure, and every measure defined on $\mathcal{P}(X)$ is an outer measure.

Theorem 42.7.11: Restriction of a measure

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $Y \in \mathcal{A}$. Then if we define

$$\mu|_Y(A \cap M) := \mu(A \cap M),$$

$\mu|_Y$ defines a measure on $\mathcal{A}|_Y$.¸

Theorem 42.7.12: Image measure

Let X, Y be sets. Let $f : X \to Y$ be a function. Let μ be an outer measure on X. then the function

$$\begin{aligned} f(\mu) : \mathcal{P}(X) &\to [0, \infty] \\ f(\mu)(B) &= \mu\big(f^{-1}(B)\big) \end{aligned}$$

is a measure on Y, which is known as the **image measure of μ under f**.

Proof. 1. First, note that we have:

$$\begin{aligned} f(\mu)(\emptyset) &= \mu\big(f^{-1}(\emptyset)\big) \\ &= \mu(\emptyset) \\ &= 0 \end{aligned}$$

2. Furthermore, for $B \subset \bigcup_{i \in \mathbb{N}} B_i$, we have $f^{-1}(B) \subset \bigcup_{i \in \mathbb{N}} B_i$. By the

subadditivity of μ, we get:

$$\begin{aligned} f(\mu)(B) &= \mu(f^{-1}(B)) \\ &\leq \sum_{i\in\mathbb{N}} \mu(f^{-1}(B_i)) \\ &= \sum_{i\in\mathbb{N}} f(\mu)(B_i). \end{aligned}$$

Therefore, $f(\mu)$ is also subadditive and thus an outer measure.

□

Lemma 42.7.13. Let X, Y be sets. Let $f : X \to Y$ be a function. Let μ be an outer measure on X and let $f(\mu)$ be the corresponding image measure. Then $B \subset Y$ is $f(\mu)$ measurable if $f^{-1}(B)$ is μ-measurable.

Proof. By the definition of measurability:

1. $B \subset Y$ is $f(\mu)$-measurable if and only if for all $T \subset Y$, we have:

   $$f(\mu)(T) \geq f(\mu)(T \cap B) + f(\mu)(T \setminus B)$$

   i.e.

   $$\mu\big(f^{-1}(T)\big) \geq \mu\big(f^{-1}(T \cap B)\big) + \mu(f^{-1}(T \setminus B))$$

   Since $f^{-1}(T\cap B) = f^{-1}(T)\cap f^{-1}(B)$ and $f^{-1}(T\setminus B) = f^{-1}(T)\setminus f^{-1}(B)$, this is equivalent to:

   $$\mu\big(f^{-1}(T)\big) \geq \mu(f^{-1}(T) \cap f^{-1}(B)) + \mu(f^{-1}(T) \setminus f^{-1}(B))$$

2. $f^{-1}(B)$ is μ-measurable if and only if for all $S \subset X$, we have:

   $$\mu(S) \geq \mu(S \cap f^{-1}(B)) + \mu(S \setminus f^{-1}(B)).$$

   In particular, if $f^{-1}(B)$ is μ-measurable, this equation must hold for $S = f^{-1}(T)$, implying that B is $f(\mu)$-measurable.

□

Definition 42.7.14: (σ)-finite measure

Let $(X, \mathcal{A}, \mu)$ be a measure space. Then:

1. We call μ **finite** if $\mu(X) < \infty$.
2. We call μ **σ-finite** if X can be written as a countable union of sets of finite measure.

Theorem 42.7.15: Measures are σ-subadditive

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $(A_i)_{i\in\mathbb{N}} \subset \mathcal{A}$ be a family of (not necessarily disjoint!) measurable subsets of X. Then we have:

$$\mu\left(\bigcup_{i\in\mathbb{N}} A_i\right) \leq \sum_{i\in\mathbb{N}} \mu(A_i)$$

Theorem 42.7.16: Continuity from above and below

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $(A_i)_{i\in\mathbb{N}} \subset \mathcal{A}$. Then we have:

1. **Continuity from below:** If $A_i \subset A_{i+1}$ for every $i \in \mathbb{N}$, we have
$$\lim_{i\to\infty} \mu(A_i) = \mu\left(\bigcup_{i\in\mathbb{N}} A_i\right)$$
2. **Continuity from above:** If $\mu(A_k) < \infty$ for any $k \in \mathbb{N}$ and $A_i \supset A_{i+1}$ for every $i \in \mathbb{N}$, we have
$$\lim_{i\to\infty} \mu(A_i) = \mu\left(\bigcap_{i\in\mathbb{N}} A_i\right)$$

Definition 42.7.17. Let $(X, \mathcal{A}, \mu)$. We call a set $A \in \mathcal{A}$ a **μ-zero set**, or simply zero set, iff $\mu(A) = 0$. We denote the set of all μ-zero sets by $\mathcal{N}(\mu)$.

Definition 42.7.18: μ-almost everywhere

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let P be some property. We say that P holds **μ-almost everywhere on M**, or that P holds for **μ-almost all $x \in M$**, iff the set of $x \in M$ such that $P(x)$ is false is a zero set.

Definition 42.7.19: Complete measure

We call a measure μ **complete** if any subset of a μ-zero set is also a μ-zero set.

Note that, in particular, any subset of a μ-zero set needs to be μ-measurable in the first place.

Lemma 42.7.20. Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $\mathcal{Z}_\mu$ be the system of all subsets of X that are also subsets of a μ-zero set. Then μ is complete if and only if $\mathcal{Z}_\mu \subset \mathcal{A}$.

Corollary 42.7.21. Let μ be an outer measure. Then $\mu|_{\mathcal{M}(\mu)}$ is complete.

Theorem 42.7.22: Completion of a Measure

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let

$$\overline{\mathcal{A}} = \left\{ A \cup N \;\middle|\; A \in \mathcal{A}, N \in \mathcal{Z}_\mu \right\}.$$

and

$$\overline{\mu}(A \cup N) := \mu(A).$$

Then $\overline{\mathcal{A}}_\mu$ is a σ-algebra and $\overline{\mu}$ is a complete measure on $\overline{\mathcal{A}}_\mu$.

Theorem 42.7.23. (Uniqueness of Completion): Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $(X, \mathcal{B}, \nu)$ be a complete measure space such that $\mathcal{A} \subset \mathcal{B}$ and $\mu(A) = \nu(A)$ for all $A \in \mathcal{A}$. Then we have $\overline{\mathcal{A}}_\mu \subset \mathcal{B}$ and $\overline{\mu} = \nu$ on $\overline{\mathcal{A}}_\mu$.

42.8 Carathéodory Extension

We can construct outer measures from a very large class of functions by the following construction:

Theorem 42.8.1: Carathéodory Extension

Let $\mathcal{S}$ be a system of subsets of a set X containing the empty set. Let $\lambda : \mathcal{S} \to [0, \infty]$ be a function such that $\lambda(\emptyset) = 0$. Then the function

$$\mu(E) := \inf \left\{ \sum_{i=1}^{\infty} \lambda(P_i) \,\middle|\, P_i \in \mathcal{S}, E \subset \bigcup_{i=1}^{\infty} P_i \right\}$$

is an outer measure on X.

Definition 42.8.2: Regular Measure

Let μ be an outer measure on X. Then we call μ **regular** iff. for every $M \subset X$, we can find a μ-measurable set $D \supset M$ such that $\mu(M) = \mu(D)$.

Lemma 42.8.3. The Carathéodory extension of a premeasure is regular.

Theorem 42.8.4: Characterization of measurable sets

Let $\lambda : \mathbb{R} \to [0, \infty]$ be a σ-finite premeasure on a ring $\mathcal{R} \subset \mathcal{P}(X)$. Let $\overline{\lambda}$ be its Carathéodory extension. Then a set $D \subset X$ is μ measurable if and only if either of the following hold:

1. There exists an $E \supset D$ such that $E \in \langle \mathcal{R} \rangle_\sigma$ and $\overline{\lambda}(E \setminus D) = 0$.
2. There exists a $C \subset D$ such that $C \in \langle \mathcal{R} \rangle_\sigma$ and $\overline{\lambda}(D \setminus C) = 0$.

42.9 Measures on $\mathbb{R}^n$

42.9.1 Jordan Content

Definition 42.9.1. Let $E \subset \mathbb{R}^n$ be an elementary set. Then we can assign to it the **elementary volume** $\mathrm{vol}(E)$, where the volume of a cuboid is the product of its side lengths and the volume of a finite union of cuboids is the sum of the volumes of the individual cuboids making up E.

Definition 42.9.2. (Inner and outer Jordan content): Let $E \subset \mathbb{R}^n$.

1. The **inner Jordan measure** $J_*(E)$ is

$$J_*(E) := \sup_{\substack{Q_i \in \boldsymbol{Q}, \\ \bigcup_{i=1}^n Q_i \subset E}} \mathrm{vol}(Q)$$

2. The **outer Jordan measure** $J_*(E)$ is

$$J^*(E) := \inf_{\substack{Q_i \in \boldsymbol{Q}, \\ \bigcup_{i=1}^n Q_i \supset E}} \mathrm{vol}(Q)$$

Definition 42.9.3. (Jordan measurable set, Jordan content): We call E **Jordan-measurable** if $J_*(E) = J^*(E)$. Then we call $J(E) = J_*(E) = J^*(E)$ the **Jordan content**.

Theorem 42.9.4. The Jordan content is σ-additive.

Theorem 42.9.5. The following are equivalent:

1. E is Jordan measurable,
2. For every $\varepsilon > 0$, there exist elementary sets $A \subset E \subset B$ such that $\mathrm{vol}(B \setminus A) \leq \varepsilon$,
3. For every $\varepsilon > 0$, there exists an elementary set A such that $J^*(A \vartriangle E) \leq \varepsilon$.

Theorem 42.9.6. The collection of subsets of $\mathbb{R}^n$ that are either Jordan measurable or have a Jordan-measurable complement form a Boolean algebra, known as the Jordan algebra.

Theorem 42.9.7. The Jordan algebra is non-atomic.

Theorem 42.9.8. (Regions under continuous Graphs are Jordan measurable): Let B be a closed box in $\mathbb{R}^n$, and let $f : B \to \mathbb{R}$ be a continuous function. Then the set

$$\left\{(x,t) \,\middle|\, x \in B, 0 \leq t \leq f(x)\right\} \subset \mathbb{R}^{n+1}$$

is Jordan measurable.

Theorem 42.9.9. Triangles are Jordan measurable.

Theorem 42.9.10. Convex polytopes in $\mathbb{R}^n$ are Jordan measurable.

Theorem 42.9.11. Open and closed Euclidean balls are Jordan measurable.

Theorem 42.9.12. Every subset of a Jordan null set is Jordan measurable and also a Jordan null set.

Theorem 42.9.13. The sets $[0,1]^2 \setminus \mathbb{Q}^2$ and $[0,1]^2 \cap \mathbb{Q}^2$ are both not Jordan measurable.

Informally, sets with a lot of "holes" or with very messy, fractal-like boundaries are generally not Jordan-measurable.

Theorem 42.9.14. There exist countable unions, and countable intersections, of Jordan measurable sets which are not Jordan measurable.

42.9.2 Lebesgue Measure

One can extend the Jordan measure to a significantly larger number of subsets of $\mathbb{R}^n$ by simply allowing countable unions of cuboids (instead of finite unions of cuboids). The Lebesgue measure is simply this generalization of the Jordan measure. Formally:

Definition 42.9.15. Let $E \subset \mathbb{R}^n$. The **Lebesgue outer measure** of E is given by

$$\lambda^*(E) := \inf_{\substack{E \subset \bigcup_{i=1}^{\infty} Q_n, \\ Q_n \in \mathcal{Q}}} \left\{ \sum_{n=1}^{\infty} \operatorname{vol}^n Q_n \right\},$$

i.e. the Lebesgue outer measure of E is the greatest lower bound of the measures of all coverings of E by cuboids.

Theorem 42.9.16. We have $\lambda^0 = \operatorname{card}$ - the zero-dimensional Lebesgue measure is the counting measure.

Proof. $\mathbb{R}^0$ is by definition cartesian product of $\mathbb{R}$ with a set of cardinality 0 i.e. the empty set, as its index set:

$$\mathbb{R}^0 = \prod_{\emptyset} \mathbb{R}$$

By definition of cartesian products, its members must be functions $\emptyset \to \mathbb{R}$. There is exactly one such "empty function", which we will denote $f_\emptyset$. Thus, the only subsets of $\mathbb{R}^0$ are the empty set and $\{f_\emptyset\}$.

1. Since λ^0 and card are measures, we have

$$\lambda^0(\emptyset) = 0 = \operatorname{card}(\emptyset)$$

2. By definition, a 0-dimensional cuboid must be a function $\emptyset \to I$ to an interval $I \subset \mathbb{R}$. Since all functions from the empty set to any set are identical, $f_\emptyset$ qualifies as a cuboid. We thus have:

$$\begin{aligned} \lambda^0(\{f_\emptyset\}) &= \operatorname{vol}^0(\{f_\emptyset\}) \\ &= \prod_{\emptyset} \operatorname{vol}(I) \\ &= 1 \\ &= \operatorname{card}(\{f_\emptyset\}) \end{aligned}$$

Where the last step follows since empty products evaluate to the multiplicative identity, in this case the number 1.

□

Theorem 42.9.17. Let $U \subset \mathbb{R}^n$ be open. Then U is a countable union of cuboids.

Therefore, we can define Lebesgue measurability similarly to Jordan measurability - a set is Lebesgue measurable if it is "almost" an open set.

Definition 42.9.18. A set $E \subset \mathbb{R}^n$ is **Lebesgue measurable** if for every $\varepsilon > 0$, there exists an open set $U \subset \mathbb{R}^n$ such that $U \subset E$ and $\lambda^*(U \setminus E) \leq \varepsilon$. If E is Lebesgue measurable, we refer to $\lambda(E) := \lambda^*(E)$ as the **Lebesgue measure** of E. If the dimension n should be emphasized, we sometimes write $\lambda(E)$ as $\lambda^n(E)$.

Theorem 42.9.19. The Lebesgue measure fulfills:

$$E \subset \bigcup_{i \in N} E_i \implies \lambda(E) \leq \sum_{i \in \mathbb{N}} \lambda(E_i)$$

This is a weakened form of σ-additivity, which is known as σ-subadditivity.

Theorem 42.9.20. The Lebesgue measure defines a σ-additive function on the Lebesgue measurable sets.

Theorem 42.9.21. The collection of subsets of $\mathbb{R}^n$ that are either Lebesgue measurable or have a Lebesgue-measurable complement form a Boolean algebra.

Theorem 42.9.22. There exist Lebesgue measurable sets which are not Borel.

You may have heard in linear algebra that the determinant of a matrix S tells us how the matrix scales the volume of the unit cube. Since the Lebesgue measure is defined using volumes of unit cubes, this intuition also holds for the image of any set E under S:

Theorem 42.9.23: Linear Transformation Equation

Let $S \in \mathbb{R}^{n \times n}$. Then for all $E \subset \mathbb{R}^n$, we have:

$$\lambda^n(S(E)) = |\det(S)| \cdot \lambda^n(E)$$

Definition 42.9.24: Smith-Volterra-Cantor Set

The Smith-Volterra-Cantor set is the set obtained by removing the middle $1/8$ of the interval $[0, 1]$, and then iteratively removing the middle $1/8$ of any remaining intervals.

Theorem 42.9.25: Compact does not imply Jordan measurable

The Smith-Volterra-Cantor set is compact, but not Jordan measurable.

Chapter 43

Measurable Functions

Definition 43.0.1: Measurable Functions

Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces. Then we call a map $f : X \to Y$ **$\mathcal{A}$-$\mathcal{B}$-measurable**, or just **measurable**, if the preimage of every measurable set is again measurable, i.e.

$$B \in \mathcal{B} \implies f^{-1}(B) \in \mathcal{A}$$

Once again, note the similarities between this definition and the topological definition of a continuous function.

If the σ-algebra on one of the sets is supposed to be clear from context, many authors only specify one of the the two sigma algebras. For example, for a function $f : X \to \overline{\mathbb{R}}$, many authors talk about $\mathcal{A}$-measurability when they implicitly mean $\mathcal{A} - \mathcal{B}(\overline{\mathbb{R}})$-measurability.

Corollary 43.0.2

The composition of two measurable functions is again measurable.

Lemma 43.0.3. Let $(X, \mathcal{A})$, $(Y, \mathcal{B})$ be measurable spaces and $f : X \to Y$. Let $\mathcal{E} \subset \mathcal{B}$. Then

$$f^{-1}(\langle \mathcal{E} \rangle_\sigma) = \left\langle f^{-1}(\mathcal{E}) \right\rangle_\sigma$$

Corollary 43.0.4

Let $\mathcal{E}$ be a base of $\mathcal{B}$, i.e. $\langle \mathcal{E} \rangle_\sigma = \mathcal{B}$. Then $f : X \to Y$ is $\mathcal{A}$-$\mathcal{B}$ measurable if and only if

$$E \in \mathcal{E} \implies f^{-1}(E) \in \mathcal{A},$$

This means we don't need to check the preimage of every single set in $\mathcal{B}$ to show that f is measurable - it suffices to check a base.

Corollary 43.0.5. Every continuous function between topological spaces is measurable in the corresponding Borel-σ-algebras.

Proof. Let $f : X \to Y$ be continuous. Then the preimage of every open set of Y is open in X, i.e. contained in the Borel σ-algebra on X, and the open sets of Y form a base of the Borel σ-algebra on Y. □

Theorem 43.0.6: Simple criteria for measurability

Let $(X, \mathcal{A})$ be a measureable space. Let $f : X \to \overline{\mathbb{R}}$. Then the following are equivalent:

1. f is $\mathcal{A}$-$\mathcal{B}(\overline{\mathbb{R}})$ measurable,
2. $\forall c \in \mathbb{R} : \{f > c\} \in \mathcal{A}$,
3. $\forall c \in \mathbb{R} : \{f \geq c\} \in \mathcal{A}$,
4. $\forall c \in \mathbb{R} : \{f < c\} \in \mathcal{A}$,
5. $\forall c \in \mathbb{R} : \{f \leq c\} \in \mathcal{A}$,

Proof.

1. The latter four conditions are equivalent to each other, since:
   (a) $\{f \geq c\} = \bigcap_{k \in \mathbb{N}} \{f > c - \frac{1}{k}\}$,
   (b) $\{f > c\} = \bigcup_{k \in \mathbb{N}} \{f \geq c + \frac{1}{k}\}$,
   (c) $\{f < c\} = X \setminus \{f \geq c\}$,
   (d) $\{f \leq c\} = X \setminus \{f > c\}$.
2. The intervals $[c, \infty]$ form a base of $\mathcal{B}(\overline{\mathbb{R}})$, since:
   (a) $\{\infty\} = \bigcap_{k \in \mathbb{N}} [k, \infty]$
   (b) $\{-\infty\} = \bigcap_{k \in \mathbb{N}} [-\infty, -k]$
   (c) $(a, b) = [a, \infty] \setminus ([b, \infty] \cap [-\infty, a])$

□

Theorem 43.0.7

Let $f, g : X \to \overline{\mathbb{R}}$ be $\mathcal{A}$-$\mathcal{B}(\overline{\mathbb{R}})$ measurable. Then the following sets are contained in $\mathcal{A}$:

1. $\{f > g\}$,
2. $\{f \geq g\}$,
3. $\{f = g\}$,
4. $\{f \neq g\}$.

Theorem 43.0.8. Let $f, g : X \to \overline{\mathbb{R}}$ be $\mathcal{A}$-$\mathcal{B}[\overline{\mathbb{R}}]$ measurable. Then the following functions are also $\mathcal{A}$-measurable:

1. cf, for all $c \in \mathbb{R}$,
2. $|f|^p$, for all $p \in \mathbb{R}_{>0}$,
3. $f + g$, assuming the sum is defined everywhere on X, i.e. there exists no $x \in X$ such that $f(x) = \infty$ and $g(x) = -\infty$ or vice versa,
4. $f \cdot g$.

Theorem 43.0.9. Let $(X, \mathcal{A})$ be a measurable space and $\mathbb{1}_E : X \to \mathbb{R}$ be the indicator function of a set $E \subset X$. Then $\mathbb{1}_E$ is $\mathcal{A}$-$\mathcal{B}[\mathbb{R}]$ measurable if and only if $E \in \mathcal{A}$.

Proof. We have $\{1\} = (-\infty, 1) \cup (1, \infty) \in \mathcal{B}[\mathbb{R}]$ and $\mathbb{1}_E^{-1}(\{1\}) = E$, so E has to be in $\mathcal{A}$ for $\mathbb{1}_E$ to be measurable.

$E \in \mathcal{A}$ is also a sufficient condition for $\mathbb{1}_E$ to be measurable, since the only other possible preimages are $\emptyset \in \mathcal{A}$, $X \in \mathcal{A}$, $X \setminus E \in \mathcal{A}$ □

Theorem 43.0.10. Let $(X, \mathcal{A})$ be a measurable space, let $D \in \mathcal{A}$, and let $f_k : D \to \mathbb{R}$ be $\mathcal{A}$ measurable. Then the following functions are $\mathcal{A}$-measurable:

1. $\inf_{n\in\mathbb{N}} f_n$
2. $\sup_{n\in\mathbb{N}} f_n$
3. $\liminf_{n\to\infty} f_n$
4. $\limsup_{n\to\infty} f_n$

Proof. For $s \in \mathbb{R}$, we have:

1. $\left\{(\inf_{n\in\mathbb{N}} f_n) \geq s\right\} = \bigcap_{k=1}^{\infty} \{f_k \geq s\} \in \mathcal{A}$
2. $\left\{(\sup_{n\in\mathbb{N}} f_n) \leq s\right\} = \bigcap_{k=1}^{\infty} \{f_k \leq s\} \in \mathcal{A}$

Therefore $\inf_{n\in\mathbb{N}} f_n$ and $\sup_{n\in\mathbb{N}} f_n$ are measurable. The same argument holds for $\sup$ and $\inf$ over subsets of $\mathbb{N}$. Therefore, the compositions

$$\liminf_{n\to\infty} f_n = \sup_{n\in\mathbb{N}} \left(\inf_{k\geq n} f_k\right)$$

and

$$\limsup_{n\to\infty} f_n = \inf_{n\in\mathbb{N}} \left(\sup_{k\geq n} f_k\right)$$

are also measurable. □

Corollary 43.0.11

Let f_n be a sequence of $\mathcal{A}$-measurable functions with a pointwise limit f. Then f is $\mathcal{A}$-measurable.

Definition 43.0.12: μ-measurable

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $D \in \mathcal{A}$. Then we call a function $f : D \to \overline{\mathbb{R}}$ μ-measurable iff:

1. $X \setminus D$ is a μ-zero set, and
2. f is $\mathcal{A}|_D$-measurable.

Lemma 43.0.13. Let $f : D \to \overline{\mathbb{R}}$ be μ-measurable. Then the function

$$g(x) = \begin{cases} f(x) & x \in D \\ 0 & x \notin D \end{cases}$$

is $\mathcal{A}$-measurable.

Lemma 43.0.14

Let $(X, \mathcal{A}, \mu)$ be a **complete** measure space. Let f be μ-measurable. Then every function g such that $g = f$ μ-almost everywhere is μ-measurable.

Theorem 43.0.15: μ-measurable pointwise limits

Let $(X, \mathcal{A}, \mu)$ be a **complete** measure space. Let $(f_k)_{k \in \mathbb{N}}$ be a family of μ-measurable functions. Then if f_k converges pointwise to a function f μ-almost everywhere, f is μ-measurable.

Chapter 44

Lebesgue Integration

44.1 Step Functions

Definition 44.1.1. Let $(X, \mathcal{A})$ be a measurable space. Then we call a function $f : Y \to \mathbb{R}$ a **step function** if it can be represented as a finite linear combination of indicator functions of sets in $\mathcal{A}$, i.e there exist $\alpha_i \in \mathbb{R}$ and $A_i \in \mathcal{A}$ such that:

$$f = \sum_{i \leq k} \alpha_i \cdot \mathbb{1}_{A_i}$$

Proposition 44.1.2. Step functions form a vector space over $\mathbb{R}$.

Proposition 44.1.3. Step functions can only take finitely different values.

Proposition 44.1.4. Every step function is (extensionally) equal to a step function over pairwise disjoint sets.

Proof. Let

$$s = \sum_{i \leq k} \alpha_i \cdot \mathbb{1}_{A_i}$$

Then if $\beta_1, \ldots, \beta_m$ are all of the finitely many different values that s can take, s is by definition equal to

$$t := \sum_{i \leq m} \beta_i \cdot \mathbb{1}_{\{s=\beta_i\}}$$

And since a function cannot take two values at once, the sets $\{s = \beta_i\}$ are disjoint. □

Lemma 44.1.5. (Sum of Step Functions): Let s_1 and s_2 be step functions

$$s_1 = \sum_{i=0}^{m} \alpha_i \mathbb{1}_{A_i}, \qquad s_2 = \sum_{j=0}^{n} \beta_j \mathbb{1}_{B_j}$$

defined on pairwise disjoint sets. Then we have:

$$s_1 + s_2 = \sum_{i=0}^{m} \sum_{j=0}^{n} (\alpha_i + \beta_j) \mathbb{1}_{A_i \cap B_j}$$

Proof. 1. Assume s_1 and s_2 are nonnegative step functions

$$s_1 = \sum_{i=0}^{m} \alpha_i \cdot \mathbb{1}_{A_i}, \qquad s_2 = \sum_{j=0}^{n} \beta_j \cdot \mathbb{1}_{B_j}$$

defined on pairwise disjoint sets.

Then:

$$s_1(x) + s_2(x) = \sum_{i=0}^{m} \alpha_i \cdot \mathbb{1}_{A_i}(x) + \sum_{j=0}^{n} \beta_j \cdot \mathbb{1}_{B_j}(x)$$

Since the A_i and B_j are disjoint coverings of X, for every $x \in X$, there exists exactly one i such that $x \in A_i$ and exactly one j such that $x \in B_j$. Therefore, every x contributes exactly one α_i for $x \in A_i$ and one β_j for $x \in B_j$, i.e. $x \in A_i \cap B_j$ and $s_1(x) + s_2(x) = \alpha_i + \beta_j$. Since this is the only intersection which contains x, we can put everything together to get our desired formula:

$$s_1(x) + s_2(x) = \sum_{i=0}^{m} \sum_{j=0}^{n} (\alpha_i + \beta_j) \cdot \mathbb{1}_{A_i \cap B_j}(x)$$

□

The integral of an indicator function is already clear from our intuition: The points contained in the area under an indicator function $\mathbb{1}_A$ are exactly the cartesian product of A with the interval $[0, 1]$, forming a "rectangle with gaps" whose side lengths are 1 and $\mu(A)$. Therefore the integral should be:

Definition 44.1.6. (Lebesgue integral of an indicator function):

$$\int_X \mathbb{1}_A \, d\mu = \mu(A)$$

The definition for the integral of a step function should follow naturally - a step function is just a finite sum of scaled characteristic functions, therefore the integral of a step function is the sum of the scaled integrals of the step functions.

Definition 44.1.7. (Lebesgue integral of a step function): Let $D \subset X$, $A_i \subset X$ pairwise disjoint, and $\alpha_i \geq 0$. Then:

$$\begin{aligned} \int_D s \, d\mu &= \int_D \left(\sum_{i \leq k} \alpha_i \cdot \mathbb{1}_{A_i} \right) d\mu \\ &:= \sum_{i \leq k} \left(\alpha_i \cdot \int_D \mathbb{1}_{A_i} \, d\mu \right) \\ &= \sum_{i \leq k} \left(\alpha_i \cdot \mu(D \cap A_i) \right) \end{aligned}$$

Theorem 44.1.8. (Linearity of the step function integral):

Theorem 44.1.9. (Monotonicity of the step function integral):

44.2 Defining the Lebesgue Integral

We can now use the step function integral to define the integral of more general functions:

Theorem 44.2.1: Approximation via step functions

Let $(X, \mathcal{A})$ be a measurable space. Let $f : X \to \overline{\mathbb{R}}$ be a nonnegative $\mathcal{A}$-measurable function. Then there exists a monotonically increasing sequence s_n of nonnegative step functions whose pointwise limit is f.

Construction 1. For $n \in \mathbb{N}$ and $k \in \{0, \dots, n \cdot 2^n\}$, we set:

$$F_{n,k} := \left\{ x \in X \;\middle|\; \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n} \right\}.$$

Then

$$s_n(x) = \begin{cases} \frac{k}{2^n} & x \in F_{n,k} \\ n & \text{otherwise} \end{cases}$$

are step functions converging to f. □

Construction 2. Define $s_0 = 0$. Then we can inductively define

$$E_n := \left\{ s_{n-1} + \frac{1}{n} \leq f \right\}$$

and

$$s_n := s_{n-1} + \frac{1}{n} \cdot \mathbb{1}_{E_n},$$

meaning we have

$$s_n = \sum_{k=1}^{n} \frac{1}{k} \cdot \mathbb{1}_{E_i}.$$

We now need to show that this series converges pointwise to f. Let $n \in \mathbb{N}$.

1. Assume $x \in E_n$. Then, by definition,

$$f_n(x) = f_{n-1} + \frac{1}{n} \leq f(x)$$

2. Assume $x \notin E_n$. Then, by induction, we have

$$f_n(x) = f_0(x) = 0 \leq f(x).$$

Therefore, we have $f_0 \leq f_1 \leq \dots$ and $f_n \leq f$ for all n, meaning that

$$\lim_{n \to \infty} f_n(x) \leq f(x).$$

If $f(x) = \infty$, then $x \in E_n$ for every n, and we have

$$\lim_{n\to\infty} f_n(x) = \sum_{k=1}^{\infty} \frac{1}{k} = \infty$$

If $f(x) < \infty$, then there must be an infinite number of $n \in \mathbb{N}$ such that $f_{n-1}(x) > f_n(x) - \frac{1}{n}$, implying

$$\lim_{n\to\infty} f_n(x) \geq f(x)$$

□

Definition 44.2.2: Lebesgue integral of a positive function

Let $f : D \to [0, \infty]$ be $\mathcal{A}$-measurable. Then:

$$\int_D f \, d\mu = \sup_{\substack{s \text{ is a step function,} \\ 0 \leq s \leq f}} \left\{ \int_D s \, d\mu \right\}$$

Definition 44.2.3. (Lebesgue integral of an arbitrary measurable function): Let f be $\mathcal{A}$-mesurable and

$$f^+ = f \cdot \mathbb{1}_{f \geq 0}, \qquad f^- = -f \cdot \mathbb{1}_{f<0}$$

Note that $f^+ \geq 0$ and $f^- > 0$. Then, as long as f^+ or f^- have a finite integral, we can define:

$$\int_D f \, d\mu \int_D f^+ \, d\mu - \int_D f^- \, d\mu$$

Corollary 44.2.4. (Integrating over subsets):

$$\int_M f \, d\mu = \int_X f \cdot \mathbb{1}_M \, d\mu = \int_M f \, d\mu|_M$$

It is common to derive from this corollary a slight abuse of notation: Assume that f is not $\mathcal{A}$-measurable, but it is μ-measurable, i.e. $f : D \to \overline{\mathbb{R}}$ such that $\mu(X \setminus D) = 0$ and f is $\mathcal{A}|_D$-measurable. Then it is common to implicitly expand the domain of D to X by setting $f(x) = 0$ on $X \setminus D$, and to therefore write:

$$\int_X f \, d\mu := \int_D f \, d\mu$$

Corollary 44.2.5. (Integrating over zero sets): Let N be a set such that $\mu(N) = 0$. Then

$$\int_N f \, d\mu = 0.$$

Definition 44.2.6: Integrable function

We call a function $f : X \to \overline{\mathbb{R}}$ **integrable** with regards to a measure μ if it is μ-measurable and

$$\int_X f \, d\mu \in \mathbb{R}$$

Proposition 44.2.7. (Integrating with the counting measure): Let X be an arbitrary set. Let card be the counting measure on $\mathcal{P}(X)$. Let $f : X \to \overline{\mathbb{R}}$. Then f is integrable with respect to card if and only if $\sum_{x\in X} f(x)$ is absolutely convergent, and we have

$$\int_X f \, d\,\mathrm{card} = \sum_{x\in X} f(x)$$

Theorem 44.2.8: Monotonicity of the Lebesgue integral

Let μ be an outer measure on X such that $f, g : X \to \overline{\mathbb{R}}$ are μ-measurable. Then if $f \leq g$ μ-almost everywhere and $\int_X f \, d\mu > -\infty$, then the integral of g exists and we have

$$\int_X f \, d\mu \leq \int_X g \, d\mu$$

Proof. 1. Assume f, g are nonnegative. Then if s is a step function such that $s \leq f$. Then if we define $t := \mathbb{1}_{f\leq g} \cdot s$, we have $t \leq g$. Furthermore, for all $c \geq 0$, we have:

$$\mu(\{s = c\}) = \mu(\{s = c\}\cap)$$

2.

□

Theorem 44.2.9

Let $f, g : X \to \overline{\mathbb{R}}$ and let f be μ-measurable. Then if $g = f$ μ-almost everywhere, then g is μ-measurable, and

$$\int_X g \, d\mu = \int_X f \, d\mu$$

as long as the right integral exists.

Lemma 44.2.10. (Chebyshev Inequality): Let $f : X \to [0, \infty]$ be μ-measurable with $\int_X f \, d\mu < \infty$. Let $s \in (0, \infty]$. Then:

$$\mu(\{x : f(x) \geq s\}) \leq \begin{cases} \frac{1}{s}\int_X f \, d\mu & s \in (0, \infty) \\ 0 & s = \infty \end{cases}$$

Corollary 44.2.11. Let $f : X \to \overline{\mathbb{R}}$ be μ-measurable. Then:

1. If $\int_X f \, d\mu < \infty$, $\{x : f(x) = \infty\}$ is a μ-zero set.
2. If $f \geq 0$ and $\int_X f \, d\mu = 0$, $\{x : f(x) > 0\}$ is a μ-zero set.

Theorem 44.2.12: Monotone Convergence Theorem

Let $f_n : X \to [0, \infty]$ be μ-measurable such that $f_i \leq f_{i+1}$. Let $\lim_{n\to\infty} f_n = f$. Then

$$\int_X \lim_{n\to\infty} f_n \, d\mu = \lim_{n\to\infty} \int_X f_n \, d\mu = \sup_{n\in\mathbb{N}} \int_X f_n \, d\mu$$

Theorem 44.2.13: Linearity of the Lebesgue integral

Let μ be a measure on X. Let $f, g : X \to \overline{\mathbb{R}}$ and $\alpha, \beta \in \mathbb{R}$. Then we have:

$$\int_X (\alpha f + \beta g) \, d\mu = \alpha \int_X f \, d\mu + \beta \int_X g \, d\mu$$

Corollary 44.2.14. Let $f : X \to \overline{\mathbb{R}}$ be μ-measurable. Then f is integrable if and only if $\left|f\right|$ is integrable.

Corollary 44.2.15. Let $f : X \to \overline{\mathbb{R}}$ be μ-measurable. Then, assuming the integral of f exists, we have

$$\left|\int_X f \, d\mu\right| \leq \int_X \left|f\right| d\mu$$

Corollary 44.2.16: Dominated Convergence Theorem

Let $g : X \to [0, \infty]$ be μ-measurable such that $\left|f\right| \leq g$ μ-almost everywhere. Then if

$$\int_X g \, d\mu < \infty,$$

f is integrable.

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44.3 Comparing Riemann Integration and Lebesgue Integration

Theorem 44.3.1. The Dirichlet function $\mathbb{1}_{\mathbb{Q}}$ is Lebesgue integrable, but not Riemann integrable.

Theorem 44.3.2

A Lebesgue-integrable function is not necessarily "almost Riemann-integrable". More formally, there exists a function $f : \mathbb{R} \to \mathbb{R}$ with the following properties:

1. f is Lebesgue-integrable.
2. There exists no Riemann-integrable function g such that $f = g$ almost everywhere.

Proof. Consider the indicator function $\mathbb{1}_{SVC}$ of the Smith-Volterra-Cantor set. □

Theorem 44.3.3

Let $f : [a, b] \to \mathbb{R}$. Then f is Riemann integrable if and only if it is the limit of a uniformly convergent sequence of step functions over Jordan measurable sets.

44.4 Convergence Theorems

Theorem 44.4.1: Levi's Stronger Monotone Convergence Theorem

The monotone convergence theorem continues to hold if $f_n \leq f_{n+1}$ and $f_n \nearrow f$ only hold μ-almost everywhere.

Corollary 44.4.2: Exchange of Integrals and Sums

Let f_n be nonnegative and μ-measurable. Then

$$\int_X \left(\sum_{n=1}^{\infty} f_n \right) d\mu = \sum_{n=1}^{\infty} \left(\int_x f_n \, d\mu \right)$$

Theorem 44.4.3: Fatou's Lemma

Let (f_n) be a sequence of μ-measurable functions $X \to \overline{\mathbb{R}}$. Let $g \in \mathcal{L}^1$. Then if we have $f_n \geq g$ μ-almost everywhere for all $n \in \mathbb{N}$, we have

$$\int_X \left(\liminf_{n\to\infty} f_n \right) d\mu \leq \liminf_{n\to\infty} \int_X f_n \, d\mu$$

Theorem 44.4.4: Lebesgue's Dominated Convergence Theorem

Let f, f_k be μ-measurable functions such that $f_k \to f$ μ-almost everywhere. Assume there exists a nonnegative Function $g \in \mathcal{L}^1$ such that $\sup_k |f_k| \leq g$ μ-almost everywhere. Then we have $f \in \mathcal{L}^1$ and

$$\int_X f \, d\mu = \lim_{k\to\infty} \int_X f_k \, d\mu$$

Lemma 44.4.5. (Continuity of Parametrized Integrals): Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $U \subset \mathbb{R}^n$ be open. Let $f : U \times X \to \mathbb{R}$. Then

$$\varphi : U \to \mathbb{R}$$
$$\varphi(y) = \int_X f(y, x) \, d\mu(x)$$

is continuous if the following hold:

1. $f(\cdot, x)$ is continuous in U for μ-almost all $x \in X$,
2. $f(y, \cdot)$ is μ-measurable for all $y \in U$,
3. there exists a function $g \in \mathcal{L}^1(\mu)$ such that for all $y \in U$ and μ-almost all $x \in X$, we have

$$\|f(y, x)\| \leq g(x)$$

Theorem 44.4.6: Differentiation under the Integral Sign

Let $(X, \mathcal{A}, \mu)$ be a measure space, $U \subset \mathbb{R}^n$ open, and $f : U \times X \to \mathbb{R}$. Then

$$\varphi(y) := \int_X f(y, x)\, d\mu(x)$$

is continuously differentiable with

$$\frac{\partial}{\partial y_i} \int_X f(y, x)\, d\mu(x) = \int_x \frac{\partial}{\partial y_i} f(y, x)\, d\mu(x)$$

if the following hold:

1. $f(\cdot, x)$ is continuously differentiable in U for almost all x,
2. $f(y, \cdot)$ is μ-measurable for almost all $y \in U$,
3. There exists a function $g \in \mathcal{L}^1(\mu)$ such that

$$\left| \frac{\partial}{\partial y_i} f(y, x) \right| \leq g(x)$$

for all $y \in U$, μ-almost all $x \in X$ and $i \in \{1, \ldots, n\}$.

Theorem 44.4.7. (Absolute continuity of the Lebesgue integral): Let $f \in \mathcal{L}^1(\mu)$. Let $\varepsilon > 0$. Then there exists a $\delta > 0$ such that, for all $A \in \mathcal{A}$:

$$\mu(A) < \delta \implies \int_A |f|\, d\mu < \varepsilon$$

44.5 L^p-Spaces

Definition 44.5.1: $\mathcal{L}^p$

Let $1 \leq p < \infty$. We denote by $\mathcal{L}^p = \mathcal{L}^p(X, \mathcal{A}, \mu)$ the set of all μ-measurable functions f such that

$$\int_X |f|^p \, d\mu < \infty$$

Definition 44.5.2: L^p Norm

Let $1 \leq p < \infty$ and $f \in \mathcal{L}^p(X, \mathcal{A}, \mu)$. We call

$$\|f\|_p = \sqrt[p]{\int_X |f|^p \, d\mu}$$

The L^p-Norm, or just p-Norm of f.

Note that, if we view integration as the natural "continuous analogue of summation"[1] and f as a vector with uncountably many components, this is a direct generalization of the familiar p-norm of vectors, such as the euclidean norm

$$\|\vec{x}\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}$$

Theorem 44.5.3: L^∞-Norm

We have

$$\|f\|_\infty := \lim_{p \to \infty} \|f\|_p = \inf_{s>0} \mu\{|f| > s\}$$

Lemma 44.5.4. (Young Inequality): Let $p, q \in (1, \infty)$ such that $\frac{1}{p} + \frac{1}{q} = 1$. Then, for nonnegative $a, b \geq 0$, we have:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

Proof. From the concavity of the logarithm, we get:

$$\ln\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \geq \frac{1}{p}\ln(a^p) + \frac{1}{q}\ln(b^q)$$

Which we can immediately simplify to get:

$$\frac{1}{p}\ln(a^p) + \frac{1}{q}\ln(b^q) = \ln(a) + \ln(b) = \ln(ab)$$

Exponentiating both sides gives us our desired result. □

[1] We can actually formalize this by going the other way - summation is just integration over a discrete space, using the counting measure as the canonical measure.

Comment 44.5.5. Keep in mind that:

$$\begin{aligned}
&\frac{1}{p} + \frac{1}{q} = 1 \\
\iff &\frac{1}{q} = 1 - \frac{1}{p} \\
&= \frac{p-1}{p} \\
\iff &q = \frac{p}{p-1}
\end{aligned}$$

Now, letting $\frac{1}{\infty} = 0$, we get:

Corollary 44.5.6: Hölder Inequality

Let $p, q \in [1, \infty]$ such that $\frac{1}{p} + \frac{1}{q} = 1$. Let $f \in \mathcal{L}^p$ and $g \in \mathcal{L}^q$. Then we have $fg \in \mathcal{L}^1$ and, in particular,

$$\begin{aligned}
&\left| \int_X f(x) \cdot g(x) \, d\mu(x) \right| \\
\leq &\|fg\|_1 \leq \|f\|_p \cdot \|g\|_q
\end{aligned}$$

Proof.

$p = 1$: In this case, we have $q = \infty$. We have

$$g(x) < \inf_{s>0} \{\mu(|g| > s) = 0\} = \|g\|_\infty$$

μ-almost everywhere, and thus

$$\begin{aligned}
\int_X |f(x) \cdot g(x)| \, d\mu(x) &\leq \int_X |f(x)| \cdot \|g\|_\infty \, d\mu(x) \\
&= \int_X |f(x)| \, d\mu(x) \cdot \|g\|_\infty \\
&= \|f\|_1 \cdot \|g\|_\infty
\end{aligned}$$

$p > 1$: Assume that $\|f\|_p \cdot \|g\|_q > 0$. Then, applying the Young inequality with $a = \dfrac{|f(x)|}{\|f\|_p}$ and $b = \dfrac{|g(x)|}{\|g\|_q}$, we get

$$\begin{aligned}
\frac{f(x)g(x)}{\|f\|_p \|g\|_q} &\leq \frac{|f(x)||g(x)|}{\|f\|_p \|g\|_q} \\
&\leq \frac{|f(x)|^p}{p\|f\|_p^p} + \frac{|g(x)|^q}{q\|g\|_q^q}
\end{aligned}$$

Integrating both sides gives:

$$\begin{aligned}
\frac{\|fg\|_1}{\|f\|_p\|g\|_q} &= \frac{1}{\|f\|_p\|g\|_q} \cdot \int_X f(x)g(x)\, d\mu(x) \\
&\leq \frac{1}{\|f\|_p\|g\|_q} \cdot \int_X |f(x)||g(x)|\, d\mu(x) \\
&\leq \frac{1}{p\|f\|_p^p} \cdot \int_X |f(x)|^p\, d\mu(x) + \frac{1}{q\|g\|_q^q} \cdot \int_X |g(x)|^q\, d\mu(x) \\
&= \frac{1}{p\|f\|_p^p} \cdot \|f\|_p^p + \frac{1}{q\|g\|_q^q} \cdot \|g\|_q^q \\
&= \frac{1}{p} + \frac{1}{q} \\
&= 1
\end{aligned}$$

Which gives us our desired inequality.

□

Lemma 44.5.7. $\mathcal{L}^p$ is closed under addition.

Proof.

1. $p = 1$: Follows from the triangle inequality on $\mathbb{R}$:

$$\begin{aligned}
\|f + g\|_1 &= \int_X |f(x) + g(x)|\, d\mu(x) \\
&\leq \int_X |f(x)| + |g(x)|\, d\mu(x) \\
&= \int_X |f(x)|\, d\mu(x) + \int_X |g(x)|\, d\mu(x) \\
&= \|f\|_1 + \|g\|_1
\end{aligned}$$

2. $p \in (1, \infty)$: Since the function $t \mapsto t^p$ is convex on $[0, \infty)$, we have:

$$\begin{aligned}
|f + g|^p &= 2^p \left|\frac{f + g}{2}\right|^p \\
&\leq 2^{p-1}(|f|^p + |g|^p)
\end{aligned}$$

3. $p = \infty$: We have $|f(x)| \leq \|f\|_\infty$ and $|g(x)| \leq \|g\|_\infty$ for μ-almost all x, which immediately gives us

$$|f(x) + g(x)| \leq |f(x)| + |g(x)| \leq \|f\|_\infty + \|g\|_\infty$$

μ-almost everywhere and thus

$$\|f + g\|_\infty = \inf_{s>0} \{|f + g| \leq s \ \mu\text{-almost everywhere}\} \leq \|f\|_\infty + \|g\|_\infty$$

□

We now want to show that the L^p norm defines a seminorm, which we will then

easily be able to turn into a proper norm.

Theorem 44.5.8: Minkowski Inequality

Let $f, g \in \mathcal{L}^p$. Then we have

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p$$

Proof. We already showed the cases $p = 1$ and $p = \infty$ while proving $\mathcal{L}^p$ is closed under addition. Thus, let $p \in (1, \infty)$. Then the triangle inequality on $\mathbb{R}$ gives:

$$\begin{aligned}
|f(x) + g(x)|^p &= |f(x) + g(x)| \cdot |f(x) + g(x)|^{p-1} \\
&\leq (|f(x)| + |g(x)|) \cdot |f(x) + g(x)|^{p-1} \\
&= |f(x)||f(x) + g(x)|^{p-1} + |g(x)||f(x) + g(x)|^{p-1}
\end{aligned}$$

Applying the Hölder inequality with $q = \frac{p}{p-1}$ gives:

$$\begin{aligned}
&\int_X |f(x)||f(x) + g(x)|^{p-1} \, d\mu(x) \\
&\leq \|f\|_p \cdot \left\| |f + g|^{p-1} \right\|_q \\
&= \|f\|_p \cdot \left(\int_X |f(x) + g(x)|^{(p-1)q} \, d\mu(x) \right)^{\frac{1}{q}} \\
&= \|f\|_p \cdot \left(\int_X |f(x) + g(x)|^p \, d\mu(x) \right)^{\frac{p-1}{p}} \\
&= \|f\|_p \cdot \|f + g\|_p^{p-1}
\end{aligned}$$

Switching the roles of f and g similarly gives:

$$\int_X |g(x)||f(x) + g(x)|^{p-1} \, d\mu(x) \leq \|g\|_p \cdot \|f + g\|_p^{p-1}$$

And putting everything together, we get:

$$\begin{aligned}
\|f + g\|_p^p &= \int_X |f(x) + g(x)|^p \, d\mu(x) \\
&\leq \int_X |f(x)||f(x) + g(x)|^{p-1} \, d\mu(x) + \int_X |g(x)||f(x) + g(x)|^{p-1} \, d\mu(x) \\
&\leq \|f\|_p \cdot \|f + g\|_p^{p-1} + \|g\|_p \cdot \|f + g\|_p^{p-1} \\
&= \|f + g\|_p \cdot \|f + g\|_p^{p-1}
\end{aligned}$$

From which we get our desired result by dividing out the factor of $\|f + g\|_p^{p-1}$. □

Corollary 44.5.9

The $\mathcal{L}^p$-norm is a seminorm on $\mathcal{L}^p$. In particular, we have:

1. $\|f\|_p \geq 0$, with $\|f\|_p = 0$ iff. $f = 0$ μ-almost everywhere
2. $\|\lambda f\|_p = |\lambda| \cdot \|f\|_p$
3. $\|f + g\|_p \leq \|f\|_p + \|g\|_p$

Corollary 44.5.10: L^p-space

Define L^p-space to be the quotient space of $\mathcal{L}^p$ under μ-almost-everywhere equality. Then $\|\cdot\|_p$ forms a proper norm on L^p.

Lemma 44.5.11. Let $1 \leq p < \infty$. Let $u_j \in L^p$ and

$$f_k = \sum_{j=1}^{k} u_j$$

Then, if $\sum_{j=1}^{\infty} \|u_j\| p < \infty$, we have:

1. The pointwise limit $f(x) = \lim_{k\to\infty} f_k(x)$ exists for μ-almost all $x \in X$,
2. $f \in L^p$,
3. $\|f - f_k\|_p \to 0$

Lemma 44.5.12: Convergence in L^p

If we have $f_k \to f$ in L^p Norm for $1 \leq p \leq \infty$, then there exists a subsequence f_{k_i} such that $f_{k_i} \to f$ pointwise μ-almost everywhere.

Theorem 44.5.13: Riesz-Fischer

For $1 \leq p \leq \infty$, $(L^p, \|\cdot\|_p)$ is a Banach space.

Definition 44.5.14. Let $1 \leq p < \infty$. Let the sequence $(f_n) \subset L^p$ converge pointwise μ-almost everywhere against the μ-measurable function f. Let

$$\nu(A) := \limsup_{n\to\infty} \int_A |f_n|^p \, d\mu$$

Then we call f_k **uniformly integrable** iff, for any $A \in \mathcal{A}$, we have:

1. $\forall \varepsilon > 0 : \exists \delta > 0 : \forall A \in \mathcal{A} : \mu(A) < \delta \implies \nu(A) < \varepsilon$
2. $\forall \varepsilon > 0 : \exists E \in \mathcal{A} : \mu(E) < \infty, \nu(X \setminus E) < \varepsilon$

Theorem 44.5.15: Vitali Convergence Theorem

Let $1 \leq p < \infty$. Let the sequence $(f_n) \subset L^p$ converge pointwise μ-almost everywhere against the μ-measurable function f. Then the following are equivalent:

1. $f \in L^p$ and $\left\|f_n - f\right\|_p \to 0$,
2. f_k is uniformly integrable.

Definition 44.5.16. Let $\Omega \subset \mathbb{R}^n$ be open. Then the **support of a function** $f : \Omega \to \mathbb{R}$ is the set

$$\operatorname{spt} f := \overline{\left\{x \in \Omega : f(x) \neq 0\right\}}$$

The space of k times continuously differentiable functions with compact support in Ω is denoted $C_c^k(\Omega)$.

Theorem 44.5.17: $C_c^0(\Omega)$ is dense in $L^p(\Omega)$

Let $\Omega \subset \mathbb{R}^n$ be open. Let $1 \leq p < \infty$. Then, for every $f \in L^p(\Omega)$, there exists a sequence f_k in $C_c^0(\Omega)$ such that $\left\|f - f_k\right\|_p \to 0$.

44.6 Density Functions

Theorem 44.6.1

Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $\theta : X \to \overline{\mathbb{R}}$ be nonnegative and μ-measurable. Then the map

$$\nu : \mathcal{A} \to \overline{\mathbb{R}}$$
$$A \mapsto \int_A \theta \, d\mu$$

is a measure, which we denote μ_θ. We call θ the **density of ν with respect to μ**.

Corollary 44.6.2. The following hold for μ_θ:

1. $\mu(A) = 0$ implies $\mu_\theta(A) = 0$.
2. For every nonnegative μ-measurable function f, we have
$$\int_X f \, d\mu_\theta = \int_X f \cdot \theta \, d\mu.$$
3. θ is unique up to equality μ-almost everywhere.

Definition 44.6.3

Let μ and ν be measures on $(X, \mathcal{A})$. Then we call ν **absolutely continuous with respect to μ**, which we denote $\nu \ll \mu$, if $\mu(A) = 0$ implies $\nu(A) = 0$.

Lemma 44.6.4. Let σ and ν be finite measures on $(X, \mathcal{A})$ such that $\nu(A) \leq \sigma(A)$ for every $A \in \mathcal{A}$. Then there exists a density function θ such that $\nu = \sigma_\theta$.

This lemma is actually a special case of the significantly more general Riesz representation theorem for Hilbert spaces, which I sadly do not have the time to go into at this point (but look it up, it's really neat!).

Lemma 44.6.5. Let μ, ν be measures on $(X, \mathcal{A})$. Let $\sigma := \mu + \nu$. Then if $f \in \mathcal{L}^*(\sigma)$, then $f \in \mathcal{L}^*(\mu)$ and $f \in \mathcal{L}^*(\nu)$ and we have

$$\int_X f \, d\sigma = \int_X f \, d\mu + \int_X f \, d\nu$$

Theorem 44.6.6. (Mini-Radon-Nikodym): Let μ, ν be finite measures on $(X, \mathcal{A})$ such that $\nu \ll \mu$. Then there exists a density function $\theta \in L^1(\mu)$ such that $\nu = \mu_\theta$.

In this case, θ is sometimes also called the **Radon-Nikodym derivative of ν with respect to μ** and denoted $\frac{d\nu}{d\mu}$.

Theorem 44.6.7. Let μ and ν be finite measures on $(X, \mathcal{A})$. Then the following are equivalent:

1. $\nu \ll \mu$,
2. There exists a density function $\theta \in L^1(\mu)$ such that $\nu = \mu_\theta$,
3. For all $\varepsilon > 0$, there exists a $\delta > 0$ such that $\mu(A) < \delta$ implies $\nu(A) < \varepsilon$.

Proof. (i) $\implies$ (ii) is our mini-Radon-Nikodym theorem. (ii) $\implies$ (iii) follows from absolute continuity of the Lebesgue integral. (iii) $\implies$ (i) follows immediately from the definition of $\ll$. □

Theorem 44.6.8: Radon-Nikodym

Let μ, ν be σ-finite measures on $(X, \mathcal{A})$ such that $\nu \ll \mu$. Then there exists a density function $\theta \in L^1(\mu)$ such that $\nu = \mu_\theta$, i.e.

$$\int_X f \, d\nu = \int_X f \cdot \theta \, d\mu$$

Definition 44.6.9

Let μ and ν be measures on $(X, \mathcal{A})$. Then we call μ and ν **singular with respect to each other**, which we denote $\mu \perp \nu$, if there exists a set $M \in \mathcal{A}$ such that

$$\mu(M) = \nu(X \setminus M) = 0.$$

Theorem 44.6.10: Lebesgue's decomposition theorem

Let μ and ν be measures on $(X, \mathcal{A})$, and let ν be σ-finite. Then there exists a unique decomposition $\nu = \nu_a + \nu_s$ such that $\nu_a \ll \mu$ and $\nu_a \perp \mu$.

Chapter 45

Integration in $\mathbb{R}^n$

45.1 Products of Measure Spaces

In this section, we will be interested in finding a canonical σ-Algebra on cartesian products of measurable spaces. Category theory tells us the correct solution directly: The product σ-algebra of an indexed family of σ-algebras should be the sigma algebra fulfilling the universal property of product objects.

We are working in the category of measurable spaces, where morphisms are measurable functions. Therefore, our definition simply amounts to defining product sigma algebras as the σ-algebras induced by the projection functions, which is in turn the smallest σ-algebra such that all projections are measurable.

Definition 45.1.1: Product σ-Algebra

Let $(X_i, \mathcal{A}_i)_{i\in I}$ be a family of measurable spaces. Then the product σ-algebra on the cartestian product $\prod_{i\in I} X_i$ is the smallest σ-Algebra $\bigotimes_{i\in I} \mathcal{A}_i$ such that every projection π_j is $\bigotimes_{i\in I} \mathcal{A}_i - \mathcal{A}_j$-measurable.

Explicitly, this means we have:

$$\bigotimes_{i\in I} \mathcal{A}_i = \left\langle \bigcup_{i\in I} \pi_i^{-1}(\mathcal{A}_i) \right\rangle_\sigma$$

Note that this is also entirely analogous to the definition of the product topology, which is the coarsest (i.e. "smallest") topology such that projections are continuous.

If the cardinality of I is a small finite number, we may sometimes write out the product σ-algebra as:

$$\bigotimes_{i\in I} \mathcal{A}_i = \mathcal{A}_1 \otimes \mathcal{A}_2 \otimes \dots$$

Theorem 45.1.2

Let $(X, \mathcal{A})$ and $(Y_i, \mathcal{B}_i)$ be measurable spaces. Let $g : X \to \prod_{i\in I} Y_i$. Then g is $\mathcal{A} - \bigotimes_{i\in I} \mathcal{B}_i$-measurable if and only if all maps $\pi_i \circ g$ are $\mathcal{A} - \mathcal{B}_i$-measurable.

Proof. 1. Assume g is measurable. We know that π_i are measurable by definition, and we know that compositions of measurable functions are measurable. Therefore, $\pi_i \circ g$ are measurable.

2. Assume $\pi_i \circ g$ are measurable. Then we have:

$$\begin{aligned} g^{-1}\left(\bigcup_{i\in I} \pi_i^{-1}(\mathcal{A}_i)\right) &= \bigcup_{i\in I} g^{-1}(\pi_i^{-1}(\mathcal{A}_i)) \\ &= \bigcup_{i\in I} (\pi_i \circ g)^{-1}(\mathcal{A}_i)) \subset \mathcal{A} \end{aligned}$$

Where the last step follows from the measurability of $\pi_i \circ g$. Therefore, since $\bigcup_{i\in I} \pi_i^{-1}(\mathcal{A}_i)$ forms a basis of $\bigotimes_{i\in I} \mathcal{A}_i$, we have that g is measurable.

□

As a corollary, we get that our product space actually fulfills the universal property of product objects:

Corollary 45.1.3. Let $(X, \mathcal{A})$ and $(Y_i, \mathcal{B}_i)_{i\in I}$ be measurable spaces. Let $g_i : X \to Y_i$ be a family of $\mathcal{A} - \mathcal{B}_i$-measurable functions. Then there exists a $\mathcal{A} - \mathcal{B}_i$-measurable function $g : X \to \prod_{i\in I} Y_i$ such that, for each $i \in I$, $g_i = \pi_i \circ g$.

Proof. Define g as the cartesian product of all g_i:

$$\begin{aligned} g : I &\to \left(X \to \prod_{i\in I} Y_i\right) \\ i &\mapsto g_i \end{aligned}$$

Then our previous lemma tells us that, since all $g_i = \pi_i \circ g$ are measurable, g is measurable as well. □

Corollary 45.1.4. Let $(X_i, \mathcal{A}_i)_{i\in I}$ be measurable spaces, and let $J \subset I$ be nonempty. Then the projection π_J is $\bigotimes_{i\in I} \mathcal{A}_i - \bigotimes_{j\in J} \mathcal{A}_j$-measurable.

Definition 45.1.5. (Embedding into a larger product space): Let $p_j \in X_j$. Then we define the **embedding of $\prod_{i\in I\setminus j} X_i$ into $\prod_{i\in I} X_i$ by x_j** to be the map e_{x_j} which takes a product $(x_i)_{i\in I\setminus j}$ and "adds in x_j as a new component", i.e. it maps $(x_i)_{i\in I\setminus j}$ to the product $(x'_i)_{i\in I}$ with $x'_j = x_j$ and $(x'_i)_{i\in I\setminus j} = (x_i)_{i\in I\setminus j}$.

Definition 45.1.6: Cuts of subsets

Let $M \subseteq \prod_{i\in I} X_i$, and let $x_j \in X_j$. Then we define the **cut of M through x_j** to be the set $M^{x_j} = e_{x_j}^{-1}(M)$.

More explicitly, M^{x_j} is the set such that for for every $y \in M$, we have $y|_{I\setminus\{j\}} \in M^{x_j}$ if and only if $y_j = x_j$.

Corollary 45.1.7. Let $(X_i, \mathcal{A}_i)_{i\in I}$ be measurable spaces, let $M \in \bigotimes_{i\in I} \mathcal{A}_i$, and

let $x_j \in X_j$ for $j \in I$. Then we have

$$M^{x_j} \in \bigotimes_{i \in I \setminus j} \mathcal{A}_i$$

Proof. Let $x_j \in X_j$, $j \in I$. We want to show that e_{x_j} is measurable, which would follow immediately if we could show that all projections $\pi_i^I \circ e_{x_j}$ are measurable.

1. Let $i \in I \setminus \neq j$. Then $\pi_i^I \circ e_{x_j} = \pi_i^{I \setminus j}$, which is measurable.
2. Let $i = j$. Then $\pi_i^I \circ e_{x_j} = \pi_j^I \circ e_{x_j}$, which is just the constant map onto x_j, which is measurable.

Since all projections are measurable, e_{x_j} itself must also be measurable. Therefore, since M is measurable, the preimage $M^{x_j} = e_{x_j}^{-1}(M)$ must also be measurable. □

Note that, in particular, this tells us that if $(X_1, \mathcal{A}_1)$ and $(X_2, \mathcal{A}_2)$ are measurable spaces, then for any $M \in \mathcal{A}_1 \otimes \mathcal{A}_2$, $x_1 \in X_1$, and $x_2 \in X_2$, we have

$$M^{x_2} = \{x \in X_1 \mid (x, x_2) \in M\} \in \mathcal{A}_1$$

and

$$M^{x_1} = \{x \in X_2 \mid (x_1, x) \in M\} \in \mathcal{A}_2$$

Theorem 45.1.8: A basis of the product σ-Algebra

Let $(X_i, \mathcal{A}_i)_{i \in I}$ be a family of measurable spaces. For each $i \in I$, let $\mathcal{E}_i$ be a basis of $\mathcal{A}_i$. Then we have

$$\bigotimes_{i \in I} \mathcal{A}_i = \left\langle \bigcup_{i \in I} \pi_i^{-1}(\mathcal{E}_i) \right\rangle_\sigma .$$

Proof. Recall that, for any map f and any system of subsets $\mathcal{S} \subset \mathcal{A}$ of a σ-algebra $\mathcal{A}$, we have

$$f^{-1}(\langle \mathcal{S}_i \rangle_\sigma) = \left\langle f^{-1}(\mathcal{S}_i) \right\rangle_\sigma$$

and

$$\left\langle \bigcup_{i \in I} \mathcal{S}_i \right\rangle_\sigma = \left\langle \bigcup_{i \in I} \langle \mathcal{S}_i \rangle_\sigma \right\rangle_\sigma ,$$

which gives us:

$$\begin{aligned}\bigotimes_{i\in I}\mathcal{A}_i &= \left\langle \bigcup_{i\in I}\pi_i^{-1}(\mathcal{A}_i)\right\rangle_\sigma \\ &= \left\langle \bigcup_{i\in I}\pi_i^{-1}(\langle\mathcal{E}_i\rangle_\sigma)\right\rangle_\sigma \\ &= \left\langle \bigcup_{i\in I}\left\langle\pi_i^{-1}(\mathcal{E}_i)\right\rangle_\sigma\right\rangle_\sigma \\ &= \left\langle \bigcup_{i\in I}\pi_i^{-1}(\mathcal{E}_i)\right\rangle_\sigma\end{aligned}$$

□

If I is finite, we get the following representation of the product σ-algebra:

Corollary 45.1.9. For each $i \in \{1,\dots,n\}$, let $(X_i, \mathcal{A}_i)$ be a measurable space such that $\mathcal{E}_i$ is a basis of $\mathcal{A}_i$ and such that X_i can be represented as a countable union of basis elements $E_i^k \in \mathcal{E}_i$. Then

$$\mathcal{Q}_0 := \left\{ \prod_{i=1}^n E_i \;\middle|\; E_i \in \mathcal{E}_i \right\}$$

is a basis of $\bigotimes_{i=1}^n \mathcal{A}_i$.

Proof. We need to show $\langle\mathcal{Q}_0\rangle_\sigma = \bigotimes_{i=1}^n \mathcal{A}_i$.

1. Let $E_i \in \mathcal{E}_i$. We know that

   $$\pi_i^{-1}(E_i) = E_i \times \prod_{j\neq i} X_j,$$

   which gives us the following representation for the basis $\mathcal{Q}$ we just established in the last theory:

   $$\mathcal{Q} = \bigcup_{i=1}^n \pi_i^{-1}(\mathcal{E}_i) = \bigcup_{i=1}^n \left\{ E_i \times \prod_{j\neq i} X_j \;\middle|\; E_i \in \mathcal{E}_i \right\}$$

   Furthermore, $\cap$-stability of σ-algebras gives us that any element

   $$\prod_{i=1}^n E_i = \bigcap_{i=1}^n \pi_i^{-1}(E_i)$$

   of $\mathcal{Q}_0$ must be contained in $\langle\mathcal{Q}\rangle_\sigma$.

   Therefore, we have

   $$\text{↑}\langle\mathcal{Q}_0\rangle_\sigma \subset \langle\mathcal{Q}\rangle_\sigma = \bigotimes_{i=1}^n \mathcal{A}_i$$

2. Now, let $E_i \in \mathcal{E}_i$ and $k \in \mathbb{N}$. By definition of $\mathcal{Q}_0$, we have

   $$E_i \times \prod_{j\neq i} E_j^k \in \mathcal{Q}_0,$$

which gives us:

$$\pi_i^{-1}(E_i) = \bigcup_{k=1}^{\infty} \left(E_i \times \prod_{j \neq i} E_j^k \right) \in \langle \mathcal{Q}_0 \rangle_\sigma ,$$

and by extension:

$$\mathcal{Q} = \bigcup_{i=1}^{n} \pi_i^{-1}(E_i) \subset \langle \mathcal{Q}_0 \rangle_\sigma$$

which means we also have

$$\bigotimes_{i=1}^{n} \mathcal{A}_i = \langle \mathcal{Q} \rangle_\sigma \subset \sigma(\mathcal{Q}_0).$$

□

Corollary 45.1.10. The Borel σ-algebra $\mathcal{B}(\mathbb{R}^n)$ is the product of the Borel σ-algebra $\mathcal{B}(\mathbb{R})$, i.e. we have:

$$\mathcal{B}(\mathbb{R}^n) = \bigotimes_{i=1}^{n} \mathcal{B}(\mathbb{R})$$

Proof. We know that the set $\mathcal{I}$ of real intervals forms a basis of $\mathcal{B}(\mathbb{R})$. Given this basis, the set $\mathcal{Q}_0$ is exactly the set of n-dimensional real cuboids, which forms a basis of $\mathcal{B}(\mathbb{R}^n)$. Therefore, we have:

$$\mathcal{B}(\mathbb{R}^n) = \langle \mathcal{Q}_0 \rangle_\sigma = \bigotimes_{i=1}^{n} \mathcal{B}(\mathbb{R})$$

□

For the final part of this chapter, we establish some notation for commonly encountered subsets of product σ-algebras.

Definition 45.1.11. Let I be a set. Then we denote by $\mathcal{P}_0(I)$ the finite, nonenmpty subsets of I.

Definition 45.1.12: Measurable Cuboids

Let $(X_i, \mathcal{A}_i)_{i \in I}$ be a family of measurable spaces. Then we define the set of **measurable cuboids in $\prod_{i \in I} X_i$** as:

$$\mathcal{Q} := \bigcup_{J \in \mathcal{P}_0(I)} \left\{ \prod_{j \in J} A_j \times \prod_{I \setminus J} X_i \;\middle|\; A_j \in \mathcal{A}_j \right\},$$

i.e. a finite number of the sides of our cuboid can be arbitrary measurable sets, and the remaining sides must correspond to the remaining domain sets $(X_i)_{i \in I \setminus J}$.

Definition 45.1.13: Cylindrical Sets

Let $(X_i, \mathcal{A}_i)_{i\in I}$ be a family of measurable spaces. Then we define the set of **cylinder sets in $\prod_{i\in I} X_i$** as:

$$\begin{aligned}\mathcal{Z} &:= \bigcup_{J\in\mathcal{P}_0(I)} \left\{ A_J \times \prod_{i\in I\setminus J} X_i \;\middle|\; A_J \in \bigotimes_{j\in J} \mathcal{A}_j \right\} \\ &= \bigcup_{J\in\mathcal{P}_0(I)} \pi_J^{-1}\left(\bigotimes_{j\in J} \mathcal{A}_j \right),\end{aligned}$$

i.e. the base of the cylinder is a measurable set $A_j \subset \prod_{i\in I} X_i$ and all other sides are once again the remaining domain sets $(X_i)_{i\in I\setminus J}$.

Note that if we view an everyday geometric cylinder $Z \subset \mathbb{R}^3$ as the product of of the circle at its base with a finite real interval, it doesn't actually qualify as a cylindrical set in $\mathbb{R}^3$, since it would have to be a product of the base circle with all of $\mathbb{R}$, i.e. only "infinite" cylinders immediately qualify.

By technicality, normal cylinders are still cylindrical sets though, since we can set $J = I$ and have view the entire cylinder as the base, without adding any additional "sides".

Therefore, this definition is not very useful if I is finite, since every measurable set is a cylindrical with itself as its base and every cylindrical set is a product of measurable sets, i.e. measurable.

Theorem 45.1.14

Let $(X_i, \mathcal{A}_i)_{i\in I}$ be a family of measurable spaces. Then the measurable cuboids in $\prod_{i\in I} X_i$ and the cylindrical sets in $\prod_{i\in I} X_i$ form bases of $\bigotimes_{i\in I} \mathcal{A}_i$.

Proof. We will show $\mathcal{Q} \subset \mathcal{Z} \subset \bigotimes_{i\in I} \mathcal{A}_i \subset \langle \mathcal{Q} \rangle_\sigma$, which implies $\langle \mathcal{Q} \rangle_\sigma \subset \langle \mathcal{Z} \rangle_\sigma \subset \bigotimes_{i\in I} \mathcal{A}_i \subset \langle \mathcal{Q} \rangle_\sigma$, proving the theorem.

1. $\mathcal{Q} \subset \mathcal{Z}$: Let $A \in \mathcal{Q}$. Then A is given as

$$\begin{aligned}A &= \prod_{j\in J}^{n} A_j \times \prod_{i\notin J} X_i \\ &= \pi_J^{-1}\left(\prod_{j\in J} A_j \right).\end{aligned}$$

   Therefore, since $\prod_{j\in J} A_j \in \bigotimes_{j\in J} \mathcal{A}_j$, A is a cylinder set.

2. $\mathcal{Z} \subset \bigotimes_{i\in I} \mathcal{A}_i$: Recall that π_J is $\bigotimes_{i\in I} \mathcal{A}_i - \mathcal{A}_J$-measurable. Therefore, for any $J \in \mathcal{P}_0(I)$, we have that $\pi_J^{-1}(\mathcal{A}_j) \subset \bigotimes_{i\in I} \mathcal{A}_i$, which means any cylinder set is contained in $\bigotimes_{i\in I} \mathcal{A}_i$.

3. $\bigotimes_{i\in I} \mathcal{A}_i \subset \langle \mathcal{Q} \rangle_\sigma$: For any $i \in I$ and $A_i \in \mathcal{A}_i$, we have:

$$\pi_i^{-1}(A_i) = A_i \times \prod_{j\neq i} X_j \in \mathcal{Q},$$

i.e. $\pi_i^{-1}(\mathcal{A}_i) \subset \mathcal{Q}$ for all $i \in I$, which means we also have:

$$\bigcup_{i\in I} \pi_i^{-1}(\mathcal{A}_i) \subset \mathcal{Q}.$$

By definition of $\bigotimes_{i\in I} \mathcal{A}_i$, we get:

$$\bigotimes_{i\in I} \mathcal{A}_i = \left\langle \bigcup_{i\in I} \pi_i^{-1}(\mathcal{A}_i) \right\rangle_\sigma \subset \langle \mathcal{Q} \rangle_\sigma .$$

□

45.2 Product Measures and Fubini's Theorem

In this section, given two measure spaces $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$, we want to derive a canonical measure $\mu \otimes \nu$ on $\mathcal{A} \otimes \mathcal{B}$. Our intuition already tells us what the measure of a cuboid should be: Given a cuboid

$$Q = A \times B \subset X \times Y,$$

where $A \in \mathcal{A}$ and $B \in \mathcal{B}$, we want the measure of the cuboid to be the product of the measures of its sides:

$$(\mu \otimes \nu)(M) = \mu(A) \cdot \nu(B)$$

Recall that, for any $M \in X \times Y$, the cuts through M by $x \in X$ and $y \in Y$ are simply defined as

$$M^x = \left\{ y \in Y \;\middle|\; (x, y) \in M \right\},$$
$$M^y = \left\{ x \in X \;\middle|\; (x, y) \in M \right\}.$$

Theorem 45.2.1

Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces. Let $M \in \mathcal{A} \otimes \mathcal{B}$. Then:

1. The function $x \mapsto \nu(M^x)$ is $\mathcal{A}$-measurable,
2. The function $y \mapsto \mu(M^y)$ is $\mathcal{B}$-measurable,
3. We have

$$\int_X \nu(M^x)\, d\mu(x) = \int_Y \mu(M^y)\, d\nu(y)$$

Proof. Let $\mathcal{S}$ be the system of all sets $M \in \mathcal{A} \otimes \mathcal{B}$ for which this theorem holds.

Let $\mathcal{Q}$ be the system of all measurable cuboids in $X \times Y$, which we have just established are simply given by $Q = A \times B$ such that $A \in \mathcal{A}$ and $B \in \mathcal{B}$.

Let $\mathcal{F}$ be the system of finite unions of cuboids (which by our definition of cuboids inherently includes complements of cuboids). Since we count $X \times Y$ as a cuboid, $\mathcal{F}$ forms a set algebra.

We will show that $\mathcal{S}$ is a monotone class containing $\mathcal{F}$, which will give us

$$\mathcal{A} \otimes \mathcal{B} = \langle \mathcal{Q} \rangle_\sigma = \langle \mathcal{F} \rangle_\sigma = \langle \mathcal{F} \rangle_M \subseteq \mathcal{S}$$

since $\mathcal{S} \subseteq \mathcal{A} \otimes \mathcal{B}$ by definition, this implies $\mathcal{A} \otimes \mathcal{B} = \mathcal{S}$.

The proof the $\mathcal{S}$ is a monotone class is however left as an exercise to the reader :) □

Theorem 45.2.2: Product measure

Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu$ be σ-finite measure spaces. Then there exists exactly one measure $\mu \otimes \nu$ on $\mathcal{A} \otimes \mathcal{B}$ such that for any $A \in \mathcal{A}$ and $B \in \mathcal{B}$, we have

$$(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B)$$

This measure is given by

$$(\mu \otimes \nu)(M) = \int_X \nu(M^x)\, d\mu(x) = \int_Y \mu(M^y)\, d\mu(y)$$

Corollary 45.2.3: Cavalieri's Principle

Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces. Then, for all $M \in \mathcal{A} \otimes \mathcal{B}$, we have

$$\begin{aligned}(\mu \otimes \nu)(M) &= \int_X \nu(M^x)\, d\mu(x) \\ &= \int_X \int_Y \chi_M(x, y)\, d\nu(y)\, d\mu(x) \\ &= \int_Y \mu(M^Y)\, d\nu(x) \\ &= \int_Y \int_X \chi_M(x, y)\, d\mu(x)\, d\nu(y)\end{aligned}$$

Corollary 45.2.4. Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces. Then, for any $M \in \mathcal{A} \otimes \mathcal{B}$, the following are equivalent:

1. $(\mu \otimes \nu)(M) = 0$,
2. $\mu(M^y) = 0$ ν-almost everywhere,
3. $\nu(M^x) = 0$ μ-almost everywhere.

Lemma 45.2.5. Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces. Let $f : X \times Y \to \mathbb{R}$ be $\mathcal{A} \otimes \mathcal{B}$ measurable. Then the function $f(x, \cdot) : Y \to \mathbb{R}$ is $\mathcal{B}$-measurable for all $x \in X$.

Theorem 45.2.6: Fubini's Theorem

Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces. Let $f \in \mathcal{L}^*(\mu \otimes \nu)$. Then:

$$\begin{aligned}\int_{X \times Y} f\, d(\mu \otimes \nu) &= \int_X \int_Y f(x, y)\, d\nu(y)\, d\mu(x) \\ &= \int_Y \int_X f(x, y)\, d\mu(x)\, d\nu(y)\end{aligned}$$

Corollary 45.2.7. The volume of the open ball

$$B_r^n(0) = \left\{\vec{x} \in \mathbb{R}^n \;\middle|\; \left\|\vec{x}\right\|_2 < r\right\}.$$

is given by:

$$\lambda^n(B_r^n(0)) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2}+1)} \cdot r^n = \begin{cases} \frac{\pi^k}{k!} \cdot r^n & n = 2k \\ \frac{\pi^k}{\prod_{i=1}^k \frac{1}{2}+k-i} \cdot r^n & n = 2k+1 \end{cases}$$

In particular:

1. The 1-dimensional unit ball is simply the interval $(-1, 1)$ and thus has length 2,
2. The 2-dimensional unit ball is a circle of radius 1 and has area π
3. The 3-dimensional unit ball has volume $\frac{4\pi}{3}$
4. The 4-dimensional unit ball has volume $\frac{\pi^2}{2}$
5. The 5-dimensional unit ball has volume $\frac{8\pi^2}{15}$
6. ...

Proof. By Cavalieri's principle and the linear transformation formula for the Lebesgue measure, we get:

$$\begin{aligned}
\lambda^n(B_r^n(0)) &= \int_{\mathbb{R}} \lambda^{n-1}(B_r^n(0)^y)\, dy \\
&= \int_{(-r,r)} \lambda^{n-1}(B_r^n(0)^y)\, dy \\
&= \int_{-1}^{1} = \det\left(\sqrt{r^2-y^2} \cdot I_n\right) \cdot \lambda^{n-1}\left\{x \in \mathbb{R}^{n-1} \;\middle|\; \|x\|_2 < 1\right\} dy \\
&= r^n \cdot \int_{-1}^{1} \left(\sqrt{r^2-y^2}\right)^{n-1} \cdot \lambda^{n-1}\left\{x \in \mathbb{R}^{n-1} \;\middle|\; \|x\|_2 < 1\right\} dy \\
&= r^n \cdot \lambda^{n-1}(B_1^{n-1}(0)) \cdot \int_{-1}^{1} \left(\sqrt{1-y^2}\right)^{n-1} dy
\end{aligned}$$

We can thus get our result by solving the integral

$$I_n := \int_{-1}^{1} \left(\sqrt{1-y^2}\right)^{n-1} dy$$

Substituting $\cos(\theta) = y$ gives:

$$\begin{aligned}
I_n &= \int_{-1}^{1} \left(\sqrt{1-y^2}\right)^{n-1} dy \\
&= \int_{-1}^{1} \left(\sqrt{1-\cos(\theta)^2}\right)^{n-1} \cdot \sin(\theta)\, d\theta \\
&= \int_0^{\pi} \sin^n(\theta)\, d\theta
\end{aligned}$$

Now, partial integration gives:

$$\begin{aligned}
I_n &= \int_0^\pi \sin^n(\theta)\, d\theta \\
&= \int_0^\pi \sin(\theta) \cdot \sin^{n-1}(\theta)\, d\theta \\
&= \left[-\cos(\theta) \cdot \sin^{n-1}(\theta)\right]_0^\pi + (n-1) \cdot \int_0^\pi \cos^2(\theta) \cdot \sin^{n-2}(\theta)\, d\theta \\
&= (n-1) \cdot \int_0^\pi \cos^2(\theta) \cdot \sin^{n-2}(\theta)\, d\theta \\
&= (n-1) \cdot \int_0^\pi (1 - \sin^2) \cdot \sin^{n-2}(\theta)\, d\theta \\
&= (n-1) \cdot (I_{n-2} - I_n),
\end{aligned}$$

which means:

$$\begin{aligned}
I_n &= (n-1) \cdot I_{n-2} - (n-1) \cdot I_n \\
\implies n I_n &= (n-1) \cdot I_{n-2} \\
\implies I_n &= \frac{n-1}{n} \cdot I_{n-2}
\end{aligned}$$

Unrolling this recursive formula, we get:

$$\begin{aligned}
I_{2k} &= I_0 \cdot \prod_{i=1}^{n} \frac{2i-1}{2i} \\
I_{2k+1} &= I_1 \cdot \prod_{i=1}^{n} \frac{2i}{2i+1}
\end{aligned}$$

We can easily calculate I_0 and I_1 explicitly:

$$\begin{aligned}
I_0 &= \int_0^\pi \sin^0(\theta)\, d\theta \\
&= \int_0^\pi 1\, d\theta \\
&= \pi
\end{aligned}$$

$$\begin{aligned}
I_1 &= \int_0^\pi \sin(\theta)\, d\theta \\
&= (-\cos(\pi)) - (-\cos(0)) \\
&= 2
\end{aligned}$$

Thus, we have:

$$\begin{aligned}
I_{2k} &= \pi \cdot \prod_{i=1}^{k} \frac{2i-1}{2i} \\
I_{2k+1} &= 2 \cdot \prod_{i=1}^{k} \frac{2i}{2i+1}
\end{aligned}$$

This gives us:

$$\begin{aligned} I_{2k+1}I_{2k} &= 2\pi \cdot \prod_{i=1}^{k} \frac{2i-1}{2i} \cdot \prod_{i=1}^{k} \frac{2i}{2i+1} \\ &= 2\pi \cdot \prod_{i=1}^{k} \frac{2i-1}{2i+1} \\ &= \frac{2\pi}{2k+1} \\ &= \frac{\pi}{k+\frac{1}{2}} \end{aligned}$$

$$\begin{aligned} I_{2k}I_{2k-1} &= 2\pi \cdot \prod_{i=1}^{k} \frac{2i-1}{2i} \cdot \prod_{i=1}^{k-1} \frac{2i}{2i+1} \\ &= 2\pi \cdot \frac{2k-1}{2k} \cdot \prod_{i=1}^{k-1} \frac{2i-1}{2i} \cdot \prod_{i=1}^{k-1} \frac{2i}{2i+1} \\ &= 2\pi \cdot \frac{2k-1}{2k} \cdot \frac{1}{2k-1} \\ &= \frac{\pi}{k} \end{aligned}$$

We know that

$$\lambda(B_1^1(0)) = \lambda((-1,1)) = 2$$

And, recalling the fact that λ^0 is the counting measure, we also know that

$$\lambda^0(B_1^0(0)) = \lambda^0(\{()\}) = 1$$

Thus, at long last, we can calculate the volumes of our unit balls:

$$\begin{aligned} \lambda^{2k}(B_1^{2k}(0)) &= (I_{2k}I_{2k-1}) \cdot (I_{2k-2}I_{2k-3}) \cdot \ldots \cdot (I_2 I_1) \cdot \lambda^0(B^0(0)) \\ &= \left(\prod_{i=1}^{k} \frac{\pi}{i} \right) \cdot 1 \\ &= \frac{\pi^k}{k!} = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)} \end{aligned}$$

$$\begin{aligned}
\lambda^{2k+1}(B_1^{2k+1}(0)) &= (I_{2k+1}I_{2k}) \cdot (I_{2k-1}I_{2k-2}) \cdot \ldots \cdot (I_3 I_2) \cdot \lambda^1(B^1(0)) \\
&= \left(\prod_{i=1}^{k} \frac{\pi}{i + \frac{1}{2}} \right) \cdot 2 \\
&= \frac{\pi^k}{\prod_{i=1}^{k} i + \frac{1}{2}} \cdot 2 \\
&= \frac{\pi^k}{\prod_{i=0}^{k} i + \frac{1}{2}} \\
&= \frac{\pi^k \cdot \sqrt{\pi}}{\left(\prod_{i=0}^{k} i + \frac{1}{2} \right) \cdot \sqrt{\pi}} \\
&= \frac{\pi^{\frac{n}{2}}}{\left(\prod_{i=0}^{k} i + \frac{1}{2} \right) \cdot \Gamma\left(\frac{1}{2}\right)} \\
&= \frac{\pi^{\frac{n}{2}}}{\left(\prod_{i=1}^{k+1} i - \frac{1}{2} \right) \cdot \Gamma\left(\frac{1}{2}\right)} \\
&= \frac{\pi^{\frac{n}{2}}}{\Gamma\left(k + \frac{1}{2} + 1\right)} \\
&= \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2} + 1\right)}
\end{aligned}$$

Which gives us

$$\lambda^n(B_r^n(0)) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)} \cdot r^n$$

as desired. □

Proposition 45.2.8. The 5-dimensional unit ball has the highest volume out of any unit ball. As the dimension increases, the volume of the unit ball converges to zero.

Proof Sketch. Plugging small natural numbers into our formula, we see that the formula is monotonically increasing for $n \leq 5$ and monotonically decreasing for $n \geq 5$. Furthermore, we have $\Gamma(\frac{n}{2} + 1) \geq \Gamma(k + 1) = k!$, which grows asymptotically faster than $\pi^{n/2} \leq \sqrt{\pi} \cdot \pi^k$. "□"

Lemma 45.2.9

Let $j \in \{1, \ldots, n\}$. Let $f : \mathbb{R}^n \to \mathbb{R}$ be continously differentiable with $f, \partial_j f \in L^1(\mathbb{R}^n)$. Then

$$\int_{\mathbb{R}^n} \partial_j f(x) \, d\lambda^n(x) = 0$$

Proof. 1. First, assume there is an $R > 0$ such that $f(x) = 0$ for all $\|x\|_2 \geq R$. Let $x = (y, x_n) \in \mathbb{R}^{n-1} \times \mathbb{R}$. Then by Fubini we get:

$$\begin{aligned}\int_{\mathbb{R}^n} \partial_j f(x)\, d\lambda^n(x) &= \int_{\mathbb{R}^n} \partial_j f(y, x_n)\, d\lambda^n(y, x_n) \\ &= \int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}} \partial_j f(y, x_n)\, d\lambda^1(x_n)\, d\lambda^{n-1}(y).\end{aligned}$$

For $j = n$, we apply the fundamental theorem of calculus and our assumption that $f(x) = 0$ for all $\|x\|_2 \geq R$ and get:

$$\begin{aligned}\int_{\mathbb{R}} \partial_j f(y, x_n)\, d\lambda^1(x_n) &= \lim_{x_n \to \infty} f(y, x_n) - f(y, -x_n) \\ &= 0\end{aligned}$$

By induction, the same holds for all j.

2. Now, let $\varphi : \mathbb{R} \to \mathbb{R}$ be a smooth function fulfilling:

$$\varphi(t) = \begin{cases} 1 & |t| \leq 1, \\ 0 & |t| \geq 2. \end{cases}$$

And let

$$\begin{aligned}\varphi_R : \mathbb{R}^n &\to \mathbb{R} \\ \varphi_R(x) &= \varphi\left(\frac{\|x\|_2}{R}\right)\end{aligned}$$

Then we have

$$(\varphi_R f)(x) = 0$$

for all $\|x\|_2 \geq 2R$, meaning that by the product rule of differentiation and step 1 we get:

$$\begin{aligned}\int_{\mathbb{R}^n} \varphi_R(x) \cdot \partial_j f(x)\, d\lambda^n(x) + \int_{\mathbb{R}^n} \partial_j \varphi_R(x) \cdot f(x)\, d\lambda^n(x) \\ = \int_{\mathbb{R}^n} \partial_j(\varphi_R(x) \cdot f(x))\, d\lambda^n(x) \\ = 0,\end{aligned}$$

i.e.

$$\int_{\mathbb{R}^n} \varphi_R(x) \cdot \partial_j f(x)\, d\lambda^n(x) = -\int_{\mathbb{R}^n} \partial_j \varphi_R(x) \cdot f(x)\, d\lambda^n(x)$$

Now, notice we also have $\varphi_R(x) = 1$ for $\|x\| \leq R$ - in particular, we have $\varphi_R \to 1$ and $\partial_j \varphi_R \to 0$ pointwise on $\mathbb{R}^n$ for $R \to \infty$, which means we have

$$\int_{\mathbb{R}^n} \partial_j f(x)\, d\lambda^n(x) = \int_{\mathbb{R}^n} \lim_{R \to \infty} \varphi_R(x) \cdot \partial_j f(x)\, d\lambda^n(x)$$

and

$$\int_{\mathbb{R}^n} \lim_{R \to \infty} \partial_j \varphi_R(x) \cdot f(x)\, d\lambda^n(x) = 0$$

Finally, we also have:

$$\left|\varphi_R \cdot (\partial_j f)\right| \leq \left|\partial_j f\right| \in L^1(\mathbb{R}^n)$$

and

$$\begin{aligned}\left|(\partial_j \varphi_R) \cdot f\right| &\leq \frac{\max\left|\varphi'\right|}{R} \cdot \left|f\right| \\ &\leq (\max\left|\varphi'\right|) \cdot \left|f\right| \in L^1(\mathbb{R}^n)\end{aligned}$$

Thus, we can apply Lebesgue's dominated convergence theorem to pull out the limits and connect the two equalities, giving us:

$$\begin{aligned}\int_{\mathbb{R}^n} \partial_j f(x)\, d\lambda^n(x) &= \int_{\mathbb{R}^n} \lim_{R\to\infty} \varphi_R(x) \cdot \partial_j f(x)\, d\lambda^n(x) \\ &= \lim_{R\to\infty} \int_{\mathbb{R}^n} \varphi_R(x) \cdot \partial_j f(x)\, d\lambda^n(x) \\ &= -\lim_{R\to\infty} \int_{\mathbb{R}^n} \partial_j \varphi_R(x) \cdot f(x)\, d\lambda^n(x) \\ &= -\int_{\mathbb{R}^n} \lim_{R\to\infty} \partial_j \varphi_R(x) \cdot f(x)\, d\lambda^n(x) \\ &= 0\end{aligned}$$

□

Corollary 45.2.10: Partial Integration-ish Formula

Let $f, g : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable such that $f, \partial_j f \in L^p(\mathbb{R}^n)$ and $g, \partial_j g \in L^q(\mathbb{R}^n)$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then we have

$$\int_{\mathbb{R}^n} (\partial_j f(x)) \cdot g(x)\, d\lambda^n(x) = 0 - \int_{\mathbb{R}^n} f(x) \cdot (\partial_j g(x))\, d\lambda^n(x)$$

Proof. Applying the product rule of differentiation, and then the last lemma, we get:

$$\begin{aligned}\int_{\mathbb{R}^n} (\partial_j f(x)) \cdot g(x)\, d\lambda^n(x) + \int_{\mathbb{R}^n} f(x) \cdot (\partial_j g(x))\, d\lambda^n(x) \\ = \int_{\mathbb{R}^n} \partial_j (f(x) \cdot g(x))\, d\lambda^n(x) \\ = 0\end{aligned}$$

□

45.3 Change of Variables

Lemma 45.3.1. Let $U \subset \mathbb{R}^n$ and $x_0 \in U$. Let $\varphi : U \to \mathbb{R}^n$ be a function such that $D\varphi(x_0)$ is invertible.

Then for a sequence $Q_j = Q(x_j, r_j) \subset U$ of cuboids of sidelength r_j with center x_j such that $r_j \to 0$ and $x_0 \in Q_j$, we have:

$$\limsup_{j\to\infty} \frac{\lambda^n(\varphi(Q_j))}{\lambda^n(Q_j)} \leq \left|\det D\varphi(x_0)\right|$$

Proof. We can assume $x_0 = 0$ and $\varphi(0) = 0$, since otherwise we can translate space as needed before doing any calculations without breaking any of our assumptions.

1. Assume $D(\varphi(0)) = E_n$. Then, by definition of differentiablity and equivalence of norms on finite-dimensional vector spaces, we have

$$\begin{aligned} 0 &= \lim_{x\to 0} \frac{\left\|\varphi(x) - \varphi(0) - D\varphi(0)x\right\|\infty}{\|x\|_\infty} \\ &= \lim_{x\to 0} \frac{\left\|\varphi(x) - x\right\|_\infty}{\|x\|_\infty}, \end{aligned}$$

   Let $\varepsilon > 0$. Then, by the definition of convergence, for every x with a sufficiently small norm, we have:

$$\frac{\left\|\varphi(x) - x\right\|_\infty}{\|x\|_\infty} \leq \varepsilon,$$

   which means

$$\left\|\varphi(x) - x\right\|_\infty \leq \varepsilon\|x\|_\infty.$$

   Furthermore, for $x \in Q_j$, we have:

$$\begin{aligned} \|x\|_\infty &= \left\|x - \vec{0}\right\|_\infty \\ &= \|x - x_0\|_\infty \\ &\leq \left\|x - x_j\right\| + \left\|x_j - x_0\right\| \\ &\leq 2r_j \end{aligned}$$

   For sufficiently large j, these imply:

$$\begin{aligned} \left\|\varphi(x) - x\right\|_\infty &\leq \varepsilon\|x\|_\infty \\ &\leq 2\varepsilon r_j. \end{aligned}$$

   Further applying the triangle inequality, we get:

$$\begin{aligned} \left\|\varphi(x) - \varphi(x)\right\| &\leq \left\|\varphi(x) - x\right\|_\infty \\ &\quad + \left\|x - x_j\right\|_\infty \\ &\quad + \left\|x_j - \varphi(x_j)\right\|_\infty \\ &\leq 2\varepsilon r_j + r_j + 2\varepsilon r_j \\ &\leq (1 + 4\varepsilon)r_j, \end{aligned}$$

which means that φ increases the side length of our cube by a factor of at most $(1+4\varepsilon)$. Therefore, it increases the volume by a factor at most $(1+4\varepsilon)^n$, i.e:

$$\frac{\lambda^n(\varphi(Q_j))}{\lambda^n(Q_j)} \leq (1+4\varepsilon)^n$$

Letting $j \to \infty$ and $\varepsilon \searrow 0$, we have

$$\begin{aligned}\limsup_{j\to\infty} \frac{\lambda^n(\varphi(Q_j))}{\lambda^n(Q_j)} &\leq \lim_{\varepsilon\to 0}(1+4|\varepsilon|)^n \\ &= 1 \\ &= |\det E_n|\end{aligned}$$

2. Now, let $S := D\varphi(0)$ and $\varphi_0 := S^{-1} \circ \varphi$, i.e. $\varphi = S \circ \varphi_0$. Then $D\varphi_0(0) = E_n$. By the linear transformation equation $\lambda^n(S(E)) = |\det(S)|\lambda^n(E)$ (42.9.2), we have:

$$\begin{aligned}\limsup_{j\to\infty} \frac{\lambda^n(\varphi(Q_j))}{\lambda^n(Q_j)} &= \limsup_{j\to\infty} \frac{\lambda^n(S(\varphi_0(Q_j)))}{\lambda^n(Q_j)} \\ &= |\det S| \limsup_{j\to\infty} \frac{\lambda^n(\varphi_0(Q_j))}{\lambda^n(Q_j)} \\ &\leq |\det S| \\ &= \left|\det D\varphi(0)\right|\end{aligned}$$

□

Theorem 45.3.2: Multivariable Substitution Formula

Let $U \subset \mathbb{R}^n$ be open. Let $\varphi : U \to \mathbb{R}^n$ be C^1. Then if $f : V \to \overline{\mathbb{R}}$ is λ^n-measurable, we have:

$$\int_V f(y)\, dy = \int_{\varphi^{-1}(V)} f(\varphi(x)) \cdot \left|\det D\varphi(x)\right| dx.$$

Corollary 45.3.3

Let $U, V \subset \mathbb{R}^n$ be open. Let $\varphi : U \to \mathbb{R}^n$ be C^1. Then if $A \subset U$ is λ^n-measurable, so is $\varphi(A)$, and we have

$$\lambda^n(\varphi(A)) = \int_A \left|\det D\varphi(x)\right| dx.$$

Proof. Apply the previous equation to $f = \mathbb{1}_{\varphi(A)}$. □

Definition 45.3.4: Gram Matrix of a Diffeomorphism

Let $\varphi : U \to V$ be a C^k-diffeomorphism between open subsets of $\mathbb{R}^n$. Then the **Gram matrix** $g(\varphi) \in C^{k-1}(U, \mathbb{R}^{n\times n})$ is given by:

$$g(\varphi)(x) = D\varphi(x)^\top D\varphi(x),$$

or equivalently:

$$g(\varphi)_{ij}(x) = \left\langle \frac{\partial}{\partial x_i}\varphi(x), \frac{\partial}{\partial x_j}\varphi(x) \right\rangle$$

Since $g(\varphi)(x)$ is the product of a matrix with its own transpose, it is symmetric and strictly positive definite. This lets us alternatively write the multivariable substitution formula as:

$$\int_V f(y)dy = \int_{\varphi^{-1}(V)} f(\varphi(x))\sqrt{\det g(\varphi)(x)}\, dx$$

45.4 Alternative coordinate systems on $\mathbb{R}^2$ and $\mathbb{R}^3$

45.4.1 Polar Coordinates

A commonly used alternative coordinate system on $\mathbb{R}^2$ is given by **polar coordinates**, which describes a point $x \in \mathbb{R}^2$ based on its distance r from the origin and the *polar angle* ω between x and $(0, r)^\top$. These might be familiar from dealing with complex numbers, where any $z \in \mathbb{C}$ can be represented as

$$z = r \cdot e^{\omega i}$$

Definition 45.4.1: Polar Coordinates on $\mathbb{R}^2$

Let $U = (0, \infty) \times (0, 2\pi)$ and $V = \mathbb{R}^2 \setminus \{(x, 0)^\top : x \geq 0\}$. The **polar coordinate transformation** is the map:

$$\begin{aligned} \varphi : U &\to V \\ \begin{pmatrix} r \\ \omega \end{pmatrix} &\mapsto \begin{pmatrix} r\cos\omega \\ r\sin\omega \end{pmatrix} \end{aligned}$$

which maps a point given in polar coordinates to its corresponding point in the cartesian coordinate system.

The Jacobian of the polar coordinate transform is:

$$\begin{aligned} D\varphi(r, \omega) &= \begin{pmatrix} \frac{\partial}{\partial r} r\cos\omega & \frac{\partial}{\partial \omega} r\cos\omega \\ \frac{\partial}{\partial r} r\sin\omega & \frac{\partial}{\partial \omega} r\sin\omega \end{pmatrix} \\ &= \begin{pmatrix} \cos\omega & -r\sin\omega \\ \sin\omega & r\cos\omega \end{pmatrix} \end{aligned}$$

Which means the determinant is:

$$\begin{aligned} \det D\varphi(r, \omega) &= \det \begin{pmatrix} \cos\omega & -r\sin\omega \\ \sin\omega & r\cos\omega \end{pmatrix} \\ &= r\cos(\omega)^2 + r\sin(\omega)^2 \\ &= r \end{aligned}$$

Corollary 45.4.2: Polar Coordinate Integral Transform

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be integrable. Then we have:

$$\int_{\mathbb{R}^2} f(x, y)\, d\lambda(x, y) = \int_0^{2\pi} \int_0^{\infty} r \cdot f(r\cos(\omega), r\sin(\omega))\, dr\, d\omega$$

Proposition 45.4.3. The inverse of the polar coordinate transform φ is:

$$\varphi^{-1}(x, y) = \begin{cases} \left(r,\ \arccos\left(\frac{x}{r}\right)\right)^\top & y \geq 0 \\ \left(r,\ 2\pi - \arccos\left(\frac{x}{r}\right)\right)^\top & y < 0 \end{cases}$$

where $r = \sqrt{x^2 + y^2}$.

Example 45.4.4. (The Gaussian Integral): We want to find the area under the Gaussian bell curve pre-normalization, i.e.

$$\int_{\mathbb{R}} e^{-x^2}\, dx$$

To do this, we add an additional dimension and exploit the resulting rotational symmetry. By Fubini's Theorem, we have:

$$\begin{aligned}
\int_{\mathbb{R}^2} e^{-(x^2+y^2)}\, d\lambda^2(x,y) &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} e^{-(x^2+y^2)}\, d\lambda(y) \right) d\lambda(x) \\
&= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} e^{-x^2} \cdot e^{-y^2}\, d\lambda(y) \right) d\lambda(x) \\
&= \int_{\mathbb{R}} e^{-x^2} \cdot \left(\int_{\mathbb{R}} e^{-y^2}\, d\lambda(y) \right) d\lambda(x) \\
&= \int_{\mathbb{R}} e^{-x^2}\, d\lambda(x) \cdot \int_{\mathbb{R}} e^{-y^2}\, d\lambda(y) \\
&= \left(\int_{\mathbb{R}} e^{-x^2}\, d\lambda(x) \right)^2
\end{aligned}$$

Therefore, we have

$$\int_{\mathbb{R}} e^{-x^2}\, d\lambda(x) = \sqrt{\int_{\mathbb{R}^2} e^{-(x^2+y^2)}\, d\lambda(x,y)^2}$$

We can now calculate the two-dimensional integral using multivariable substitution to transform to polar coordinates:

$$\begin{aligned}
\int_{\mathbb{R}^2} e^{-(x^2+y^2)}\, d\lambda^2(x,y) &= \int_{(0,\infty)\times(0,2\pi)} r e^{-r^2}\, d\lambda^2(r,\omega) \\
&= \int_0^\infty \left(\int_0^{2\pi} r e^{-r^2}\, d\omega \right) dr \\
&= \int_0^\infty 2\pi r e^{-r^2}\, dr \\
&= 2\pi \int_0^\infty r e^{-r^2}\, dr \\
&= 2\pi \int_0^\infty \frac{1}{2} e^{-r^2}\, 2r dr \\
&= 2\pi \int_{-\infty}^0 \frac{1}{2} e^s\, ds \\
&= \pi \int_{-\infty}^0 e^s\, ds \\
&= \pi
\end{aligned}$$

Which means that the area under the bell curve is $\sqrt{\pi}$.

Example 45.4.5. (Beta function): For $p, q \in \mathbb{R}_{>0}$, the beta function B is defined as:

$$B(p,q) = \int_0^1 t^{p-1} \cdot (1-t)^{q-1}\, dt$$

The beta function is of interest primarily because of its close relationship to the gamma function:

Theorem 45.4.6. We have:

$$B(p,q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$$

Proof. Substituting $t = g(\omega) := \sin(\omega)^2$ into the definition of the beta function gives us:

$$\begin{aligned} g'(\omega) &= \frac{\mathrm{d}}{\mathrm{d}\omega} \sin(\omega)^2 \\ &= \sin(\omega)\cos(\omega) + \cos(\omega) + \sin(\omega) \\ &= 2\sin(\omega)\cos(\omega) \end{aligned}$$

and:

$$1 = \sin(\omega)^2 \Leftrightarrow \omega = \frac{\pi}{2},$$

which means we have:

$$\begin{aligned} B(p,q) &= \int_0^1 t^{p-1} \cdot (1-t)^{q-1}\, dt \\ &= \int_0^{\pi/2} \sin(\omega)^{2p-2} \cdot (1-\sin(\omega)^2)^{q-1} \cdot 2\sin(\omega)\cos(\omega)\, d\omega \\ &= \int_0^{\pi/2} \sin(\omega)^{2p-2} \cdot \cos(\omega)^{2q-2} \cdot 2\sin(\omega)\cos(\omega)\, d\omega \\ &= 2 \cdot \int_0^{\pi/2} \sin(\omega)^{2p-1} \cdot \cos(\omega)^{2q-1}\, d\omega \end{aligned}$$

Meanwhile, performing a reverse substitution with $t := g(r) = r^2$ in the definition of the gamma function gives:

$$\begin{aligned} \Gamma(m) &= \int_0^\infty t^{m-1} e^{-t}\, dt \\ &= \int_0^\infty (r^2)^{m-1} e^{-r^2} \cdot 2r\, dr \\ &= 2\int_0^\infty r^{2m-1} e^{-r^2}\, dr \end{aligned}$$

Now, we get:

$$\begin{aligned} B(p,q)\Gamma(p+q) &= 2\int_0^\infty B(p,q) \cdot r^{2p+2q-1} \cdot e^{-r^2}\, dr \\ &= 4\int_0^\infty \int_0^{\pi/2} (\sin\omega)^{2p-1} \cdot r^{2p-1} \cdot (\cos\omega)^{2q-1} \cdot r^{2q-1} \cdot r \cdot e^{-r^2}\, d\omega\, dr \\ &= 4\int_0^\infty \int_0^{\pi/2} (r\sin\omega)^{2p-1} \cdot (r\cos\omega)^{2q-1} \cdot r \cdot e^{-r^2}\, d\omega\, dr \end{aligned}$$

By applying the polar coordinate integral transform in reverse, we get:

$$\begin{aligned}
B(p,q)\Gamma(p+q) &= 4\int_0^\infty \int_0^{\pi/2} (r\sin\omega)^{2p-1} \cdot (r\cos\omega)^{2q-1} \cdot r \cdot e^{-r^2}\ d\omega\ dr \\
&= 4\int_0^\infty \int_0^\infty x^{2p-1} \cdot y^{2q-1} \cdot e^{-\sqrt{x^2+y^2}^2}\ dy\ dx \\
&= 4\int_0^\infty \int_0^\infty x^{2p-1} \cdot y^{2q-1} \cdot e^{-x^2-y^2}\ dy\ dx \\
&= \left(2\int_0^\infty x^{2p-1}e^{-x^2}\ dx\right) \cdot \left(2\int_0^\infty y^{2q-1}e^{-y^2}\ dy\right) \\
&= \Gamma(p)\Gamma(q)
\end{aligned}$$

□

45.4.2 Spherical Coordinates

The most natural generalization of polar coordinates to $\mathbb{R}^3$ are *spherical coordinates*, in which $\theta \in (0, \pi)$ describes the *polar angle* between a point P and the z-Axis and $\omega \in (0, 2\pi)$ describes the *azimuthal angle* between the x-axis and the projection P' of P onto the xy-plane. We can derive spherical coordinates from two dimensional coordinates as follows:

1. Take a point $P = (x, y, z) \in \mathbb{R}^3$. Notice that $z = r\cos(\theta)$.
2. Project P to a point $P' = (x, y)$. The distance of P' from $(0, 0)$ is exactly $\rho := r\sin\theta$. Therefore, P' has polar coordinates (ρ, ω), which gives us:

$$P = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \rho\cos\omega \\ \rho\sin\omega \\ r\cos\theta \end{pmatrix} = \begin{pmatrix} r\sin\theta\cos\omega \\ r\sin\theta\sin\omega \\ r\cos\theta \end{pmatrix}$$

Definition 45.4.7: Spherical Coordinates on $\mathbb{R}^3$

Let $U = (0, \infty) \times (0, \pi) \times (0, 2\pi)$ and $V = \mathbb{R}^3 \setminus \{(x, 0, z)^\top : x \geq 0\}$. The **spherical coordinate transformation** is the map:

$$\begin{aligned} \varphi : U &\to V \\ \begin{pmatrix} r \\ \theta \\ \omega \end{pmatrix} &\mapsto \begin{pmatrix} r\sin\theta\cos\omega \\ r\sin\theta\sin\omega \\ r\cos\theta \end{pmatrix} \end{aligned}$$

which maps a point given in spherical coordinates to its corresponding point in the cartesian coordinate system.

The Jacobian of the spherical coordinate transform is:

$$\begin{aligned} D\varphi(r, \theta, \omega) &= \begin{pmatrix} \frac{\partial}{\partial r} r\sin\theta\cos\omega & \frac{\partial}{\partial\theta} r\sin\theta\cos\omega & \frac{\partial}{\partial\omega} r\sin\theta\cos\omega \\ \frac{\partial}{\partial r} r\sin\theta\sin\omega & \frac{\partial}{\partial\theta} r\sin\theta\sin\omega & \frac{\partial}{\partial\omega} r\sin\theta\sin\omega \\ \frac{\partial}{\partial r} r\cos\theta & \frac{\partial}{\partial\theta} r\cos\theta & \frac{\partial}{\partial\omega} r\cos\theta \end{pmatrix} \\ &= \begin{pmatrix} \sin\theta\cos\omega & r\cos\theta\cos\omega & -r\sin\theta\sin\omega \\ \sin\theta\sin\omega & r\cos\theta\sin\omega & r\sin\theta\cos\omega \\ \cos\theta & -r\sin\theta & 0 \end{pmatrix} \end{aligned}$$

And a long, tedious and unenlightening calculation eventually gives the determinant as:

$$\det D\varphi(r, \theta, \omega) = r^2\sin(\theta)$$

Which gives us:

Corollary 45.4.8: Spherical Coordinate Integral Transform

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be integrable. Then we have:

$$\begin{aligned} &\int_{\mathbb{R}^3} f(x, y, z)\, d\lambda^3(x, y, z) \\ &= \int_0^{2\pi}\int_0^{\pi}\int_0^{\infty} r^2\sin\theta \cdot f(r\sin\theta\sin\omega, r\sin\theta\cos\omega, r\cos\theta)\, dr\, d\theta\, d\omega \end{aligned}$$

Example 45.4.9. (Another derivation of the volume of B_R^3):

$$\begin{aligned}
\lambda^3(B_R^3) &= \int_{\mathbb{R}^3} \mathbb{1}_{B_R^3}(x,y,z)\, d\lambda^3(x,y,z) \\
&= \int_{\mathbb{R}^3} \mathbb{1}_{\left\{\sqrt{x^2+y^2+z^2} < R\right\}}(x,y,z)\, d\lambda^3(x,y,z) \\
&= \int_0^{2\pi}\int_0^{\pi}\int_0^{\infty} \mathbb{1}_{\{r<R\}}(r)\cdot r^2\sin\theta\, dr\, d\theta\, d\omega \\
&= \int_0^{2\pi}\int_0^{\pi}\int_0^{\infty} \mathbb{1}_{(0,R)}(r)\cdot r^2\sin\theta\, dr\, d\theta\, d\omega \\
&= \left(\int_0^{\infty} \mathbb{1}_{(0,R)}(r)\cdot r^2\, dr\right)\cdot\left(\int_0^{\pi}\sin\theta\, d\theta\right)\cdot\left(\int_0^{2\pi} 1\, d\omega\right) \\
&= \left(\int_0^{R} r^2\, dr\right)\cdot\left(\int_0^{\pi}\sin\theta\, d\theta\right)\cdot\left(\int_0^{2\pi} 1\, d\omega\right) \\
&= \frac{1}{3}R^3\cdot(-\cos(\pi)+\cos(0))\cdot 2\pi \\
&= \frac{4}{3}\pi R^3
\end{aligned}$$

Example 45.4.10. (Gravitational Potential of a Sphere:): The gravitational potential that a body $B \subset \mathbb{R}^3$ with mass distribution $m : B \to \mathbb{R}$ exerts on a point $x \in \mathbb{R}^3 \setminus B$ is given by:

$$V(x) = -G\cdot\int_B \frac{m(r)}{\|x-y\|_2}\, d\lambda^3(y)$$

We will derive a closed form expression for the case $B = B_R^3$. If we assume that the mass distribution is rotationally symmetric, we can replace $m : \mathbb{R}^3 \to \mathbb{R}$ with a function $\widetilde{m} = m(\|y\|_2) : \mathbb{R} \to \mathbb{R}$. We will commit a slight abuse of notation here and let m be this new function $\widetilde{m}$ from now on. Because of this rotational symmetry, we also have $V(x) = V(0,0,\|x\|)$.

Now, if y has spherical coordinates (r,θ,ω), then assuming y lies outside of our ball we can use the Pythagorean theorem to get:

$$\begin{aligned}
\|x-y\|_2^2 &= (\|x\|_2 - r\cos\theta)^2 + r^2\sin^2\theta \\
&= \|x\|_2^2 + r^2 - 2r\|x\|_2\cos\theta
\end{aligned}$$

Thus, we have:

$$\begin{aligned}
V(x) &= -G\cdot\int_{B_R^3}\frac{m(r)}{\|x-y\|_2}\, d\lambda^3(y) \\
&= -G\int_0^{2\pi}\int_0^{\pi}\int_0^{R}\left(\frac{m(r)}{\sqrt{\|x\|_2^2+r^2-2r\|x\|_2\cos\theta}}\cdot r^2\sin\theta\right) dr\, d\theta\, d\omega \\
&= -2\pi G\int_0^{R} r^2\cdot m(r)\cdot\left(\int_0^{\pi}\frac{\sin\theta}{\sqrt{\|x\|_2^2+r^2-2r\|x\|_2\cos\theta}}\, d\theta\right) dr
\end{aligned}$$

Substituting $t = g(\theta) = -\cos\theta$ into the inner integral, we get:

$$\begin{aligned}
&\int_0^\pi \frac{\sin\theta}{\sqrt{\|x\|_2^2 + r^2 - 2r\|x\|_2 \cos\theta}}\, d\theta \\
=&\int_{-1}^1 \frac{1}{\sqrt{\|x\|_2^2 + r^2 + 2r\|x\|_2 t}}\, dt \\
=&\left[\frac{1}{r\cdot\|x\|_2}\cdot\sqrt{\|x\|_2^2 + r^2 + 2r\|x\|_2 t}\right]_{t=-1}^{t=1} \\
&= \frac{1}{r\cdot\|x\|_2}\cdot\left(\sqrt{\|x\|_2^2 + r^2 + 2r\|x\|_2} - \sqrt{\|x\|_2^2 + r^2 - 2r\|x\|_2}\right) \\
&= \frac{1}{r\cdot\|x\|_2}\cdot((\|x\|_2 + r) - \|x\|_2 - r) \\
&= \frac{2}{\|x\|_2}
\end{aligned}$$

Therefore, we have:

$$V(x) = \frac{4\pi}{\|x\|_2}\cdot\int_0^R m(r)r^2\, dr$$

Now, we can once again use the rotational symmetry of m to describe the total mass M of our body in terms of spherical coordinates:

$$\begin{aligned}
M &= \int_{B_R^3(0)} m(y)\, d\lambda^3(y) \\
&= \int_0^{2\pi}\int_0^\pi\int_0^R m(r)r^2\sin\theta\, dr\, d\theta\, d\omega \\
&= 4\pi\int_0^R m(r)r^2\, dr
\end{aligned}$$

Thus, the total mass distribution of our sphere is simply:

$$V(x) = -\frac{GM}{\|x\|}$$

which is exactly the same mass distribution we would get if we concentrated the entire mass of our sphere at its center. This relation is part of a theorem in classical mechanics known as the **shell theorem**.

45.4.3 Cylindrical Coordinates

Alternatively, one can trivially extend polar coordinates into three dimensions by simply leaving the z coordinate unchanged, leading to *cylindrical coordinates*:

Definition 45.4.11: Cylindrical Coordinates on $\mathbb{R}^3$

Let $U = (0,\infty)\times(0,2\pi)\times\mathbb{R}$ and $V = \mathbb{R}^3\setminus\{(x,0,z)^\top : x \geq 0\}$. The **spherical coordinate transformation** is the map:

$$\begin{aligned}\varphi : U &\to V\\ \begin{pmatrix} r\\ \omega\\ z\end{pmatrix} &\mapsto \begin{pmatrix} r\cos\omega\\ r\sin\omega\\ z\end{pmatrix}\end{aligned}$$

which maps a point given in cylindrical coordinates to its corresponding point in the cartesian coordinate system.

Since the cylindrical coordinate transform leaves the z coordinate unchanged, its Jacobian and its determinant end up being almost identical:

$$\begin{aligned}D\varphi(r,\omega,z) &= \begin{pmatrix} \frac{\partial}{\partial r}r\cos\omega & \frac{\partial}{\partial \omega}r\cos\omega & \frac{\partial}{\partial z}r\cos\omega\\ \frac{\partial}{\partial r}r\sin\omega & \frac{\partial}{\partial \omega}r\sin\omega & \frac{\partial}{\partial z}r\sin\omega\\ \frac{\partial}{\partial r}z & \frac{\partial}{\partial \omega}z & \frac{\partial}{\partial z}z\end{pmatrix}\\ &= \begin{pmatrix}\cos\omega & -r\sin\omega & 0\\ \sin\omega & r\cos\omega & 0\\ 0 & 0 & 1\end{pmatrix}\end{aligned}$$

$$\begin{aligned}\det D\varphi(r,\omega,z) &= \det\begin{pmatrix}\cos\omega & -r\sin\omega & 0\\ \sin\omega & r\cos\omega & 0\\ 0 & 0 & 1\end{pmatrix}\\ &= \det\begin{pmatrix}\cos\omega & -r\sin\omega\\ \sin\omega & r\cos\omega\end{pmatrix}\\ &= r\cos(\omega)^2 + r\sin(\omega)^2\\ &= r\end{aligned}$$

Corollary 45.4.12: Cylindrical Coordinate Integral Transform

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be integrable. Then we have:

$$\int_{\mathbb{R}^3} f(x,y,z)\, d\lambda(x,y,z) = \int_{\mathbb{R}}\int_0^{2\pi}\int_0^{\infty} r\cdot f(r\cos(\omega), r\sin(\omega), z)\, dr\, d\omega\, dz$$

Proposition 45.4.13. The inverse of the cylindrical coordinate transform φ is:

$$\varphi^{-1}(x,y,z) = \begin{cases}\left(r,\ \arccos\left(\frac{x}{r}\right),\ z\right)^\top & y \geq 0\\ \left(r,\ 2\pi - \arccos\left(\frac{x}{r}\right),\ z\right)^\top & y < 0\end{cases}$$

where $r = \sqrt{x^2+y^2}$.

Theorem 45.4.14: Volume of a Solid of Revolution

Let S be a solid constructed by rotating a function $\rho(z)$ around the z axis, i.e.

$$S = \left\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 < \rho(z)^2\right\}$$

Then we have:

$$\lambda^3(S) = \pi \int_{\mathbb{R}} \rho(z)^2 \, d\lambda^1(z)$$

Proof.

$$\begin{aligned}
\lambda^3(S) &= \int_{\mathbb{R}^3} \mathbb{1}_{\{x^2+y^2 \,<\, \rho(z)^2\}}(x, y, z) \, d\lambda^3(x, y, z) \\
&= \int_{\mathbb{R}} \int_0^{2\pi} \int_0^{\infty} r \cdot \mathbb{1}_{\{r^2 \,<\, \rho(z)^2\}}(r, \omega, z) \, dr \, d\omega \, dz \\
&= \int_{\mathbb{R}} \int_0^{2\pi} \int_0^{\infty} r \cdot \mathbb{1}_{\{r \,<\, \rho(z)\}}(r, \omega, z) \, dr \, d\omega \, dz \\
&= \int_{\mathbb{R}} \int_0^{2\pi} \int_0^{\rho(z)} r \, dr \, d\omega \, dz \\
&= \int_{\mathbb{R}} \int_0^{2\pi} \frac{1}{2}\rho(z)^2 \, d\omega \, dz \\
&= \int_{\mathbb{R}} \pi\rho(z)^2 \, dz \\
&= \pi \int_{\mathbb{R}} \rho(z)^2 \, dz
\end{aligned}$$

□

45.5 The Surface Area Measure

Definition 45.5.1. Let $U \supset \mathbb{R}^n$ be open. We call a map $f : U \to \mathbb{R}^{n+k}$ an *immersion* iff f is continuously differentiable and $Df(x)$ is invertible.

Definition 45.5.2. Let $M \subset \mathbb{R}^{n+k}$ be an n-dimensional C^1 submanifold. Then a *local parametrization of* M is an injective immersion $f : \mathbb{R}^n \supset U \to M \subset \mathbb{R}^{n+k}$

Definition 45.5.3: Gram Matrix

Let $U \subset \mathbb{R}^n$ and $f : U \to \mathbb{R}^{n+k}$. Then the *Gram matrix* $g_f(x)$ is the matrix

$$g_f(x) = Df(x)^\top Df(x)$$

Our goal now is to find a sensible definition of *surface area*, i.e. of an n-dimensional measure associated with an object embedded into $n + k$-dimensional space.

Corollary 45.5.4: Area of an Embedded Diffeomorphism of a Set

Let $f = S \circ \varphi$, where $\varphi : U \to V$ is a diffeomorphism with $U, V \subset \mathbb{R}^n$ and S is a linear isometry. Then we have

$$\lambda^n(\varphi(E)) = \int_E \sqrt{\det g_f}$$

This theorem suggests that it is sensible to use the same definition for arbitrary immersions:

Definition 45.5.5: Area of an Immersion of a Set

Let $f \in C^1(U, \mathbb{R}^{n+k})$ be an n-dimensional immersion. Let $E \subset U$ be λ^n-measurable. Then the n-dimensional area of $f(E)$ is given by:

$$A(f, E) = \int_E \sqrt{\det g_f}$$

$\sqrt{\det g_f}$ is known as the *Jacobian of* f, and often denoted Jf, but I dislike notation that hides the actual calculation being done behind multiple layers of nested notation and will thus be sticking to $\sqrt{\det g_f}$.

Corollary 45.5.6: Length of a Regular Curve

An Immersion $f : I = (a, b) \to \mathbb{R}^n$ is known as a *regular curve*. The one-dimensional area of the curve, i.e. its length, is given by

$$L(f) := A(f, I) = \int_a^b \left\| Df(t) \right\|_2 dt$$

Proof. We have:

$$Df(t) = \begin{pmatrix} f_1'(t) \\ \vdots \\ f_n'(t) \end{pmatrix}$$

and thus

$$g_f(t) = Df(t)^\top Df(t) = \left\| Df(t) \right\|_2^2$$

□

Note that if some areas of the curve are covered multiple times, then this length formula counts them multiple times as well.

Corollary 45.5.7: Area of a Regular Surface

An immersion $f : U \supset \mathbb{R}^2 \to \mathbb{R}^3$, with U open, is known as a *regular surface*. Its area is given by

$$\begin{aligned} A(f) := A(f, U) &= \int_U \sqrt{\left\| \frac{\partial}{\partial x} f \right\|_2^2 \cdot \left\| \frac{\partial}{\partial y} f \right\|_2^2 - \left\langle \frac{\partial}{\partial x} f, \frac{\partial}{\partial y} f \right\rangle^2} \, d\lambda^2(x, y) \\ &= \int_U \left\| \frac{\partial}{\partial x} f \times \frac{\partial}{\partial y} f \right\| d\lambda^2(x, y) \end{aligned}$$

Proof. We have

$$Df(x, y) = \begin{pmatrix} \frac{\partial}{\partial x} f(x, y) & \frac{\partial}{\partial y} f(x, y) \end{pmatrix},$$

therefore our Gram matrix is

$$g_f = \begin{pmatrix} \left\| \frac{\partial}{\partial x} f \right\|_2^2 & \left\langle \frac{\partial}{\partial x} f, \frac{\partial}{\partial y} f \right\rangle_2 \\ \left\langle \frac{\partial}{\partial x} f, \frac{\partial}{\partial y} f \right\rangle & \left\| \frac{\partial}{\partial y} f \right\|_2^2 \end{pmatrix},$$

giving our desired result for $\sqrt{\det g_f}$.´ □

Theorem 45.5.8: Area of a Graph

Let $U \subset \mathbb{R}^n$ be open, $u : U \to \mathbb{R}^k$ and

$$\begin{aligned} f : U &\to \mathbb{R}^{n+k} \\ x &\mapsto (x, u(x)) \end{aligned}$$

Then we have:

$$A(f, U) = \int_U \sqrt{\det(E_n + Du(x)^\top Du(x))}\, d\lambda^n(x)$$

If $k = 1$, i.e. $U \subset \mathbb{R}^n$, $u : U \to \mathbb{R}$, then we can simplify significantly and get:

$$A(f, U) = \int_U \sqrt{1 + \|Du(x)\|^2}\, d\lambda^n(x)$$

Theorem 45.5.9: Surface Measure

Let M be an n-dimension C^1-submanifold of $\mathbb{R}^{n+k}$ locally parametrized by $f : U \to M$. Let $E \subset M$. Then there exists a measure ω_M such that

$$\omega_M(E) = A(f, E)$$

Theorem 45.5.10: Surface of Revolution

Let S be a solid of revolution generated by $\rho(z)$. Then the surface area of S is:

$$\omega_S(S) = 2\pi \int_{\mathbb{R}} \rho(z) \sqrt{1 + \left(\frac{\mathrm{d}}{\mathrm{dz}}\rho(z)\right)^2}\, dz$$

Proof. This follows from our formula for the surface area of a regular surface. However, as additional practice, let us derive the formula from scratch. We can parametrize S up to a zero set via:

$$f : \begin{pmatrix} \theta \\ z \end{pmatrix} \mapsto \begin{pmatrix} \rho(z)\cos(\theta) \\ \rho(z)\sin(\theta) \\ z \end{pmatrix}$$

Which gives us:

$$\begin{aligned} Df(\theta, z) &= \begin{pmatrix} \frac{\partial}{\partial\theta}\rho(z)\cos(\theta) & \frac{\partial}{\partial z}\rho(z)\cos(\theta) \\ \frac{\partial}{\partial\theta}\rho(z)\sin(\theta) & \frac{\partial}{\partial z}\rho(z)\sin(\theta) \\ \frac{\partial}{\partial\theta}z & \frac{\partial}{\partial z}z \end{pmatrix} \\ &= \begin{pmatrix} -\rho(z)\sin(\theta) & \cos(\theta)\cdot\frac{\partial}{\partial z}\rho(z) \\ \rho(z)\cos(\theta) & \sin(\theta)\cdot\frac{\partial}{\partial z}\rho(z) \\ 0 & 1 \end{pmatrix} \end{aligned}$$

This gives us the Gram matrix:

$$\begin{aligned}
g_f(\theta, z) &= \begin{pmatrix} -\rho(z)\sin(\theta) & \rho(z)\cos(\theta) & 0 \\ \cos(\theta)\cdot\frac{\partial}{\partial z}\rho(z) & \sin(\theta)\cdot\frac{\partial}{\partial z}\rho(z) & 1 \end{pmatrix} \cdot \begin{pmatrix} -\rho(z)\sin(\theta) & \cos(\theta)\cdot\frac{\partial}{\partial z}\rho(z) \\ \rho(z)\cos(\theta) & \sin(\theta)\cdot\frac{\partial}{\partial z}\rho(z) \\ 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} \rho(z)^2\cdot(\sin^2(\theta)+\cos^2(\theta)) & \begin{matrix} -\rho(z)\sin(\theta)\cos(\theta)\frac{\mathrm{d}}{\mathrm{d}z}\rho(z) \\ +\rho(z)\sin(\theta)\cos(\theta)\frac{\mathrm{d}}{\mathrm{d}z}\rho(z) \end{matrix} \\ \begin{matrix} -\rho(z)\sin(\theta)\cos(\theta)\frac{\mathrm{d}}{\mathrm{d}z}\rho(z) \\ +\rho(z)\sin(\theta)\cos(\theta)\frac{\mathrm{d}}{\mathrm{d}z}\rho(z) \end{matrix} & (\cos^2(\theta)+\sin^2(\theta)\left(\frac{\partial}{\partial z}\rho(z)\right)^2+1 \end{pmatrix} \\
&= \begin{pmatrix} \rho(z)^2 & 0 \\ 0 & \left(\frac{\partial}{\partial z}\rho(z)\right)^2+1 \end{pmatrix}
\end{aligned}$$

and thus:

$$\begin{aligned}
\sqrt{\left|\det g_f(\theta, z)\right|} &= \sqrt{\rho(z)^2\left(\frac{\partial}{\partial z}\rho(z)\right)^2+\rho(z)^2} \\
&= \rho(z)\sqrt{1+\frac{\partial}{\partial z}\rho(z))^2}
\end{aligned}$$

Which leaves us with:

$$\begin{aligned}
\omega_S(S) &= \int_0^{2\pi}\int_{\mathbb{R}} \rho(z)\sqrt{1+\frac{\partial}{\partial z}\rho(z))^2}\,dz\,d\theta \\
&= 2\pi\int_{\mathbb{R}} \rho(z)\sqrt{1+\frac{\partial}{\partial z}\rho(z))^2}\,dz
\end{aligned}$$

□

Example 45.5.11. (Gabriel's Horn): We want to show that the solid of rotation G generated by $z \in [1, \infty)$ and $\rho(z) = \frac{1}{z}$ has finite volume, but infinite surface area.

1. We already showed in the section on cylindrical coordinates that the volume is simply:

$$\begin{aligned}
\lambda^3(G) &= \pi\cdot\int_1^\infty \rho(z)^2\,dz \\
&= \pi\cdot\int_1^\infty \frac{1}{z^2}\,dz \\
&= -\frac{\pi}{2}\cdot\left[\frac{1}{z}\right]_{z=1}^{z=\infty} \\
&= -\frac{\pi}{2}\cdot(0-1) \\
&= \frac{\pi}{2}
\end{aligned}$$

2. We have just shown that the surface area is given by:

$$\begin{aligned}\omega_S(S) &= 2\pi \int_{\mathbb{R}} \rho(z)\sqrt{1 + \frac{\partial}{\partial z}\rho(z))^2}\, dz \\ &= 2\pi \int_1^\infty \frac{1}{z}\sqrt{1 - \frac{1}{z^2}}\, dz \\ &> 2\pi \int_1^\infty \frac{1}{z}\, dz \\ &= 2\pi(\ln(\infty) - \ln(1)) \\ &= \infty\end{aligned}$$

Theorem 45.5.12: Surface Integral

Let $M \subset \mathbb{R}^{n+k}$ be an n-dimensional C^1 submanifold of $\mathbb{R}^{n+k}$, and let $M = \bigsqcup_{i\in I} M_i$ be a decomposition of M into countably many pairwise disjoint measure sets such that $M_i \subset V_i$ for a local parametrization $f_i : U_i \to V_i$. Then for any measurable function $u : M \to \overline{\mathbb{R}}$, we have:

$$\int_M u(x)\, d\omega_M(x) = \sum_{i\in I} \int_{f_i^{-1}(M_i)} u(f_i(x)) \cdot \left|\sqrt{\det g_{f_i}(x)}\right| d\lambda^n(x)$$

In particular, if we can find a local parametrization $f : U \to V$ such that $M \setminus V$ is a zero set, we have:

$$\int_M u(x)\, d\omega_M(x) = \int_{f^{-1}(M)} u(f(x)) \cdot \left|\det Df(x)\right| d\lambda^n(x)$$

Theorem 45.5.13: Onion Formula

Integrating a function f over $\mathbb{R}^n$ is the same thing as integrating over all surface integrals of the function over all balls B_r^n, i.e. we have:

$$\int_{\mathbb{R}^n} f(x)\, d\lambda^n(x) = \int_0^\infty \left(\int_{\partial B_r^n} f(y)\, d\omega_{\partial B_r^n}(y)\right) dr$$

Note that via multivariable substitution we also get:

$$\int_0^\infty \left(\int_{\partial B_r^n} f(y)\, d\omega_{\partial B_r^n}(y)\right) dr = \int_0^\infty \left(\int_{\partial B_1^n} r^n f(ry)\, d\omega_{\partial B_1^n}(y)\right) dr$$

Corollary 45.5.14. We have:

$$\omega_{\partial B_1^n}(\partial B_1^n) = (n+1) \cdot \lambda^{n+1}(B_1^{n+1}),$$

i.e. the surface area of the n-dimensional unit ball is $(n+1)$ times the volume of the $n+1$-dimensional unit ball.

Proof. Applying the onion formula and then Fubini twice, we get:

$$\begin{aligned}
\lambda^{n+1}(B_1^{n+1}) &= \int_{\mathbb{R}^n} \mathbb{1}_{B_1^n}(x)\, d\lambda^n(x) \\
&= \int_0^\infty \int_{\partial B_r^n} \mathbb{1}_{B_1^n}(y)\, d\omega_{\partial B_r^n}(y)\, dr \\
&= \int_{\partial B_r^n} \int_0^\infty \mathbb{1}_{B_1^n}(y)\, dr\, d\omega_{\partial B_r^n}(y) \\
&= \int_{\partial B_r^n} \int_0^1 1\, dr\, d\omega_{\partial B_r^n}(y) \\
&= \int_0^1 \int_{\partial B_r^n} 1\, d\omega_{\partial B_r^n}(y)\, dr \\
&= \int_0^1 \omega_{\partial B_r^n}(\partial B_r^n)\, dr \\
&= \int_0^1 r^n \cdot \omega_{\partial B_1^n}(\partial B_1^n)\, dr \\
&= \frac{\omega_{\partial B_1^n}(\partial B_1^n)}{n+1}
\end{aligned}$$

□

45.6 The Divergence Theorem

Definition 45.6.1. Let $\Omega \in \mathbb{R}^n$. We say Ω has a C^1 boundary if $\partial\Omega$ is an $n-1$ dimensional C^1-manifold such that Ω locally lies on only one side of $\partial\Omega$.

More precisely, Ω has a C^1 boundary iff for every point $p \in \partial\Omega$, there exists an open set $U \subset \mathbb{R}^n$, an open interval $I \subset \mathbb{R}$, an a C^1-function $u : U \to I$ such $p \in U \times I$ and

$$\Omega \cap (U \times I) = \left\{(x, y) \in U \times I : y < u(x)\right\}$$

Lemma 45.6.2. Let $\Omega \in \mathbb{R}^n$ be open with C^1 boundary. Then we have:

1. $\partial\Omega \cap (U \times I) = \left\{(x, y) \in U \times I : y = u(x)\right\}$
2. $(\mathbb{R}^n \setminus \overline{\Omega}) \cap (U \times I) = \left\{(x, y) \in U \times I : y \geq u(x)\right\}$

In particular, $\partial\Omega$ is an $n-1$ dimensional manifold given via the graph criterion as $(x, u(x))$.

Theorem 45.6.3: Outer Normal Field

Let $\Omega \subset \mathbb{R}^n$ be open with C^1 boundary. Then there exists exactly one map $\nu_\Omega : \partial\Omega \to \mathbb{R}^n$ such that:

1. $\nu_\Omega(p) \perp T_p(\partial\Omega)$,
2. $\left\|\nu_\Omega(p)\right\|_2 = 1$,
3. $\exists \varepsilon > 0 : t \in (0, \varepsilon) \implies p + t\nu_\Omega(p) \notin \Omega$.

This map is continuous and unique and is known as the *outer normal field of* Ω. If u is given as in the definition of C^1 boundary, and $p = (x, u(x))$, then we have:

$$\nu_\Omega(p) = \frac{(-Du(x), 1)^\top}{\sqrt{1 + \|Du(x)\|_2^2}}$$

Lemma 45.6.4: Partition of Unity

Let $W_i, i \in I$ be an open cover of a compact set $K \subset \mathbb{R}^n$. Then there exists a finite family of smooth functions $\chi_j : \mathbb{R}^n \to \mathbb{R}^n$ with compact support such that

1. For all $x \in K$, we have $\sum_{j \in J} \chi_j(x) = 1$
2. For all $j \in J$, there exists an i such that $\mathrm{spt}\chi_j \subset W_i$.

Theorem 45.6.5: Divergence Theorem / Gauss's Theorem

Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with C^1 boundary and outer normal $\nu_\Omega : \partial\Omega \to \mathbb{R}^n$. Let $f : \overline{\Omega}\mathbb{R}$ be continuously differentiable. Then we have:

$$\int_\Omega \operatorname{div} f(x)\, d\lambda^n(x) = \int_{\partial\Omega} \left\langle f(x), \nu(x) \right\rangle \, d\omega(x).$$

Theorem 45.6.6: Multidimensional Partial Integration

Let $\Omega \subset \mathbb{R}^n$ be a bounded open set with C^1 boundary and outer normal $\nu_\Omega : \partial\Omega \to \mathbb{R}^n$. Let $f : \mathbb{R}^n \to \mathbb{R}$ and $G : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable. Then we have:

$$\begin{aligned} &\int_\Omega f \cdot \operatorname{div}(G)\, d\lambda^n(x) \\ &= \int_{\partial\Omega} f \cdot \langle G, \nu_\Omega \rangle \, d\omega(x) - \int_\Omega \left\langle \vec{\nabla} f, G \right\rangle \, d\lambda^n(x) \end{aligned}$$

Proof. By the product rule for divergence, we have:

$$\begin{aligned} &f(x) \cdot \operatorname{div}(G(x)) + \left\langle \vec{\nabla} f(x), G(x) \right\rangle \\ &= \operatorname{div}(f(x) \cdot G(x)) \end{aligned}$$

Now, simply apply the divergence theorem, together with basic properties of scalar products, to get:

$$\begin{aligned} &\int_\Omega f(x) \cdot \operatorname{div}(G(x)) + \left\langle \vec{\nabla} f(x), G(x) \right\rangle \, d\lambda^n(x) \\ &= \int_\Omega \operatorname{div}(f(x) \cdot G(x))\, d\lambda^n(x) \\ &= \int_{\partial\Omega} \left\langle f(x) \cdot G(x), \nu_\Omega(x) \right\rangle \, d\omega(x) \\ &= \int_{\partial\Omega} f(x) \cdot \langle G(x), \nu_\Omega(x) \rangle \, d\omega(x), \end{aligned}$$

which we can rearrange to get our desired identity. □

If we plug $G(x) := \vec{\nabla} g(x)$ into this formula, we get:

Corollary 45.6.7: Green's First Identity

Let $\Omega \subset \mathbb{R}^n$ be open and bounded with C^1 boundary. Let $f : \overline{\Omega} \to \mathbb{R}$ be continuously differentiable and let $g : \overline{\Omega} \to \mathbb{R}$ be twice continuously differentiable. Then we have:

$$\begin{aligned}
&\int_\Omega f \cdot \Delta g + \left\langle \vec{\nabla} f, \vec{\nabla} g \right\rangle \, d\lambda^n \\
&= \int_{\partial\Omega} f \cdot \left\langle \nu_\Omega, \vec{\nabla} g \right\rangle \, d\omega \\
&\left(= \int_{\partial\Omega} f \cdot \frac{\partial}{\partial \nu_\Omega} g \, d\omega(x) \right)
\end{aligned}$$

Theorem 45.6.8: Green's Second Identity

Let $\Omega \subset \mathbb{R}^n$ be open and bounded with C^1 boundary. Let $f : \overline{\Omega} \to \mathbb{R}$ and $g : \overline{\Omega} \to \mathbb{R}$ be twice continuously differentiable. Then we have:

$$\begin{aligned}
&\int_\Omega f \cdot \Delta g - g \cdot \Delta f \, d\lambda^n \\
&= \int_{\partial\Omega} f \cdot \left\langle \nu_\Omega, \vec{\nabla} g \right\rangle - g(x) \cdot \left\langle \nu_\Omega(x), \vec{\nabla} f(x) \right\rangle \, d\omega(x)
\end{aligned}$$

Proof. This is just two instances of Green's first identity subtracted from each other:

$$\begin{aligned}
&\int_\Omega f(x) \cdot \Delta g(x) - g(x) \cdot \Delta f(x) \, d\lambda^n \\
&= \int_\Omega f(x) \cdot \Delta g(x) + \left\langle \vec{\nabla} f(x), \vec{\nabla} g(x) \right\rangle \, d\lambda^n \\
&- \int_\Omega g(x) \cdot \Delta f(x) + \left\langle \vec{\nabla} f(x), \vec{\nabla} g(x) \right\rangle \, d\lambda^n \\
&= \int_{\partial\Omega} f(x) \cdot \left\langle \nu_\Omega(x), \vec{\nabla} g(x) \right\rangle \, d\omega(x) \\
&- \int_{\partial\Omega} g(x) \cdot \left\langle \nu_\Omega(x), \vec{\nabla} f(x) \right\rangle \, d\omega(x) \\
&= \int_{\partial\Omega} f(x) \cdot \left\langle \nu_\Omega(x), \vec{\nabla} g(x) \right\rangle - g(x) \cdot \left\langle \nu_\Omega(x), \vec{\nabla} f(x) \right\rangle \, d\omega(x)
\end{aligned}$$

□

Chapter 46

Convolutions

Convolution is a way of creating a function by taking a weighted average of two functions. Among other things, it can be used to smooth a given function.

Lemma 46.0.1. $\tau_h : \mathbb{R}^n \to \mathbb{R}^n$ be the function that translates a vector by the vector h, i.e. $\tau_h(x) = x + h$. Let $f \in L^p(\mathbb{R}^n)$ for $1 \leq p < \infty$. Then the following hold:

1. $f \circ \tau_h \in L^p(\mathbb{R}^n)$
2. $\left\|f \circ \tau_h\right\|_p = \left\|f\right\|_p$
3. $\lim_{h \to 0} \left\|f \circ \tau_h - f\right\|_p = 0$

Definition 46.0.2: Convolution

Let $1 \leq p < \infty$, $f \in L^p(\mathbb{R}^n)$ and $g \in L^1(\mathbb{R}^n$. The *convolution of f and g* is defined to be:

$$f * g : \mathbb{R}^n \to \overline{\mathbb{R}}$$
$$(f * g)(x) = \int_{\mathbb{R}^n} f(x - y) g(y) \, d\lambda^n(y)$$

Theorem 46.0.3. Let $1 \leq p < \infty$, $f \in L^p(\mathbb{R}^n)$ and $g \in L^1(\mathbb{R}^n$. Then we have:

1. $f * g \in L^p(\mathbb{R}^n)$
2. $\left\|f * g\right\|_p \leq \left\|f\right\|_p \left\|g\right\|_1$
3. $f * g = g * f$

Theorem 46.0.4. (Approximation through convolution): Let $\eta \in L^1(\mathbb{R}^n)$ such that

$$\int_{\mathbb{R}^n} \eta(x)\, d\lambda^n(x).$$

Let $\rho > 0$ and

$$\eta_\rho(x) := \rho^{-n} \cdot \eta\left(\frac{x}{\rho}\right).$$

Let $1 \leq p < \infty$ and $f \in L^p(\mathbb{R}^n)$. Then we have:

1. $f * \eta_\rho \in L^p(\mathbb{R}^n)$
2. $\|f * \eta_\rho\|_p \leq \|f\|_p \cdot \|\eta\|_1$
3. $\|(f * \eta_\rho) - f\|_p \to 0$

Definition 46.0.5: Multi-index Notation

Let $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}_0^n$. Then we define:

1. $|\alpha| = \sum_{i=1}^{n} \alpha_i$
2. $\partial^\alpha f(x) = \frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}} \ldots \frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}} f(x)$

Theorem 46.0.6: Smoothing

Let $\eta \in C^k(\mathbb{R}^n)$ such that for $|\alpha| \leq k$, we have

$$\|\partial^\alpha \eta\|_{C^0(\mathbb{R}^n)} \leq K.$$

Let $f \in L^1(\mathbb{R}^n)$. Then we have:

1. $f * \eta \in C^k(\mathbb{R}^n)$,
2. $\partial^\alpha (f * \eta) = f * (\partial^\alpha \eta)$,
3. $\|\partial^\alpha (f * \eta)\|_{C^0(\mathbb{R}^n)} \leq K \|f\|_1$

Smoothing lets us strengthen our result on the density of continous functions in $L^p(\Omega)$:

Corollary 46.0.7: Density of C_0^∞ in $L^p(\Omega)$

Let $\Omega \subset \mathbb{R}^n$ be open. Let $1 \leq p < \infty$. Then, for every $f \in L^p(\Omega)$, there exists a sequence $f_k \in C_0^\infty(\Omega)$ such that $\|f - f_k\|_{L^p(\Omega)} \to 0$.

Definition 46.0.8: Locally integrable Function

Let $\Omega \subset \mathbb{R}^n$ open and $1 \leq p \leq \infty$. Then we call a function $f : \Omega \to \mathbb{R}$ *locally integrable* iff for any compact set $K \subset \Omega$, we have $\mathbb{1}_K \cdot f \in L^p(\Omega)$. We denote the set of all locally integrable functions $f : \Omega \to \mathbb{R}$ by

$$L^p_{\text{loc}}(\Omega)$$

Lemma 46.0.9: Fundamental Lemma of the Calculus of Variations

Let $\Omega \subset \mathbb{R}^n$ be open. Let $f \in L^1_{\text{loc}}(\Omega)$ such that

$$\begin{aligned} &\forall \varphi \in C_0^\infty(\Omega) : \\ &\quad \varphi \geq 0 \implies \int_\Omega f\varphi \, d\lambda^n(x) \geq 0 \end{aligned}$$

Then we have $f(x) \geq 0$ for λ^n-almost all $x \in \Omega$.

Corollary 46.0.10: There is no identity of convolution.

There exists no function $\eta \in L^1(\mathbb{R}^n)$ such that for all $f \in C_0^\infty(\mathbb{R}^n)$, we have $f * n = f$.